A zero-free half-plane from finite ordinary source moments

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For every modulus $q \geq 1$ and every presented Dirichlet character χ mod q, the theorem is

$$\Re s > \theta = \frac{8747729897}{10000000000}, \qquad \neg(\chi = \mathbf{1} \text{ and } s = 1) \quad \Longrightarrow \quad L(s,\chi) \neq 0.$$

Thus $c = 1 - \theta = 1252270103/10000000000$. The boundary is strict. The argument first treats finite-order Hecke characters over $\mathbb{Q}(\sqrt{-3})$, then passes to every ordinary presented character, including the zeta function with its principal pole excluded.

Let β be the full finite-order Hecke zero supremum defined in Section 9, with principal poles excluded. If $\beta \leq 17/20$, both endpoint conclusions below follow immediately. In the remaining case $\beta > 17/20$, the lower-bound hypotheses of the moment and buffered-zero arguments hold throughout.

The proof uses one finite dependency chain. The accepted two-level affine theorem at submitted revision `0db497edc397d7076b1652b060e49c76b683e901` supplies the Hecke bound $174963413/200000000$. We weaken it to $87481891/100000000$ and hence may use the natural cost $K_0 = 4.49782692$. The first 64 derived global source faces and their complete return give $\beta \leq 8747759/10000000$. Only then do we use $K = 6(2\beta - 1) \leq 11243277/2500000 = 4.4973108$ to prove the selected finite feedback graph. Its two final ordinary moments enter the final same-zero count and physical return. No source premise uses the endpoint it is intended to prove.

The arithmetic coefficients, natural zeros, common punctures, six units, root multiplicities, independent heights and common profiles are retained throughout. The following foundational arguments make these conventions and the actual transforms explicit. The public catalogue QRH-Catalogue is a directory for finding and contributing reusable QRH mathematics and formal components; its source and evidence pointers specify the scope of each reused result.

1 Arithmetic, reflection and polynomial moments

Lattice Poisson summation transforms a sextic row into a finite Gauss coefficient; cubic-theta automorphy reflects its completed cubic coefficient. After fixing phases and zeros, we prove completed energy, inverse and mixed plain moments, and the probe's exact Poisson factorization, without assuming a zero-free boundary.

These specialized arguments reconstruct Hailey Collet's *A zero-free half-plane from a single compensating prime*, revision `0b97bfdae9f026d9446d077952b2b4980700e742`, arithmetic conventions and Appendices A–D (with the source's attribution to OpenAI's GPT-6 Pro). The probe and moment framework originates in OpenAI's manuscripts of 30 September and 5 October 2026; their terminal zero-free claims are not inputs.

1.1 Classical inputs and their scope Classical sieve inputs are Goldmakher–Louvel, *A quadratic large sieve inequality over number fields*, Math. Proc. Cambridge Philos. Soc. 154 (2013), 193–212, arXiv:1112.1642v2, Theorem 1.1 and Corollary 1.2, specialized and verified in (27), and Heath-Brown, *Kummer's conjecture for cubic Gauss sums*, Israel J. Math. 120 (2000), 97–124, Theorem 2, DOI:10.1007/s11856-000-1273-y, specialized in (36). Both require squarefree **ideals**, not squarefree rational norms. The quadratic sieve requires a fixed ray modulus and row-independent coefficients.

The cubic theta function, Kubota multiplier and cusp expansions are the unconditional formulas in Dunn–Radziwiłł, *Bias in cubic Gauss sums: Patterson's conjecture*, Ann. of Math. 200 (2024), 967–1057, arXiv:2109.07463v3, ğ5.1, (5.1)–(5.17) and cusp tables. Their needed coefficients, three cusp sequences and archimedean factors appear in the reflection proof. We use neither its conditional prime asymptotics nor conditional improved sieve.

For a primitive finite-order Hecke character χ of $K = \mathbf{Q}(\sqrt{-3})$, of conductor norm Q, Hecke's theorem continues $L_K(s,\chi)$ (entire unless principal) and gives

$$\Lambda(s,\chi) = C_\chi^{s/2}\Gamma(s)L_K(s,\chi), \qquad \Lambda(s,\chi) = \epsilon_\chi \Lambda(1-s,\overline{\chi}), \qquad C_\chi = \frac{3Q}{(2\pi)^2}, \quad |\epsilon_\chi| = 1. \tag{1}$$

For principal χ, Λ has only simple poles at $0, 1$. Hecke's theorem also gives finite order, polynomial growth on closed vertical strips after removing poles, and absolute Euler convergence for $\Re s > 1$.

The normalization is Goldmakher–Louvel, ğ3, (3.1)–(3.2), with $(r_1, r_2, |d_K|) = (0, 1, 3)$ and conductor norm absorbed into C_χ. Other classical inputs are cubic reciprocity and supplementary laws, finite-field Gauss–Jacobi identities, Fourier inversion, self-dual Poisson summation, Mellin inversion, Stirling and Phragmén–Lindelöf. Their uses, including moving masks, are explicit below and assume no sought zero-free boundary.

Additional classical inputs used below. The fixed-ray prime ideal theorem is used with the field and ray modulus fixed before the asymptotic scale. For every integral ray modulus $\mathfrak{m}$ of K, every class C of its finite ray-class group, and every fixed finite excluded prime set S,

$$\#\{\mathfrak{p} \notin S : (\mathfrak{p}, \mathfrak{m}) = 1, [\mathfrak{p}] = C,\ N\mathfrak{p} \le X\} \sim \mathrm{Li}(X)/h_{\mathfrak{m}}.$$

A fixed union B of classes has density $|B|/h_{\mathfrak{m}}$. No uniformity in a moving modulus is asserted. For fixed nonnegative, nonzero $w \in C_c^\infty((1,2))$, partial summation consequently gives

$$\sum_{\mathfrak{p}\in B} w(N\mathfrak{p}/P)(N\mathfrak{p})^{-5/6} \sim \frac{|B|}{h_{\mathfrak{m}}} \frac{P^{1/6}}{\log P} \int_0^\infty w(t) t^{-5/6}\, dt.$$

Indeed Stieltjes partial summation against the stated counting asymptotic, followed by $x = Pt$, gives this formula uniformly on the fixed compact support of w. It supplies the fixed-ray amplifier pool in ğ7 and the nonzero principal normalizer in ğ14.3; the latter's all-good-prime pool has density one. Finite excluded primes do not change the asymptotic.

We also use classical Dirichlet nonvanishing on the line $\Re s = 1$: for every modulus $q \ge 1$ and Dirichlet character $\chi \bmod q$, $L(s,\chi) \ne 0$ when $\Re s = 1$, excluding the principal pole at $s = 1$. Together with absolute Euler convergence for $\Re s > 1$, this is the boundary input in ğ14.13. An imprimitive character's deleted factors $1-\chi^*(p)p^{-s}$ are nonzero on $\Re s \ge 1$ because $|\chi^*(p)p^{-s}| \le p^{-1} < 1$. Boundary nonvanishing is a separate classical theorem; it is not deduced here from the functional equation.

1.2 Arithmetic conventions Put $\mathcal{O} = \mathbf{Z}[\omega]$, $\omega = e^{2\pi i/3}$, $\lambda = 1 + 2\omega$, $q_a = Na$. This norm-Euclidean ring has six units; every ideal prime to 3 has a unique primary generator $\equiv 1 \pmod{3\mathcal{O}}$. Ideal sums count it once; element sums retain units. Ideal and lattice counts through norm $X \ge 1$ are $O(X)$.

For each target character fix, before varying scales or rows, a finite S containing conductor primes, primes above 6, and a small-prime cutoff. Columns avoid S unless specified; primes outside S are good. Superscript S deletes these Euler factors, a star means squarefree ideals, and $\sum^{(6)}$ means element rows with valuations $0, \dots, 5$.

At a good prime $\mathfrak{p}$, $\chi_{\mathfrak{p}}(v)$ is the sextic residue symbol, the sixth root of unity $\equiv v^{(q_{\mathfrak{p}}-1)/6}$ at units, extended multiplicatively in the denominator. **All integer character powers vanish at nonunits**, including zero and negative powers; only an absent modulus-one factor is identically one. Write

$$\psi_u(n) = \chi_n(u), \qquad \alpha(a) = a/|a|, \qquad e(z) = \exp(2\pi i \operatorname{Tr}(z/\lambda)), \qquad d\mu(z) = \frac{2}{\sqrt{3}}\, dx\, dy. \tag{2}$$

Here $\operatorname{Tr} z = z + \bar{z}$; $e(zy)$ is self-dual on $\mathcal{O}$, with Fourier kernel $e(-zy)$. For a zero-extended character χ, put

$$g_\chi(c,h) = \sum_{d \bmod c} \chi(d) e(hd/c), \qquad \gamma_j(c) = q_c^{-1/2} g_{\chi_c^j}(c,1) \quad (c \text{ squarefree}). \tag{3}$$

The inverse-character sum γ_{-1} does not mean conjugation. For odd $a + b\omega$, define

$$\Gamma_{\mathrm{q}}(a+b\omega) = \tfrac{1}{2}(1 + i^{-b} + i^a + i^{b-a}), \quad G(c) = \overline{\chi_c(4)}\Gamma_{\mathrm{q}}(c), \quad R(v,w) = \frac{\Gamma_{\mathrm{q}}(vw)}{\Gamma_{\mathrm{q}}(v)\Gamma_{\mathrm{q}}(w)}. \tag{4}$$

Lemma 2.1 proves that G is a finite-ray unit phase, R a symmetric sign-valued bicharacter, and $\chi_b(a) = R(a,b)\chi_a(b)$ for coprime primary indices. Expand finite-ray functions into genuine characters before using multiplicativity.

The ideal character ψ_u is the arithmetic-Frobenius Kummer character of $K(u^{1/6})/K$, retaining even redundant zeros at primes of u. At a good prime with valuation j, tame inertia has exponent $\pm j \bmod 6$ and conductor exponent one unless $6 \mid j$. At S, Hensel makes deep principal units sixth powers. Thus fixed finite-ray twists of $\psi_u^{\pm 1}$ have conductor norm $O_S(q_u)$; every nonunit sixth-power-free row prime to S is nonprincipal.

1.3 Smooth weights and uniformity An annular weight is smooth and compactly supported in $(0,\infty)$, with $\widehat{W}(s)=\int_0^\infty W(y)y^s\,dy/y$. Constants may depend on fixed characters, supports and positive losses, never moving arithmetic labels. Real exponents lie in fixed bounded sets. All estimates use finitely many smooth seminorms; multiplying a test by y^{b+it}, for bounded b, costs $(1+|t|)^{O(1)}$.

Profiles are common; arithmetic coefficients are row-independent. Integration by parts gives a compact joint logarithmic profile a Fourier expansion with finite weighted L^1 norm. Minkowski and Cauchy reduce its row norm to separated unit-modulus norm powers. Thus if $C_j(W)=\sum_n c_{j,n}W(x_{j,n})$ satisfies $\sum_j|C_j(W)|^2\le M\|W\|^2_{C^m(I')}$ for every common test in I', then kernels supported in fixed $I^2\Subset(I')^2$ satisfy

$$\left|\sum_j b_j\sum_{n_1,n_2}c_{j,n_1}\overline{c_{j,n_2}}K_j(x_{j,n_1},x_{j,n_2})\right|\ll M\sup_j\|K_j\|_{C^{2m+4}},\qquad |b_j|\le 1. \tag{5}$$

Fourier decay $(1+|t_1|)^{-m-2}(1+|t_2|)^{-m-2}$ and the common-test bounds leave integrable powers -2. Sobolev on unit intervals in log scale, real power and height similarly permits rowwise smooth parameters after summing common-profile bounds. Choose all internal smooth orders before the external height cutoff.

2 Finite arithmetic for both transforms

2.1 Gauss phases **Lemma 2.1** (Finite Gauss phases). *For squarefree n, and coprime squarefree a,b, the conventions of Section 1.2 give*

$$\begin{gathered}\chi_b(a)=R(a,b)\chi_a(b),\qquad R(a,b)=R(b,a)\in\{\pm1\},\quad (a,b)=1,\\ \gamma_2(n)^3=\mu(n)\alpha(n),\qquad \gamma_1(n)\gamma_2(n)=\mu(n)\alpha(n)G(n),\\ \mu(n)\gamma_{-1}(n)=\chi_n(-1)G(n)^{-1}\overline{\alpha(n)}\gamma_2(n),\\ a_\nu(ab)=a_\nu(a)a_\nu(b)\chi_b(a)^4,\qquad a_\nu(n)=\overline{\alpha(n)}\gamma_2(n)\nu(n),\quad (a,b)=1.\end{gathered} \tag{6}$$

Proof. At a prime p, put $J(A,B)=\sum_x A(x)B(1-x)$ with zero extensions. Substitution $u=2x-1$ gives $\chi_p(4)J(\chi_p,\chi_p)=J(\chi_p,\chi_p^3)$; Gauss–Jacobi and $\gamma_2(p)\gamma_4(p)=1$ give

$$\gamma_1(p)\gamma_2(p)=\overline{\chi_p(4)}\gamma_3(p)\gamma_2(p)^3.$$

The Jacobi sum $J(\chi_p^2,\chi_p^2)$ has norm q_p and is divisible by p, by expanding $x^{(q_p-1)/3}(1-x)^{(q_p-1)/3}$ modulo p and summing powers. Its unit follows modulo three: $\prod_{x\ne0,1}x(1-x)=1$ implies $\sum j_x\equiv0\pmod3$ for character values ω^{j_x}, so the sum is $-1\pmod3$. Since $p\equiv1\pmod3$, it equals $-p$, giving $\gamma_2(p)^3=-\alpha(p)$.

For odd c, Gaussian-regularized Poisson summation gives

$$|c|^{-1}\sum_{x\bmod c}e(x^2/c)=\tfrac12\sum_{y\bmod 2\mathcal{O}}e(-cy^2/4)=\Gamma_{\mathrm{q}}(c).$$

For regulator $e^{-\pi\eta|z|^2}$, the transformed factor has $r_\eta=1+3q_c\eta^2/16$. Replacing r_η by one costs $O_c(1)$; divide by Gaussian mass $\asymp\eta^{-1}$ and let $\eta\downarrow0$. The square classes $1,-1,\lambda,-\lambda$ have values $1,1,i,-i$, whose quotient is the symmetric bicharacter

$$R((-1)^e\lambda^f,(-1)^g\lambda^h)=(-1)^{eh+fg+fh}.$$

Grouping by $y=x^2$ gives $\Gamma_{\mathrm{q}}(p)=\gamma_3(p)$; CRT and cubic reciprocity give sextic reciprocity with R. Also $\chi_c(4)=(-2/c)_3=(c/(-2))_3$ is a fixed character modulo two. Hence $G(c)=\overline{\chi_c(4)}\Gamma_{\mathrm{q}}(c)$ is a fixed ray phase, and

$$\begin{gathered}G(ab)=G(a)G(b)R(a,b),\qquad R(a,a)=\chi_a(-1),\\ G(ba^{-1})=\chi_a(-1)\overline{G(a)}G(b)R(a,b).\end{gathered} \tag{7}$$

CRT extends the prime identities to squarefree indices; $\gamma_2(ab)=\gamma_2(a)\gamma_2(b)\chi_a(b)^2\chi_b(a)^2$ gives the a_ν identity. Finally $\gamma_1(n)\gamma_{-1}(n)=\chi_n(-1)$ gives inverse conversion in (6). All equalities retain nonunit zeros. □

2.2 Finite Fourier correlations For primary moduli distinguish the normalized Gauss sum from G:

$$\mathfrak{g}(a,h) = q_a^{-1/2} \sum_{x \bmod a} \chi_a(x) e(hx/a), \qquad \mathfrak{g}(1,h) = 1.$$

For a good prime p, $P = q_p$, $j \geq 1$, lifting $x = x_0 + py$ gives

$$|\mathfrak{g}(p^j,h)| = P^{(j-1)/2} \mathbf{1}_{v_p(h)=j-1} \quad (6 \nmid j),$$

$$\mathfrak{g}(p^j,h) = P^{j/2} \mathbf{1}_{p^j|h} - P^{j/2-1} \mathbf{1}_{p^{j-1}|h} \quad (6 \mid j). \tag{8}$$

Here $v_p(0) = \infty$; a sixth-power character retains its coprimality mask.
Define the full correlation

$$F(u,v;j) = \sum_{\substack{x \bmod u,\ y \bmod v \\ vx - uy \equiv j \pmod{uv}}} \chi_u(x)\overline{\chi_v(y)}. \tag{9}$$

Additive orthogonality gives, with negative Fourier sign,

$$F(u,v;j) = \frac{1}{\sqrt{q_u q_v}} \sum_{h \bmod uv} \mathfrak{g}(u,h)\overline{\mathfrak{g}(v,h)} e(-jh/(uv)). \tag{10}$$

At $j = 0$, both nonzero residues are units, forcing $u \mid v$ and $v \mid u$. Hence

$$F(u,v;0) = \mathbf{1}_{u=v}\phi(u). \tag{11}$$

For complete common-support extraction, let

$$u = Da, \quad v = Eb, \quad (a,b) = 1, \quad (ab, DE) = 1,$$

where every prime of D also divides E, and conversely. Then

$$F(Da,Eb;j) = F(D,E;j)R(a,E)R(b,D)R(a,b)\chi_a(j)\overline{\chi_b(-j)}. \tag{12}$$

At primes of a, b, reduction gives $\chi_a(j)\overline{\chi_a(Eb)}$ and $\overline{\chi_b(-j)}\chi_b(Da)$. At common primes substitute $x' = bx$, $y' = ay$, leaving $\overline{\chi_D(b)}\chi_E(a)$. The three unit quotients give the three R factors. All inversions are at units; nonunit j gives the same zero on both sides.

From (9), at common multiplicities i, j the correlation vanishes unless $p^{\min(i,j)} \mid h$. For divided frequency k and $i = j$, its local value is

$$P^{i-1} \begin{cases} P-1, & p \mid k, \\ -1, & p \nmid k,\ 6 \nmid i, \\ P-2, & p \nmid k,\ 6 \mid i. \end{cases} \tag{13}$$

For $i > j$, it vanishes unless $6 \mid j$ and $p \nmid k$, in which case it is

$$P^{j-1}(P-1)\chi_p(k)^{i-j}. \tag{14}$$

For $j > i$, conjugate and negate the frequency. Indeed dividing the common power fixes one variable when multiplicities differ; equal multiplicities give a character sum over $\mathbb{F}_P^\times \setminus \{1\}$.

Artificial extension rule. (12) need not hold on noncoprime residual pairs. Extend its right side to all individually allowed a, b and first insert

$$\sum_{(a,b)=1} c(a,b)F(Da,Eb;j) = \sum_t \mu(t) \sum_{t \mid a,b} c(a,b)\widetilde{F}_{D,E}(a,b;j). \tag{15}$$

The complete Möbius sum removes artificial pairs before factorwise estimates. Its extra primes are outside genuine common support and inherit none of its valuation restrictions.

2.3 The completed coefficient and the two-transform plan For a fixed finite-ray character ν set

$$\mathcal{C}_V(N;k,\nu)=\sum_{c\ \mathrm{sf},b}\frac{\overline{\alpha(cb^3)}\gamma_2(c)\nu(cb^3)\chi_{cb^3}(k)}{\sqrt{q_c}q_b}V(q_cq_b^3/N). \tag{16}$$

The ideals c,b may share primes. Exact cube inversion removes the completing b-sum for the inverse moment. The plain moment instead closes a two-factor family with fixed displayed moving radical: its width includes that entire radical, even at exponent zero. Later marked coefficients must use the appropriate closure.

3 Cubic-theta reflection and completed energy

3.1 Cubic reflection For $k=u_0k_Sk_{\mathrm{good}}$, the unit, S-valuations modulo six and good row's ray class give finitely many characters $\nu(A)\chi_A(u_0k_S)R(A,k_{\mathrm{good}})$. Remaining dependence is in local good-prime exponents; the reflection's fixed modulus contains no moving prime.

Let ν_0 lie in a fixed finite set of ray characters with prescribed zeros. At finitely many good primes choose $j_{\mathfrak{p}}\in\{0,\dots,5\}$ and set $\Psi(A)=\nu_0(A)\prod_{\mathfrak{p}}\chi_{\mathfrak{p}}(A)^{j_{\mathfrak{p}}}$. Define

$$\mathcal{C}_V(X;\Psi)=\sum_{n\ \mathrm{sf},b}\frac{\gamma_2(n)\overline{\alpha(nb^3)}\Psi(nb^3)}{\sqrt{q_n}\,q_b}V(q_nq_b^3/X). \tag{17}$$

Proposition 3.1 (Cubic reflection). *For annular V, or the Gaussian used in the probe, this sum is a divisor-bounded finite sum of branches*

$$\zeta\sum_{\mu\neq0}\frac{d(\mu)\alpha(\mu)\vartheta(\lambda^4\mu)}{\sqrt{q_\mu}}\prod_{\mathfrak{p}\in A}B_{\mathfrak{p},j_{\mathfrak{p}}}(\lambda^4\mu)V^\sharp\!\left(\frac{q_\mu X}{q_{c_F}^2q_r^2}\right),\quad r=\prod_{\mathfrak{p}\in A}\mathfrak{p}, \tag{18}$$

where c_F has finitely many fixed possibilities and the local factors are

$$B_{\mathfrak{p},j}(x)=\begin{cases}\chi_{\mathfrak{p}}(x)^{-j-2}, & j\notin\{0,4\},\\ q_{\mathfrak{p}}^{-1/2}(-1+q_{\mathfrak{p}}1_{\mathfrak{p}|x}), & j=4,\\ q_{\mathfrak{p}}^{-1/2}\chi_{\mathfrak{p}}(x)^{-2}, & j=0\text{ active.}\end{cases} \tag{19}$$

Every nonzero exponent is active. An exponent-zero prime additionally has an inactive branch, scalar $1-q_{\mathfrak{p}}^{-1}$, omitting it from r.

The branch scalar, needed when the prime varies, is

$$\zeta=-\frac{i}{81}\overline{\alpha(c)}^2\widehat{\phi}(h_0)\overline{\kappa_F}\prod_{\mathfrak{p}\in I}(1-q_{\mathfrak{p}}^{-1})\prod_{\mathfrak{p}\in A}\chi_{\mathfrak{p}}(\sigma_{\mathfrak{p}})^{-2}\omega_{\mathfrak{p},j_{\mathfrak{p}}},\qquad c=c_Fr, \tag{20}$$

Here I is the inactive set, $\widehat{\phi}$ the normalized fixed-modulus Fourier coefficient below, and

$$\sigma_{\mathfrak{p}}=\lambda^2c/\mathfrak{p},\qquad \epsilon_{\mathfrak{p}}=-\big((\lambda^3c/\mathfrak{p})\sigma_{\mathfrak{p}}\big)^{-1}\pmod{\mathfrak{p}},$$
$$\omega_{\mathfrak{p},j}=\begin{cases}\chi_{\mathfrak{p}}(-1)^j\gamma_j(\mathfrak{p})\gamma_{j+2}(\mathfrak{p})\chi_{\mathfrak{p}}(\epsilon_{\mathfrak{p}})^{-j-2}, & j\notin\{0,4\},\\ \gamma_4(\mathfrak{p}), & j=4,\\ -\gamma_2(\mathfrak{p})\chi_{\mathfrak{p}}(\epsilon_{\mathfrak{p}})^{-2}, & j=0.\end{cases}$$

The inverse is in the residue field at a unit. Each $|\omega_{\mathfrak{p},j}|=1$, hence $|\zeta|\le1/81$. The three fixed cusp sequences satisfy

$$\mu=u\lambda^mnb^3,\qquad m\ge-4,\qquad |d(u\lambda^mnb^3)|\le27\,3^{m/6}\sqrt{q_b}. \tag{21}$$

The dual primary ideals n,b avoid λ; n is squarefree. They may share primes and meet S, without the primal S-exclusion. Fixing the active radical modulo one fixed ray modulus makes d,ϑ,c_F independent of its moving primes; only a bounded scalar and the displayed local characters vary.

With $\widehat{V}(t)=\int_0^\infty V(x)x^t\,dx/x$, the exact transform is

$$V^\sharp(y)=\frac{1}{2\pi i}\int_{(0)}\widehat{V}(-t)\prod_{\pm}\frac{\Gamma(1+t\pm1/6)}{\Gamma(1-t\pm1/6)}\left(\frac{(2\pi)^4y}{27}\right)^{-t}dt. \tag{22}$$

For $V(y) = y^{1/2}W(y)$, with annular W, every fixed A, j has a finite J such that

$$|(y\partial_y)^j V^\sharp(y)| \ll_{A,j} \|W\|_{C^J} \min(y^{1/4}, y^{-A}). \tag{23}$$

The same bound holds for V_G, with a fixed constant in place of the annular seminorm.

Proof. Use the classical cubic-theta automorphy and cusp expansions of ğ1.1. Put $\check{e}(z) = e(\lambda z)$ and $L(t) = \left(\begin{smallmatrix} 1 & 0 \\ t & 1 \end{smallmatrix}\right)$. On $\Gamma_1(3) = \{g \in \mathrm{SL}_2(\mathcal{O}) : g \equiv I \pmod 3\}$, the theta multiplier is $\kappa(\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)) = (c/a)_3$ for $c \neq 0$, otherwise one; extend to $\mathrm{SL}_2(\mathbf{Z})\Gamma_1(3)$ with value one on $\mathrm{SL}_2(\mathbf{Z})$. The conjugate theta coefficient at $\lambda^{-3}nb^3$ is $3^{5/2}|b|\chi_n(\lambda)^{-2}\gamma_2(n)$, its constant term $3^{5/2}v^{2/3}/2$. At $I, L(\lambda), L(-\lambda)$ the expansions are

$$\sigma_H v^{2/3} + \sum_{\mu \neq 0} d_H(\mu) v K_{1/3}(4\pi|\mu|v)\check{e}(\mu z). \tag{24}$$

These cusp expansions give (21); their twisted reflection below uses no prime asymptotic.

Fix L containing primary and supplementary conditions and fixed conductors. Set $\phi(x) = \chi_x(\lambda)^2\nu_0(x)$ with prescribed primary and nonunit zeros, and $\widehat{\phi}(h_0) = q_L^{-1}\sum_{x \bmod L}\phi(x)e(-h_0x/L)$. Expand modulo L. At a moving prime,

$$q^{-1}\sum_{x \bmod \mathfrak{p}} \chi_{\mathfrak{p}}(x)^j e(-hx/\mathfrak{p}) = \begin{cases} q^{-1/2}\tau^-_{\mathfrak{p},j}\chi_{\mathfrak{p}}(h)^{-j}, & j \neq 0, h \neq 0, \\ 0, & j \neq 0, h = 0, \\ -q^{-1}, & j = 0, h \neq 0, \\ 1 - q^{-1}, & j = 0, h = 0, \end{cases} \tag{25}$$

Here $|\tau^-_{\mathfrak{p},j}| = 1$. An active branch translates by $z_h = \lambda^2(h_0/L + \sum_{\mathfrak{p}|r} h_{\mathfrak{p}}/\mathfrak{p})$, all $h_{\mathfrak{p}} \neq 0$.

Reduce $\lambda^2 h_0/L = a_F/c_F$, unit-normalized with $a_F \equiv 1 \pmod 3$ if $\lambda \mid c_F$, otherwise $c_F \equiv 1 \pmod 3$. Then

$$c = c_F r, \qquad a = a_F r + \lambda^2 c_F \sum_{\mathfrak{p}|r}(r/\mathfrak{p})h_{\mathfrak{p}}, \qquad z_h = a/c.$$

Fix $r \bmod M^2$, $M = \lambda^{12}L^4$, and lift $h_{\mathfrak{p}} \equiv 0 \pmod{M^2}$. Complete (a, c) to $g = \left(\begin{smallmatrix} a & b_0 \\ c & \delta \end{smallmatrix}\right)$ with $\delta \equiv a^{-1}$ to the required depth at primes of c_F, $\delta \equiv 0$ at primes of M outside c_F, and $\delta \equiv a^{-1} \pmod{\mathfrak{p}}$ for $\mathfrak{p} \mid r$. Choose

$$H = \begin{cases} I, & & 3 \mid c, \\ L(u_0), & u_0 \in \{\lambda, -\lambda\},\ u_0 \equiv c \pmod 3, & v_\lambda(c) = 1, \\ EL(-u_0), & u_0 \equiv a \pmod 3, & (c, \lambda) = 1, \end{cases} \qquad E = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

In the last case u_0 ranges over fixed representatives of $\mathcal{O}/3\mathcal{O}$, not merely $\{\pm\lambda\}$. These congruences give $gH^{-1} \equiv I \pmod 3$; trivial multipliers on $\mathrm{SL}_2(\mathbf{Z})$ and translations by $\mathbf{Z} + 3\mathcal{O}$ reduce to the same three cusps.

Multiplier reciprocity gives $\kappa(gH^{-1}) = \kappa_F\prod_{\mathfrak{p}|r}\chi_{\mathfrak{p}}(a)^2$. The fixed-sector factors are respectively $(c_F/a)_3$, $(-u_0/(a - u_0b_0))_3((c_F/u_0)/a)_3$, $(a/c_F)_3$, independent of the residual column.

The coordinate formula

$$g^{-1}(a/c + z, v) = \left(-\frac{\delta}{c} - \frac{\bar{z}}{c^2(|z|^2 + v^2)}, \frac{v}{q_c(|z|^2 + v^2)}\right) \tag{26}$$

shows that $\partial_{\bar{z}}$ at zero kills the constant term and equals $-(cv)^{-2}\partial_{z'}$. For $x = \lambda^4\mu$, $f = \lambda^3 c_F$, additive CRT gives

$$\check{e}(-\delta\mu/c) = \vartheta(x)\prod_{\mathfrak{p}|r} e(\epsilon_{\mathfrak{p}}h_{\mathfrak{p}}^{-1}x/\mathfrak{p}), \quad \vartheta(x) = e(-\delta_F r^{-1}x/f), \quad \epsilon_{\mathfrak{p}} \equiv -(\lambda^5 c_F^2(r/\mathfrak{p})^2)^{-1} \pmod{\mathfrak{p}}.$$

The conjugate multiplier contributes $\chi_{\mathfrak{p}}(h_{\mathfrak{p}})^{-2}$; substituting $y = h_{\mathfrak{p}}^{-1}$ gives exponent $j+2$ and exactly the Gauss/Ramanujan factors (19), with their zeros. Their normalized scalars are $\omega_{\mathfrak{p},j}$ and (20), recording the entire active conductor's dependence.

To check the archimedean normalization, define

$$\mathcal{D}_\Psi(s) = \sum_{n\ \mathrm{sf},b} \gamma_2(n)\overline{\alpha(nb^3)}\Psi(nb^3)q_n^{-s}q_b^{1/2-3s}.$$

Multiply the conjugate theta μ-coefficient by $\phi(\lambda^3\mu)\prod_{\mathfrak{p}}\chi_{\mathfrak{p}}(\lambda^3\mu)^{j_{\mathfrak{p}}}$, obtaining Θ_Ψ. Differentiation supplies $2\pi i\overline{\mu}$, and

$$\int_0^\infty v^{2s}K_{1/3}(4\pi|\mu|v)\,dv = \frac{2^{2s-1}\Gamma(s+1/3)\Gamma(s+2/3)}{(4\pi|\mu|)^{2s+1}}.$$

Consequently, initially for $\Re s > 1$,

$$\int_0^\infty \partial_{\bar z}\Theta_\Psi(0,v)v^{2s-1}\,dv = \frac{3^{5/2}}{4}\left(\frac{27}{(2\pi)^2}\right)^s \Gamma(s+1/3)\Gamma(s+2/3)\mathcal{D}_\Psi(s).$$

Apply (26) and change height to $1/(q_cv)$, contributing $-c^{-2}q_c^{1-2t}$, with $|c^{-2}q_c| = 1$. Division by the direct Mellin factor and $t \mapsto -t$ give (22) and scalar $1/81$. Differentiated cusp expansions decay at both endpoints, so their Mellin integrals are entire. Absolute convergence on the direct/reflected half-planes, Stirling and Phragmén–Lindelöf give intervening polynomial vertical bounds. Rapid test decay permits shifting to $\Re t = -1/4$ without gamma poles, giving small-y decay; arbitrary positive lines give rapid large-y decay. This proves (18)–(23), including finite seminorm control. □

3.2 Unmarked energy, including a frozen label We use the quadratic large sieve (Goldmakher–Louvel, Theorem 1.1 and Corollary 1.2) in the form

$$\sum_{q_R\le K}^{*}\left|\sum_{q_n\le N}^{*} c_n\chi_R(n)^3\right|^2 \ll_\varepsilon (KN)^\varepsilon(K+N)\sum_n |c_n|^2. \tag{27}$$

Coefficients are R-independent. The primary quadratic symbol has primitive conductor $R\lambda^{e(R)}$, $e(R) = (q_R-1)/2 \pmod 2$; primary adjustment contributes that power of the character modulo λ. A primary square class modulo four fixes $e(R)$. Coprime indices in it have primitive product conductor their product and reciprocity factor $R(\cdot,\cdot)$, verifying the sieve's Hecke-family hypotheses. Finite ray partition and transpose duality give (27). Row restrictions decrease its positive sum; column restrictions are zero coefficients.

For a squarefree ideal f, set

$$\Psi_k(a) = \nu(a)\chi_a(k)\chi_a(f)^4,$$

and, with $V(y) = y^{1/2}W(y)$, define

$$T(X;k,f) = \sum_{n\ \mathrm{sf},b} \frac{\overline{\alpha(n)}\gamma_2(n)\Psi_k(n)\overline{\alpha(b)}^3\Psi_k(b)^3}{\sqrt{q_n}\,q_b}V(q_nq_b^3/X). \tag{28}$$

Proposition 3.2 (Completed energy with a frozen label). *For every $\varepsilon > 0$,*

$$\sum_{0<q_k\le H}|T(X;k,f)|^2 \ll (HXq_f)^\varepsilon\|W\|_{C^J}^2\left(H + \frac{H^2q_f}{X}\right), \qquad H, X \ge 1. \tag{29}$$

The constants and smooth order are uniform in f.

Proof. Allow an arbitrary ideal g in place of f and squarefree rows u_0s. Write $s = tk_0$, $t \mid gS$, $H_0 = H/q_t$, $(k_0, gS) = 1$. Residual exponent one turns (19) into $\chi_{\mathfrak{p}}(x)^3$. Frozen exponents are $j_{\mathfrak{p}} \equiv 1_{\mathfrak{p}\mid t} + 4v_{\mathfrak{p}}(g) \pmod 6$, including every collision; absorb fixed primes in the finite sector.

After fixing unit, λ-valuation and a finite ray class, a reflected branch is

$$S_m[A,Y](k_0) = 3^{-m/3}\sum_{n\ \mathrm{sf},b}\frac{A(n,b)\chi_{k_0}(nb)^3}{\sqrt{q_n}\,q_b}V^\sharp\left(\frac{3^mq_nq_b^3H_0^2}{Yq_{k_0}^2}\right), \tag{30}$$

times a bounded row scalar, with k_0-independent A, initially $|A| \le a = 1$, $Y \asymp H_0^2/X$. For $j = 4$ set $A_n = 1_{\mathfrak{p}\nmid n}A(\mathfrak{p}n,b)$, $A_b = 1_{\mathfrak{p}\nmid n}A(n,\mathfrak{p}b)$. Splitting $1_{\mathfrak{p}\mid nb^3} = 1_{\mathfrak{p}\mid n} + 1_{\mathfrak{p}\nmid n,\mathfrak{p}\mid b}$ gives, for $q = q_{\mathfrak{p}}$,

$$S_m[AB_{\mathfrak{p},4}, q^2Y] = -q^{-1/2}S_m[A, q^2Y] + \chi_{k_0}(\mathfrak{p})^3 S_m[A_n, qY] + q^{-1/2}\chi_{k_0}(\mathfrak{p})^3 S_m[A_b, Y/q]. \tag{31}$$

This preserves squarefreeness and shared primes. The branch updates of (a^2, Y) are (a^2, q^2Y) for $j \notin \{0,4\}$; $(a^2/q, q^2Y)$ for active $j = 0$; $(a^2/q, q^2Y), (a^2, qY), (a^2/q, Y/q)$ for $j = 4$. Inactive primes increase neither entry. Consequently,

$$a^2 \le 1, \qquad a^2Y \ll H_0^2 q_t^2 q_g / X. \tag{32}$$

A prime of t costs at most q^2; a valuation-one prime of g outside t has $j = 4$, costing q; higher valuations cost at most $q^2 \le q^{v_{\mathfrak{p}}(g)}$.

Dual stars impose squarefreeness without S-exclusion. Split the finitely many squarefree S-parts of n into bounded row factors before (27). Retain the actual row cutoff in ğ1.3's joint profile; on $q_n \asymp U, q_b \asymp B$, separate into

$$G_{b,\tau}(k_0) = \sideset{}{^*}\sum_{q_n \asymp U} \frac{A(n,b)}{\sqrt{q_n}} (q_n/U)^{i\tau} \chi_{k_0}(n)^3.$$

Their Fourier densities are $\ll (1 + 3^m UB^3/Y)^{-A}(1+|\tau|)^{-2}$. Cauchy and (27) give

$$\sideset{}{^*}\sum_{q_{k_0} \le H_0} |G_{b,\tau}(k_0)|^2 \ll a^2 (H_0 U)^{\varepsilon}(H_0 + U).$$

Weighted Cauchy–Schwarz in b, using $\sum_{q_b \asymp B} q_b^{-1} \ll 1$, now yields

$$\|S_{m;U,B}\|_2 \ll a\mathcal{D}^{\varepsilon} 3^{-m/3}(\sqrt{H_0} + \sqrt{U})(1 + 3^m UB^3/Y)^{-2},$$

For $\mathcal{D} \ge 2 + H + X + q_g$, use $\sum_{U,B}(1+UB^3/Z)^{-2} \ll \log^2(2+Z)$ and $\sum_{U,B} \sqrt{U}(1+UB^3/Z)^{-2} \ll \sqrt{Z}$, then sum $m \ge -4$, obtaining $\|\text{branch}\|_2^2 \ll a^2\mathcal{D}^{\varepsilon}(H_0 + Y)$. Equation (32) and divisor-many branches give

$$\sideset{}{^*}\sum_{q_s \le H} |T(X; u_0 s, g)|^2 \ll (HXq_g)^{\varepsilon} \|W\|_{C^J}^2 \left(H + H^2 q_g / X\right).$$

Finally write uniquely $k = u_0 s v^2$, s squarefree, allowing $(s, v) > 1$. Multiplicativity with zeros and $8 \equiv 2 \pmod 6$ give $T(X; u_0 s v^2, f) = T(X; u_0 s, f v^2)$. The last bound at $(H/q_v^2, g = fv^2)$ has main expression $q_v^{-2}(H + H^2 q_f / X)$. Sum $\sum_v q_v^{-2} < \infty$ and six units to obtain (29). □

Taking $f = 1$, with $V(y) = y^{1/2} W(y)$, gives

$$\sum_{0 < q_k \le H} |\mathcal{C}_V(N; k, \nu)|^2 \ll (HN)^{\varepsilon}(H + H^2/N)\|W\|_{C^J}^2. \tag{33}$$

A log-annular partition extends this to V_G, whose seminorms are summable against every fixed exponential; the row-norm triangle sum converges.

3.3 A quadratic–cubic norm for one prime Let k, n be squarefree, $q_k \ll K$, $q_n \asymp N$, $q_b \asymp B$ with arbitrary b. Rows avoid S; dual indices may meet $S \setminus \{\lambda\}$. For k-independent $\beta(n,b)$ write

$$b = gt^2, \quad c = (n,g), \quad n = cm, \quad g = ch, \qquad q_c \asymp C, \quad q_g \asymp G, \quad q_t \asymp T, \quad B \asymp GT^2.$$

Here g is squarefree, c, m, h are pairwise coprime squarefree, and g, t may share primes. Define the dyadic quadratic row norm

$$\mathcal{Q}_{C,G,T}(\beta) = \sum_{\substack{k \text{ sf},\ (k,S)=1 \\ q_k \ll K}} \left| \sum_{\substack{n \text{ sf},\ q_n \asymp N,\ q_b \asymp B \\ b = gt^2,\ g \text{ sf},\ q_g \asymp G,\ q_t \asymp T \\ c = (n,g),\ q_c \asymp C}} \beta(n,b)\chi_k(nb)^3 \right|^2.$$

All column restrictions are k-independent. Then

$$\mathcal{Q}_{C,G,T}(\beta) \ll (KNB)^{\varepsilon} T \sum_{q_t \asymp T} \sum_{\substack{r \text{ sf, } (r,S)=1 \\ q_r \ll \min(K,C)}} \left(\frac{K}{q_r} + \frac{NG}{C^2}\right) \frac{C}{q_r} \sum_{c',m,h} |\beta(rc'm, rc'ht^2)|^2. \tag{34}$$

The last sum retains the support, $c = rc'$. Fixed row and coefficient restrictions are allowed.

Proof. The zero-preserving identity is

$$\chi_k(nb)^3 = \chi_k(mh)^3 \mathbf{1}_{(k,ct)=1}.$$

Cauchy over $O(T)$ choices of t supplies T. Fixing t, keep $(k,t) = 1$ as a row restriction and expand $1_{(k,c)=1} = \sum_{r|(k,c)} \mu(r)$. Divisor Cauchy costs $(KNB)^{\varepsilon}$. With $k = rk'$, $c = rc'$, squarefreeness imposes $(k', r) = (c', r) = 1$; no (k', c')-mask is reinserted termwise.

Absorb $\chi_r(mh)^3$ into column coefficients; split finite squarefree S-parts of m, h into bounded row factors. Their residual product is squarefree and good, of norm $O(NG/C^2)$. Apply (27) at K/q_r, group by product at divisor cost, and use Cauchy over $O(C/q_r)$ choices of c'. This proves (34); restore S-parts inside its positive sum. Empty quotients give zero; in nonempty bounded ranges replace scales by $\max(1, \text{scale})$ at constant cost. □

For $|\beta| \le 1$, ideal counting and dyadic summation in (34) give

$$\sum_k \left| \sum_{n,b} \beta(n,b) \chi_k(nb)^3 \right|^2 \ll (KNB)^{\varepsilon}(K + NB)NB. \tag{35}$$

Indeed for $q_r \asymp r_0$, coefficient count times C/q_r is $O(NG/r_0^2)$; $O(r_0)$ choices of r and both T-factors give NB/r_0.

The cubic sieve (Heath-Brown, Theorem 2) for squarefree primary Eisenstein ideals and fixed coefficients is

$$\sum_{q_a \le U}^{*} \left| \sum_{q_b \le V}^{*} c_b \chi_a(b)^2 \right|^2 \ll (UV)^{\varepsilon} \{U + V + (UV)^{2/3}\} \sum_b |c_b|^2. \tag{36}$$

Rational norms need not be squarefree; reciprocity and conjugation allow either cubic orientation.

Lemma 3.3 (One-prime hybrid norm). *Let p run over good primes with $q_p \asymp L$. Suppose $|a(p)| \le 1$ is independent of k, n, b, and $|\beta(n,b)| \le 1$ is independent of k, p. With the other variables as above,*

$$\sum_k \left| \sum_p \frac{a(p)}{\sqrt{q_p}} \mathbf{1}_{p\nmid k} \sum_{n,b} \beta(n,b) \chi_k(nb)^3 \chi_p(n)^{-2} \mathbf{1}_{p\nmid b} \right|^2 \tag{37}$$
$$\ll (KNBL)^{\varepsilon}(K + NB)B\{N + L + (NL)^{2/3}\}.$$

Separate fixed restrictions on the row, prime and dual supports, including ray sectors, are allowed.

Proof. First omit $1_{p\nmid k}$. Insert the prime sum into (34)'s column coefficient. Its decomposition gives

$$\chi_p(rc'm)^{-2} \mathbf{1}_{p\nmid rc'ht^2} = \chi_p(r)^{-2} \chi_p(c'm)^{-2} \mathbf{1}_{p\nmid ht}.$$

The character retains zeros at r, c', m. Fix r, h, t and absorb $\chi_p(r)^{-2} 1_{p\nmid ht}$ into $a(p)$, whose central squared mass is $\sum_{q_p \asymp L} q_p^{-1} \ll 1$. In the positive sum discard $|\beta|^2 \le 1$, group by squarefree $j = c'm = n/r$ at divisor cost, and apply (36), giving

$$(KNBL)^{\varepsilon} \left\{ \frac{N}{q_r} + L + \left(\frac{NL}{q_r}\right)^{2/3} \right\}.$$

The $O(G/C)$ choices of h, both T-factors and C/q_r give B/q_r; the dyadic r-count cancels its denominator. Since $NG/C^2 \ll NB$, summing gives (37) without the mask.

For collisions put $k = pk'$. Divisor Cauchy and (35), fixing p and absorbing its characters into β, give

$$\ll (KNBL)^{\varepsilon} \sum_{q_p \asymp L} \frac{1}{q_p} \left(\frac{K}{q_p} + NB\right) NB \ll (KNBL)^{\varepsilon}(K/L + NB)NB.$$

This is bounded by (37), whose brace is $\ge N$. Triangle restores the mask. An absent prime uses (35), equivalent to $L = 1$. □

3.4 Whole-index compensation and the required energy Insert $1_{p|nb^3}$ in (17) for a selected good prime p. If p is already a base local prime, its original zero kills the sum. Otherwise write

$$\mathbf{1}_{p|nb^3} = 1 - \chi_p(nb^3)^0$$

and reflect both terms. Inactive coefficients combine to $1-(1-q_p^{-1}) = q_p^{-1}$, with p absent from the conductor. The active dual factor is

$$q_p^{-1/2}\chi_p(\lambda^4\mu)^{-2}, \qquad (38)$$

with scalar (20) replacing $\omega_{p,0}$ by $-\omega_{p,0}$. Identical fixed Fourier term and active radical give identical cusp data, so subtract before estimating, even when primal squarefree and cube indices share p.

For distinct active r, p, scalar (20) has pair factor

$$\chi_r(p)^{2j_r+2}\chi_p(r)^{2j_p+2} = (p/r)_3^{j_r+j_p+2}.$$

For a residual row prime $j_r = 1$ and marked prime $j_p = 0$, this equals one. Freeze other active primes, including every $j = 4$ prime. Separately fixing row product R and $P = p$ modulo ğ3.1's full M^2 fixes cusp and additive data without pair restrictions. The remaining scalar separates into bounded row and prime factors.

For dual index a fixed factor times nb^3, the moving characters are

$$\chi_R(nb^3)^3 = \chi_R(nb)^3, \qquad \chi_P(nb^3)^{-2} = \chi_P(n)^{-2}\mathbf{1}_{(P,b)=1}.$$

Primal row–prime collisions retain $1_{(P,R)=1}$; frozen collisions give separate row, prime or dual restrictions. Thus Lemma 3.3's two independence hypotheses hold. Logarithmic Fourier separation handles joint completed-norm/prime-norm weights with finite weighted seminorms.

For a bounded row-independent coefficient $a(\mathfrak{p})$, define

$$\mathcal{T}(m) = \sum_{q_\mathfrak{p} \asymp Z^{5/33}} a(\mathfrak{p}) \sum_{c \text{ sf},n} \frac{\overline{\alpha(cn^3)}\gamma_2(c)\nu(cn^3)\chi_{cn^3}(m)}{\sqrt{q_c}\, q_n} \mathbf{1}_{\mathfrak{p}|cn^3} V\left(\frac{q_c q_n^3}{Z q_\mathfrak{p}}\right). \qquad (39)$$

Including a common joint norm profile, the estimate is

$$\sum_{0<q_m \ll Z^{28/33}} |\mathcal{T}(m)|^2 \ll Z^{28/33+\varepsilon}. \qquad (40)$$

The prime stays inside the norm throughout the proof.

Proposition 3.4 (Whole-index one-prime energy). *The sum (39) satisfies (40).*

Proof. Locally put $M = 28/33$, $z = 5/33$, $N_* = 1+z = 38/33$. Freeze the row's maximal powerful part, S-supported valuation-one part and unit. Their norm dyad has exponent O, the remaining squarefree row exponent $H \le M - O + O(1/\log Z)$. Apply ğ3.1's finite sector reduction.

For active frozen good primes, let A_0 be radical exponent; S_0 count small Ramanujan and active zero-mask factors; N_0, B_0 count forced squarefree/cube factors. Splitting as in (31) leaves shared cube occurrences in b on the squarefree branch, or extracts one cube occurrence with factor $q_p^{-1/2}$. Powerful support gives

$$2A_0 \le O, \qquad 0 \le N_0 \le A_0, \qquad S_0, B_0 \ge 0.$$

Inactive marked primes have mass $\sum_{q_p \asymp Z^z} q_p^{-1} \ll Z^\varepsilon$; remove by triangle and set $z_a = 0$. Active primes stay inside the norm with $z_a = z$.

For residual dual squarefree, cube and ramified exponents v, b, e_λ, the central coefficient and dual length are

$$Z^{-v/2-b-e_\lambda/3-(S_0+B_0+z_a)/2}, \qquad T_d = 2H + 2A_0 + 2z_a - N_* - N_0 - 3B_0. \qquad (41)$$

The actual kernel argument is a fixed factor times $Z^{v+3b+e_\lambda-T_d} y_n y_b^3 y_R^{-2} y_P^{-2}$, each y normalized by its dyadic center, omitting y_P when inactive. By (23), extract $Z^{-(T_d-v-3b-e_\lambda)_+/4}$ from the **linear** vector before Fourier separation; the normalized profile has bounded seminorms. Then enlarge the positive row norm to $q_R \ll Z^H$.

Apply (37) at $(K,N,B,L)=(Z^H,Z^v,Z^b,Z^{z_a})$, or (35) when inactive. Its prime inverse square root is already included. Multiply the remaining squared coefficient and count powerful rows; with $u_h=\min\{v,z_a,(v+z_a)/3\}$ the exponent is

$$\begin{aligned}E = O/2+\max(H,v+b)-S_0-B_0+z_a-u_h-b-2e_\lambda/3\\ -\tfrac12(T_d-v-3b-e_\lambda)_+.\end{aligned}\tag{42}$$

Powerful ideals number $O(Z^{O/2})$: write x^2y^3, y squarefree, and sum $Z^{O/2}q_y^{-3/2}$. Fixed-supported parts and allocations have divisor-many choices. Their Cauchy cost is the single $O/2+\varepsilon$ already in (42).

Retain $v+3b+e_\lambda\le T_d$, suppressing tiny buffers. Since

$$T_d-H\le M+2z-N_*=0,$$

we have $\max(H,v+b)=H$. If $u_h=z_a$, then $E\le M$; otherwise $u_h\ge v/2$. Put $y_d=v+3b+e_\lambda$; then

$$u_h+b+2e_\lambda/3+\tfrac12(T_d-y_d)_+\ge y_d/4+\tfrac12(T_d-y_d)_+\ge T_d/4.$$

Substitution in (42), with $\Delta_H=M-O-H\ge0$, gives

$$4(E-M)\le N_*+2z_a-2M-2\Delta_H-(2A_0-N_0+B_0+4S_0)\le N_*+2z-2M=-\frac{8}{33}.\tag{43}$$

Hence all retained blocks have $E\le M$.

Discard whole tail dyads outside a small retention buffer before separation; their kernel argument exceeds a positive power of Z. For $|K(y)|\le C_Ay^{-A}$, $A>1$, lattice shells give

$$\sum_{\substack{0\ne\mu\in\lambda^{-4}\mathcal{O}\\ aq_\mu>U}}|K(aq_\mu)|\ll_A C_A(1+a^{-1})U^{1-A},\qquad U\ge1.\tag{44}$$

Indeed each dyadic shell has $O(1+2^jU/a)$ points. Raw coefficients, rows, primes and branches have fixed polynomial cost; choose A afterwards for an arbitrarily small tail. The lower bound $e_\lambda\ge-4\log_Z3$ and fixed support shifts cost $O(1/\log Z)$. Buffers, dyads and finite profile orders fit every requested loss, proving (40). □

4 The inverse second moment

The two Poisson transforms recover the essential Möbius coefficients through the cubic Gauss identities.

Theorem 4.1 (Inverse second moment). *Fix $C,\kappa,\varepsilon>0$. For $Z\ge2$, $1\le D,H\le Z^C$, and $H\ge DZ^\kappa$,*

$$\sum_{0<q_u\le H}\left|D^{-1/2}\sum_n\mu(n)\nu(n)\chi_n(u)^{\pm1}W(q_n/D)\right|^2\ll HZ^\varepsilon\|W\|_{C^J}^2.\tag{45}$$

The fixed ν includes its presentation's Euler deletions. The order depends only on C,κ,ε, support and fixed arithmetic data; test norm twists cost a fixed height power.

Proof. We prove it through the positive energy

$$E(H,X,F;\xi,W)=\frac{1}{XF}\sideset{}{^*}\sum_{F\le q_f<2F}\sum_{0<q_k\le H}\left|\sideset{}{^*}\sum_n a_\xi(n)\chi_n(k)\chi_n(f)^4W(q_n/X)\right|^2.\tag{46}$$

The label f is averaged. Write $E^{(r)}$ for an added inner mask $1_{(n,r)=1}$; T is (28).

4.1 Poisson, smooth separation, and punctures For a primitive character ψ modulo m, masked Poisson summation is

$$\sum_{k\in\mathcal{O}}\psi(k)1_{(k,r)=1}\Phi(q_k/H)=\frac{H\gamma(\psi)}{\sqrt{q_m}}\sum_{d\mid\mathrm{rad}\,r}\frac{\mu(d)\psi(d)}{q_d}\sum_h\overline{\psi(h)}\widehat{\Phi}\left(\frac{Hq_h}{q_dq_m}\right).\tag{47}$$

Here Φ is radial Schwartz and $\gamma(\psi)=q_m^{-1/2}\sum_{x\bmod m}\psi(x)e(x/m)$. Expand the mask as $\sum_{d\mid(r,k)}\mu(d)$, substitute $k=d\ell$, and Poisson-sum modulo m. The finite transform is $\sqrt{q_m}\gamma(\psi)\overline{\psi(h)}$, with nonunit zeros.

Primitive principal modulus one has value one at $h=0$, contributing $H\widehat{\Phi}(0)\prod_{\mathfrak{p}\mid r}(1-q_{\mathfrak{p}}^{-1})$. Shared primes of m,r vanish through $\psi(d)$.

Smooth quadratic kernels are separated by (5).

Möbius inversion and (6) remove a puncture without changing XF:

$$E^{(r)}(H,X,F;\xi,W)\le\tau(\operatorname{rad} r)\sum_{d\mid\operatorname{rad} r}E(H,X/q_d,Fq_d;\xi,W). \tag{48}$$

Set $n=dm$, changing f to df; $\chi_m(d)^4$ retains $(m,d)=1$, and $(d,f)>1$ vanishes. Cauchy in d and positive enlargement of the injective image $f\mapsto df$ give (48), preserving XF.

4.2 Canonical estimate and cube completion We claim that, for every fixed $C_0\ge 1,\kappa>0$,

$$E(H,X,F;\xi,W)\ll Z^{\varepsilon}\Sigma\|W\|_{C^J}^2,\quad \Sigma=XF\le Z^{C_0},\quad H,X,F\ge 1,\quad H\le\Sigma Z^{-\kappa}. \tag{49}$$

The derivative order is independent of the location of the fixed support interval.

For $\Psi_{k,f}(n)=\xi(n)\chi_n(k)\chi_n(f)^4$, cube inversion gives the exact identity

$$X^{-1/2}\sideset{}{^*}\sum_n a_\xi(n)\chi_n(k)\chi_n(f)^4W(q_n/X)=\sum_h\frac{\mu(h)\overline{\alpha(h)}^3\Psi_{k,f}(h)^3}{q_h}T(X/q_h^3;k,f). \tag{50}$$

Opening T, each total cube b has divisor factor $\sum_{h\mid b}\mu(h)=1_{b=1}$, by complete multiplicativity with zeros.

Put $H_c^3=\min(X,X^2/H^2)$, $L_b=X/q_b^3$. Weighted Cauchy, (29) and the f-average bound $q_h\le H_c$ by

$$Z^{\varepsilon}\|W\|_{C^J}^2\left(H+\frac{H^2FH_c^3}{X}\right)\ll Z^{\varepsilon}\Sigma\|W\|_{C^J}^2\quad(H\le\Sigma).$$

Reopen the remaining T terms. A total cube b has coefficient $\overline{\alpha(b)}^3\xi(b)^3 1_{(b,f)=1}\sum_{h\mid b,q_h>H_c}\mu(h)$, bounded by $\tau(b)$, and bounded row multiplier $\chi_b(k)^3$. A second weighted Cauchy uses $\sum_{q_b\le R}\tau(b)/q_b\ll\log^2(2R)$ to give

$$E(H,X,F;\xi,W)\ll Z^{\varepsilon}\left\{\Sigma\|W\|_{C^J}^2+\sup_{q_b>H_c,\ L_b>1}E(H,L_b,F;\xi,W)\right\}. \tag{51}$$

For $L_b\le 1$, nonempty annular support has boundedly many columns, costing $O(H)$. The supremum is empty when $H^2\le X$.

4.3 Two Poisson transformations and the local projector For $1\le H,L,F,\Sigma\le Z^{C_0}$ and $\max(H,LF)\le\Sigma$, we prove

$$E(H,L,F;\xi,W)\ll Z^{\varepsilon}\Sigma\|W\|_{C^{4m+12}}^2\left(1+\sup\frac{E(H',X',F';\xi',U)}{\Sigma'}\right), \tag{52}$$

where $\Sigma'=X'F'$, $H',X',F'\ge 1$, $\|U\|_{C^m}\le 1$ on a fixed enlargement of the support, and

$$H'\le\frac{HL}{\Sigma F},\qquad\frac{H'}{\Sigma'}\le\frac{H}{\Sigma},\qquad\Sigma'\le L. \tag{53}$$

Square a rescaled real Schwartz function of compact Fourier support to obtain radial $\Phi\ge 1$ on $[0,1]$, $\Phi\ge 0$, with $\operatorname{supp}\widehat{\Phi}\subset[0,C_\Phi]$. For $\operatorname{supp} W\subset[u,v]$, fix $c_I\ge\max(1,4C_\Phi v^2)$ and enlarged support $[u/2,2v]$.

Apply (47) to the square. Write $n_i=Cu_i$, $C=(n_1,n_2)$; the primitive row conductor is u_1u_2, residual mask C, divisor $d\mid C$. Zero frequency costs $O(H\|W\|_\infty^2)$. Use $a_\xi(n)\gamma_1(n)=\mu(n)G(n)\xi(n)$, expand its fixed ray factor, and remove $(u_1,u_2)=1$ by complete Möbius inversion, $u_i=tx_i$. A fixed ray component is

$$P_{C,d,t}(y;U)=\sideset{}{^*}\sum_n\mu(n)\xi_1(n)1_{(n,t)=1}\overline{\chi_n(y)}\chi_n(C)^4\chi_n(d)U(q_n/\ell),\qquad\ell=\frac{L}{q_Cq_t}. \tag{54}$$

Here C,t are coprime squarefree, $d \mid C$, $q_C q_t \le 2vL$. Combine frequency and label as $y = hf^2$, retaining $(f, Ct) = (h,t) = 1$. This map has divisor-bounded fibres; enlarge its nonzero image to $q_y \le Y_{C,d}$, where

$$w_{C,d} = \frac{Hq_C}{q_d L^2 F}, \qquad Y_{C,d} = \frac{c_I \Sigma LF q_d}{Hq_C^2}.$$

Since $\Sigma \ge LF$, Cauchy and smooth separation give the positive form

$$Q_{\xi_1}(U) = \sum_{\substack{C,t \text{ sf} \\ (C,t)=1 \\ q_C q_t \le 2vL}} \sum_{d \mid C} w_{C,d} \sum_{y} \Phi(q_y / Y_{C,d}) |P_{C,d,t}(y;U)|^2. \tag{55}$$

If $Q_{\xi_1}(U) \le M\|U\|_{C^j}^2$ for all common U, then (5) gives $E \ll Z^\varepsilon(\Sigma + M)\|W\|_{C^{2j+4}}^2$. Kernels combine annular weights, inverse roots and $\widehat{\Phi}(A/(x_1x_2))$ on bounded support, with uniform derivatives.

Apply (47) to y, write columns gz_i, take $e \mid g$, and remove $(z_1,z_2) = 1$ by $z_i = wn_i$. Nonzero terms have

$$(g, Ct) = (w, gCth) = 1, \qquad e \mid g, \qquad h \ne 0,$$

and sign $\mu(e)\mu(w)$. Inverse conversion (6) restores cubic coefficients. The ray phase is $\xi_1(z_1 z_2^{-1}) G(z_2 z_1^{-1})$; expanding $\xi_1(z)G(z^{-1}) = \sum_{\xi'} c_{\xi'} \xi'(z)$ gives $a_{\xi'}(n)$ independent of reconstruction choices.

Reconstruct the puncture, label and row by

$$r = tg/e, \qquad f' = Cew, \qquad k' = deh, \qquad a^\sharp(n) = a_{\xi'}(n) 1_{(n,r)=1} \chi_n(k') \chi_n(f')^4. \tag{56}$$

The ideals r, f' are coprime squarefree. Locally,

$$\chi_n(dh)\overline{\chi_n(e)}\chi_n(C)^4 = \chi_n(deh)\chi_n(Ce)^4; \tag{57}$$

Both sides have unit exponent five at e and vanish at nonunits; w contributes (6)'s multiplicative correction.

For $U_0(x) = x^{-1/2}U(x)$, the scale, scalar and kernel are

$$X_0' = \frac{\ell}{q_g q_w} = \frac{L}{q_r q_{f'}}, \qquad w_{C,d} \frac{Y_{C,d} q_g}{q_e \ell} = \frac{c_I \Sigma q_r}{L^2},$$

$$K_{r,f',k'}(x_1,x_2) = U_0(x_1)\overline{U_0(x_2)}\widehat{\Phi}\left(\frac{c_I \Sigma F q_{k'} q_r^2}{HL x_1 x_2} \right). \tag{58}$$

These depend only on (r, f', k'). Support forces $q_r q_{f'} \le 2vL$ and $0 < q_{k'} \le HL/(\Sigma F q_r^2)$.

At fixed (r, f', k'), all preimages are

$$t \mid r, \quad f' = Cew, \quad e \mid k', \quad (w, k') = 1, \quad d \mid (C, k'), \quad g = e(r/t), \quad h = k'/(de),$$

with disjoint $f' = Cew$. For $\mathfrak{p} \mid f'$, write $I_\mathfrak{p} = 1_{\mathfrak{p} \mid k'}$; allocation to C, e, w gives

$$\underbrace{(1 + I_\mathfrak{p})}_{\mathfrak{p} \mid C} + \underbrace{(-I_\mathfrak{p})}_{\mathfrak{p} \mid e} + \underbrace{(-(1 - I_\mathfrak{p}))}_{\mathfrak{p} \mid w} = I_\mathfrak{p}. \tag{59}$$

The C terms distinguish $\mathfrak{p} \nmid d$ and $\mathfrak{p} \mid d$. Shared kernel, scalar and character permit this signed cancellation before absolute values: it forces $f' \mid k'$, while $t \mid r$ contributes $\tau(r)$.

Writing Z_Q for the second zero frequency, we have

$$Q_{\xi_1}(U) - Z_Q = \frac{c_I \Sigma}{L^2} \sum_{\xi'} c_{\xi'} \sum_{\substack{r,f' \text{ sf} \\ (r,f')=1}} q_r \tau(r) \sum_{\substack{k' \ne 0 \\ f' \mid k'}} \sideset{}{^*}\sum_{n_1,n_2} a^\sharp(n_1)\overline{a^\sharp(n_2)} K_{r,f',k'}(q_{n_1}/X_0', q_{n_2}/X_0'). \tag{60}$$

By (47), $|Z_Q| \ll Z^\varepsilon \Sigma \|U\|_\infty^2$, since its absolute divisor sum is $\ll \Sigma \sum_C \tau(C) q_C^{-2} \sum_{q_t \ll L} q_t^{-1}$.

On dyads $q_r \asymp R, q_{f'} \asymp F'$, set

$$X' = \frac{L}{RF'}, \qquad \Sigma' = \frac{L}{R}, \qquad H' = \frac{HL}{\Sigma F R^2}. \tag{61}$$

Nonempty support and the projector give $F' \le q_{k'} \le HL/(\Sigma F q_r^2)$, hence $H' \ge 1$. For $j \mid r$,

$$\frac{X'}{q_j} \ge \frac{\Sigma F q_r^2}{HRq_j} \ge \frac{\Sigma F}{H} \ge 1. \tag{62}$$

Puncture removal therefore preserves admissibility and Σ'. Since $1 \le X'/X_0' < 4$, rescaled kernels have fixed enlarged support and uniform C^{2m+4} norm $\ll \|U\|^2_{C^{2m+4}}$.

Writing S_m for (52)'s supremum, remove punctures and positively enlarge (f', k') to bound each retained r's common-test sum by $Z^\varepsilon(\Sigma')^2 S_m \|V\|^2_{C^m}$. Apply (5) and count $O(R)$ values of r; the scalar is

$$\frac{\Sigma R}{L^2} R(\Sigma')^2 = \Sigma.$$

Absorb divisor and dyadic counts in Z^ε. This proves (52); (61) gives all inequalities (53).

4.4 Closing the finite-depth induction Inductively prove (49) for $H \le Z^{j\kappa}$. At $j = 0$, counting gives $E \ll HX\|W\|^2_\infty \le \Sigma\|W\|^2_\infty$. Next use (51). A nonempty supremum has $H^2 > X$, $H_c^3 = X^2/H^2$, $q_b > H_c$. Apply (52) at $L = L_b$, keeping $\Sigma = XF$; every child satisfies

$$H' \le \frac{HL_b}{\Sigma F} = \frac{H}{q_b^3 F^2} < H\left(\frac{H}{\Sigma}\right)^2 \le Z^{-2\kappa} H \le Z^{j\kappa}, \quad H'/\Sigma' \le Z^{-\kappa}, \quad \Sigma' \le \Sigma.$$

Induction applies. Distribute losses among completion, transfer and child, propagating orders backwards by $m \mapsto 4m + 12$ and retaining the completed-energy order. Depth $\le \lceil C_0/\kappa \rceil$ gives finite orders and support enlargements, fixed before external heights. This proves (49).

4.5 Return to the inverse polynomial Majorize the u-ball by $\Phi(q_u/H)$, expand and apply (47). Write columns gz_i, mask divisor $e \mid g$, frequency $h \ne 0$; zero frequency costs $O(H\|W\|^2_\infty)$. Inverse conversion (6) and fixed ray expansion restore cubic coefficients. Remove $(z_1, z_2) = 1$ by $z_i = vn_i$. Retaining zeros, the bijection is

$$b = g/e, \qquad f = ev, \qquad k = eh; \qquad e = (f, k), \quad v = f/e, \quad g = be, \quad h = k/e. \tag{63}$$

Here b, f are coprime squarefree. The identities $gv = bf$, $he^5v^4 = kf^4$, $\mu(e)\mu(v) = \mu(f)$ and $q_g/q_e = q_b$ give each nonzero ray component

$$\begin{aligned}\frac{H}{D^2} \sum_{\substack{b,f \text{ sf} \\ (b,f)=1}} \mu(f) q_b \sum_{k\ne 0} \sideset{}{^*}\sum_{n_1,n_2} a_\xi(n_1)\overline{a_\xi(n_2)}\chi_{n_1}(kf^4)\overline{\chi_{n_2}(kf^4)} 1_{(n_1n_2,b)=1} \\ \times W_0(q_{bfn_1}/D)\overline{W_0(q_{bfn_2}/D)}\widehat{\Phi}\left(\frac{Hq_k}{q_f^2 q_{n_1} q_{n_2}}\right),\end{aligned} \tag{64}$$

with $W_0(x) = x^{-1/2}W(x)$ and support $q_b q_f \ll D$, $q_k \ll D^2/(Hq_b^2)$. On $q_b \asymp B, q_f \asymp F$ put

$$X = \frac{D}{BF}, \qquad \mathcal{H} = \frac{CD^2}{HB^2}, \qquad \Sigma = XF = \frac{D}{B},$$

Fixed C covers supports. Since $\mathcal{H}/\Sigma = CD/(HB) \ll Z^{-\kappa}$, apply (49) with gap $\kappa/2$, then (5), (48). Kernels have uniform derivatives, giving dyad bound

$$Z^\varepsilon \frac{HB}{D^2} B(XF)^2 \|W\|^2_{C^J} = HZ^\varepsilon \|W\|^2_{C^J}.$$

Subunit column scales are empty or have boundedly many columns; count them. Subunit row scales are empty. Sum dyads and fixed ray components, including zero frequency, to prove (45). Conjugation with $\nu \mapsto \bar{\nu}$ gives the other orientation. Finite seminorms of $W(y)y^{it}$ cost $(1 + |t|)^{O(1)}$.

For (95), take $Z = \max(2, D, H)$. If $D \ge 2$ and $H \ge D^{1+\vartheta}$, choose $\kappa = \vartheta/(1 + \vartheta)$; rename the loss to get $H(HD)^\varepsilon$. Count bounded columns. All element rows, both orientations and common-profile control follow. □

5 The plain moment with moving radical

5.1 Equal-product rectangle cancellation Two Poisson transforms decrease width except for fixed-character rows; equal-product rectangle subtraction cancels their principal volume.

Fix a finite group Θ of S-supported ray characters closed under reciprocity, conjugation and $n \mapsto \chi_n(-1)$. For $\vartheta \in \Theta$, squarefree $q_{R_*} \le Z^B$ and fixed-annular W_1, W_2, put

$$L_i(X,t) = \sum_n \vartheta(n)\mathbf{1}_{(n,R_*)=1} q_n^{it} W_i(q_n/X).$$

Poisson summation gives, uniformly for $X > 0$,

$$L_i(X,t) = c_{\vartheta,R_*} X^{1+it} I_i(t) + O\big(Z^\epsilon (1+|t|)^J\big), \qquad |L_i(X,t)| \ll X, \tag{65}$$

Here $I_i(t)$ is a fixed lattice-measure multiple of $\int W_i(y) y^{it}\, dy$, and c_{ϑ,R_*} is X,t-independent.

Without the mask, Poisson on fixed lattice cosets for $X^{it} f_t(z/\sqrt{X})$ gives the zero-frequency main term. Integration by parts bounds other frequencies by $(1+|t|)^J (1+|\xi|)^{-J}$; $J > 4$ handles $X \ge 1$, counting $X < 1$. Insert

$$L_i(X,t) = \sum_{d \mid R_*} \mu(d)\vartheta(d) q_d^{it} L_i^0(X/q_d, t).$$

The main coefficient is

$$c_{\vartheta,R_*} = c_\vartheta \sum_{d \mid R_*} \frac{\mu(d)\vartheta(d)}{q_d}.$$

Norm powers cancel in this coefficient; divisor-many errors cost Z^ϵ. Counting gives $|L_i| \ll X$.

Consequently, if

$$U_1 U_2 = V_1 V_2 = T, \qquad U_i, V_i \ge Z^r \quad (i = 1,2), \quad r \ge 0,$$

then

$$T^{-1/2} \left| L_1(U_1,t) L_2(U_2,t) - L_1(V_1,t) L_2(V_2,t) \right| \ll Z^\epsilon (1+|t|)^{2J} T^{1/2} Z^{-r}. \tag{66}$$

Main terms equal $c_{\vartheta,R_*}^2 T^{1+it} I_1(t) I_2(t)$. Error scales are $\le T Z^{-r}$; the error product also fits since $T \ge Z^{2r}$. Without a positive lower scale, use volume $O(T^{1/2})$.

The character, mask and norm power must agree across both factors and rectangles.

5.2 The family closed under both transforms Moving twists and a common exclusion close the family under both transforms; no selected prime or zero-free hypothesis is used.

Fix bounded nonnegative real ranges for m, q, n_1, n_2, B. A fixed twist within a row sum has the form

$$\tau(n) = \nu(n) \prod_{p \mid A} \chi_n(p)^{e_p}, \qquad \nu \in \Theta, \quad q_{\mathrm{rad}\, A} \le Z^q,$$

Retain all displayed zeros, including exponents 0 mod 6, and count their moving support in q. Reciprocity reduces other residue products to this form. A squarefree puncture $q_R \le Z^B$ is fixed across the row sum and all plain factors. Define

$$\psi_k(n) = \tau(n)\chi_n(k)\mathbf{1}_{(n,R)=1}, \qquad S_k(N;W) = N^{-1/2} \sum_n \psi_k(n) W(q_n/N).$$

Theorem 5.1 (Moving-radical mixed plain moment). *For every fixed $\epsilon > 0$,*

$$\sum_{\substack{0 < q_k \le Z^m \\ \psi_k^* \ne 1}} |S_k(Z^{n_1}; W_1) S_k(Z^{n_2}; W_2)|^2 \ll Z^{m+q+\epsilon}. \tag{67}$$

No conductor lower bound or condition on $n_1 + n_2$ is imposed. The bound carries $\|W_1\|_{C^J}^2 \|W_2\|_{C^J}^2$, uniformly over moving supports, punctures and bounded lengths, with ğ1.3's real-power and height control. Increase J for any fixed list of Euler/profile derivatives. With no moving twist and two copies of one polynomial,

$$\sum_{\substack{0<q_k\le H\\(\nu\chi_\bullet(k))^*\neq 1}} |S_k(N;W)|^4 \ll H^{1+\epsilon} \qquad (68)$$

for bounded polynomial ranges of N relative to H.

Proof. Descend in effective width $M = m + q$; fixed-conductor twists alone do not close under the first transform.

5.3 Reflection and the centered comparison A natural presentation has exactly fixed, displayed moving and row zeros. Remove extra punctures by

$$S_{k,R}(N;W) = \sum_{d\mid R} \mu(d)\psi_k^0(d)q_d^{-1/2} S_k^0(N/q_d;W). \qquad (69)$$

Its coefficient mass is Z^ϵ. At each width prove all natural lengths before removing extra punctures; recursive children have smaller width.

Let ψ_k^* have conductor norm Q_k, and put $C_k = 3Q_k/(2\pi)^2$. If $R_{0,k}$ is the redundant natural radical, then

$$C_k q_{R_{0,k}} \ll Z^M. \qquad (70)$$

Each good prime is counted once in primitive conductor or redundant radical, whose union lies in row radical and displayed moving support. Fixed primes cost a constant.

The primitive functional equation gives exactly

$$S_{\psi_k^*}(N;W) = \epsilon_k S_{\overline{\psi_k^*}}(C_k/N;W^\sharp), \qquad \mathcal{M}W^\sharp(s) = \mathcal{M}W(1-s)\frac{\Gamma(s)}{\Gamma(1-s)}. \qquad (71)$$

Central normalization adds no C_k, N power. Inside an absolute product, conjugate the entire reflected factor and profile to restore orientation. Delete redundant Euler factors before reflection, restore geometrically afterwards: scales are $C_k q_d/(Nq_h)$, $d \mid R_{0,k}$, $\operatorname{rad} h \mid R_{0,k}$, with coefficients $\epsilon_k\mu(d)\psi_k^*(d)\overline{\psi_k^*(h)}/\sqrt{q_d q_h}$. Their mass is Z^ϵ and largest scale $\ll Z^M/N$.

The gamma quotient gives bounded Euler derivatives at zero and rapid decay at infinity, hence a centrally summable annular partition with polynomial height cost. With padding $\xi > 0$, retain dual lengths $\le M - \log_Z N + \xi$ and arbitrarily small tails. Apply ğ1.3's Sobolev common-profile bounds to rowwise scales.

Reflect any original factor longer than $M/2$. It remains to prove a padded core with

$$n_i \le M/2 + \xi, \qquad A := n_1 + n_2 \le M + 2\xi. \qquad (72)$$

At each width first prove the uncentered range $A \le 5M/6$. Otherwise put $L = M/4$. If one factor is shorter than L, reflect the other: total becomes $\le 3M/2 - A + \xi < 2M/3 + \xi$, already covered. If both are $\ge L$, compare

$$S_k(X_1;W_1)S_k(X_2;W_2) \quad \text{with} \quad S_k(Y_1;W_1)S_k(Y_2;W_2),$$

where $Y_1 = Z^L$, $Y_2 = X_1X_2/Y_1$. Reflecting the latter's second factor again gives total $\le 3M/2 - A + \xi$, so the comparison is known at this width. Retain the difference as one coefficient; its positive squared norm may include all rows. Subtraction controls added principal rows.

For fixed extracted ideals b_1, b_2, the coefficient to preserve is

$$\mathcal{D}_b(l_1,l_2) = W_1(q_{b_1}q_{l_1}/X_1)W_2(q_{b_2}q_{l_2}/X_2) - W_1(q_{b_1}q_{l_1}/Y_1)W_2(q_{b_2}q_{l_2}/Y_2). \qquad (73)$$

All four formal scales are $\ge Z^L$ and extracted products stay equal. Retain empty rectangles until cancellation.

5.4 First Poisson transform Set $H = Z^m$, $X = Z^A$ and majorize rows by fixed nonnegative radial Schwartz Φ_1; central squared normalization is X^{-1}. Zero frequency requires every prime's net sextic exponent to vanish, hence no prime occurs once. The combined ideal is powerful, norm $O(X^2)$: $O(X^{1+\epsilon})$ possibilities with divisor-many four-factor allocations, costing $O(HZ^\epsilon) \le Z^{M+\epsilon}$.

For nonzero frequency extract full common support as Ca, Db, $(a,b) = (ab, CD) = 1$, $\operatorname{rad} C = \operatorname{rad} D$. Put

$$c = \log_Z q_C,\ d = \log_Z q_D,\ p = \log_Z q_{\operatorname{rad} C}.$$

Let r contain common primes of nonzero net exponent modulo six, $R = \log_Z q_r$, with primitive common character ξ_r. Expand the remaining common mask by $e \mid \operatorname{rad} C/r$, $E = \log_Z q_e$; thus $R + E \le p$.

For the allocated convolution coefficients $A_C(a), A_D(b)$, residue-class Poisson gives, up to a frozen scalar of modulus at most one,

$$\frac{H}{Xq_e\sqrt{q_r}} \sum_{h\neq 0} \mathfrak{g}_{\xi_r}(r,h) \sum_{(a,b)=1} \frac{A_C(a)\overline{A_D(b)}}{\sqrt{q_a q_b}} \tau_C(a)\overline{\tau_D(b)} R(a,b)\mathfrak{g}(a,h)\overline{\mathfrak{g}(b,-h)}\widehat{\Phi}_1\left(\frac{Hq_h}{q_e q_r q_a q_b}\right). \tag{74}$$

Here $\mathfrak{g}_{\xi_r}$ is the normalized primitive Gauss sum, of modulus at most one, and one consistent orientation is

$$\tau_C(a) = \tau(a)\chi_a(er)\xi_r(a), \qquad \tau_D(b) = \tau(b)\chi_b(er)\overline{\xi_r(b)}.$$

Indeed $k = ek'$ and Poisson modulo rab give $H/(q_e q_r q_a q_b)$; three Gauss sums restore $\sqrt{q_r q_a q_b}$ and central normalization supplies X^{-1}. Reciprocity gives $R(a,b)$. Full displayed moving length becomes

$$\widetilde{q} = q + R + E. \tag{75}$$

Old moving zeros puncture both products, so discard colliding allocations first; new common support is disjoint even when exponents cancel.

The natural first dual scale has logarithmic length

$$K_0 = 2A - c - d + R + E - m. \tag{76}$$

Retain full smooth dyads $q_h \asymp Z^K$, $K \le K_0 + \xi$. Separate only row and two **whole-product** norms with one Fourier measure; expand $R(a,b)$ on its ray group. Before Cauchy insert (15)'s full $\sum_{s|a,b}\mu(s)$, and put $s_0 = \log_Z q_s$.

The two resulting positive Gauss norms have the form

$$\mathcal{N}_C = \sum_h \omega_K(q_h/Z^K)\left|Z^{-(A-c)/2}\sum_{s|a} A_C(a)\tau'_C(a)q_a^{it}\mathfrak{g}(a,h)\right|^2, \tag{77}$$

and similarly for D. Within each norm the profiles and masks are common; row dependence is a bounded scalar. Coefficient (73) retains a common whole-product character and norm power.

A sufficient estimate is

$$\mathcal{N}_C \ll Z^{A-c+\widetilde{q}+B_c-s_0+\epsilon}, \qquad B_c = \frac{(3c-5d-R)_+}{6}. \tag{78}$$

The complete exponent balance is

$$\underbrace{m - A - R/2 - E}_{\text{Poisson prefactor}} + \underbrace{p + s_0}_{\text{common-support and divisor counts}} + \frac{1}{2}\{A - c + \widetilde{q} + B_c - s_0 + A - d + \widetilde{q} + B_d - s_0\}$$

$$= M + \frac{1}{2}\{B_c + B_d - (c + d - 2p - R)\} \le M. \tag{79}$$

Locally at multiplicities $i \ge j \ge 1$, put $r_p = 1_{6\nmid i-j}$. The contribution is $\le (3i - 5j - r_p)_+/6$, compared with nonnegative $i + j - 2 - r_p$. When the former is positive, six times their difference is $3i + 11j - 12 - 5r_p \ge 0$. Subadditivity of positive parts proves (79). For fixed radical, Rankin and divisor bounds give Z^ϵ allowed powers and allocations, verifying all label counts.

5.5 Second transform and width decrease Set $a_0 = A - c$, $D_0 = d + (K_0 - K)$, and

$$J = D_0 - c, \qquad g = J_+ + \sigma, \tag{80}$$

Choose small width step $\sigma > 0$. Enlarge the positive Gauss norm to a smooth ball of length $K + g$, adding its zero row. By (8), only sixth-power columns contribute: $O(Z^{a_0/6+\epsilon})$ columns, Gauss magnitude $\le Z^{a_0/2}$, hence centrally normalized square $Z^{a_0/3+\epsilon}$. Nonempty $s \mid a$ forces $s_0 \le a_0/6 + O(1/\log Z)$, so (78) covers it.

For the remaining norm, (10) and lattice Poisson give the exact expansion

$$Z^{K+g-a_0} \sum_{u,v} \frac{B(u)\overline{B(v)}\tau_1(u)\overline{\tau_1(v)} q_u^{it} q_v^{-it}}{\sqrt{q_u q_v}} \sum_j F(u,v;j) \widehat{\Phi}_2\left(\frac{Z^{K+g} q_j}{q_u q_v} \right), \tag{81}$$

Here B retains the whole rectangle difference and $s \mid u, v$. At $j = 0$, (11) forces $u = v$, giving exponent $K + g - s_0$ and excess over (78)

$$(A - M) - D_0 - B_c + g \le (A - M) + \sigma + O(\xi). \tag{82}$$

Since $D_0 \ge -\xi$, $-D_0 + (D_0 - c)_+ \le O(\xi)$; thus padded $A \le M + 2\xi$ suffices.

For $j \ne 0$, retain full-kernel dyads through length $2a_0 - K - g + \xi$, then extract common support $u = D_2 a, v = E_2 b$. Put

$$c_2 = \log_Z q_{D_2}, \ d_2 = \log_Z q_{E_2}, \ b_2 = (c_2 + d_2)/2,$$

$$p_2 = \log_Z q_{\operatorname{rad} D_2}, \quad g_2 = \log_Z q_{(D_2, E_2)}.$$

At equal non-six-divisible multiplicities, split divided frequencies into units and nonunits, whose radical lengths are t_2, V. By (13)–(14),

$$j = (D_2, E_2) V_{\mathrm{id}} h', \qquad |F(D_2, E_2; j)| \le Z^{g_2 - t_2},$$

where V_{id} is the product of the nonunit primes. Set the scalar to zero off its partition before enlarging rows. New row and displayed moving lengths are

$$m' = 2a_0 - K - g - g_2 - V, \quad q' = \widetilde{q} + t_2 + V,$$

$$M' = m' + q' = M + J - g - g_2 + t_2 \le M - \sigma. \tag{83}$$

Here $g_2 \ge t_2$. Sixth-divisible fixed factors give a common puncture with retained ray phase, without a new pair-dependent mask.

Since $s \mid u, v$ lies in the second common radical, its count is $Z^{p_2 - s_0 + \epsilon}$. The ledger is

$$\underbrace{K + g - a_0}_{\text{Poisson and roots}} + \underbrace{p_2 - s_0}_{\text{common supports}} + \underbrace{g_2 - t_2}_{\text{correlation}} - \underbrace{b_2}_{\text{remaining central conversion}}, \tag{84}$$

so the inner residual products are allowed exponent

$$M' + \Delta_{\mathrm{child}}, \qquad \Delta_{\mathrm{child}} = b_2 - p_2 + B_c \ge 0. \tag{85}$$

Separate (81)'s kernel and inverse roots only in row and whole-product norms. Remove residual coprimality by (15); for each prime of its divisor t, use

$$\mathbf{1}_{p \mid l_1 l_2} = \mathbf{1}_{p \mid l_1} + \mathbf{1}_{p \mid l_2} - \mathbf{1}_{p \mid l_1} \mathbf{1}_{p \mid l_2}. \tag{86}$$

In each selected factor put $l_i = p l_i'$ without $(l_i', p) = 1$. The common character supplies a scalar, dividing corresponding scales in both rectangles by P; mask and norm power remain common. On $q_t \asymp Z^{t_-}$, each side extracts total length $r_j \ge t_- + O(1/\log Z)$. Central factors $Z^{-r_j/2}$ cancel the Z^{t_-} count; allocations cost divisors.

Both children are the original whole-product twist times $\chi_n((D_2, E_2) V_{\mathrm{id}} h')$, differing by Θ, so either both induce a member of Θ or neither. In the latter case apply Cauchy, then split centered children, discard empty annuli and rescale bounded subunit annuli. The completed theorem at smaller width $M' + O(\xi)$ gives $Z^{M'+\epsilon}$, within (85), with common polynomial masks. It remains to treat fixed-character children.

5.6 Exceptional children Outside fixed support, exceptionality prescribes $v_p(h') \bmod 6$, so $h' = h_0 v^6$ for fixed sixth-power-free h_0. Let v_1 be radical length of nonunit common multiplicity-one primes, $f = 2v_1$. Their fixed row valuation is two, forcing $v_p(h') \equiv 4 \pmod 6$. Old moving support is disjoint by its zeros, hence $q_{h_0} \geq Z^{4v_1}$. The weaker sufficient count is

$$\#\{h' \text{ exceptional}\} \ll Z^{(m'-f)/6+\epsilon}. \tag{87}$$

This covers all Θ and positively added rows; a forced h_0 that does not fit gives an empty range.

After (86), paired normalized convolution volume is $\ll Z^{a_0-b_2+\epsilon}$. Its excess, with (87), over (85) is

$$\frac{m'-f}{6} + a_0 - b_2 - (M' + \Delta_{\text{child}}) = A - \frac{5}{6}M - F_1 - F_2, \tag{88}$$

where

$$F_1 = \frac{c}{6} + \frac{5D_0}{6} + \frac{\widetilde{q}}{6} + B_c - \frac{5g}{6},$$

$$F_2 = 2b_2 - \frac{5}{6}g_2 - p_2 + t_2 + \frac{V}{6} + \frac{f}{6}. \tag{89}$$

There are the two local lower bounds

$$F_1 \geq \frac{2}{3}c - \frac{5}{6}\sigma - O(\xi), \qquad F_2 \geq \frac{2}{3}b_2. \tag{90}$$

If $D_0 \geq c$, substitution gives $F_1 = c + \widetilde{q}/6 + B_c - 5\sigma/6$. Otherwise $\widetilde{q} \geq R$ implies $\widetilde{q} + 6B_c \geq (3c - 5D_0)_+ - O(\xi)$; use $c + 5D_0 + (3c - 5D_0)_+ \geq 4c$.

For F_2, the local contributions, in units of $\log_Z P$, are:

Common multiplicities; divided frequency	F_2	b_2
$i = i,\ 6 \nmid i,$ unit	$7i/6$	i
$i = i,\ 6 \nmid i,$ nonunit	$(7i-5)/6 + \frac{1}{3}\mathbf{1}_{i=1}$	i
$i = i,\ 6 \mid i$	$7i/6 - 1$	i
$i > j,\ 6 \mid j,$ unit	$i + j/6 - 1$	$(i+j)/2$

The middle column is always $\geq 2/3$ of the last, with equality at nonunit multiplicity one; these exhaust (13)–(14).

For $A \leq 5M/6$, (88)–(90) finish the uncentered stage, with terminal loss $5\sigma/6 + O(\xi)$.

For the centered stage, the common-character calculation gives, on an exceptional row, a coefficient of the form

$$\vartheta(l_1)\vartheta(l_2)\mathbf{1}_{(l_1 l_2, R_*)=1} q_{l_1 l_2}^{it} \mathcal{D}_b(l_1, l_2). \tag{91}$$

The common mask contains all redundant natural zeros. Before final extraction, one paired side has all four formal scales at least

$$Z^{L-c-\min(c_2,d_2)}.$$

Set $r = (L - c - \min(c_2, d_2))_+$. If that side extracts $r_1 < r$, (66) saves $Z^{-(r-r_1)}$; otherwise use volume. Divisor counts satisfy

$$t_- - r_1 - r_2 - (r - r_1)_+ \leq -r \tag{92}$$

so all allocated paired products cost $Z^{a_0-b_2-r+\epsilon}$. Triangle before the full divisor sum loses this saving.

With $v = c + \min(c_2, d_2)$ and $b_2 \geq \min(c_2, d_2)$, the remaining deficit is at most

$$A - \frac{5}{6}M - \frac{2}{3}v - (L - v)_+ + \frac{5}{6}\sigma + O(\xi) \leq A - M + \frac{5}{6}\sigma + O(\xi). \tag{93}$$

The maximum over $v \geq 0$ occurs at $v = L = M/4$; padded $A \leq M + 2\xi$ leaves only a small terminal loss.

5.7 Closing the finite descent Choose small floor $\rho > 0$, step $\sigma > 0$, then padding $\xi > 0$. At $M \le \rho$, reflection and counting in (72) lose at most $\rho + 2\xi$. Above it, (83) decreases width by σ before enclosure; $\xi \ll \sigma$ leaves decrease $\ge \sigma/2$. Depth is at most $D = 2 + \lceil 2M_{\max}/\sigma \rceil$.

Within each band prove: uncentered range, centered core, reflection to all lengths, then extra punctures by (69). Comparison uses the already proved uncentered range with positive margin; other children use strictly smaller completed bands.

Make every terminal loss (floor, diagonal, exceptional) $< \epsilon/4$ by choosing ρ, σ. Choose padding and local divisor/Schwartz losses $\ll \epsilon/D$. Triangle takes the maximum of child exponents after counting mass; Cauchy averages them. Each branch incurs one terminal loss plus edge losses.

Measure powers before dyadic rounding. Use joint log boxes, selecting whole frequency dyads from the outer box's scale supremum with padding $Z^{\xi/2}$, independently of live columns. Discarded genuine Poisson arguments are $\ge Z^{\xi/4}$. Bounds $|\mathfrak{g}(a,h)| \le \sqrt{q_a}$, $|F(u,v;j)| \le q_u q_v$ and polynomial counts make tails arbitrarily small by Schwartz decay **before** separation.

Retained $\widehat{\Phi}(R\exp L(x))$ has uniform Euler derivatives for $R > 0$ and a common Fourier measure of finite weighted L^1 norm. Remove central powers before inverse-root separation. Finite-depth support enlargements cost $O(1/\log Z)$; rescale subunit annuli only in positive child norms, preserving formal centered scales. Formulas (74), (12), (86) preserve character/common-mask closure with polynomial norms. Choose seminorm and height orders backwards, then thresholds and external tail orders. This proves (67). □

No moving twist and two identical profiles give (96); restricting the positive row sum to sixth-power-free rows decreases it. Conjugation gives the inverse orientation.

5.8 Coefficient-specific corollaries and sixth-power amplification For a fixed finite-ray character ν, let $\varphi_u = \nu\psi_u^{\pm 1}$, with its original zeros, and define

$$M_u(D;W) = D^{-1/2}\sum_n \mu(n)\varphi_u(n)W(q_n/D), \qquad S_u(N;W) = N^{-1/2}\sum_n \varphi_u(n)W(q_n/N). \tag{94}$$

The preceding theorems imply the following two coefficient-specific estimates:

$$\sum_{0<q_u\le H} |M_u(D;W)|^2 \ll H(HD)^{\varepsilon}\|W\|_{C^J}^2, \qquad H \ge D^{1+\vartheta}, \quad \vartheta > 0; \tag{95}$$

$$\sum_{\substack{q_u \asymp U \\ \text{sixth-power-free} \\ \varphi_u^* \ne 1}} |S_u(U^m;W)|^4 \ll U^{1+\varepsilon}\|W\|_{C^J}^4. \tag{96}$$

Here $m \ge 0$ is bounded and φ_u^* is the primitive inducing character. Equations (95), (96) follow from Theorems 4.1 and 5.1, with ğ1.3's uniformity and no zero-free assumption; (95) includes sixth-power rows.

Sixth-power amplification extends (95) to longer inverse polynomials. The exact reconstruction identity is

$$M_u(D;W) = \sum_{d\mid \mathrm{rad}(a)} \mu(d)\varphi_u(d)q_d^{-1/2}M_{ua^6}(D/q_d;W). \tag{97}$$

Separate squarefree column primes dividing a to prove (97), retaining shared u, a zeros. Its coefficient mass is $\ll q_a^{\varepsilon}$. Log-scale differentiation sends W to $-W/2 - yW'$, so Sobolev and (95) give

$$\sum_{0<q_v\ll H} \sup_{0<E\le D} |M_v(E;W)|^2 \ll H(HD)^{\varepsilon}, \qquad H \ge D^{1+\vartheta}. \tag{98}$$

Only $O(\log(2+D))$ log intervals occur; subunit scales have boundedly many columns and eventually none.

For sixth-power-free $q_u \asymp U$, put $H = \max(2U, D^{1+\vartheta})$ and average (97) over the $\gg_S (H/U)^{1/6}$ primary ideals $q_a \le (H/U)^{1/6}$ prime to S. The map $(u,a) \mapsto ua^6$ is injective: valuations modulo six recover u, primary generators its unit. Equation (98) gives $\sum^{(6)} |M_u(D)|^2 \ll U^{1/6}H^{5/6}(UD)^{\varepsilon}$. For bounded r, take $D = U^r$ and small ϑ:

$$\sum_{q_u\asymp U}^{(6)} |M_u(U^r;W)|^2 \ll U^{e(r)+\varepsilon}, \qquad e(r) = \max\{1, (1+5r)/6\}. \tag{99}$$

6 The exact Poisson series of the common probe

Here $X, Y, Z > 0$ are independent. This unmarked probe supplies the exact coefficient interface; compensated or affine probes must derive their combinations from it.

Fix primitive η and let b_* generate the product of S. Choose primitive $\xi \bmod b_*$, quadratic at odd residue characteristic and cubic at norm four; thus $\xi^6 = 1$ on units. Put

$$\tau = q_{b_*}^{-1/2} g_\xi(b_*, 1), \qquad \Xi(A) = \xi(A)\chi_A(b_*), \qquad |\tau| = 1.$$

Choose nonnegative nonzero annular weights W_0, W_1, w, and set

$$V_G(r) = \frac{e^{-(\log r)^2/4}}{2\sqrt{\pi}}, \qquad \widehat{V}_G(t) = e^{t^2}, \tag{100}$$

The ideals c, n may share primes. With $A = cn^3$, define

$$T_{m,a,\eta}(t) = \sum_{c \text{ sf},n} \gamma_2(c)\overline{\alpha(A)G(A)}\eta(A)\Xi(A)^{-1}R(A,a)\chi_A(m)q_c^{-1/2-t}q_n^{-1-3t}, \tag{101}$$

$$\begin{aligned} I_\eta(X,Y,Z) = \frac{1}{Y}\sum_a \frac{W_1(q_a/Y)\chi_a(b_*)}{\tau\xi(a)\sqrt{q_{b_*}q_aX}} \sum_m \xi(m)q_a^{-1/2} g_{\chi_a}(a,-m) W_0\left(\frac{q_m}{q_{b_*}q_aX}\right) \\ \times \frac{1}{2\pi i}\int_{(4)} Z^t e^{t^2} T_{m,a,\eta}(t)\, dt. \end{aligned} \tag{102}$$

Annular weights make a, m finite, $m \neq 0$; remaining sums/integrals converge absolutely. Expand $\overline{G}$ on its ray group and partition a into classes σ for $R(A,a)$. Then $B_{m,\sigma}(Z)$ is a fixed finite combination of (16) with V_G. Put $Q_0 = q_{b_*}XY$ and fix $\Omega = 1$ on $\operatorname{supp} W_0 \operatorname{supp} W_1$. Mellin inversion gives a finite sum of

$$\frac{Q_0^{-1/2}}{2\pi}\int_{\mathbf{R}} \widehat{W}_0(iv) \sum_{m\neq 0} \Omega(q_m/Q_0)\xi(m)(q_m/Q_0)^{-iv} A_{m,\sigma,v}(Y) B_{m,\sigma}(Z)\, dv, \tag{103}$$

where

$$A_{m,\sigma,v}(Y) = Y^{-3/2}\sum_{a\in\sigma} \frac{\chi_a(b_*)}{\tau\xi(a)} W_1(q_a/Y)(q_a/Y)^{-1+iv} g_{\chi_a}(a,-m). \tag{104}$$

The normalization follows from $q_m/(q_{b_*}q_aX) = (q_m/Q_0)/(q_a/Y)$.

Let $\widetilde{W}_0(|y|^2)$ be the planar Fourier transform of $W_0(|z|^2)$, and $\mathcal{M}(z) = \int_0^\infty \widetilde{W}_0(r) r^{z-1}\, dr$, holomorphic for $\Re z > 0$ and rapidly decreasing vertically on compact substrips.

Proposition 6.1 (Exact common probe series).

$$\begin{aligned} I_\eta(X,Y,Z) = \frac{1}{(2\pi i)^3}\int_{(3)}\int_{(3)}\int_{(2)} X^{1/2-z}Z^{s+z-1}Y^{w-1}e^{(s+z-1)^2}\mathcal{M}(z)\widehat{W}_1(w) \\ \times \sum_u^{(6)} q_u^{-z}\overline{\xi(u)}\frac{\zeta_K^S(6z)L_K^S(w,\psi_u)}{L_K^S(s,\eta\overline{\psi_u})} H_{\eta,u}(s,w,z)\, dz\, dw\, ds. \end{aligned} \tag{105}$$

Here $u \neq 0$ is prime to S, including all units.

Section 6.2 derives the local factors $H_\mathfrak{p}$ and full/prime-divisible series $P_\mathfrak{p}, P_\mathfrak{p}^*$.

Proof.

6.1 The finite transform and phase cancellation Expanding the Gauss sum in the physical row gives the finite coefficient

$$\mathcal{F}(a,A,h) = \sum_{\substack{d \bmod a \\ h\equiv b_*Ad \pmod a}} \chi_a(d)\, g_{\xi\chi_A}\left(b_*A, \frac{h - b_*Ad}{a}\right). \tag{106}$$

The finite transform of $\xi(m)\chi_A(m)q_a^{-1/2}g_{\chi_a}(a,-m)$ modulo full ab_*A equals $q_a^{1/2}\mathcal{F}(a,A,h)$: summing lifts modulo b_*A imposes the congruence, including common primes of a, A.

Poisson at scale $q_{b_*}q_aX$ gives factor X/q_A and argument Xq_h/q_A. Combining $q_a^{1/2}$ with (102)'s square root leaves $X^{1/2}/(\sqrt{q_{b_*}}q_A)$ and the character/Y^{-1} factors. Primitivity at b_* kills $(h,S)>1$, including $h=0$. Write uniquely $h=ur^6$, u sixth-power-free, r primary.

Mellin inversion of the two radial weights, followed by $s=t+1-z$, gives the row Dirichlet series

$$\mathcal{F}_{\eta,u}(s,w,z)=\sum_{c\ \mathrm{sf},n,a,r}K_\eta(c,n,a,r;u)q_c^{-1/2-s}q_n^{-1-3s}q_a^{-w}q_r^{-6z},\tag{107}$$

where

$$K_\eta=\frac{\mathcal{F}(a,A,ur^6)\chi_a(b_*)}{\sqrt{q_{b_*}}\tau\xi(a)\overline{\xi(u)}}\gamma_2(c)\overline{\alpha(A)G(A)}\eta(A)\Xi(A)^{-1}R(A,a).\tag{108}$$

The remaining row factor is $q_u^{-z}\overline{\xi(u)}$, and the global weight is

$$X^{1/2-z}Z^{s+z-1}Y^{w-1}e^{(s+z-1)^2}\mathcal{M}(z)\widehat{W}_1(w).$$

Interchanges are absolute at $(\Re s,\Re w,\Re z)=(3,3,2)$ by $|\mathcal{F}(a,A,h)|\le q_aq_{b_*A}$, counting and Mellin decay. We factor (107).

At good p, put $Q=q_p$, $e_0=v_p(c)\in\{0,1\}$, $l=v_p(n)$, $k=v_p(a)$, $m=v_p(r)$, $t=e_0+3l$, $j=v_p(u)$, $j'=j+6m$, $\epsilon_p=\chi_p(-1)$. The local convolution is

$$C_{\mathfrak{p}}(t,k,j')=\sum_{\substack{d \bmod \mathfrak{p}^k\\ \mathfrak{p}^{j'}\equiv\mathfrak{p}^td\pmod{\mathfrak{p}^k}}}\chi_{\mathfrak{p}}(d)^kg_{\chi_{\mathfrak{p}}^t}\left(\mathfrak{p}^t,\frac{\mathfrak{p}^{j'}-\mathfrak{p}^td}{\mathfrak{p}^k}\right).\tag{109}$$

At $k=0$ the absent outer character is one; at $t=0$ the Gauss factor is one. Residue lifting gives

$$C_p(t,k,j')=\begin{cases}1, & t=k=0,\\ \mathbf{1}_{j'=0}, & t=0,\ k>0,\\ g_{\chi_p^t}(p^t,p^{j'}), & t>0,\ k=0,\\ \epsilon_p^{-1}g_{\chi_p}(p,1)g_{\chi_p^{t-1}}(p^t,p^{j'-1}), & t>0,\ k=1,\ j'\ge1,\\ 0, & \text{otherwise.}\end{cases}\tag{110}$$

For case four, expanding the Gauss sum and summing d mod p changes exponent t to $t-1$. For $k\ge2$, the final d-lift preserves its character but has nontrivial additive phase, hence cancels. Principal local characters retain their nonunit zeros.

For unit parts $h_p=h/p^{j'}$, $a_p^\circ=a/p^k$, $A_p^\circ=A/p^t$, CRT and $d=h_p(b_*A_p^\circ)^{-1}d'$ give

$$\chi_p(h_p)^{k-t}\chi_p(a_p^\circ)^t\chi_p(A_p^\circ)^{t-k}\chi_p(b_*)^{t-k}.$$

Put $\rho=\chi_p(u/p^j)$. The b_* Gauss factor is $\sqrt{q_{b_*}}\tau\xi(A)\overline{\xi(h/a)}$, with $\overline{\xi(h/a)}=\overline{\xi(u)}\xi(a)$ since $\xi^6=1$. Thus (108)'s fixed ξ factors cancel; the preceding b_* factors cancel $\chi_a(b_*)\chi_A(b_*)^{-1}$.

For distinct good primes, reciprocity gives

$$R(A,a)\prod_p\chi_p(a_p^\circ)^{t_p}\chi_p(A_p^\circ)^{-k_p}=\prod_pR(p,p)^{t_pk_p}.$$

Use $\gamma_2(c)=\mu(c)\alpha(c)G(c)/\gamma_1(c)$ and CRT for $\gamma_1(c)$. At $p<q$, the remaining exponent is $t_pt_q-e_pe_q=3(e_pl_q+e_ql_p+3l_pl_q)$; its sign product is

$$\prod_{p<q}R(p,q)^{e_pl_q+e_ql_p+l_pl_q}=R(c,n)G(n^3)\prod_p\overline{G(p^3)}^{l_p}R(p,p)^{-e_pl_p-l_p(l_p-1)/2}.$$

Expand (7) to verify this. Then $G(c)R(c,n)G(n^3)=G(cn^3)$ cancels the conjugate phase; $G(p^3)=\gamma_3(p)$ and $R(p,p)=\epsilon_p$. The coefficient therefore factors at every prime without ray restriction.

6.2 Local Euler factors At $\mathfrak{p} \notin S$, put

$$Q = q_{\mathfrak{p}}, \quad j = v_{\mathfrak{p}}(u) \in \{0, \dots, 5\}, \quad \rho = \chi_{\mathfrak{p}}(u/\mathfrak{p}^j),$$

$$a_{\mathfrak{p}} = \overline{\alpha(\mathfrak{p})}^3 \eta(\mathfrak{p})^3, \quad b_{\mathfrak{p}} = \overline{\alpha(\mathfrak{p})}^2 \eta(\mathfrak{p})^2 \overline{\chi_{\mathfrak{p}}(4)}, \quad b_{\mathfrak{p}}^3 = a_{\mathfrak{p}}^2,$$

$$V = Q^{-6z}, \quad R = a_{\mathfrak{p}}^2 Q^{4-6s-6z}, \quad W^\circ = \rho Q^{-w}, \quad W = \chi_{\mathfrak{p}}(u) Q^{-w}, \quad D = \eta(\mathfrak{p})\overline{\chi_{\mathfrak{p}}(u)} Q^{-s}. \tag{111}$$

For $j > 0$, $W = D = 0$ while W° retains its unit phase. The marked and full series are

$$P_{\mathfrak{p}}^* = \frac{1}{1-R}\left\{\frac{R(1-Q^{-1}) - \eta(\mathfrak{p})(Q-1)Q^{-s-w}V^{\mathbf{1}_{j\le 1}}}{1-V} + J_j\right\},$$

$$P_{\mathfrak{p}} = \frac{1}{1-V} + \mathbf{1}_{j=0}\frac{W^\circ}{1-W^\circ} + P_{\mathfrak{p}}^*, \qquad H_{\mathfrak{p}} = P_{\mathfrak{p}}\frac{(1-V)(1-W)}{1-D}, \quad H_{\eta,u} = \prod_{\mathfrak{p}\notin S} H_{\mathfrak{p}}, \tag{112}$$

with

j	J_j
0	$-\eta(\mathfrak{p})\rho^{-1}Q^{-s} + W^\circ R$
1	$\eta(\mathfrak{p})Q^{-s-w}$
2	$a_{\mathfrak{p}}\rho^{-3}Q^{3/2-3s}$
3	$-\eta(\mathfrak{p})b_{\mathfrak{p}}\rho^{-2}Q^{2-3s-w} + b_{\mathfrak{p}}^2\rho^{-4}Q^{2-4s}$
4	$-\eta(\mathfrak{p})a_{\mathfrak{p}}\rho^{-3}Q^{5/2-4s-w}$
5	$-a_{\mathfrak{p}}^2 Q^{3-6s}$

The indicator exponent gives V for $j = 0, 1$, otherwise one.
Equivalently, using (111),

$$P_p = \sum_{\substack{e_0=0,1\\ l,k,m\ge 0}} (-1)^{e_0}\gamma_1(p)^{-e_0}\eta(p)^{e_0}\left(a_p\overline{\gamma_3(p)}\right)^l \rho^{k-t}\epsilon_p^{tk-e_0 l - l(l-1)/2} \times C_p(t,k,j+6m)Q^{-(s+1/2)e_0-(1+3s)l-wk-6zm}, \qquad t = e_0 + 3l. \tag{113}$$

Restricting to $t > 0$ gives the marked series, with families

t	$k=0$	$k=1$
$6r+6$	$j' \ge t-1$	$j' = t$
$6r+1$	$j' = t-1$	$j' \ge t$
$6r+3$	$j' = t-1$	$j' = t$
$6r+4$	$j' = t-1$	$j' = t$

$r \ge 0.$ (114)

Residue lifting supports a nonprincipal Gauss sum exactly at valuation $t-1$; a principal one has its negative boundary there and value $Q^t - Q^{t-1}$ above. In (110), use exponents t at $k = 0$ and $t-1$ at $k = 1$.

Insert $j' = j + 6m$. Increasing r and first allowed m by one multiplies by R; further m increments multiply by V. The strict tails are

$$\frac{R(1-Q^{-1})}{(1-R)(1-V)}, \qquad -\frac{\eta(p)(Q-1)Q^{-s-w}V^{\mathbf{1}_{j\le 1}}}{(1-R)(1-V)}.$$

The second starts at $m = r+1$ for $j \le 1$, otherwise $m = r$. Equality cases give J_j by (6): e.g. $j = 2$ from $(t,k,m) = (3,0,0)$ and $j = 3$ from $(3,1,0), (4,0,0)$. The restored cube phase is $\overline{\gamma_3(p)}$, retaining parity through $\gamma_3(p)^2 = \chi_p(-1)$.

For $t = 0$, $k = 0$ gives $(1-V)^{-1}$; $k > 0$ requires $j = m = 0$ and gives $W^\circ/(1-W^\circ)$. Thus (112) holds in every valuation, including collisions. Extracting $(1-V)^{-1}, (1-W)^{-1}, 1-D$ gives

$$\mathcal{F}_{\eta,u}(s,w,z) = \frac{\zeta_K^S(6z)L_K^S(w,\psi_u)}{L_K^S(s,\eta\overline{\psi_u})}H_{\eta,u}(s,w,z).$$

Together with Section 6.1 this proves (105). □

6.3 Interfaces retained for later chapters The retained contracts are: (29), arbitrary squarefree frozen f at cost q_f; (45), exact Möbius coefficients, common profile and power gap; (67), two factors with common character, natural zeros, common puncture and row-fixed moving radical paid in $m+q$. Identities (8)–(15), (47) preserve collision masks; (105) retains its quotient and all valuation factors.

Selected-prime, merged inverse–plain, relative-character and incumbent-dependent coefficients require their own contracts. In particular, (40) is only the stated-scale theta energy, not an inverse-times-prime or plain-fourth-times-prime moment.

7 Marked completed energy and the incumbent moments

Using ğ3.1's reflection and the finite identities (8)–(15), (47), we prove multi-slot completed energy, inverse-times-prime and natural marked plain moments. Original MF equation labels are retained; identical foundational formulas are aliases. Here β is the actual Hecke zero supremum, and every prime estimate assumes $2\beta-1\le\kappa$.

The original and refined bootstrap ranges remain distinct. Every Z^{ϵ} means a loss chosen before the scale. Rows are Eisenstein elements with all six units and arbitrary valuations.

7.1 The completed reflection with product marks Whole-index marks change conductors at vanishing local frequency. Retain compatible branches until row and mark phases separate, then apply the quadratic–cubic sieve.

7.1.1 The completed family and fixed conventions Use ğ1.2's arithmetic conventions, $q_a=|a|^2$, $Z\ge 2$, and fixed S containing primes above 6. Primary generators represent ideals prime to λ. In particular,

$$\chi_p(A)^0=1_{p\nmid A}. \qquad \text{(MF2.1)}$$

Exponent reduction never deletes this mask.

Fix finitely many finite-ray ν, zero outside primary S-units. A squarefree good label f and squarefree puncture r_ρ may vary between invocations but are fixed within a row norm:

$$\rho(A)=1_{(A,r_\rho)=1}, \qquad \Psi_k(A)=\nu(A)\rho(A)\chi_A(k)\chi_A(f)^4. \qquad \text{(MF2.2)}$$

The puncture may depend on frozen outer labels, but neither the current row nor a subsequently averaged f.

For a fixed number of pairwise disjoint prime lists $\mathcal{P}_i$, with $q_p\asymp Z^{z_i}$ on the i-th list and $|a_i(p)|\le 1$, put

$$d(A)=\sum_{(p_i)\in\prod_i\mathcal{P}_i}\prod_i a_i(p_i)\,1_{\prod_i p_i\mid A}, \qquad z=\sum_i z_i. \qquad \text{(MF2.3)}$$

The completed sum at length X is

$$\mathcal{T}_d(X;k;V)=\sum_{\substack{n\ \mathrm{sf},\,b\\(nb,S)=1}}\frac{\overline{\alpha(n)}\gamma_2(n)\Psi_k(n)\,\overline{\alpha(b)}^{\,3}\Psi_k(b)^3}{\sqrt{q_n}\,q_b}\,d(nb^3)V(q_nq_b^3/X). \qquad \text{(MF2.4)}$$

Primal n,b may share primes. Tests are annular smooth or rapidly decreasing with all Euler derivatives at both endpoints; include the central conversion factor $y^{1/2}$ in V.

A common smooth profile in completed norm and individual prime norms, on fixed logarithmic prime boxes, is allowed inside the tuple sum. Logarithmic Fourier inversion reduces it to an integrable common family of V; bounds use finitely many weighted Euler seminorms, uniformly in lists, bounded coefficients, k,f,r_ρ and bounded logarithmic lengths. Heights cost fixed polynomials. Product coefficients are required during assignment to extracted factors; afterwards surviving tuple coefficients may be arbitrary bounded functions independent of row and dual indices.

Classical theta and sieve inputs are exactly ğ1.1's unconditional results; moving-conductor and marked reductions follow below.

7.1.2 Exact reflection, with the local zeros retained The finite theta source

With $\check{e}(z) = \exp(2\pi i(z+\bar{z}))$ and $e(z) = \check{e}(z/\lambda)$, the theta input [DR, (5.4), (5.7), (5.9), (5.12)–(5.17), Appendix A] gives

$$\Gamma_2 = \langle \mathrm{SL}_2(\mathbb{Z}), \Gamma(3)\rangle, \qquad \theta(gw) = \kappa(g)\theta(w), \qquad \kappa\begin{pmatrix} a & b \\ c & d\end{pmatrix} = \left(\frac{c}{a}\right)_3 \quad (g \in \Gamma(3),\ c \neq 0). \tag{MF2.5}$$

The multiplier is one on $\mathrm{SL}_2(\mathbb{Z})$ and on the $c = 0$ elements of $\Gamma(3)$. Let

$$L(t) = \begin{pmatrix} 1 & 0 \\ t & 1\end{pmatrix}, \qquad \Theta_j(w) = \theta(L(t_j)w), \qquad (t_0, t_1, t_2) = (0, \omega^2, \omega).$$

Translations by $\mathbb{Z} + 3\mathcal{O}$ have multiplier one, so $L(\pm\lambda)$ give Θ_1, Θ_2. For $u_0 = s + t\omega$, $0 \le s, t < 3$, $\theta(\left(\begin{smallmatrix} u_0 & -1 \\ 1 & 0\end{smallmatrix}\right) w) = \Theta_t(w)$.

For each of these three sources define d_H by

$$\overline{\theta(H(z,v))} = C_H(v) + \sum_{\mu \neq 0} d_H(\mu)\, v K_{1/3}(4\pi|\mu|v)\check{e}(\mu z). \tag{MF2.6}$$

Their support and coefficient bound are

(MF2.7) is the cusp support and bound (21), with $m = h$, $d = d_H$, and $\sqrt{q_b} = |b|$.

The dual unit u and primary ideals n, b retain the full cusp support: n, b may share primes and meet $S \setminus \{\lambda\}$, without primal S-exclusion.

For the precise source check, the ordinary cubic sums

$$\widetilde{g}_3(n) = q_n^{-1/2} \sum_{v \bmod n} \left(\frac{v}{n}\right)_3 \check{e}(v/n)$$

have modulus one for primary squarefree n by finite Gauss and CRT, including the norm-four prime -2 using only its cubic symbol. The three source magnitudes are $3^{h/6+8/3}|b|$, $3^{h/6+3}|b|$, and $9|b|$ at $h = -4$, each within (MF2.7). Valuations uniquely determine n, b. At infinity, $\lambda^3\mu \equiv 1 \pmod 3$ selects

$$d_I(\lambda^{-3}nb^3) = 3^{5/2}|b|\widetilde{g}_3(n), \qquad \widetilde{g}_3(n) = \chi_n(\lambda)^{-2}\gamma_2(n) \quad ((n,S) = 1). \tag{MF2.8}$$

Shared n, b primes remain allowed.

All nonzero element rows use one fixed modulus

Write uniquely $k = uk_S k_{\mathrm{good}}$ with fixed generators at S and primary generators elsewhere. On primary S-units, reciprocity gives

$$\chi_A(k) = \chi_A(uk_S)R(A, k_{\mathrm{good}}) \prod_{p \mid k_{\mathrm{good}}} \chi_p(A)^{v_p(k)}. \tag{MF2.9}$$

At collisions both sides vanish, including $6 \mid v_p(k)$. Fixing u, S-valuations modulo six and the reciprocity ray class of k_{good} puts the first two factors times ν in a fixed finite S-supported family Ψ_0. Supplementary characters have only these finitely many exponents. The fourth-power label adds no sign because $R(A, f)^4 = 1$.

Fix E with support exactly S, divisible by these conductors. Choose L on the same support, $18 \mid L$, $E \mid L$, so

$$\varphi(x) = \begin{cases} \chi_x(\lambda)^2\Psi_0(x), & x \equiv 1 \pmod 3,\ (x, E) = 1, \\ 0, & \text{otherwise} \end{cases}$$

is periodic: $(\lambda/x)_3$ has modulus 9 by the supplementary law; all other characters and zeros have fixed moduli. Evaluate the zero branch first. Set

$$\widehat{\varphi}(h_0) = q_L^{-1} \sum_{x \bmod L} \varphi(x)e(-h_0x/L), \qquad \mathfrak{M} = \lambda^{12}L^4. \tag{MF2.10}$$

Keep every good prime of k, f, r_ρ in a moving local set $\mathcal{P}$, combining exponents modulo six; none enters L. Reflect $\Psi(A) = \Psi_0(A)\prod_{p\in\mathcal{P}} \chi_p(A)^{j_p}$, $0 \le j_p < 6$.

The reflection identity

For $m \not\equiv 0 \pmod 6$, let

$$\tau^{\pm}_{p,m} = q_p^{-1/2} \sum_{y \bmod p} \chi_p(y)^m e(\pm y/p), \qquad |\tau^{\pm}_{p,m}| = 1.$$

A branch chooses $h_0 \bmod L$ and an active subset containing all $j_p \neq 0$. Let r be its product. Reduce $\lambda^2 h_0/L = a_F/c_F$, making the numerator primary if $\lambda \mid c_F$, otherwise the denominator. Put $c = c_F r$ and, modulo active p, define

$$\sigma_p = \lambda^2 c/p, \qquad \epsilon_p = -\big((\lambda^3 c/p)\sigma_p\big)^{-1} = -\lambda^{-5}(c/p)^{-2},$$

$$B_{p,j}(x) = \begin{cases} \chi_p(x)^{-j-2}, & j \neq 0,4, \\ q_p^{-1/2}(-1+q_p 1_{p\mid x}), & j=4, \\ q_p^{-1/2}\chi_p(x)^{-2}, & j=0, \end{cases} \qquad \omega_{p,j} = \begin{cases} \tau^-_{p,j}\tau^+_{p,j+2}\chi_p(\epsilon_p)^{-j-2}, & j \neq 0,4, \\ \tau^-_{p,4}, & j=4, \\ -\tau^+_{p,2}\chi_p(\epsilon_p)^{-2}, & j=0. \end{cases} \tag{MF2.11}$$

Indices on τ are reduced modulo six.

Define

$$K_* = \frac{(2\pi)^4}{27}, \qquad \mathcal{R}(t) = \prod_{\pm} \frac{\Gamma(1+t\pm 1/6)}{\Gamma(1-t\pm 1/6)}, \qquad V^\sharp(x) = \frac{1}{2\pi i}\int_{(\sigma)} \widehat{V}(-t)\mathcal{R}(t)(K_* x)^{-t}\,dt, \quad \sigma \geq 0. \tag{MF2.12}$$

The integral is independent of $\sigma \geq 0$. Let $\mathcal{T}_1(X;\Psi;V)$ be (MF2.4) with no mark and character Ψ. Each branch's source d, additive character ϑ and unit κ_F depend only on $h_0, r \bmod \mathfrak{M}^2$.

(MF2.13) is (18) with scalar (20), under $\mathcal{T}_1 = \mathcal{C}_V(X;\Psi)$, $\widehat{\phi} = \widehat{\varphi}$, $c = c_F r$, and the local factors (MF2.11). The scalars agree since $\tau^-_{p,j} = \chi_p(-1)^j\gamma_j(p)$ and $\tau^+_{p,j} = \gamma_j(p)$.

Dual sums converge absolutely, $|\zeta| \leq 1/81$, with $O_L(2^{|\mathcal{P}|})$ branches, or $O_L(4^{|\mathcal{P}|})$ after splitting $j=4$. Choices are simultaneous: adding an inactive $j=0$ prime multiplies by $1-q_p^{-1}$ while preserving conductor, source, additive character, other phases and kernel.

Conductor and phase proof. The fixed-modulus construction of ğ3.1 applies; we retain its extra explicit sector checks here. Its local Fourier identity is

(MF2.14) is (25), with $q = q_p$, $j = j_p$ and $\tau^-_{\mathfrak{p},j} = \tau^-_{p,j}$.

Hence the masked theta function is

$$F(z,v) = \sum_{h_0,(h_p)} \widehat{\varphi}(h_0) \prod_p C_{p,j_p}(h_p)\,\overline{\theta(z+z_h,v)}, \qquad z_h = \lambda^2\left(h_0/L + \sum_p h_p/p\right). \tag{MF2.15}$$

At infinity $\lambda^3\mu = nb^3$, the mask first deletes $(nb,E) > 1$, leaving multiplier $\chi_n(\lambda)^2\Psi(n)\Psi(b)^3$, also at shared moving primes. Thus (MF2.8) identifies the completed direct series.

Fix h_0 and the active frequencies. Then

$$z_h = a/c, \qquad c = c_F r, \qquad a = a_F r + \lambda^2 c_F \sum_{p\mid r}(r/p)h_p. \tag{MF2.16}$$

The numerator is a unit at c_F and at $p \mid r$ (there $a \equiv \sigma_p h_p$), so the fraction is reduced even for nonunit h_0. Lift $h_p \equiv 0 \pmod{\mathfrak{M}^2}$ while retaining its p-class and fix $r \bmod \mathfrak{M}^2$; then $a \bmod \mathfrak{M}c_F$ is fixed. Choose by CRT

$$\begin{aligned} \delta' &\equiv a^{-1} \pmod{\mathfrak{M}c_F} \text{ at primes of } c_F, \\ \delta' &\equiv 0 \pmod{\mathfrak{M}} \text{ at other primes of } \mathfrak{M}, \\ \delta' &\equiv a^{-1} \pmod{p}\ (p \mid r). \end{aligned}$$

Put $b = (a\delta' - 1)/c$, $g = \left(\begin{smallmatrix} a & b \\ c & \delta' \end{smallmatrix}\right)$. At c_F, $b \equiv 0 \pmod{\mathfrak{M}}$; elsewhere in $\mathfrak{M}$, $b \equiv -c^{-1}$. Thus fixed-prime matrix residues are frequency-independent. Periodic lifts and CRT-modulus shifts of δ' avoid zero entries if needed for symbols.

If $v_\lambda(c) \geq 2$, take $H = I$: $g \in \Gamma(3)$ and $\kappa(g) = (c_F/a)_3(a/r)_3$. If $v_\lambda(c) = 1$, take $H = L(u_0)$, $u_0 \in \{\pm\lambda\}$, $u_0 \equiv c \pmod 3$, and $A = a - u_0 b$. Then $G = gH^{-1} \in \Gamma(3)$, a, A are coprime primary, and

$$a(c - u_0\delta') = cA - u_0, \qquad bc \equiv -1 \pmod a.$$

Cubic reciprocity gives

$$\kappa(G) = \left(\frac{-u_0}{A}\right)_3 \left(\frac{a}{A}\right)_3^{-1} = \left(\frac{-u_0}{A}\right)_3 \left(\frac{c_F/u_0}{a}\right)_3 \left(\frac{a}{r}\right)_3.$$

If $\lambda \nmid c$, choose $u_0 = s + t\omega \equiv a \pmod 3$, $0 \le s, t < 3$, and $H = \left(\begin{smallmatrix} u_0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. Then

$$G = gH^{-1} = \begin{pmatrix} -b & a + bu_0 \\ -\delta' & c + \delta' u_0 \end{pmatrix} \in \Gamma(3).$$

Both $-b, c$ are primary and $-bc = 1 - a\delta' \equiv 1 \pmod 9$. For $\delta' = u\lambda^h \delta_0$, supplementary laws handle unit/λ factors modulo 9; reciprocity gives numerator $1 \bmod \delta_0$, so $(\delta'/(-bc))_3 = 1$. Hence $\kappa(G) = (a/c_F)_3 (a/r)_3$. In all cases,

$$\kappa(G) = \kappa_F \prod_{p \mid r} \chi_p(a)^2, \qquad \text{(MF2.17)}$$

The sector fixes κ_F, d_H, without requiring unit-frequency support for $\widehat{\varphi}$.
For $x = \lambda^4 \mu$ and $D_F = \lambda^3 c_F$, additive CRT gives

$$\check{e}(-\delta'\mu/c) = \vartheta(x) \prod_{p \mid r} e(\epsilon_p h_p^{-1} x/p), \qquad \vartheta(x) = e(-\delta'_F r^{-1} x/D_F). \qquad \text{(MF2.18)}$$

Fixed congruences determine $\delta'_F \bmod D_F$, and the sector fixes r^{-1}. Conjugating (MF2.17) contributes $\overline{\kappa_F} \prod_p \chi_p(\sigma_p)^{-2} \chi_p(h_p)^{-2}$. Multiply (MF2.14) and set $y = h_p^{-1}$: Gauss exponent $j + 2$, or $-1 + q_p 1_{p|x}$ at $j = 4$, gives exactly $\omega_{p,j} B_{p,j}(x)$ with zeros. Inactive and absent zero-exponent primes have identical active radical, proving simultaneous compatibility.

Proof of the Mellin normalization and continuation. Put

$$D(s, \Psi) = \sum_{n \text{ sf}} \overline{\alpha(n)} \gamma_2(n) \Psi(n) q_n^{-s}, \quad L_0(s, \Psi) = \sum_b \overline{\alpha(b)}^3 \Psi(b)^3 q_b^{-3s+1/2}, \quad T = L_0 D.$$

These primary S-unit sums converge for $\Re s > 1$. Put $J(s) = \int_0^\infty \partial_{\bar{z}} F(0, v) v^{2s-1}\, dv$; the Bessel integral is

$$\int_0^\infty K_{1/3}(y) y^{w-1}\, dy = 2^{w-2} \Gamma((w - 1/3)/2) \Gamma((w + 1/3)/2).$$

Using $\partial_{\bar{z}} \check{e}(\mu z) = 2\pi i \bar{\mu} \check{e}(\mu z)$ and $\overline{\alpha(\lambda^{-3})} = -i$,
(MF2.19) is the direct Mellin identity, with $\Theta_\Psi = F$, $\mathcal{D}_\Psi(s) = T(s, \Psi)$ and left side $J(s)$.
The inverse cusp action is (26), with $\delta = \delta'$.
At $z = 0$ the chain rule gives $-\overline{\kappa(G)} (cv)^{-2} \partial_z$ at $(-\delta'/c, 1/(q_c v))$, deleting every cusp constant. Substitute $v \mapsto 1/(q_c v)$ and the Bessel integral: for $\Re s < 0$, $J(s)$ equals

$$-\frac{i}{4} (2\pi)^{2s-2} \overline{\alpha(c)}^2 q_c^{1-2s} \overline{\kappa(G)} \prod_{\pm} \Gamma(3/2 - s \pm 1/6) \sum_{\mu \neq 0} d_H(\mu) \alpha(\mu) \check{e}(-\delta'\mu/c) q_\mu^{-(1-s)}. \qquad \text{(MF2.20)}$$

By (MF2.7), this converges absolutely. Direct and inverse-cusp expansions give exponential endpoint decay; log-height integration by parts makes J entire and rapidly decreasing on fixed strips. Equation (MF2.19) makes T entire, bounded right of 1 and polynomial left of 0 via (MF2.20). Its preliminary intervening bound is $(1 + |s|)^C e^{\pi |\Im s|}$. Apply the maximum principle to $T(s)(B + s)^{-C'} e^{\epsilon s^2}$, $-B$ outside the strip, then let $\epsilon \downarrow 0$, obtaining polynomial strip growth.
Mellin inversion gives $(2\pi i)^{-1} \int_{(a)} \widehat{V}(s - 1/2) X^{s-1/2} T(s, \Psi)\, ds$, $a > 1$. Shift left of zero, use (MF2.20)/(MF2.19) and set $t = 1/2 - s$. The gamma quotient is $\mathcal{R}(t)$; the remaining factor is

$$-\frac{i}{81} \overline{\alpha(c)}^2 q_\mu^{-1/2} (K_* q_\mu X / q_c^2)^{-t}.$$

Absolute convergence and Mellin decay justify interchanges and shifts. The finite factors give (MF2.13), also for the rapidly decreasing tests admitted here.

7.1.3 Common profiles and whole-series tails For $A, B \ge 0$, with B an integer, set

$$\mathcal{M}_{A,B}(V) = \sup_{-A\le\eta\le 1/4} \int_{\mathbb{R}} (1+|t|)^B |\widehat{V}(\eta+it)|\,dt.$$

The first negative numerator pole is $-5/6$ and $|\mathcal{R}(\sigma+it)| \ll_A (1+|t|)^{4\sigma}$ for $-1/4 \le \sigma \le A$. Shifts to $-1/4, 0, A$ give

$$|(x\partial_x)^j V^\sharp(x)| \ll_{A,j} \min\{x^{1/4}, (1+x)^{-A}\}\mathcal{M}_{A,\lceil 4A\rceil+j+2}(V). \qquad \text{(MF2.21)}$$

These are finite seminorms: $B+2$ integrations by parts bound $\mathcal{M}_{A,B}$ by

$$\ll_{A,B} \int_0^\infty (1 + x^{-A} + x^{1/4})\big(|V(x)| + |(x\partial_x)^{B+2}V(x)|\big)\frac{dx}{x}.$$

Replacing V by $x^{it_0}V$ costs at most $(1+|t_0|)^B$.

On a fixed compact log box let $F_Y(\mathbf{y}) = W(\mathbf{y})V^\sharp(Y\prod_i y_i^{a_i})$, Y fixed per block. Equation (MF2.21) bounds each fixed derivative by

$$m_A(Y) = \min\{Y^{1/4}, (1+Y)^{-A}\}$$

times fixed seminorms. Logarithmic Fourier integration by parts gives the same weighted L^1 scalar. For a linear row vector, set $F_Y = m_A(Y)\widetilde{F}_Y$ and apply one common Fourier inversion and Minkowski:

$$\|U(F_Y)\|_{\ell^2(I)} \le m_A(Y) \int |\widehat{\widetilde{F}_Y}(\mathbf{t})|\,\|B(\mathbf{t})\|_{\ell^2(I)}\,\frac{d\mathbf{t}}{(2\pi)^d}. \qquad \text{(MF2.22)}$$

Then enlarge the positive row set using the separated definition of B on added rows, without asserting kernel bounds there. Squaring gives $m_A(Y)^2$, in particular small-argument energy gain $Y^{1/2}$. Row-dependent Fourier measures would invalidate this step.

For $\Lambda = \mathcal{O}$ or $\lambda^{-4}\mathcal{O}$, $|K(x)| \le C_A x^{-A}$ $(x \ge 1,\ A > 1)$ gives

(MF2.23) is the shell bound (44), valid for both lattices with $a > 0$, $U \ge 1$.

Its $O(1 + 2^j U/a)$ shell count controls the whole discarded series independently of retained-index bounds.

7.1.4 The quadratic–cubic norm, including both collision masks Imported sieves and their domains

For squarefree primary good ideals, use the quadratic sieve

(MF2.24) is (27) under $(R, n, K, N) = (a, b, U, V)$, with the same row-independent coefficients.

For clarity, the primitive ideal character in ğ3.2's family check is

$$\vartheta_a((x)) = \epsilon_\lambda(x)^{e(a)}(x/a)_2, \qquad e(a) = (q_a - 1)/2 \bmod 2,$$

Here ϵ_λ is the nontrivial character modulo λ. Primary adjustment uses square powers of ω; its sign contributes $\epsilon_\lambda^{e(a)}$. CRT gives conductor $a\lambda^{e(a)}$, trivial infinite type, omitting λ when $e(a) = 0$. Fixed primary square classes modulo four make $e(a)$ equal; coprime product conductors are exactly products, and reciprocity is a fixed-ray bicharacter. These are [GL, Definition 1]'s three conditions; apply its theorem and transpose duality.

Heath-Brown's cubic sieve gives

(MF2.25) is (36), since $(b/a)_3 = \chi_a(b)^2$, with squarefree primary ideals, lengths U, V and row-independent c_b.

Neither sieve requires squarefree rational norms or exclusion of rational prime factors. Conjugation/reciprocity changes cubic orientation. Fixed row restrictions decrease the positive norm; fixed column restrictions are zero coefficients. Resolve mixed collision masks first.

Quadratic reduction for a squarefree index times a cube

Use ğ3.3's variables: squarefree good $q_k \ll K$, squarefree primary $q_n \asymp N$, primary $q_b \asymp B$, allowing dual $S \setminus \{\lambda\}$ primes and arbitrary k-independent $\beta(n, b)$ with fixed supports.

Write uniquely

$$b = gt^2, \qquad c = (n, g), \qquad n = cm, \qquad g = ch,$$

Here g is squarefree; c, m, h are pairwise coprime squarefree, allowing $(g, t) > 1$. On $q_c \asymp C, q_g \asymp G, q_t \asymp T$, $B \asymp GT^2$, and

$$\chi_k(nb)^3 = \chi_k(mh)^3\, 1_{(k,ct)=1}. \qquad \text{(MF2.26)}$$

Apply the quadratic reduction proved in ğ3.3: Cauchy in t, retain $(k,t)=1$, expand $1_{(k,c)=1}$ completely, and set $k=rk'$, $c=rc'$. Keep $(k',r)=(c',r)=1$; $(r,t)>1$ vanishes. No (k',c')-mask is restored termwise.

The same fixed-S split, squarefree product grouping and Cauchy over $O(C/q_r)$ choices of c' give, with $\mathcal{D}_r(t)$ retaining the original support,

(MF2.27) is the positive quadratic bound (34), with $Q_{C,G,T} = \mathcal{Q}_{C,G,T}$ and final sum restricted to $\mathcal{D}_r(t)$. All its row/column independence and common-prime conventions are unchanged.

This positive form preserves tuple independence. For $|\beta| \le 1$, the triple count is $O(NG/(q_rC))$; on $q_r \asymp r_0$, $O(r_0)$ choices of r and $O(T)$ choices of t give

$$\ll (KNB)^\epsilon \frac{NB}{r_0}\left(\frac{K}{r_0}+\frac{NG}{C^2}\right) \ll (KNB)^\epsilon NB(K+NB).$$

Triangle over C, G, T dyads yields

(MF2.28) is (35), with the same K, N, B, supports and $|\beta| \le 1$.

Subunit quotients are empty; bounded ranges, including the unit, cost constants.

Hybrid norm

Let P be squarefree good with $q_P \asymp L$. Suppose $|a(P)|, |\beta(n,b)| \le 1$, with a independent of k, n, b and β independent of k, P. Then

$$\mathcal{H} = \sum_k \left| \sum_P \frac{a(P)}{\sqrt{q_P}} 1_{(P,k)=1} \sum_{n,b} \beta(n,b)\chi_k(nb)^3 \chi_P(n)^{-2} 1_{(P,b)=1} \right|^2 \qquad \text{(MF2.29)}$$
$$\ll_\epsilon (KNBL)^\epsilon (K+NB)B\{N+L+(NL)^{2/3}\}.$$

Fixed supports and ray sectors may be imposed while preserving both independence conditions.

Expand $1_{(P,k)=1}$ completely over $D \mid (P,k)$ and apply divisor Cauchy. Put $P = DP'$, $k = Dk'$ and

$$a_D(P') = a(DP')1_{(D,P')=1}, \qquad \beta_D(n,b) = \beta(n,b)\chi_D(nb)^3\chi_D(n)^{-2}1_{(D,b)=1}.$$

Energy gains q_D^{-1}, with lengths $K/q_D, L/q_D$. Do not restore $(P',k')=1$. Apply (MF2.27) to k' with column coefficient

$$\beta_D(n,b)\sum_{P'} a_D(P') q_{P'}^{-1/2} \chi_{P'}(n)^{-2} 1_{(P',b)=1}$$

as its row-independent coefficient. In its notation the positive output for one C, G, T-block is bounded by

$$(KNBL)^\epsilon \sum_D \frac{1}{q_D} T \sum_{q_t \asymp T} \sum_r \left(\frac{K}{q_Dq_r}+\frac{NG}{C^2}\right)\frac{C}{q_r} \sum_{c',m,h} |\beta_D(rc'm, rc'ht^2)|^2$$
$$\cdot \left| \sum_{P'} \frac{a_{D,r}(P')}{\sqrt{q_{P'}}} \chi_{P'}(c'm)^{-2} 1_{(P',ht)=1} \right|^2, \qquad a_{D,r}(P') = a_D(P')\chi_{P'}(r)^{-2}. \qquad \text{(MF2.30)}$$

The identity used to obtain it, with zeros included, is

$$\chi_{P'}(rc'm)^{-2}1_{(P',rc'ht^2)=1} = \chi_{P'}(r)^{-2}\chi_{P'}(c'm)^{-2}1_{(P',ht)=1}.$$

The character supplies the zeros at r, c', m.

Only in this positive sum discard $|\beta_D|^2 \le 1$. For fixed h, t, D, r, coefficients are independent of squarefree primary $j = c'm = n/r$, with mass $\sum_{P'} q_{P'}^{-1}|a_{D,r}(P')|^2 1_{(P',ht)=1} \ll (KNBL)^\epsilon$. Group by j, retaining its S-parts and norm $O(N/q_r)$; (MF2.25) and $O(G/C)$ values of h give on $q_D \asymp D_0, q_r \asymp r_0$

$$\ll (KNBL)^\epsilon \frac{G}{C}\left\{\frac{N}{r_0}+\frac{L}{D_0}+\left(\frac{NL}{r_0D_0}\right)^{2/3}\right\}.$$

The t-sum and C/q_r give $T^2(C/r_0)(G/C) \asymp B/r_0$; $O(r_0)$ values of r cancel its denominator and $\sum_{q_D \asymp D_0} q_D^{-1} \ll 1$. Replace D_0, r_0 by one in increasing sieve factors, use $NG/C^2 \ll NB$ and sum dyads. This proves (MF2.29) with all original masks.

7.1.5 Whole-index marks and separation of their phases For marks $\mathcal{Q}$ outside the base local set, expand $\prod_q 1_{q|A} = \prod_q (1 - \chi_q(A)^0)$ and apply compatible (MF2.13). Each inactive contribution is

$$1 - (1 - q_q^{-1}) = q_q^{-1}.$$

The subtracted active branch contributes $q_q^{-1/2}\chi_q(\lambda^4\mu)^{-2}$, scalar $-\omega_{q,0}$. Thus

$$\begin{aligned}\mathcal{T}_{\mathcal{Q}}(X;\Psi;V) &= \sum_{\mathcal{B}} \zeta_{\mathcal{Q}} \sum_{\mu\neq 0} \frac{d(\mu)\alpha(\mu)\vartheta(\lambda^4\mu)}{\sqrt{q_\mu}} \prod_{p\in\mathcal{A}_0} B_{p,j_p}(\lambda^4\mu) \prod_{q\in A_{\mathcal{Q}}} q_q^{-1/2}\chi_q(\lambda^4\mu)^{-2} V^\sharp(q_\mu X/q_c^2),\\ \zeta_{\mathcal{Q}} &= -\frac{i}{81}\overline{\alpha(c)}^2 \widehat{\varphi}(h_0)\overline{\kappa_F} \prod_{p\in\mathcal{I}_0}(1-q_p^{-1}) \prod_{q\in I_{\mathcal{Q}}} q_q^{-1} \prod_{p\in\mathcal{A}_0} \chi_p(\sigma_p)^{-2}\omega_{p,j_p} \prod_{q\in A_{\mathcal{Q}}} \chi_q(\sigma_q)^{-2}(-\omega_{q,0}).\end{aligned} \tag{MF2.31}$$

Here base sets are $\mathcal{A}_0, \mathcal{I}_0$, marks split as $\mathcal{Q} = A_{\mathcal{Q}} \sqcup I_{\mathcal{Q}}$, and $c = c_F \prod_{p\in\mathcal{A}_0\cup A_{\mathcal{Q}}} p$. Inactive marks are absent from conductor and kernel. A mark already in the base set conflicts with its primal zero, so delete that term first.

The phase at an active prime has the exact form

$$\chi_p(\sigma_p)^{-2}\omega_{p,j} = \xi_{p,j}\chi_p(c/p)^{2j+2}, \qquad |\xi_{p,j}| = 1, \tag{MF2.32}$$

Here $\xi_{p,j}$ depends only on p, j. Substitute $\epsilon_p = -\lambda^{-5}(c/p)^{-2}$ in (MF2.11), using $2j+2 \equiv -2 \pmod 6$ at $j = 4$. Distinct active primes therefore give

$$\chi_p(q)^{2j_p+2}\chi_q(p)^{2j_q+2} = \left(\frac{q}{p}\right)_3^{j_p+j_q+2}. \tag{MF2.33}$$

Residual row exponent one and mark exponent zero give phase one. Exponent four would retain a cubic row interaction, so freeze every nonresidual base prime first.

Let R be residual row, P active mark product, F_{act} frozen active product. Separately fix $R, P \bmod \mathfrak{M}^2$, determining d, ϑ, κ_F without pair restrictions. Equation (MF2.33) assigns row–row/frozen-row phases to a row scalar and mark–mark/frozen-mark phases to a tuple scalar; $\overline{\alpha(c)}^2$ separates. Primal collisions retain $1_{(P,R)=1}$.

After fixing the dual unit, ramified power, and extracted frozen factors, the remaining dual part is nb^3. Its moving columns are exactly

$$\chi_R(nb^3)^3 = \chi_R(nb)^3, \qquad \chi_P(nb^3)^{-2} = \chi_P(n)^{-2}1_{(P,b)=1}. \tag{MF2.34}$$

Source, additive character and frozen columns depend only on the full extracted dual index. Frozen collisions are fixed row, tuple or dual restrictions. Common separation therefore gives tuple coefficients independent of R, n, b, dual coefficients independent of R, P, and exactly (MF2.29)'s mixed masks.

7.1.6 Reflected energy and its two consequences Block parameters and estimate

Fix $q_f \asymp Z^V$, $Q = \log_Z q_{r_\rho}$, $X = Z^{N_*}$, and rows $0 < q_k \ll Z^M$. Split the ideal of k as

$$(k) = k_{\mathrm{pow}}k_{\mathrm{sup}}R.$$

The maximal powerful part contains valuations ≥ 2; k_{sup} contains valuation-one primes in $S \cup \mathrm{supp}(fr_\rho)$; residual R is squarefree, with disjoint supports. Freeze both actual parts, unit and sectors. On dyads $q_{k_{\mathrm{pow}}} \asymp Z^O$, $q_R \asymp Z^H$, retain the actual row cutoff:

$$0 \leq H \leq M - O + O(1/\log Z). \tag{MF2.35}$$

Equality need not hold.

Choose the active zero-exponent base primes and split every $j = 4$ factor. For a large $j = 4$ term use the disjoint assignment

$$1_{p|nb^3} = 1_{p|n} + 1_{p\nmid n}1_{p|b}.$$

Let A_0 be active non-slot radical length outside R; S_0 counts small $j = 4$ and active base-zero factors; N_0, B_0 count large factors assigned to squarefree/cube indices. Let $z_a = \sum_{i\,\mathrm{active}} z_i \leq z$, so $q_P \asymp Z^{z_a}$. All are nonnegative.

For actual frozen products $F_{\mathrm{act}}, N_F, B_F$ of logarithmic norms A_0, N_0, B_0, write a dual index as

$$\mu = u\lambda^h N_F n (B_F b)^3, \quad h \ge -4, \quad q_n \asymp Z^v, \quad q_b \asymp Z^{\ell_b}, \quad e_\lambda = h \log_Z 3,$$

Here $v, \ell_b \ge 0$; retain source restrictions and allowed S-parts. Define

$$T_d = 2H + 2A_0 + 2z_a - N_* - N_0 - 3B_0, \qquad u_h = \min\{v, z_a, (v+z_a)/3\}. \tag{MF2.36}$$

For any fixed $\tau_{\mathrm{ref}} > 0$, all dyads outside

$$v + 3\ell_b + e_\lambda \le T_d + \tau_{\mathrm{ref}} \tag{MF2.37}$$

have arbitrarily small total size. A retained dyad, including the sum over active tuples, satisfies

$$\begin{aligned} E_{\mathrm{ref}} = {} & \frac{O}{2} + \max(H, v+\ell_b) - S_0 - B_0 + z_a - u_h - \ell_b - \frac{2}{3}e_\lambda \\ & - \frac{1}{2}(T_d - v - 3\ell_b - e_\lambda)_+. \end{aligned} \tag{MF2.38}$$

For each frozen pair, energy over R is $\ll Z^{E_{\mathrm{ref}} - O/2 + \epsilon}$; summing the O-dyad and local choices gives $Z^{E_{\mathrm{ref}}+\epsilon}$. Uniformity is as in ğ7.1.1. Row restrictions are independent of active tuples/dual indices except $1_{(P,R)=1}$; character, puncture and original product mark are independent of subsequently averaged f. Thresholds depend only on bounded log ranges, profiles, fixed moduli, $\epsilon, \tau_{\mathrm{ref}}$; negative $e_\lambda = O(1/\log Z)$.

Proof. Each inactive slot has mass $\sum_{p \in \mathcal{P}_i} q_p^{-1} \ll Z^\epsilon$, including bounded lists. Freeze it by row-space triangle, making collisions fixed row restrictions; active tuples stay inside the norm.

Equation (MF2.7), before the local factors, gives

$$\frac{|d(\mu)|}{\sqrt{q_\mu}} \le 27\, q_{\lambda^h}^{-1/3} q_{N_F n}^{-1/2} q_{B_F b}^{-1}.$$

Small $j = 4$ and active zero factors contribute $q_p^{-1/2}$. Large $j = 4$ contributes $q_p^{1/2}$: squarefree assignment cancels its root denominator, cube assignment leaves $q_p^{-1/2}$. Shared N_F, b cube occurrences stay in b. Hence the common coefficient per tuple has exponent

$$-v/2 - \ell_b - e_\lambda/3 - (S_0 + B_0 + z_a)/2. \tag{MF2.39}$$

The denominator is $c = c_F F_{\mathrm{act}} RP$. With actual annular coordinates

$$y_R = q_R/Z^H, \quad y_n = q_n/Z^v, \quad y_b = q_b/Z^{\ell_b}, \quad y_i = q_{p_i}/Z^{z_i},$$

the kernel argument is exactly

$$\frac{q_\mu X}{q_c^2} = Y\, y_n y_b^3 y_R^{-2} \prod_{i \text{ active}} y_i^{-2}, \qquad Y = q_{c_F}^{-2} Z^{v + 3\ell_b + e_\lambda - T_d}. \tag{MF2.40}$$

This remains exact at shared primes. The annular monomial is bounded above/below, so (MF2.22) extracts

$$m_{\mathrm{ker}} = Z^{-(T_d - v - 3\ell_b - e_\lambda)_+/4}$$

to be factored from the whole linear row vector, with bounded normalized seminorms.

Before separation remove tails: violating (MF2.37) forces argument $> Z^{\tau_{\mathrm{ref}}/2}$ for large Z. Use $|d(\mu)|/\sqrt{q_\mu} \le 27\, 3^{4/3}$ and local factors $\le q_p^{1/2}$. Rows, tuples, branches and $1 + q_c^2/X$ cost $Z^{B_{\mathrm{tail}}}$, fixed before kernel order. Apply (MF2.23) at $a = X/q_c^2$, $U = Z^{\tau_{\mathrm{ref}}/2}$ with

$$A > 1 + 2(B_{\mathrm{tail}} + D)/\tau_{\mathrm{ref}}.$$

This bounds all discarded μ by $O(Z^{-D})$ times a finite seminorm and fixed height power.

Separate each retained (MF2.40) once, then enlarge the positive row norm to $q_R \ll Z^H$. Phase separation gives (MF2.29) with

$$K = Z^H, \qquad N = Z^v, \qquad B = Z^{\ell_b}, \qquad L = Z^{z_a}.$$

Leave $q_P^{-1/2}$ inside that norm. The outside squared coefficient is $Z^{-v - 2\ell_b - 2e_\lambda/3 - S_0 - B_0 + \epsilon}$. Since

$$\max\{v, z_a, 2(v+z_a)/3\} = v + z_a - u_h,$$

the hybrid norm gives (MF2.38)'s first line without $O/2$; m_{ker}^2 supplies its last term.

The powerful count proved after (42) is $O(Z^{O/2})$; the supported part divides $\mathrm{rad}(Sfr_\rho)$, giving Z^ϵ choices, as do local branches. Divisor Cauchy and the positive frozen-part sum add $O/2+\epsilon$ once; active tuples are already included. This proves (MF2.38).

General label and puncture width

At an active non-slot prime, the coefficient of its log-norm in $2A_0 - N_0 - S_0 - 4B_0$ is

Local choice: j odd or $j=2$; *Coefficient*: 2
Local choice: $j=0$, active; *Coefficient*: 1
Local choice: $j=4$, small or assigned to n; *Coefficient*: 1
Local choice: $j=4$, assigned to b; *Coefficient*: -2
Local choice: $j=0$, inactive; *Coefficient*: 0

Residual primes contribute two. Primewise these coefficients are bounded by valuations of $(q_k^2/q_{k_{\mathrm{pow}}})q_f q_{r_\rho}$: powerful valuation $e \geq 2$ supplies e, simple rows two, label/puncture-only primes one; overlaps increase availability. Thus

$$q_R^2 Z^{2A_0 - N_0 - S_0 - 4B_0} \leq C_S \frac{q_k^2}{q_{k_{\mathrm{pow}}}} q_f q_{r_\rho},$$

and therefore

$$T_d - S_0 - B_0 \leq 2M - O + Q + V - N_* + 2z_a + O(1/\log Z). \qquad \text{(MF2.41)}$$

This remains uniform in actual f, r_ρ when completion gives $N_* = N - 3\ell_h$, without replacing H by $M - O$.

With no slots, $z_a = u_h = 0$. On a retained dyad, (MF2.38) gives

$$E_{\mathrm{ref}} \leq O/2 + \max\{H, T_d - S_0 - B_0\} + O(1/\log Z) + \tau_{\mathrm{ref}}.$$

Drop nonpositive terms for the row branch; for the column branch use $v \leq T_d - 3\ell_b - e_\lambda + \tau_{\mathrm{ref}}$. Apply (MF2.35), (MF2.41), choose small buffer/losses and sum dyads:

$$\sum_{\substack{0<q_k \ll Z^M \\ q_{k_{\mathrm{pow}}} \asymp Z^O}} |\mathcal{T}_1(Z^{N_*}; k; V)|^2 \ll_\epsilon Z^{O/2 + \max\{M-O,\, 2M-O+Q+V-N_*\}+\epsilon}. \qquad \text{(MF2.42)}$$

In particular, with $f = r_\rho = 1$, summing O gives the unmarked completed bound

(MF2.43) is the unmarked bound (33), with $(H, N) = (Z^M, Z^{N_*})$ and equivalent power notation $Z^{\max(M, 2M-N_*)+\epsilon}$, for the tests admitted in ğ7.1.1. The preceding argument extends its annular/Gaussian statement to that test class.

This includes the balanced energy $Z^{1+\epsilon}$ at $M = N_* = 1$.

Short marked energy without label or puncture

When $f = r_\rho = 1$, every active non-slot prime outside R divides the powerful part. Thus

$$2A_0 \leq O + O(1/\log Z), \qquad 0 \leq N_0 \leq A_0.$$

Theorem 7.1 (Short marked completed energy). *If $M + 2z \leq N_*$, then for every $\epsilon > 0$,*

$$\sum_{0<q_k \ll Z^M} |\mathcal{T}_d(Z^{N_*}; k; V)|^2 \ll_\epsilon Z^{M + (N_* + 2z - 2M)_+/4 + \epsilon}. \qquad \text{(MF2.44)}$$

Proof. Use (MF2.38), suppressing $O(1/\log Z)$ and tiny τ_{ref}. Since

$$T_d - H \leq M + 2z - N_* \leq 0,$$

retention gives $\max(H, v + \ell_b) = H + O(\tau_{\mathrm{ref}} + 1/\log Z)$. If $u_h = z_a$, then $E_{\mathrm{ref}} \leq M$ up to this error; otherwise $u_h \geq v/2$. Put $y_d = v + 3\ell_b + e_\lambda$; $\ell_b \geq 0$, $e_\lambda \geq -O(1/\log Z)$ imply

$$u_h + \ell_b + \frac{2}{3}e_\lambda + \frac{1}{2}(T_d - y_d)_+ \geq \frac{1}{4}y_d + \frac{1}{2}(T_d - y_d)_+ - O(1/\log Z) \geq \frac{1}{4}T_d - O(1/\log Z).$$

With $\Delta_H = M - O - H \geq -O(1/\log Z)$, substitution into (MF2.38) gives

$$\begin{aligned} 4(E_{\mathrm{ref}} - M) &\leq N_* + 2z_a - 2M - 2\Delta_H - (2A_0 - N_0 + B_0 + 4S_0) + O(\tau_{\mathrm{ref}} + 1/\log Z) \\ &\leq N_* + 2z - 2M + O(\tau_{\mathrm{ref}} + 1/\log Z). \end{aligned}$$

Restore the row branch, tails and losses to obtain (MF2.44), independently of moments or zero-free conclusions. □

7.2 The inverse and plain moment inductions We prove inverse moments and unconditional zero-slot/canonical common-profile positive-slot plain moments from the finite identities and reflected energy above. Inverse descent terminates at reflection; plain descent is independent of the inverse moment.

7.2.1 Common conventions and analytic bookkeeping Retain ğ7.1's conventions: primary ideal columns avoid S; all element rows and displayed zeros remain. Fixed ray moduli lie in S, moving punctures are separate. Put

$$a_0(n) = \overline{\alpha(n)}\gamma_2(n) \qquad (n \text{ squarefree}).$$

For disjoint prime lists $\mathcal{P}_i$, fixed throughout the current row and label averages, define a **product mark**

(MF3.1) is the product mark (MF2.3) on slot subset I, with lists and coefficients fixed throughout the current row and label averages.

Slot count is fixed; coefficients need not be multiplicative and include individual smooth weights. Product form, beyond a divisor bound, allows

$$\mathbf{1}_{p|D_1\cdots D_h n} = \sum_{j=1}^{h} \mathbf{1}_{p|D_j} \prod_{a<j} \mathbf{1}_{p\nmid D_a} + \mathbf{1}_{p|n} \prod_{a\leq h} \mathbf{1}_{p\nmid D_a} \qquad \text{(MF3.2)}$$

to assign each slot first to one extracted factor or retain it on the residual column. Assigned coefficients are divisor-bounded; survivors retain product form. Allocate the two copies of a squared mark independently.

Common-profile lemma. For finitely many compact normalized-norm profiles, radial Schwartz kernels and fixed-ray Fourier expansions, keep nonsmooth arithmetic restrictions in outer weights. Multiply a coupled $H(y_1, \dots, y_d)$ by larger individual cutoffs equal to one on current supports and use

$$H(y) = (2\pi)^{-d} \int_{\mathbf{R}^d} \widehat{H}(t) \prod_j y_j^{it_j}\, dt, \qquad \int |\widehat{H}(t)|\langle t\rangle^J dt \ll_{J,d} \max_{|\beta|\leq 2r} \|D^\beta H\|_\infty, \quad 2r > J + d, \qquad \text{(MF3.3)}$$

Here $D_j = y_j\partial_{y_j}$, $\langle t\rangle = 1 + |t|$. Define the measure on the ambient box before choosing row or live label; norm relations restrict evaluation points only. Include inverse roots once in the coupled profile; individual column tests get only cutoffs and Fourier powers.

Logarithmic integration by parts with $(1-\Delta)^r$ proves the bound. For $K(A\prod y_j^{e_j})$, Euler derivatives are fixed multiples of $(x\partial_x)^r K(x)$ uniformly in $A > 0$, so fixed derivative orders add no Z power; norm heights cost polynomials. Also,

(MF3.4) is (44) on $\mathcal{O}$, with $(U, A) = (Y, B)$ and the kernel seminorm absorbed in the constant. It holds for $a > 0$, $Y \geq 1$, $B > 1$.

Discard tails on the genuine expression before coprime extension or absolute Fourier integration. Choose a sector-fixed outer ball whose complement forces kernel argument $> Z^\tau$, retaining the full kernel inside. If raw factors and $1+a^{-1}$ cost Z^{B_0}, choose $B > 1+(B_0+T)/\tau$ in (MF3.4) for $O(Z^{-T})$. Polynomial scale bounds fix B_0 before B.

For uniformity choose terminal profile/height orders, then propagate backwards through the bounded-depth acyclic graph using (MF3.3). Normalize discrete counting measures after extracting their Z^b mass once. Cauchy averages child exponents; Minkowski takes their maximum after coefficient mass, so numerical loss grows only with path length. Fix quotient/separation cutoffs and bounded clipping families before the final Z threshold. A nonempty nominal negative-length window has $1 \leq Z^{-n} \leq B$ (fixed upper endpoint), hence clips to length zero by bounded dilation.

Masked primitive Poisson. For primitive zero-extended ψ mod $\mathfrak{m}$ and radial Schwartz Φ, use **(MF3.5)** is (47), under $(H,m,r)=(K,\mathfrak{m},R)$, $\gamma(\psi)=\gamma(\psi;\mathfrak{m})$ and $\widehat{\Phi}=\mathcal{F}\Phi$.

The proof is ğ4.1's complete mask expansion and primitive finite transform; Parseval gives $|\gamma|=1$. For the primitive principal character use modulus one, $\gamma=1$, including $h=0$. Its nonzero frequencies cost $O_\Phi(d(\operatorname{rad} R))$ since $A\sum_{h\neq 0}|\mathcal{F}\Phi(Aq_h)|\ll 1$; restore them with the original outer mask at diagonal cost. Expand fixed rays into characters first, keeping deleted conductor zeros in R.

7.2.2 A canonical marked energy Fix squarefree $\rho(n)=1_{(n,r_\rho)=1}$ independent of row k and label f, with $Q=\log_Z q_{r_\rho}$. Let $q_f\asymp Z^V$ and label-only $0\le w(f)\le Cd(f)^C$. For $F=N+V$, define

$$\mathcal{E}(N,V,M)=Z^{-V}\sum_f w(f)\sum_{0<q_k\ll Z^M}\left|Z^{-N/2}\sum_{n\ \mathrm{sf}}a_0(n)\nu(n)\rho(n)\chi_n(k)\chi_n(f)^4d_I(n)W(q_n/Z^N)\right|^2. \tag{MF3.6}$$

Here ν is fixed finite-ray, lengths are bounded, slot sum is $\le z_0$, and no additional n-coefficient is allowed.

Proposition 7.2 (Canonical marked energy). *For any fixed $c_*>0$,*

$$F-M-Q-z_0\ge c_*,\qquad 4F-3M-6z_0\ge c_*\quad\Longrightarrow\quad \mathcal{E}(N,V,M)\ll Z^{F+\epsilon}. \tag{MF3.7}$$

Uniformity includes moving punctures, the stated label weights and every surviving slot subset, always with the original cap z_0.

Proof. **Terminal completion.** Choose $0<d\le c_*/200$. Set $\Psi_k(n)=\nu(n)\rho(n)\chi_n(k)\chi_n(f)^4$ multiplicatively on all ideals. For $V_*(y)=y^{1/2}W(y)$ define

$$T_{d_*}(X;k)=\sum_{n\ \mathrm{sf},b}\frac{a_0(n)\Psi_k(n)\overline{\alpha(b)}^3\Psi_k(b)^3}{\sqrt{q_n}\,q_b}\,d_*(nb^3)V_*(q_nq_b^3/X).$$

The exact cube inversion is

$$Z^{-N/2}\sum_{n\ \mathrm{sf}}a_0(n)\Psi_k(n)d_I(n)W(q_n/Z^N)=\sum_h\frac{\mu(h)\overline{\alpha(h)}^3\Psi_k(h)^3}{q_h}T_{d_h}(Z^N/q_h^3;k),\quad d_h(A)=d_I(h^3A). \tag{MF3.8}$$

Opening the completion and setting $c=hb$ leaves mark $d_I(nc^3)$ independent of h; $\sum_{h|c}\mu(h)=1_{c=1}$ verifies the identity, normalization and masks.

Use fixed dyadic windows with unit center zero. If $V<d$ and the h-center $\ell_1<d$, freeze h, assign slots by (MF3.2) and apply reflected energy (RE) at $N_*=N-3\log_Z q_h$, surviving cap $z_a\le z_0$. The factor $\Psi_k(h)^3$ contracts the joint (f,k) norm and $\sum_{q_h\asymp Z^{\ell_1}}q_h^{-1}\ll 1$. If c is the current margin in (MF3.7) and η bounds window errors, RE gives

$$H\le M-O+\eta,\qquad T_d-S_0-B_0\le M-O+2z_a-z_0-c+5d+4\eta. \tag{MF3.9}$$

For the second, apply (MF2.41), the first margin, and $2V+3\log_Z q_h<5d+3\eta$.

Retention gives $v+3b+e_\lambda\le T_d+\tau_{\mathrm{ref}}$, $e_\lambda\ge-\eta$. The row branch yields $E\le M+z_a+5\eta/3\le F-c+5\eta/3$. The column branch gives $E\le O/2+T_d-S_0-B_0+5\eta/3+\tau_{\mathrm{ref}}$ if $u=z_a$; otherwise $u\ge v/2$ gives $E\le O/2+(T_d-S_0-B_0)/2+z_a+7\eta/6+\tau_{\mathrm{ref}}/2$. Substitution and the respective margins give

$$E\le F-2c+5d+17\eta/3+\tau_{\mathrm{ref}},\qquad E\le F-M/4-3c/4+5d/2+19\eta/6+\tau_{\mathrm{ref}}/2. \tag{MF3.10}$$

These are $<F$ for $c\ge c_*/2$ and η,τ_{ref} and reflected divisor loss $\le c_*/1000$. Summing labels costs $Z^{V+o(1)}$, canceled by Z^{-V}.

Nonterminal block. If $V<d$, $\ell_1\ge d$, reopen (MF3.8), combine its cube with h into b, center ℓ, and put $r=N-3\ell$. If $V\ge d$, take $b=1$, $\ell=0$, $r=N$. Joint Hilbert-space triangle and common-profile separation reduce to

$$Z^{-r-2\ell-V}\sum_f w(f)\sum_k\Phi(q_k/Z^M)\left|\sum_{q_b\asymp Z^\ell}\beta(b,f)\chi_b(k)^3\sum_{n\ \mathrm{sf}}a_0(n)\nu(n)\rho(n)\chi_n(k)\chi_n(f)^4d_I(nb^3)W(q_n/Z^r)\right|^2, \tag{MF3.11}$$

Here $V+\ell\ge d$, β is divisor-bounded, independent of k,n, and includes $(b,f)=1$; b,n may share primes. We return this to (MF3.6) with shorter rows.

First Poisson transformation. In the square write

$$\mathfrak{B} = \operatorname{rad}(b_1 b_2), \qquad n_i = A_i C u_i, \quad A_i = (n_i, \mathfrak{B}), \quad C = (n_1/A_1, n_2/A_2).$$

Then u_1, u_2 are coprime squarefree and avoid $C\mathfrak{B}$. At $p \mid \mathfrak{B}$, put $a_i = 1_{p|A_i}$, $\pi = v_p(b_1 b_2) \bmod 2$, $t_p = a_1 - a_2 + 3\pi \bmod 6$, and $R_1 = \prod_{t_p \neq 0} p$. Primitive character and residual mask are

$$\chi_{u_1}\overline{\chi_{u_2}} \prod_{t_p \neq 0} \chi_p^{t_p}, \qquad \mathfrak{m}_1 = u_1 u_2 R_1, \qquad R_{\mathrm{mask}} = C\mathfrak{B}/R_1. \tag{MF3.12}$$

Apply (MF3.5), with $d_k \mid C\mathfrak{B}/R_1$ and frequency h. Also denote the centers of A_i, C, R_1, d_k by A_i, B, R, δ. Actual conductor length is $\log_Z(q_{n_1} q_{n_2}/(q_{A_1} q_{A_2} q_C^2)) + \log_Z q_{R_1}$. The principal case forces $u_1 = u_2 = 1$, $\pi = 0$, $a_1 = a_2$ everywhere, hence $n_1 = n_2$ and square $b_1 b_2$. Counting even all cube pairs gives normalized $O(Z^{M+o(1)})$, also covering restored principal nonzero frequencies with the same mask.

CRT, $\gamma_1(u)\gamma_2(u) = \mu(u)\alpha(u)G(u)$ and reciprocity give inverse residual coefficients. Expand their joint fixed-ray factor $G([u_1 u_2^{-1}])R([u_1 u_2^{-1}], [\prod p^{t_p}])$ before coprime extension. Set $E = \prod_{\pi=1} p^{a_1+a_2}$, $y = hf^2E$. The residual exponents $4a_i + 1_{t_p \neq 0}(1 \pm t_p) + \pi(a_1 + a_2) \bmod 6$ agree on both sides by this exhaustive table:

π: 0;	a_1: 0;	a_2: 0;	t_p: 0;	*Residual exponent*: 0
π: 0;	a_1: 1;	a_2: 0;	t_p: 1;	*Residual exponent*: 0
π: 0;	a_1: 0;	a_2: 1;	t_p: 5;	*Residual exponent*: 0
π: 0;	a_1: 1;	a_2: 1;	t_p: 0;	*Residual exponent*: 4
π: 1;	a_1: 0;	a_2: 0;	t_p: 3;	*Residual exponent*: 4
π: 1;	a_1: 1;	a_2: 0;	t_p: 4;	*Residual exponent*: 4
π: 1;	a_1: 0;	a_2: 1;	t_p: 2;	*Residual exponent*: 4
π: 1;	a_1: 1;	a_2: 1;	t_p: 3;	*Residual exponent*: 4

An exponent zero still means a puncture. Write $b_1 b_2 = q^2 s$, s squarefree, and put

$$J_2 = \prod_{\pi=0,\ a_1=a_2=1} p, \quad J = sJ_2, \quad q = J_2 q_0.$$

Then J is squarefree and the residual factor is $\chi_n(J)^4 1_{(n,\operatorname{rad} q_0)=1}$. For center j of J, the identities

$$\log_Z q_{q_0} = \tfrac{1}{2}\log_Z(q_{b_1} q_{b_2}) - \tfrac{1}{2}\log_Z q_s - \log_Z q_{J_2} \geq 0, \qquad \log_Z \frac{q_{R_1} q_E}{q_{A_1} q_{A_2}} = \log_Z q_s - 2\log_Z q_{J_2} \tag{MF3.13}$$

give the retained row enclosure

$$H_c = 2r - 2B - M + 4\ell + 2V + \delta - j, \qquad H_+ = H_c + 12\eta + \tau, \qquad 0 < q_y \leq Z^{H_+}. \tag{MF3.14}$$

The actual argument is $Z^M q_h/(q_{d_k} q_{\mathfrak{m}_1})$; if $\leq Z^\tau$, (MF3.13) gives $\log_Z q_y \leq H_c + 12\eta + \tau$. Discard only this outer ball's complement by (MF3.4); $H_+ < 0$ leaves no nonzero rows.

Extend the quotient-free expression to individually allowed u_1, u_2, retaining full kernel, formal product conductor and outer ball. Insert $1_{(u_1,u_2)=1} = \sum_{t' \mid (u_1,u_2)} \mu(t')$ and write $u_i = t' x_i$, center t. The complete Möbius sum removes artificial noncoprime conductors; squarefreeness retains $(x_i, t') = 1$. With

$$s_i = r - A_i - B - t, \qquad \varkappa_i = M - 2r - 2\ell - V - \delta + A_i + B - R/2, \tag{MF3.15}$$

the separated inverse polynomial on side i is

$$P_i(y) = \sum_{x \text{ sf}} \mu(x)\nu_i(x)\rho(x)\overline{\chi_x(y)}\chi_x(CJ)^4\chi_x(d_k)\mathbf{1}_{(x,\operatorname{rad}(q_0 t'))=1} d_{I_i}(x) W_i(q_x/Z^{s_i}). \tag{MF3.16}$$

The ray character ν_i is fixed. The f factor became $\overline{\chi_x(f^2)}$, explaining row y and conjugation. Priority allocation to $(b_i, A_i, C, t'; x_i)$ leaves original slot subsets; CJ, q_0, t' supply all earlier survival zeros.

The first common profile uses independent normalized norms

$$(A_1, A_2, C, d_k, R_1, t', h, x_1, x_2).$$

Apart from its larger cutoffs, its real root and kernel are

$$y_d^{-1} y_R^{-1/2} y_t^{-1} (y_1 y_2)^{-1/2} W(y_{A_1} y_C y_t y_1) \overline{W(y_{A_2} y_C y_t y_2)} K_1\left(A_1^{\mathrm{ker}} \frac{y_h}{y_d y_R y_t^2 y_1 y_2} \right), \quad A_1^{\mathrm{ker}} = Z^{M+h_1-\delta-R-2t-s_1-s_2}, \tag{MF3.17}$$

Here h_1 centers bare frequency. Multiplying the profile by $Z^{-9\eta/2}$ leaves outside side factors $Z^{(\varkappa_i+9\eta/2)/2}$, exactly separating prefactor and kernel. Fresh supports satisfy $|\log_Z q_{x_i} - s_i| \le 4\eta$; profile bounds are uniform for $A_1^{\mathrm{ker}} > 0$.

Apply weighted Cauchy to the **whole** outer tuple before bounding weights. At fixed reconstruction and $y \neq 0$,

$$\sum_{f^2 E | (y)} w(f) \ll d((y))^{C+1} \ll Z^{o(1)}. \tag{MF3.18}$$

Thus f, h disappear together through $h = y/(f^2 E)$, retaining $-V$ in $\varkappa_i$. Only now smoothly majorize the positive row ball at Z^{H_+}. Average J, freeze q_0. If r_{diff} is the radical length of even-parity primes with $a_1 \neq a_2$, the table gives $\log_Z q_{R_1} = \log_Z q_s + r_{\mathrm{diff}}$; hence

$$\log_Z q_{q_0} \le \ell + R/2 - j + 5\eta/2. \tag{MF3.19}$$

Counting q_0 and J costs $Z^{\ell+R/2+7\eta/2+o(1)}$ once.

Second Poisson. On one first-Cauchy side expand $|P_i(y)|^2$, write $x_a = g' z_a$, $g' = (x_1, x_2)$ with center g, and apply (MF3.5): character $\overline{\chi_{z_1}} \chi_{z_2}$, modulus $z_1 z_2$, mask g', divisor $d_2 \mid g'$ with center θ, frequency k''. Put

$$\lambda_c = H_c - \theta - (s_i - g), \qquad M_c = 2(s_i - g) - H_c + \delta + 2\theta = M - 4\ell - 2A_i - 2t - 2g + 2\theta - 2V + j, \quad M' = M_c + \eta. \tag{MF3.20}$$

Retain $0 < q_{d_k d_2 k''} \le Z^{M'}$. Conductor error is 10η; two d_2 and one d_k contribute 3η. Subtracting $H_+ = H_c + 12\eta + \tau$ cancels the same τ, making the complement an actual-ratio tail under (MF3.4).

The principal case $z_1 = z_2 = 1$ has nominal total exponent

$$\varkappa_i + H_c + s_i + B + t + \ell + R/2 = F - B - j. \tag{MF3.21}$$

Prefactor, enclosure, diagonal window and outer counts add $\le 26\eta + \tau + o(1)$, within target, including restored principal nonzero frequencies with the same mask.

Use $\mu(z)\gamma_{-1}(z) = a_0(z)\chi_z(-1)\overline{G(z)}$. The paired ray factor $G([z_2 z_1^{-1}])$ survives common-factor removal. Extend separate columns/full kernel off coprimality, insert the full Möbius sum $z_a = v' n_a$, v' squarefree with center v. Retain $(v', g'CJ) = (g', CJ) = (v', k'') = 1$. Set

$$\gamma = (q_0, t', r_g), \quad r_g = g'/d_2, \qquad k_{\mathrm{new}} = d_k d_2 k'', \qquad f_{\mathrm{new}} = JCd_2 v', \qquad \rho_\gamma(n) = \rho(n) \mathbf{1}_{(n, \mathrm{rad}(q_0 t' r_g))=1}. \tag{MF3.22}$$

Zeros make J, C, d_2, v' pairwise coprime squarefree, hence f_{new} squarefree, and

$$\chi_n(d_k)\chi_n(k'')\overline{\chi_n(d_2)}\chi_n(CJ)^4\chi_n(v')^4 = \chi_n(k_{\mathrm{new}})\chi_n(f_{\mathrm{new}})^4. \tag{MF3.23}$$

Use $\overline{\chi_n(d_2)} = \chi_n(d_2)^5$ with zeros and $a_0(v'n) = a_0(v')a_0(n)\chi_n(v')^4$. Expand $G([n_2 n_1^{-1}])$ to get the same fixed ν' on both sides. Priority allocation leaves original slot subsets, with survival zeros supplied by $f_{\mathrm{new}}, \rho_\gamma$. Freeze only γ; average all four label factors.

The formal child lengths and their actual enclosing windows are

$$\begin{aligned} N_c &= r - A_i - B - t - g - v, & V_c &= B + \theta + v + j, \\ F_c &= N_c + V_c, & N' &= \max(0, N_c), \\ F' &= N' + V_c, & 0 \le \delta_N &= N' - N_c \le 6\eta. \end{aligned} \tag{MF3.24}$$

Fresh column and label errors are $6\eta, 4\eta$. Omit empty windows; other clipping is bounded dilation. Put $D_c = \ell + A_i + t + g - \theta + V$; then

$$\begin{aligned} F_c &= F - D_c - 2\ell + j, & M_c &= M - 2D_c - 2\ell + j, \\ F_c - M_c &= F - M + D_c, & 4F_c - 3M_c &= 4F - 3M + 2(A_i + t + g - \theta + V) + j. \end{aligned} \tag{MF3.25}$$

Project fixed triples from the genuine reconstruction set **before** positive enlargement. Each has a source witness, giving $g-\theta \ge -2\eta$, $j \le 2\ell+3\eta$, $\log_Z q_{q_0} \le \ell+\eta$, hence $Q' \le Q+\ell+t+g-\theta+4\eta \le Q+D_c+4\eta$. Therefore

$$F'-M'-Q'-z_0 \ge c-5\eta, \quad 4F'-3M'-6z_0 \ge c-7\eta, \quad M-M' \ge 2(V+\ell)-8\eta, \quad F' \le F+11\eta. \tag{MF3.26}$$

These hold on structural triples, not every point of a containing ball.

The second common profile uses $(g', d_2, v', k'', n_1, n_2)$, root $y_d^{-1}y_v^{-1}(y_1y_2)^{-1/2}$, windows $W_i(y_gy_vy_1)\overline{W_i(y_gy_vy_2)}$, and kernel $K_2(A_2^{\mathrm{ker}}y_k/(y_dy_v^2y_1y_2))$, $A_2^{\mathrm{ker}} = Z^{H_++h_2-\theta-2v-2N_c}$. Multiplying by $Z^{-6\eta}$ leaves exact outside factor $Z^{\lambda_c+18\eta+\tau}$. Fresh tests $\omega_a(Z^{\delta_N}y)y^{\pm it_a}$ and density are independent of $k_{\mathrm{new}}, f_{\mathrm{new}}, \gamma$. Retain both extracted squared magnitudes, priority masks, $\mu(d_2)\mu(v')$ and frequency mask in signed outer weights. Fourier inversion restores terms exactly; then weighted Cauchy over all outer tuples permits bounding common zeros by one.

At fixed $\gamma, f_{\mathrm{new}}, k_{\mathrm{new}}$, with at most K_0 original slots, split $f_{\mathrm{new}} = JCd_2v'$, $J = sJ_2$, reconstruct $b_1b_2 = q_0^2J_2^2s$, and choose A_i, d_k. Multiplicity is at most

$$D(f_{\mathrm{new}})C(\gamma), \qquad D(f) = d(f)^{9+4K_0}, \quad C(\gamma) = d(q_0)^{5+2K_0}d(t')^{2K_0}d(r_g)^{2K_0}. \tag{MF3.27}$$

Indeed $d_k \mid f_{\mathrm{new}} \operatorname{rad} q_0$, and extracted slot primes divide reconstructed factors, giving those divisor powers. The frequency is then determined; old f, h are absent. By (MF3.19), distinct triples have weighted count

$$\ll Z^{C_c+11\eta/2+o(1)}, \qquad C_c = \ell + R/2 - j + t + g - \theta. \tag{MF3.28}$$

Extract that count once and normalize its measure. The unnormalized squared child is $Z^{F'}\mathcal{E}$; a child bound $Z^{F'+\epsilon_{\mathrm{child}}}$ therefore contributes $Z^{2F'+\epsilon_{\mathrm{child}}}$. The exact balance

$$\varkappa_i + \lambda_c + C_c + 2F_c = F \tag{MF3.29}$$

gives exponent $F+40\eta+\tau+\pi+\epsilon_{\mathrm{child}}$: profiles and triple count contribute $9\eta/2+18\eta+11\eta/2 = 28\eta$, clipping $2\delta_N \le 12\eta$. Here π combines freely chosen divisor, ray, dyadic and coefficient losses. Cauchy takes geometric means; neither exponent nor triple mass doubles.

Closure. Put $D = \lceil (M_{\max}+2)/d \rceil + 1$. At depth h use margin $c_h = c_* - 7h\eta$ and cap $M \le M_{\max} - hd$. Choose

$$\eta \le \min\{d/16, c_*/(14D), \epsilon/(160D), c_*/1000\}, \qquad \tau, \pi \le \epsilon/(4D), \qquad \tau \le c_*/1000, \tag{MF3.30}$$

with the reflected losses above. By (MF3.26), rows decrease by $\ge d$, margins stay $\ge c_*/2$, and F remains bounded through depth D; nonnegative row lengths preclude a strict call there. Allocate losses within π, fix raw tails by (MF3.4), propagate finite smooth/height orders backwards, then take Z large for window errors. Path loss is $< \epsilon$. Completion, principal counts and strict descent prove (MF3.7). □

7.2.3 The marked inverse estimate and inverse amplification Let $\varphi_u(n) = \nu(n)\chi_n(u)^{\varepsilon_\chi}$, with one common sign $\varepsilon_\chi \in \{\pm 1\}$, and put

$$M_u(Z^r) = Z^{-r/2}\sum_n \mu(n)\varphi_u(n)W(q_n/Z^r), \qquad Q_u = \prod_i \left(Z^{-z_i/2} \sum_{p \in \mathcal{P}_i} a_i(p)\varphi_u(p) \right).$$

Lists are fixed disjoint, $|a_i(p)| \le 1$, $q_p \asymp Z^{z_i}$; individual annular weights are absorbed at fixed supremum cost. Retain natural zeros. Length ranges and slot count are bounded/fixed; $c_1, c_2 > 0$ are fixed.

Theorem 7.3 (Marked inverse moment). *The full-width inverse estimate is*

$$r + 2z \le m - c_1, \qquad 2r + 8z \le 3m - c_2 \quad \Longrightarrow \quad \sum_{0 < q_u \ll Z^m} |M_u(Z^r)Q_u|^2 \ll Z^{m+\epsilon}. \tag{MF3.31}$$

The prime coefficients are arbitrary bounded row-independent coefficients, and no mesh is required.

Proof. Set $P=\prod_i p_i$, $j=(n,P)$ and fix its assigned slots. Write $n=jn_0$, $P=jP_0$, $c=n_0P_0$; c is squarefree, $(c,j)=1$. The identities $\mu(n_0)=\mu(c)\mu(P_0)$ and $\varphi_u(n)\varphi_u(P)=\varphi_u(j)^2\varphi_u(c)$ retain zeros. For assigned length G, put

$$z_0=z-G,\qquad D_0=r+z-2G,\qquad Z^{-(r+z)/2}=Z^{-G}Z^{-D_0/2}. \tag{MF3.32}$$

The $O(Z^{G+o(1)})$ assigned tuples have triangle mass canceled by Z^{-G}; $\varphi_u(j)^2$ contracts. For fixed j, the profile argument is $(q_j/Z^G)(q_c/Z^{D_0})/\prod_{i\,\text{surviving}}(q_{p_i}/Z^{z_i})$. Separate it commonly before choosing survivors, absorb $\mu(p_i)$ into coefficients, and conjugate the whole polynomial if needed. It remains to bound

$$R_0(u)=Z^{-D_0/2}\sum_{c\ \mathrm{sf}}\mu(c)\nu_0(c)\mathbf{1}_{(c,j)=1}\overline{\chi_c(u)}d_0(c)W_0(q_c/Z^{D_0}). \tag{MF3.33}$$

The Fourier measure is uniform over the fixed q_j/Z^G interval.

Square in a smooth row ball, write $c_i=Cz_i$, $(z_1,z_2)=1$, and apply (MF3.5) with $\overline{\chi_{z_1}}\chi_{z_2}$, $d'\mid C$, frequency h'. For centers B,θ of C,d' put $P_1=B-\theta\ge -2\eta$ and retain

$$M_c=2D_0-m-2P_1,\qquad M_{\mathrm{init}}=M_c+6\eta+\tau_{\mathrm{init}},\qquad 0<q_{d'h'}\le Z^{M_{\mathrm{init}}}. \tag{MF3.34}$$

Conductor error is 4η and two d' appearances add 2η, so the complement is an actual-ratio tail. Principal diagonal costs $O(Z^{m+o(1)})$, including bounded nonempty negative-D_0 windows.

As in the second canonical transform, Gauss conversion gives $a_0(z)\nu_*(z)\overline{\chi_z(d')}\chi_z(h')$, with common ν_* after expanding $G([z_2z_1^{-1}])$. Insert full Möbius $z_a=sn_a$, retaining $(C,j)=(s,Cj)=(s,h')=1$. Set

$$t=C/d',\qquad k=d'h',\qquad f=d's,\qquad \rho_{t,j}(n)=\mathbf{1}_{(n,\mathrm{rad}(tj))=1}. \tag{MF3.35}$$

The zero-preserving identity

$$a_0(sn)\nu_*(sn)\overline{\chi_{sn}(d')}\chi_{sn}(h')=a_0(s)\nu_*(s)\overline{\chi_s(d')}\chi_s(h')a_0(n)\nu_*(n)\chi_n(k)\chi_n(f)^4 \tag{MF3.36}$$

holds for $(s,n)=1$ by $\overline{\chi_n(d')}=\chi_n(d')^5$, including zeros. The squarefree f and priority allocation to $(t,d',s;n)$ preserve original slot subsets, with survival zeros supplied by $\rho_{t,j}$ and $\chi_n(f)^4$.

If v centers s, the formal child parameters are

$$N_c=D_0-B-v,\quad V_c=\theta+v,\quad F_c=D_0-P_1,\quad N'=\max(0,N_c),\quad 0\le\delta_N=N'-N_c\le 3\eta,\quad F'=F_c+\delta_N. \tag{MF3.37}$$

For normalized (C,d',s,h',n_1,n_2) use root $y_d^{-1}y_s^{-1}(y_1y_2)^{-1/2}$, windows $W_0(y_Cy_sy_1)\overline{W_0(y_Cy_sy_2)}$, kernel $K_{\mathrm{init}}(A^{\mathrm{ker}}y_h/(y_dy_s^2y_1y_2))$, $A^{\mathrm{ker}}=Z^{m+H-\theta-2(D_0-B)}$ (H the bare-frequency center). Multiplying the profile by $Z^{-3\eta}$ leaves $Z^{\varkappa_c+3\eta}$, $\varkappa_c=m-2D_0+P_1$. Fresh tests are bounded clipped cutoffs times Fourier powers. Inversion restores the full kernel before Cauchy, retaining (MF3.34)'s mask and all three zeros outside it.

At fixed t,f,k, choose $d'\mid f$; then $s=f/d'$, $C=td'$, $h'=k/d'$. With at most K_0 slots, multiplicity is $\le d(f)^{1+2K_0}d(t)^{2K_0}$. Project distinct t once from the genuine outer set: $\log_Z q_t\le P_1+2\eta$, giving weighted count $Z^{P_1+2\eta+o(1)}$; extract it and normalize. The actual puncture has $Q'\le P_1+G+3\eta$. Then

$$\begin{aligned}F'-M_{\mathrm{init}}-Q'-z_0&\ge c_1-9\eta-\tau_{\mathrm{init}},\\ 4F'-3M_{\mathrm{init}}-6z_0&\ge c_2-22\eta-3\tau_{\mathrm{init}},\\ \varkappa_c+P_1+2F_c&=m.\end{aligned} \tag{MF3.38}$$

The formal differences are $m-r-2z+2G$ and $3m-2r-8z+10G+2P_1$. Choose $\eta,\tau_{\mathrm{init}}$ small and apply canonical margin $\min(c_1,c_2)/2$. The last identity and $2\delta_N\le 6\eta$ give exponent $m+11\eta+\pi_{\mathrm{init}}+\epsilon_{\mathrm{canonical}}$. Allocate these within ϵ, then choose tails and backward seminorm orders as above. This proves (MF3.31), hence MI at $m=1$. □

Unmarked consequence (UM). With no slots, $Z=H$, $m=1$, $r=\log_H D$, the condition $H\ge D^{1+\vartheta}$ gives $1-r\ge\vartheta/(1+\vartheta)$ and $3-2r\ge 1$; count bounded scales. Thus

(MF3.39) is (95), with $\varphi_u=\nu\chi_\bullet(u)^{\varepsilon_\chi}$, both signs and every element row, under $H\ge D^{1+\vartheta}$.

All original zeros, finite-seminorm and polynomial-height control remain.

7.2.4 The enlarged plain-moment family Use $\psi_k(n) = \tau(n)\chi_n(k)$ on $0 < q_k \ll Z^m$, where τ combines a fixed ray character and displayed moving symbols. Their union **before** cancellations/exponent reduction has norm $\le Z^q$; width is $M = m + q$. A squarefree common mask $q_R \le Z^B$ may depend on frozen labels but is fixed over the row sum and applies to every plain and prime factor. Canceled row-character primes retain natural zeros; they do not become freely chosen row masks.

For positive slots fix nonzero integral ray modulus $\mathfrak{M}$, a subgroup containing global units, common annular profile W, anchor η_0, and ideal $Q \subseteq \mathfrak{M}$. Require $Q_0 = Q \cap \mathfrak{q}_{\eta_0}$ to be nonzero, proper and contained in (72). Each slot has $\nu_i = \theta_i\overline{\eta_0}$ for a fixed-ray-quotient character θ_i, bounded real power σ_i and height v_i:

$$Q_{k,i} = Z^{-z_i/2} \sum_{p \in \mathcal{P}_i^{\mathrm{can}}} \psi_k(p)\nu_i(p)W(q_p/Z^{z_i})(q_p/Z^{z_i})^{-\sigma_i+iv_i}.$$

Here $\mathcal{P}_i^{\mathrm{can}}$ is the genuine canonical good-prime pool at Z^{z_i}, retaining natural zeros and common puncture. MP uses this pool and common W; it asserts neither arbitrary subsets, separate slot profiles nor a finite-character-expansion extension. MI permits arbitrary subsets. Physical scales are separated annuli.

Put $z = \sum_i z_i$, $A = n_1 + n_2 + z$. The two centrally normalized plains have independent profiles and heights with polynomial costs.

Theorem 7.4 (Natural marked plain moment). *The zero-slot assertion is the unconditional Theorem 5.1. For positive slots require $0 < z_i \le \eta_{\mathrm{mesh}}$, with mesh chosen below before the slot system. Absorb zero-width bounded factors into fixed arithmetic data. Assume*

$$7/10 \le \kappa \le 1, \qquad \beta \ge 51/100, \qquad 2\beta - 1 \le \kappa.$$

The centrally normalized plain sums are

$$S_{k,i}(X) = X^{-1/2} \sum_n \psi_k(n)\mathbf{1}_{(n,R)=1}W_i(q_n/X),$$

with all integral ideals n and the fixed bad-prime zeros understood. Then

$$\begin{aligned} &z = 0: \quad \sum_{\psi_k^* \ne 1} |S_k(Z^{n_1})S_k(Z^{n_2})|^2 \ll Z^{M+\epsilon} \quad \text{for all bounded } n_1, n_2 \ge 0; \\ &z > 0: \quad n_1 + n_2 + 6\kappa z \le M \implies \sum_{k \text{ eligible}} |S_k(Z^{n_1})S_k(Z^{n_2}) \prod_i Q_{k,i}|^2 \ll Z^{M+\epsilon}. \end{aligned} \tag{MF3.40}$$

A natural state has a whole-product presentation, complete natural zeros, fixed nonzero puncture and nonnegative smooth radial measure. Its width counts row-ball exponent plus displayed moving modulus before cancellation. Keep exactly nonzero rows not induced from $\mathfrak{Q}_0 = Q \cap \mathfrak{q}_{\eta_0}$: fixed-inducing means a primitive finite Hecke character of conductor dividing $\mathfrak{Q}_0$ induces the actual row function with natural zeros. Deleting an extra puncture preserves this predicate, since additional zero Euler factors do not change the primitive inducing character. Initially,

$$\mathrm{initialKeep}(\eta, \mathfrak{Q}_0) = \{k \in \mathcal{O} : k \ne 0, \text{ the natural presentation of } n \mapsto \eta(n)\chi_n(k) \text{ is not induced from } \mathfrak{Q}_0\}.$$

The presentation retains fixed bad-prime zeros, primitive-conductor zeros, redundant moving zeros and row radical. Eligibility excludes every fixed-inducing presentation, not just principal characters or a chosen list.

Quantifier order: for a requested loss and bounded width/puncture/length ranges choose one mesh, uniformly for $\kappa \in [7/10, 1]$, before arithmetic family, slot count, profiles or target height. After fixing a finite slot index type and common W, choose finite seminorms and height order before fixed character/ideal; their constant and eventual threshold precede live subset, slot characters, lengths, powers, heights, radial state and plain profiles. Positive slots allow a common whole-product norm power; independent plain twists are controlled by seminorms. Zero slots retain independent heights, all bounded lengths and every nonprincipal inducing character, including fixed-conductor rows.

Absorb fixed conductors in arbitrarily small width padding charged to the requested loss before the eventual threshold. Moving support remains width, never a profile seminorm.

The physical initial state has scale U, puncture one, fixed source η and width $1 + \delta$ once $q_{\mathfrak{q}_\eta} \le U^\delta$. Its measure is $1_{\mathrm{initialKeep}(\eta,\mathfrak{Q}_0)}(k)\Phi(q_k/U)$, $\Re\Phi \ge 0$, with the literal natural plains and slots above, and relative character

$$\nu_i(n) = \theta_i(n)\overline{\eta_0(n)},$$

Lift θ_i from $(\mathcal{O}/\mathfrak{M})^\times/H$; both factors retain their zeros. The prime pool is the identity ray class, cut at $\max(1,b)U^{z_i}$; W selects its exact annulus. Physically $W = \overline{F.W}$, $\theta_i = 1$, real power $33/50$, height $-\Im z_i$. Fixed-inducing sixth-power-free rows have prescribed good-prime valuations modulo six, hence bounded norm and leave a growing lower annulus.

The zero-slot endpoint restores fixed nonprincipal characters by masked lattice cancellation below. Width counts full moving support but excludes the puncture. For exceptional children fix a finite conjugation-stable S-supported group Θ containing fixed phases, inverses and $n \mapsto \chi_n(-1)$; it does not replace positive-slot eligibility. Taking $m = 1, q = 0, n_1 = n_2$ gives MP in the positive family and MP0 for all nonprincipal zero-slot rows.

Proof. **Extra masks.** A natural presentation has only primitive, redundant displayed, row and S zeros. Delete its additional fixed R mask to obtain ψ_k^0. Multiplicativity gives

$$S_{k,R}(X) = \sum_{d|R} \frac{\mu(d)\psi_k^0(d)}{\sqrt{q_d}} S_k^0(X/q_d), \qquad Q_{k,i,R} = Q_{k,i}^0 - Z^{-z_i/2} \sum_{p|R} \psi_k^0(p)\nu_i(p) W(q_p/Z^{z_i})(q_p/Z^{z_i})^{-\sigma_i+iv_i}. \tag{MF3.41}$$

Total coefficient mass is $Z^{o(1)}$: $\prod_{p|R}(1-q_p^{-1/2})^{-1} \ll_\epsilon Z^\epsilon$, and a frozen slot has $Z^{-z_i/2} \ll q_p^{-1/2}$. Apply both identities and Minkowski. The affine length $A + (6\kappa-1)z$ decreases; if slots vanish the zero-slot row set contains the original eligible rows. Nonempty subunit plains clip by bounded dilation without exceeding old nonnegative lengths. Prove natural states uniformly, retaining this deletion at every child.

Whole-factor reflection. Use ğ5.3's natural reflection, with primitive ψ_k^*, conductor norm Q_k, $C_k = 3Q_k/(2\pi)^2$ and redundant radical $R_{0,k}$.

(MF3.42) is (70), $C_k q_{R_{0,k}} \ll Z^M$, with the same full-support width.

Unit and S-valuations modulo six give finite ray sectors. For nonprincipal inducing characters, the central primitive reflection has scale C_k/X, unit root number and profile

(MF3.43) is (71)'s Mellin profile formula, $\widehat{W^\sharp}(s) = \widehat{W}(1-s)\Gamma(s)/\Gamma(1-s)$.

The proof in ğ5.3 substitutes the functional equation, sets $s \mapsto 1-s$ and shifts without an L-pole. It gives bounded Euler derivatives at zero, rapid decay at infinity, and finite-seminorm/polynomial-height control by Stirling and $\widehat{Wy^{it}}(s) = \widehat{W}(s+it)$.

Deleting redundant factors before reflection and restoring geometrically gives exactly

$$S_k^0(X;W) = \varepsilon_k \sum_{d_0|R_{0,k}} \sum_{\mathrm{rad}(h_0)|R_{0,k}} \frac{\mu(d_0)\psi_k^*(d_0)\overline{\psi_k^*(h_0)}}{\sqrt{q_{d_0}q_{h_0}}} S_{\overline{\psi_k^0}}\left(\frac{C_k q_{d_0}}{X q_{h_0}}; W^\sharp\right). \tag{MF3.44}$$

Its mass is $Z^{o(1)}$. Lower annuli of $W^\sharp$ have central coefficients $O(2^{-j/2})$; upper annuli decay arbitrarily rapidly. Discard above relative scale $Z^{\xi/2}$. A main scale $Y < Z^{-\xi}$ contributes $O_B(Y^{B-1/2})$ by counting and decay, for arbitrary fixed B. Handle row-dependent coefficients by mass, then scale suprema: integrate function and log-scale derivative over $O(\log Z)$ unit boxes, in two variables for two scales. The derivative is $-W/2 - yW'$, so common-profile bounds apply. Conjugate the entire reflected factor to restore orientation. By (MF3.42), retained scales satisfy

$$n_{\mathrm{ref}} \le M - n + \xi. \tag{MF3.45}$$

Absorb fixed endpoints and boxes in the final threshold; negative right side gives an empty range. Reflecting plains longer than $M/2$ reduces zero slots to

(MF3.46) is the padded core (72), with $2\xi \le \delta$: $n_i \le M/2 + \xi$ and $A \le M + 2\xi \le M + \delta$.

Prime cost. On eligible rows each slot remains nonprincipal after expanding the canonical identity-ray projector as the average of all fixed-quotient characters. Its product with $\nu_i = \theta_i\overline{\eta_0}$ has conductor dividing Q_0; a principal resulting row would therefore be induced from Q_0. Multiplication by such a fixed character preserves eligibility by inverse multiplication. The finite average changes only constants; no arbitrary-pool/profile extension follows. Under $2\beta - 1 \le \kappa$, move the smooth von Mangoldt contour to $s_\kappa + e$, $s_\kappa = (1+\kappa)/2$. There $|L'/L| \ll_e \log(Q(3+|t|)^2)$: on the zero-free disk centered at $2+it$ of radius $2 - s_\kappa - e/4$, the Euler branch $g = \log L$ has $g(2+it) = O(1)$ and $\Re g \ll \log(Q(3+|t|)^2)$ by functional-equation strip growth. Borel–Carathéodory and then Cauchy's derivative bound on successively smaller disks containing $s_\kappa + e + it$ prove the claim; all gaps are fixed multiples of e. Prime powers of degree ≥ 2 cost $O(P^{1/2+\epsilon})$. Divide the annular weight by $\log(Py)$; redundant zeros delete only $O(\log Z)$ primes. Central normalization over the fixed slots gives

$$\left|\prod_i Q_{k,i}\right|^2 \ll Z^{\kappa z+\epsilon_1}\, p_J(W)^b \langle \mathbf{t}\rangle^h. \qquad \text{(MF3.47)}$$

Contour displacement costs $2ez$, depending on total length, not slot count. Heights translate rapidly decreasing Mellin transforms; required derivative/height orders do not affect the exponent.

7.2.5 Induction order and the centered comparison Choose floor $\rho > 0$, step $\sigma > 0$, padding $\delta \ll \min(\rho, \sigma)$ and bands of length $\sigma/4$. Use the zero-slot theorem first, then prove positive bands increasingly; within each band prove $A \le 5M/6$ before centering. The zero-slot proof in ğ5 supplies all lengths by reflection and padded-floor excess $\le \rho + \delta$. At $M \le \rho$, positive slots follow from it and (MF3.47), since

$$z \le M/(6\kappa), \qquad \kappa z \le M/6. \qquad \text{(MF3.48)}$$

Its additional terminal excess is at most $\rho/6$.

For $A > 5M/6$, put $L = M/4$. If one plain is shorter than L, reflect the other; otherwise compare lengths $L, A - z - L$ with the same profiles, reflecting the larger comparison factor only to estimate the comparison. Since $q_\kappa = 6\kappa - 1 > 2$ and $z < (M/6)/q_\kappa < M/12$, both comparison lengths are $\ge L$. Reflection gives

$$A_{\mathrm{comp}} \le 3M/2 - A + 2z + \xi, \quad \min\{5M/6 - A_{\mathrm{comp}},\, M - A_{\mathrm{comp}} - q_\kappa z\} \ge \mathfrak{m}(7/10)M - \xi, \quad \mathfrak{m}(\kappa) = \frac{2\kappa - 1}{2(6\kappa - 1)}. \qquad \text{(MF3.49)}$$

Indeed $t = M - A < M/6$, $z \le t/q_\kappa$ give margins $\ge M/3 - t(1 + 2/q_\kappa) - \xi$ and $M/2 - t(2 + 2/q_\kappa) - \xi$. Zero-slot comparison margin is $\ge M/6 - \xi$. Choose $\xi < \rho\mathfrak{m}(7/10)/2$; each comparison then uses its already proved uncentered stage.

For both plains $\ge L$, put $X_i = Z^{n_i}$, $Y_1 = Z^L$, $Y_2 = X_1X_2/Y_1$. Subtract and retain the equal-product rectangles; after extracting b_i, the coefficient is

(MF3.50) is the exact rectangle difference (73), with $D_b = \mathcal{D}_b$ and $X_1X_2 = Y_1Y_2$.

Use the same b_i even when one rectangle vanishes. The comparison is already bounded on eligible rows; enlarge the positive norm of the **whole difference** to a smooth ball of all rows. In the uncentered range enlarge the original squared norm. Never Poisson-sum an eligibility indicator.

The invariant coefficient carried through the transformations is

$$\prod_{i\in I'} \nu_i(p_i) W(q_{p_i}/Z^{z_i})(q_{p_i}/Z^{z_i})^{-\sigma_i + i v_i}\, D_b(l_1, l_2)\, \tau_1(u) q_u^{it}\, \mathbf{1}_{(u,R)=1} \mathbf{1}_{s|u}, \qquad u = \Big(\prod_{i\in I'} p_i\Big) l_1 l_2, \qquad \text{(MF3.51)}$$

Here s is fixed squarefree; uncentered inputs omit the second rectangle. The character, mask and norm power act on the entire product, preserving collisions by zero extension.

Use one aggregate tolerance. For fixed slot count N, let H_N bound sums of absolute logarithmic endpoints of every slot/plain window and all finite later enlargements; set $\theta_N = H_N/\log Z$. Every slot subset's total ratio error is $\le \theta_N$. Extract plain powers using exact norms. Stage errors are $O(\theta_N)$ with numerical coefficient independent of N, not a loss per slot.

7.2.6 First Poisson: from plain products to Gauss polynomials Set $H = Z^m$, $X = Z^A$. The zero-frequency argument of ğ5.4 applies unchanged to full products: zero net sextic exponents exclude multiplicity-one primes, giving a powerful ideal of norm $O(X^2)$, $O(X^{1+\epsilon})$ choices and divisor-bounded slot/plain allocations. Multiplying central X^{-1} and row mean $\le H$ gives $O(Z^{m+o(1)})$.

For nonzero frequency write full products Ca, Db with complete common support, $(a, b) = (ab, CD) = 1$. Fix each common prime's full plain valuations and slot occurrences, extract them, freeze supplying slots and puncture every residual factor. Apply the same allocation to both rectangles; retain any scalar zero before enlargement.

Let c, d, p be lengths of $C, D, \mathrm{rad}(CD)$. Let r contain nonzero net exponents modulo six, length R, primitive character ξ_r; remaining common-mask divisor e has $E \le p - R$. For allocated coefficients without the whole-product twist, set $\tau_C(a) = \tau(a)\chi_a(er)\overline{\xi_r(a)}$ and similarly for τ_D. Poisson/CRT give, up to bounded frozen scalar,

$$\frac{H}{Xq_e\sqrt{q_r}} \sum_{h\ne 0} \mathcal{G}_{\xi_r}(r, h) \sum_{(a,b)=1} \frac{A_C(a)\overline{A_D(b)}}{\sqrt{q_a q_b}} \tau_C(a)\overline{\tau_D(b)} R(a, b) \mathcal{G}(a, h)\overline{\mathcal{G}(b, -h)} K_1\left(\frac{Hq_h}{q_e q_r q_a q_b}\right). \qquad \text{(MF3.52)}$$

Here $\mathcal{G}$ is the normalized sum with local values (8), $|\mathcal{G}_{\xi_r}| \le 1$. As in ğ5.4, Poisson after $k = ek'$ contributes $H/(q_e q_r q_a q_b)$; Gauss sums restore $\sqrt{q_r q_a q_b}$. CRT gives $\chi_a(br)\overline{\chi_b(ar)\xi_r(ab)}$, whose a, b ratio is $R(a, b)$ by reciprocity. New moving primes puncture every residual factor even if exponents cancel; their support and frequency lengths are

(MF3.53) combines (75) and (76): $\widetilde{q} = q + R + E$, $K_0 = 2A - c - d + R + E - m$, with this chapter's full-product length $A = n_1 + n_2 + z$.

The first kernel's nominal frequency scale is Z^{K_0}.

For both Poisson transforms use one nonnegative norm-dyadic partition. In an outer sector retain whole weights meeting $q_h \le T_{\mathrm{sec}} Z^{\xi/2}$, T_{sec} the maximum nominal scale. For large Z, upper length is $K \le K_0 + \delta_1$, $0 \le \delta_1 < \xi$; discarded genuine arguments are $\ge Z^{\xi/4}$, since full-product support costs e^{2H_N}. Apply (MF3.4) with raw bounds $|\mathcal{G}| \le \sqrt{q_a}$, $|F| \le q_u q_v$ and polynomial column/count cost, fixing that cost before decay order. Retained weights/full kernels are common to rectangles and independent of live columns.

On each dyad extend the separated Gauss bridge off coprimality using formal norms, original masks and bicharacter. Insert the complete $\sum_{s|(a,b)} \mu(s)$ before absolute values; let $s_0 = \log_Z q_s$. Separate kernel/inverse roots in row and whole-product norms, adding an aggregate outer-ratio coordinate if needed. Each column receives one common norm power across its plains and rectangles. Then Cauchy and $|\mathcal{G}_{\xi_r}| \le 1$ give

(MF3.54) is (77) with $\mathfrak{g} = \mathcal{G}$, this chapter's A_C, τ'_C and dyadic $h \ne 0$ support. Its complete coefficient is (MF3.51).

The exchanged D norm has the same form. Outside exponent is $m - A - R/2 - E$; common-support and Möbius counts are $Z^{p+s_0+o(1)}$. Count radicals first, then bounded powers via $\#\{v : \operatorname{rad} v \mid c, q_v \le Z^T\} \le Z^{aT} \prod_{p|c}(1 - q_p^{-a})^{-1}$ for small $a > 0$; allocations cost divisors.

It suffices to prove

(MF3.55) is the target bound (78), here with loss ϵ_G: $\mathcal{N}_C \ll Z^{A-c+\widetilde{q}+B_c-s_0+\epsilon_G}$, $B_c = (3c - 5d - R)_+/6$.

Its exchanged counterpart and outside factors have exponent $M + \{B_c + B_d - (c + d - 2p - R)\}/2$. The local comparison is

$$B_c + B_d \le c + d - 2p - R. \qquad \text{(MF3.56)}$$

This is the primewise inequality proved after (79), with identical multiplicities and $r_p = 1_{6\nmid i-j}$; slot occurrences are already included in those multiplicities.

7.2.7 Enlarging a Gauss-row norm Extract length w from (MF3.54), adding moving radical $w_o \le w$ with squared scalar Z^{-w_o}. Put $a_0 = A - c - w$; (MF3.55) permits the unweighted norm exponent

$$\Lambda_c = a_0 + \widetilde{q} + w_o + w + B_c - s_0 + \epsilon_G. \qquad \text{(MF3.57)}$$

Define $J = d - c + (K_0 - K) - 2w + w_o$, $J_+ = \max(J, 0)$. With zero slots take $w = w_o = 0$, $\ell = 0$, and enlarge the row norm to a smooth ball of length $K + g$, $g = J_+ + \sigma$.

For positive slots choose $\ell_* = \sigma/3$, mesh $< \sigma/6$, and pool $\mathcal{P} = \{p : Z^{\ell_*}/2 < q_p \le Z^{\ell_*}, p \notin S\}$. The fixed-field prime ideal theorem gives $|\mathcal{P}| = Z^{\ell_* + o(1)}$. For large Z it avoids all live slots. For each $h \ne 0$, delete the $O(\log Z)$ primes dividing h, s or frozen supports.

For $p \nmid uh$, the prime-power Gauss identities (8) and CRT give

$$\mathcal{G}(p^i u, h) = R(p, u)^i \chi_p(u)^{2i} \mathcal{G}(p^i, h)\mathcal{G}(u, h), \qquad \mathcal{G}(u, hp^6) = \mathcal{G}(u, h). \qquad \text{(MF3.58)}$$

Replacing h by hp^6 changes only valuations $i = 1, 6, 7$. Assign powers to the two plains (pool and slots are disjoint), puncturing the residual product and updating both rectangles. Jensen over eligible primes and these finite allocations bounds the original square by amplified square and extracted errors. Each output hp^6 has boundedly many representations, since pool length is fixed positive and output length bounded; density is $Z^{-\ell_* + o(1)}$. Take $\ell = \ell_*$, $g = J_+ + 2\sigma$, enclosing amplified rows since $6\ell_* = 2\sigma$.

The extracted cases are exactly:

Column valuation i: 1;	*Squared extraction coefficient*: q_p^{-1};	w: ℓ_p;	w_o: ℓ_p
Column valuation i: 6;	*Squared extraction coefficient*: $(1 - q_p^{-1})^2$;	w: $6\ell_p$;	w_o: 0
Column valuation i: 7;	*Squared extraction coefficient*: q_p^{-1};	w: $7\ell_p$;	w_o: ℓ_p

Here $\ell_p = \log_Z q_p \in [\sigma/6, \sigma/3]$. At valuation six, $\chi_p(u)^{12}$ is the existing common puncture; retain $R(p,u)^6$. Valuations one/seven retain a moving character counted in w_o. Freeze each error's p, take $\ell = 0$, $g = J_+ + \sigma$, and transform without re-amplifying. The normalized prime average has no extra count. Always

$$0 \le w_o \le w \le 7\sigma/3. \qquad \text{(MF3.59)}$$

Coefficient (MF3.51), multiplied by $\mathcal{G}(u,h)$, retains its difference, puncture and s-divisibility.

Add zero only at the final smooth ball, never amplify it. By (8), $\mathcal{G}(u,0)$ vanishes unless u is sixth-power, and is $\le \sqrt{q_u}$. For nonempty upper bound $Y = e^{H_N} Z^{a_0}$ and $s \mid u$, $q_s^6 \le Y$. Counting $O(Y^{1/6})$ columns gives normalized square $Z^{-a_0} Y^{4/3+o(1)} \le Z^{a_0 - s_0 + 2\theta_N + o(1)}$, within (MF3.57).

7.2.8 Second transform and the actual children Restore live slot sums before squaring. If $B(u)$ is the complete convolution coefficient, with $s \mid u$, (10) gives

$$\begin{aligned} &Z^{-\ell} \sum_h \Phi_2(q_h/Z^{K+g}) \left| Z^{-a_0/2} \sum_u B(u)\tau_1(u) q_u^{it} \mathcal{G}(u,h) \right|^2 \\ &= Z^{K+g-\ell-a_0} \sum_{u,v} \frac{B(u)\overline{B(v)}\tau_1(u)\overline{\tau_1(v)} q_u^{it} q_v^{-it}}{\sqrt{q_u q_v}} \\ &\quad \times \sum_j F(u,v;j) K_2\!\left(\frac{Z^{K+g} q_j}{q_u q_v} \right). \end{aligned} \qquad \text{(MF3.60)}$$

At $j = 0$, (11) forces $u = v$ and $F(u,u;0) = \phi(u) \le q_u$. There are $O(Z^{a_0 - s_0 + \theta_N})$ supported multiples of s, giving exponent $K + g - \ell - s_0 + \theta_N + o(1)$ and nominal excess

$$(A - M) - d - (K_0 - K) - w_o - B_c + g - \ell \le A - M + 5\sigma/3 + \delta_1. \qquad \text{(MF3.61)}$$

If $J \ge 0$, $-d - (K_0 - K) - w_o + J_+ = -c - 2w$; otherwise it is $\le \delta_1$. The largest increment is $2\sigma - \ell_* = 5\sigma/3$, a terminal loss, plus δ in the zero-slot padded core.

Localize genuine nonzero j at nominal $Z^{2a_0 - K - g}$ by the whole-dyad rule before common extraction $u = D_2 a, v = E_2 b$. Put

$$c_2 = \log_Z q_{D_2}, \quad d_2 = \log_Z q_{E_2}, \quad b_2 = (c_2 + d_2)/2, \quad p_2 = \log_Z q_{\mathrm{rad}(D_2E_2)}, \quad G_c = (D_2, E_2), \quad g_2 = \log_Z q_{G_c}.$$

Apply the complete-support factorization

(MF3.62) is (12), with $(D,E) = (D_2, E_2)$ and the same residual a, b, j.

It holds only for $(a,b) = (ab, D_2E_2) = 1$, allowing arbitrary powers and slot/plain coincidences. Extracted primes puncture residual factors; freeze supplying slots. A plain p^5 and slot p therefore retain $\chi_p(j)^6$, not one.

Equations (13)–(14) force $G_c \mid j$. At equal non-six-divisible multiplicities split divided frequency into units/nonunits, radical lengths t_2, V and nonunit radical V_{id}. Set $j = G_c V_{\mathrm{id}} h'$. Extend $1_{\mathrm{part}} F(D_2, E_2; G_c V_{\mathrm{id}} h')/Z^{g_2 - t_2}$ by zero off its partition; it has modulus ≤ 1 by

Common multiplicities and divided frequency: $i = i$, $6 \nmid i$, unit;	*Bound*: P^{i-1}
Common multiplicities and divided frequency: $i = i$, $6 \nmid i$, nonunit;	*Bound*: P^i
Common multiplicities and divided frequency: $i = i$, $6 \mid i$, either;	*Bound*: P^i
Common multiplicities and divided frequency: $i > j_0$, $6 \mid j_0$, unit;	*Bound*: P^{j_0}

Other unequal cases vanish. In $v = G_c V_{\mathrm{id}}$, reduce valuations modulo six: nonzero residues retain moving characters, zeros the existing puncture. Active primes lie in $t_2 + V$. Factor only $\chi_n(v)$, preserving varying-row zeros of $\chi_n(h')$. Thus

$$m' = 2a_0 - K - g - g_2 - V, \quad q' = \widetilde{q} + w_o + t_2 + V, \qquad M' = m' + q' = M + J - g - g_2 + t_2 \le M - \sigma. \qquad \text{(MF3.63)}$$

The pulled-back retained weight has $q_{h'} \le Z^{m' + \delta_2}$, $0 \le \delta_2 < \xi$. Nonempty ranges have $m' + \delta_2 \ge 0$; declare $m'_{\mathrm{act}} = m' + \delta_2$, $M'_{\mathrm{act}} = M' + \delta_2$. Extend weight and partition scalar by zero across this ball, keeping them signed.

The condition $s \mid u, v$ forces s into the common radical, so labels cost $Z^{p_2 - s_0 + o(1)}$. The ledger is

Factor: Original squared central normalization;	*Exponent*: $-a_0$
Factor: Poisson factor and inverse roots;	*Exponent*: $K + g - a_0$
Factor: Amplifier density;	*Exponent*: $-\ell$
Factor: Common-support count with fixed s;	*Exponent*: $p_2 - s_0$
Factor: Common correlation;	*Exponent*: $g_2 - t_2$
Factor: Conversion to normalized residual products;	*Exponent*: $a_0 - b_2$

Subtracting this sum from (MF3.57) leaves the inner allowance

$$M' + \Delta, \qquad \Delta = b_2 - p_2 + w + B_c + \ell \geq w + \ell. \tag{MF3.64}$$

The last inequality uses $b_2 \geq p_2$.

Extend (MF3.62)'s right side to individually allowed pairs, retaining original common-support masks, formal full norms, roots, kernel and pulled-back row weight. Insert full $\sum_{t \mid (a,b)} \mu(t)$; only this sum restores the genuine correlation. Its artificial common primes do not enter D_2, E_2 or frequency restrictions. Separate in row/whole-product norms, then allocate primes of t by

$$\mathbf{1}_{p \mid \prod_i n_i} = \sum_{\varnothing \neq J} (-1)^{|J|+1} \prod_{i \in J} \mathbf{1}_{p \mid n_i}, \tag{MF3.65}$$

Factors are the two plains and live slots. Selected plains become $n_i = pl_i$ without residual puncture, dividing corresponding scales in both rectangles by q_p. Selected slots freeze; selecting distinct primes in one prime slot vanishes. Quotients may retain p.

Thus each child retains a whole-product character, common mask and norm power across plains/rectangles; live slots keep original ν_i without Gauss factors. Sides may have different whole-product norm powers, but their fixed-inducing predicates agree: conjugation and multiplication by fixed-modulus characters preserve conductor divisibility, including unchanged redundant zeros. At each fixed-modulus stage enlarge Θ to all primitive characters dividing that modulus plus the finitely many fixed ray twists; it remains finite. On an exceptional row inducing ϑ, freeze live slots to obtain

(MF3.66) is the exceptional rectangle coefficient (91), with $D_b = \mathcal{D}_b$, this child's ϑ, t, and its complete squarefree mask R_*.

The polynomial-size R_* is common to both variables/rectangles. Retain extracted row scalars, including zeros, until positive or absolute estimation.

Extraction normalization. Put $\alpha_1 = a_0 - c_2$, $\alpha_2 = a_0 - d_2$. Let F_j be genuinely frozen entrance slots and I_j the live slots before final allocation. Their pre-allocation plain product T_j satisfies $T_j \prod_{i \in I_j} Z^{z_i} = Z^{\alpha_j + e_j}$, $e_j = \sum_{i \in F_j} \log_Z(q_{p_i}/Z^{z_i})$. Let $d_{j,i}$ be final products selected in plain i, $J_j \subset I_j$ newly frozen slots, and put

$$a_{j,i} = \log_Z q_{d_{j,i}}, \quad r_j = a_{j,1} + a_{j,2} + \sum_{i \in J_j} z_i, \quad \omega_j = \sum_{i \in J_j} \log_Z(q_{p_i}/Z^{z_i}).$$

Use lower-endpoint dyads $Z^{t_-} \leq q_t < 2Z^{t_-}$. Every selected product is divisible by t, so

$$r_j + \omega_j \geq t_-, \qquad |\omega_j| \leq \theta_N, \qquad |e_j + \omega_j| \leq \theta_N. \tag{MF3.67}$$

The last bound uses disjoint subsets of current entrance slots, without recounting ancestors. Raw child normalizer is $Z^{-(\alpha_j - r_j)/2}$, extraction scalar $Z^{-r_j/2}$, formal scale product $Z^{\alpha_j + e_j - r_j}$. Signed raw rectangles may have negative individual lengths.

Child eligibility. Split by the same fixed-inducing predicate with complete source masks. On kept rows, (MF3.62)–(MF3.65) give exactly the whole-product character in nonzero h' with pulled-back radial weight. Frozen zeros belong to its natural presentation; the extra common mask is row-fixed. Delete it by (MF3.41), obtaining a natural state of width M'_{act} with precisely the non-fixed-inducing keep set. Each plain and surviving slot has this same character/mask, and Fourier density precedes live labels/rows. Hence the smaller-width theorem applies to the actual positive norm; complementary fixed-inducing rows remain for separate evaluation.

Nonexceptional children. Use Cauchy, then split centered children into rectangles; only now bound common weights/scalars by one and enlarge the ball. For pre-allocation plain lengths x_i and post-lengths

$y_i = x_i - a_{j,i}$, put $\pi_0 = \sum(-x_i)_+$, $\pi = \sum(-y_i)_+$. Omit rectangles whose plain upper support is below the first ideal; otherwise $0 \le \pi_0 \le \pi \le \theta_N$. The clipped reduction and total are

$$r_{\mathrm{clip},j} = r_j - (\pi - \pi_0) \ge 0, \qquad A_{\mathrm{clip},j} = \alpha_j + e_j - r_j + \pi, \tag{MF3.68}$$

and its exact scalar relative to the pre-normalizer is $Z^{(-r_{\mathrm{clip},j}+e_j+\pi_0)/2}$. For a paired divisor dyad, the count and both scalars satisfy

$$t_- - \frac{r_{\mathrm{clip},1} + r_{\mathrm{clip},2}}{2} + \frac{e_1 + \pi_{0,1} + e_2 + \pi_{0,2}}{2} \le 2\theta_N. \tag{MF3.69}$$

Substitute (MF3.68) and (MF3.67); there are at most four rectangle pairs. Do not assign a common clipped reduction to a signed difference.

With no surviving slot, (MF3.69) and the smaller-width zero-slot bound fit (MF3.64), $\Delta \ge 0$. Otherwise side one's pre-allocation total is $A' = A - c - w - c_2$, and

$$(A' - M') - (A - M) = g + w - d - c_2 - (K_0 - K) - w_o + g_2 - t_2 \le 6(w + \ell) + \delta_1. \tag{MF3.70}$$

Use $g_2 - t_2 \le c_2$. In the amplified norm, $g \le d + (K_0 - K) + 6\ell + \delta_1$; in errors, $g \le d + (K_0 - K) + \sigma + \delta_1$ and the extraction table gives $w - w_o + \sigma \le 6w$. Exchange c_2, d_2 for the other side.

Delete extra masks. For each actual natural child with live slots put $F_{\mathrm{act}} = (A_{\mathrm{act}} - M'_{\mathrm{act}} + (6\kappa - 1)z_{\mathrm{act}})_+$. Equations (MF3.68)–(MF3.70), parent capacity and aggregate endpoint bound give

$$F_{\mathrm{act}} \le 6(w + \ell) + \delta_1 + 2\theta_N. \tag{MF3.71}$$

Use (MF3.47) to remove whole slots until defect vanishes or no slot remains. A length-d slot releases $6\kappa d$ capacity and costs κd energy. A greedy prefix overshoots by at most one mesh, so

$$\kappa d_z \le F_{\mathrm{act}}/6 + \kappa\eta_{\mathrm{mesh}} \le \Delta + \delta_1/6 + \theta_N/3 + \eta_{\mathrm{mesh}}. \tag{MF3.72}$$

If slots remain use the smaller-width positive theorem; otherwise use unrestricted zero-slot bounds. Equations (MF3.69), (MF3.72) and $M'_{\mathrm{act}} - M' = \delta_2$ give edge cost $\le \delta_2 + \delta_1/6 + 7\theta_N/3 + \eta_{\mathrm{mesh}}$ plus its small-power share. Cauchy averages costs, including the single mesh.

7.2.9 Fixed-character rows and rectangle cancellation **Exceptional count.** Genuine second allocations avoid old moving support by original zeros, so no old character cancels $G_c V_{\mathrm{id}}$ there. For nonunit common multiplicity-one primes of radical length v_1, put $f = 2v_1$. Their fixed row valuation is two; unramified fixed-modulus induction forces $v_p(h') \equiv 4 \pmod 6$. More generally each good prime has a prescribed valuation class modulo six, zero outside frozen support. Units and S-valuations give finitely many sectors, with sixth powers free. Thus $(h') = h_0 v^6$, h_0 fixed sixth-power-free, $q_{h_0} \ge Z^{4v_1} = Z^{2f}$. Counting gives $(m'_{\mathrm{act}} - 2f)/6$; we use only

$$\#\{h' \text{ exceptional} : q_{h'} \ll Z^{m'_{\mathrm{act}}}\} \ll Z^{(m'-f)/6+\delta_2/6+o(1)}. \tag{MF3.73}$$

Below the forced initial norm the range is empty up to fixed endpoints. This counts principal and all other fixed-inducing rows, including positively added partition rows; no t_2 saving is used.

After full Möbius allocation, side j's normalized absolute volume is $O(Z^{(\alpha_j - r_j)/2 + e_j})$. Both extraction scalars and divisor count give paired volume $\le Z^{a_0 - b_2 + 2\theta_N + o(1)}$ by (MF3.67), since $t_- - r_1 - r_2 \le \omega_2$, $r_1 \ge 0$. Including the exceptional count, its nominal excess is

$$\frac{m' - f}{6} + a_0 - b_2 - (M' + \Delta) = A - \frac{5}{6}M - F_1 - F_2, \tag{MF3.74}$$

where

$$\begin{aligned} F_1 &= c/6 + 5\{d + (K_0 - K)\}/6 + w/3 + \widetilde{q}/6 + w_o + B_c - 5g/6 + \ell, \\ F_2 &= 2b_2 - 5g_2/6 - p_2 + t_2 + V/6 + f/6. \end{aligned} \tag{MF3.75}$$

These expressions obey

$$F_1 \ge \frac{2}{3}(c + w) - 3\sigma - 5\delta_1/6, \qquad F_2 \ge \frac{2}{3}b_2. \tag{MF3.76}$$

Initially take $g = J_+$, $\ell = 0$, $D_0 = d + K_0 - K$. For $J \ge 0$, $F_1 = c + 2w + \widetilde{q}/6 + w_o/6 + B_c$. Otherwise $\widetilde{q} \ge R$ gives $B_c + \widetilde{q}/6 \ge (3c - 5D_0)_+/6 - 5\delta_1/6$, hence $F_1 \ge 2(c + w)/3 - w/3 - 5\delta_1/6$. Restoring $g \le J_+ + 2\sigma$, $\ell \ge 0$, $w \le 7\sigma/3$ costs $< 3\sigma$. For F_2 use the exact local table of ğ5.6, with $j = j_0$:

The four pairs (F_2, b_2) are $(7i/6, i)$, $((7i-5)/6+1_{i=1}/3, i)$, $(7i/6-1, i)$, and $(i+j_0/6-1, (i+j_0)/2)$ for, respectively, equal non-six-divisible/unit, equal non-six-divisible/nonunit, equal six-divisible, and unequal $i > j_0$, $6 \mid j_0$/unit cases.

Each first entry is $\ge 2/3$ of the second; at nonunit $i = 1$, $f/6$ gives equality. Thus (MF3.76) holds and uncentered $A \le 5M/6$ costs at most $3\sigma + O(\xi + \theta_N)$, validating comparisons.

Masked lattice cancellation. Apply ğ5.1 to fixed $\vartheta \in \Theta$, polynomial squarefree R_*, annular W_i and common real t.

(MF3.77) is (65), with the same $L_i(X, t)$, c_{ϑ,R_*} and $I_i(t)$, loss ϵ_1 and $\langle t\rangle = 1 + |t|$. It holds uniformly for $X > 0$.

Its proof there is fixed-lattice Poisson followed by complete mask inversion; in particular $c_{\vartheta,R_*} = c_\vartheta \sum_{d|R_*} \mu(d)\vartheta(d)/q_d$. Primary/$S$ restrictions are fixed lattice classes, and already excluded primes can be omitted. Thus it applies to every retained exceptional mask.

For $U_i = X_i/q_{b_i}$, $V_i = Y_i/q_{b_i}$, $T = U_1U_2 = V_1V_2$, the sum of (MF3.66) is the difference of the two L_1L_2 products. If all four scales are $\ge Z^r$, $r \ge 0$, then

(MF3.78) is the normalized rectangle bound (66), with these U_i, V_i, T, loss ϵ_1 and height factor $\langle t\rangle^{2J}$.

The same row, mask and norm power make main terms equal. Without a nonnegative lower scale use the all-scale volume $O(T^{1/2})$.

For centered exceptional children, pre-allocation lower lengths are $L - c - w - c_2$ and $L - c - w - d_2$. Put $r = (L - c - w - \min(c_2, d_2))_+$ and call the saving side one. Final extraction removes at most r_1 from either plain: use (MF3.78) if $r_1 < r$, otherwise volume. Preserve raw equal-product scales without clipping. Exactly,

$$t_- - r_1 - r_2 - (r - r_1)_+ = (t_- - r_2) - \max(r, r_1) \le \omega_2 - r. \qquad \text{(MF3.79)}$$

Paired volume is therefore $Z^{a_0 - b_2 - r + 2\theta_N + o(1)}$, uniformly in live slots. Their labels never entered Fourier density; sum them now by volume with (MF3.67)'s normalization.

Put $v = c + w + \min(c_2, d_2)$. From (MF3.76), $F_1 + F_2 \ge 2v/3 - 3\sigma - 5\delta_1/6$. The remaining centered exceptional deficit is bounded by

$$A - 5M/6 - 2v/3 - (L - v)_+ \le (A - M)_+, \qquad L = M/4. \qquad \text{(MF3.80)}$$

As in ğ5.6, the maximum occurs at $v = L$ and equals $A - M$. Positive slots have $A \le M$, zero-slot padding $A \le M + \delta$; exceptional terminal cost is $\le 3\sigma + \delta$, with reserved support/frequency errors.

7.2.10 Closure and uniform common mesh Strict nominal widths decrease by σ. Actual enclosure/support corrections cost $C_*\xi$ for numerical C_* independent of slot count: they involve one aggregate θ_N, two frequency enclosures, at most two plain clippings and finitely many aggregate lengths in (MF3.53)–(MF3.80). Ledgers are affine or positive-part affine; positive capacity varies by at most $|\Delta M| + |\Delta A| + 5|\Delta z|$, since $6\kappa - 1 \le 5$. Greedy removal charges one mesh.

Choose ρ, σ, δ from the bounded real ranges and final ϵ, with $\delta \ll \min(\rho, \sigma)$, so that

$$T_{\text{term}} := \rho + \delta + \rho/6 + 3\sigma + 5\sigma/3 < \epsilon/4.$$

Put $D = 2 + \lceil 2M_{\max}/\sigma \rceil$, and, still before specifying the fixed slot system, choose

$$\xi < \min\{\delta/2, \rho\mathfrak{m}(7/10)/2, \sigma/(4C_*), \epsilon/(16C_*D)\}, \qquad \eta_{\text{mesh}} < \min\{\sigma/6, \epsilon/(16C_*D)\}, \qquad \epsilon_0 < \epsilon/(16C_*D). \qquad \text{(MF3.81)}$$

These preserve comparison margins, padding and width decrease $\ge \sigma/2$. Each strict child leaves its band; depth is $\le D - 2$. Same-band comparisons use earlier uncentered stages, each plain reflects at most once before centering, and errors are not re-amplified. Thus the induction is acyclic.

Now fix arithmetic data, profiles and finite slot count, then H_N and boxes through depth D. Eventually $\theta_N \ll \xi$, tails satisfy whole-dyad inclusions and pool avoids slots. Allocate stage losses within ϵ_0: e.g. at most $N+2$ mask masses get $\epsilon_0/(10(N+2))$ each; use global divisor bounds $C_N^{\omega(v)}, d_{N+2}(v)^C \ll_{N,C,a} q_v^a$ for one small a. Prime displacement costs $2ez$, independent of slot count. These affect constants, not the already chosen mesh.

With d strict calls remaining allow $E_d = T_{\text{term}} + (d + 1)C_*(\eta_{\text{mesh}} + \xi + \epsilon_0)$. Floor, diagonal and exception fit E_0; (MF3.69)–(MF3.72) bound a strict edge by its child plus one allowance, including

comparisons/reflection masses. Cauchy averages and triangle takes maxima, so terminal loss occurs once. Equation (MF3.81) gives $E_{D-2} < \epsilon$, simultaneously for every surviving slot subset.

Choose raw tails and finite profile/height orders backwards, then one final threshold for tails, endpoints and logarithms. Child extra masks lie on frozen columns or a frozen amplifier prime, hence stay polynomial through finite depth. Exceptional masks also include row radicals and are covered uniformly by (MF3.77). All moving support, including canceled powers, remains explicit. This proves (MF3.40).

For completeness, fixed nonprincipal zero-slot rows are also restored directly. For a primitive nonprincipal ϑ of the fixed modulus, orthogonality gives $c_\vartheta = 0$, hence $c_{\vartheta,R_*} = 0$ in (MF3.77). Each normalized nonempty $X \geq 1$ plain is $O(Z^{\epsilon_1} X^{-1/2} \langle t \rangle^J)$; bounded subunit scales use volume. Prescribed valuations modulo six give $O(Z^{m/6+\epsilon_1})$ rows by the exceptional parameterization without forced-prime savings. The fixed family is finite, so multiplying and choosing small ϵ_1 restores all such rows within $Z^{M+\epsilon}$, with independent heights and common masks. Principal inducing characters remain excluded. □

7.3 Natural moments and the physical interfaces **Proposition 7.6** (Refined incumbent moment). *The same natural moment, with each active $0 < z_i \leq \eta_{\mathrm{mesh}}$, is valid for $7/10 \leq \kappa \leq 3/4$, $\beta \geq 51/100$ and $2\beta - 1 \leq \kappa$, with its mesh chosen uniformly in this interval before the slot system, profiles and target height.*

Proof. Restrict the same proof to $[7/10, 3/4]$: $6\kappa - 1 \leq 7/2$, margin $\geq \mathfrak{m}(7/10)M > 0$, removal ratio $1/6$. Equation (MF3.81) preserves the stated quantifier order. The mesh and all profile controls remain common on this closed interval. □

Literal initial fields and positivity.

For fixed source η, take scale U, puncture one, width $1 + \delta$ once $q_{\mathfrak{q}_\eta} \leq U^\delta$, and keep set initialKeep$(\eta, \mathfrak{Q}_0)$. At $X_1 = X_2 = U^m$, Theorem 7.4 gives

$$\sum_k \mathbf{1}_{\mathrm{initialKeep}(\eta,\mathfrak{Q}_0)}(k) \Re\Phi(q_k/U)\, |S_k(U^m)^2 \prod_{i\in I} Q_{k,i}|^2 \ll \mathcal{P}_J(W_1, W_2)^2 \mathcal{D}_J(\Phi)(1+|t_1|+|t_2|+\max_i |v_i|)^h U^{1+\delta+\epsilon}$$

provided $2m + 6\kappa \sum_{i\in I} z_i \leq 1$ and $0 < z_i \leq \eta_{\mathrm{mesh}}$. Here $\mathcal{P}_J$ uses finitely many plain Euler seminorms (common slot profile fixed in the constant); $\mathcal{D}_J$ controls the nonnegative radial Schwartz profile. Choose J, h before heights and live subset. Absorb δ and local loss into the requested loss. This physical field is $|S|^4 |\prod Q_i|^2$.

Theorem 7.3 admits literal relative-character slot coefficients, fixed ray Fourier coefficients and common character conjugation, all bounded and row-independent. Shared inverse primes are allocated by (MF3.32). At $m = 1$ the gates are $r + 2z < 1$, $2r + 8z < 3$. For finitely many profiles/derivatives take maximum orders/thresholds and union seminorms. All energies use nonnegative measures and squared moduli, so retained subfamilies decrease them. Signed Möbius coefficients remain until their full identity is restored.

Every rescaled completed term.

Theorem 7.1 covers all original slot subsets. Priority allocation to frozen products preserves row-independent surviving coefficients, retaining exact central scalars and tuple counts outside. For physical rescaling by subset width d,

$$M' = M - 2d, \qquad z' = \ell - d, \qquad N' = 1 + \ell - d.$$

If $M + \ell \leq 1$, then $M' + 2z' \leq N'$ for every such subset. The energy exponent is therefore $M' + (1 + 3\ell - 2M + d)_+/4 + \epsilon$. After the physical Cauchy square root, its exact compensator and assigned tuple mass give the additional exponent

$$-d + \frac{1}{8}(1 + 3\ell - 2M + d)_+.$$

Its slopes are $-1, -7/8$, so the maximum is $d = 0$; this pays every rescaled term with its actual outside coefficient. Individual energies need not be $\leq Z^{M'}$.

Lemma 7.7 (Actual finite source rows). *Let $\mathcal{R}$ be a finite row family with an injective map $\iota : \mathcal{R} \longrightarrow \mathcal{O}$. Suppose that, for each $u \in \mathcal{R}$, its displayed character χ_u agrees at every ideal with the natural function $n \mapsto \eta(n)\chi_n(\iota(u))$, including the fixed bad-prime zero mask. Suppose also that $\iota(u) \in$ initialKeep$(\eta, \mathfrak{Q}_0)$ and $q_{\iota(u)} \leq U$. Choose a fixed smooth radial majorant $\Phi \geq 0$ with $\Phi \geq 1$ on $[0, 1]$. Then the unweighted finite-row plain times marked-prime energy is at most the initial-state positive energy in Theorem 7.4. It consequently obeys its bound $CU^{1+\delta+\epsilon}$ and the stated polynomial height control whenever $2m + 6\kappa z \leq 1$.*

Proof. The literal character identity identifies all plain and relative-slot coefficients. Injectivity reindexes without multiplicity; the radial majorant and keep indicator dominate on the image. Enlarge only

the resulting nonnegative squared-modulus sum. For profiles with at most two logarithmic derivatives, the product rule for $W(y)y^{-\sigma+it}$ gives finite derivatives of W times polynomials in bounded σ and height t. Combine with the moment's height polynomial. Degree and constant precede live rows, subset, ray characters and heights, proving the stated finite field and uniformity. □

8 Overlapping prime lists in the inverse moment

This coefficient-specific extension precedes affine arguments and is endpoint-independent.

Lemma 8.1 (Overlap adapter for the inverse moment). *The estimate of Theorem 7.3 remains valid for a fixed finite number of possibly overlapping or repeated prime lists with the same bounded annular coefficients. Both admission inequalities may use the full sum of the original slot widths.*

Proof. Partition finite prime tuples by exact equality pattern. Freeze each block of $k \geq 2$ equal primes before the row norm. Its intersecting annuli have comparable D_i, giving central mass

$$\sum_{\mathfrak{p}\in\bigcap_i \mathcal{P}_i} \prod_{i=1}^{k} D_i^{-1/2}|b_i(\mathfrak{p})| \ll (\min_i D_i) \prod_{i=1}^{k} D_i^{-1/2} \ll D^{1-k/2} \ll 1.$$

Constants depend only on fixed slot count and profile suprema. Keep $\varphi_u(\mathfrak{p})^k$ until taking the positive norm, then contract by its modulus ≤ 1 with zeros retained. The inverse polynomial is unchanged and may share that prime.

Delete frozen primes from singleton lists. For K singletons, color their union independently with K colors. Each distinct ordered tuple receives prescribed slot colors with probability K^{-K}; repeated tuples never do. Its polynomial is therefore K^K times the average over products on disjoint, row-independent colored lists. Apply Theorem 7.3, then Minkowski/Jensen, costing K^{2K} in energy. For $K = 0$ use UM. Deleting slots decreases both admission expressions.

Triangle over bounded-mass frozen choices and finitely many equality patterns proves the claim. This coloring extension applies only to MI; MP retains its canonical-pool contract and overlap allocation. □

9 The zero supremum and the natural moment range

Let $\mathcal{H}$ be the presented finite-order Eisenstein Hecke family with prescribed Euler factors and natural zeros. Its actual zero supremum is

$$\beta = \sup\left(\{\tfrac{1}{2}\} \cup \{\Re\rho : \chi \in \mathcal{H},\ \Re\rho > 0,\ L_K(\rho,\chi) = 0,\ (\chi,\rho) \neq (1,1)\}\right).$$

Absolute convergence of the Euler product in $\Re s > 1$ gives $1/2 \leq \beta \leq 1$. The principal pole at one is excluded throughout.

Lemma 9.1 (Primitive targets approaching the supremum). *The same supremum is obtained from primitive targets. If $\varepsilon > 0$ and $\beta - \varepsilon \geq 1/2$, there is a primitive target with a zero ρ satisfying $\Re\rho > \beta - \varepsilon$.*

Proof. A presentation induced by χ^* has $L_K(s,\chi^*)$ times finitely many $1 - \chi^*(\mathfrak{p})(N\mathfrak{p})^{-s}$. These are one when the value is zero and otherwise vanish only on $\Re s = 0$. Primitive and presented positive-half-plane zeros coincide. Supremum approximation then supplies the asserted primitive zero; no attainment is needed. □

Lemma 9.3 (A common-reserve Mellin contradiction). *Let $1/2 \leq \theta < \beta$, and suppose k_0 is real. Suppose there are common numbers $\omega, \sigma_0 > 0$ with $\omega < \beta - \theta$ such that each primitive target has a probe J_η and a function H_η, holomorphic on $\Re s > \theta$, satisfying*

$$|H_\eta(s) - 1| \leq \frac{1}{2}, \qquad f_\eta(Z) = \frac{1}{2\pi i}\int_{(c)} Z^{s-k_0} e^{(s-5/6)^2} \frac{H_\eta(s)}{L_K(s,\eta)}\, ds, \tag{I.3}$$

on an absolute line $c > 1$, with the stated vertical convergence. Assume that f_η decays faster than every power at zero and that, for large Z,

$$J_\eta(Z) \ll_\eta Z^{\theta-k_0+\omega}, \qquad J_\eta(Z) - f_\eta(Z) \ll_\eta Z^{\beta-k_0-\sigma_0}. \tag{I.4}$$

Then these hypotheses contradict the definition of β.

Proof. Put $m = \min(\beta - \theta - \omega, \sigma_0) > 0$. The two bounds on the same probe give $f_\eta(Z) \ll_\eta Z^{\beta-k_0-m}$, so

$$F_\eta(s) = \int_0^\infty f_\eta(Z) Z^{k_0-s} \frac{dZ}{Z}$$

is locally uniformly holomorphic for $\Re s > \beta - m$: use this bound at infinity and arbitrary decay at zero, also after logarithmic differentiation. Absolute-line Fourier–Mellin inversion identifies it with the Gaussian quotient in (I.3); meromorphic identity continues throughout this connected half-plane.

Since $\beta - m \geq \theta + \omega > \theta$, the numerator is holomorphic and nonzero ($|H_\eta| \geq 1/2$). Any denominator zero, of any multiplicity, would give an impossible pole. Euler deletions change no such zero by Lemma 9.1; the principal pole creates no reciprocal pole.

Thus every primitive target is zero-free for $\Re s > \beta - m$, with common m despite target-dependent constants. Lemma 9.1 at $\varepsilon = m/2$ supplies a forbidden zero, since $\beta - m/2 > \theta \geq 1/2$. □

10 The analytic reduction of the literal initial source

We carry the literal marked coefficient through the two Poisson formulas of Chapters 1 and 7, preserving actual pools, allocations and normalizers. Later source classes are restrictions of these expressions.

10.1 The coefficient and the reference comparison Fix $U > 1$, $H_0 = U^M$, a finite slot set I, $P_i = U^{w_i}$ and

$$b_i(\mathfrak{p}) = \nu_i(\mathfrak{p}) W(q_\mathfrak{p}/P_i)(q_\mathfrak{p}/P_i)^{-\sigma_i + iv_i}.$$

Each relative ν_i retains both defining zero sets. Pools contain all support primes in the identity ray class and may overlap. Actual affine input has $\sigma_i = 33/50$, $\operatorname{supp} W \subset [1,2]$. Put $\chi_z(n) = \eta(n)\chi_n(m^6 Az)$ with original zeros, and

$$L_z(p, X; t) = \sum_{n \neq 0} \chi_z(n) q_n^{it} p(q_n/X), \qquad V = X_1 X_2 \prod_{i \in I} P_i.$$

The energy to reduce is

$$(115) E_X = \frac{1}{V} \sum_{\substack{z \neq 0 \\ \operatorname{keep}(z)}} \Phi(q_z/H_0) \left| L_z(p_1, X_1; t) L_z(p_2, X_2; t) \prod_i \sum_{\mathfrak{p} \in \mathcal{P}_i} b_i(\mathfrak{p}) \chi_z(\mathfrak{p}) q_\mathfrak{p}^{it} \right|^2.$$

Use real nonnegative Φ (or $\Re\Phi$ in the complex-profile convention). Keep the incumbent's actual natural predicate and fixed conductor ideal dividing (72); later zero bins do not replace it.

Assume $0 < M_{\min} \leq M \leq M_{\max}$ and

$$(116) .41M \leq n_i := \log_U X_i \leq .44M, \qquad w_i \geq 0, \qquad z_I := \sum_i w_i \leq M/25.$$

Choose the reference rectangle

$$(117) Y_1 = U^{M/4}, \quad Y_2 = X_1 X_2 / Y_1, \qquad D(l_1, l_2) = p_1(q_{l_1}/X_1) p_2(q_{l_2}/X_2) - p_1(q_{l_1}/Y_1) p_2(q_{l_2}/Y_2).$$

The rectangles share product, whole character, mask, slots and norm power. Keep their difference until positive child norms form.

For actual-pool tuples $\mathbf{l} = (p_i, l_1, l_2)$ on the union of plain annuli, put $J(\mathbf{l}) = (\prod p_i) l_1 l_2$. With common puncture R and squarefree seed s, define

$$(118) \beta_{R,s}(J) = \sum_{\substack{\mathbf{l} \\ J(\mathbf{l}) = J}} \left(\prod_i b_i(p_i) \right) D(l_1, l_2) \mathbf{1}_{(J,R)=1} \mathbf{1}_{s | J}.$$

Initially $R = s = 1$. Zeros remain zero; later masks/divisors are actual extractions, not new conditions on plain quotients.

Lemma 10.1 (The original comparison). *Suppose $\beta \geq 51/100$ and $2\beta - 1 \leq 3/4$. For every $\epsilon > 0$ one common positive slot mesh gives*

$$E_Y \ll \mathcal{D}(\Phi) \mathcal{H}_J (1 + |t| + \max_i |v_i|)^J U^{M+\epsilon}, \qquad \mathcal{D}(\Phi) = 16(1+\pi)^2 \max_{0 \leq j \leq 2} \sup_x |x|^j |\Phi(x)|.$$

The mesh precedes the fixed dictionary; subsequent orders/constants/threshold cover all surviving subsets, four plain derivative pairs and (116).

Proof. Comparison lengths are $M/4, n_1+n_2-M/4$. Reflect the second whole factor if $>M/2$. By (MF3.42), conductor times redundant radical is $O(U^M)$, hence retained reflected length $\le M/2+\xi$. Then

$$M/4+(M/2+\xi)+(9/2)z_I \le .93M+\xi < M$$

For $\xi < .07M_{\min}$ with loss reserve, apply Theorem 7.4 at $\kappa = 3/4$, or unrestricted zero-slot bounds. Capacity is applied after reflection.

Use the exact natural reflection (MF3.44): it retains all holes, has $U^{o(1)}$ coefficient mass, and after whole-factor conjugation restores orientation without changing relative slots. Its profile has bounded zero-end Euler derivatives, rapid upper decay and summable central annuli; empty annuli vanish and retained subunit scales dilate boundedly.

Apply common-profile/Sobolev control to rowwise scales and Stirling to heights. After fixing the dictionary, take finite maxima over subsets and four profile pairs, before live labels/heights. □

10.2 The first exact source Let E_D be the energy of the whole difference in (117). Positivity gives

$$E_X \le 2E_D + 2E_Y. \tag{119}$$

Enlarge only E_D to all rows; never transform the keep predicate.

Fix nonnegative smooth Ψ supported in $[-1, b_\Phi+1]$, $\Psi \ge 1$ on $[0, b_\Phi]$, so $0 \le \Phi \le \mathcal{D}(\Phi)\Psi$ for $x \ge 0$. Before live subsets/labels, expand (118) in the full lattice square and extract complete common support Ca, Db, $(a,b) = (ab, CD) = 1$. Define

$$c = \log_U q_C,\ d = \log_U q_D,\ p = \log_U q_{\mathrm{rad}(CD)},\ A_0 = \log_U V.$$

Let r contain nonzero net common sextic exponents, length R_1, primitive character ξ_r. Expand the remaining mask by $e \mid \mathrm{rad}(CD)/r$, length E_1, so $R_1+E_1 \le p$. For actual allocated convolutions A_C, A_D, nonzero frequency gives

$$\frac{H_0}{Vq_e\sqrt{q_r}} \sum_{h\neq 0} \mathfrak{g}_{\xi_r}(r,h) \sum_{(a,b)=1} \frac{A_C(a)\overline{A_D(b)}}{\sqrt{q_a q_b}} \tau_C(a)\overline{\tau_D(b)} R(a,b) \times \mathfrak{g}(a,h)\overline{\mathfrak{g}(b,-h)}\widehat{\Psi}\left(\frac{H_0 q_h}{q_e q_r q_a q_b}\right). \tag{120}$$

One orientation is $\tau_C(a) = \tau(a)\chi_a(er)\overline{\xi_r(a)}$, $\tau_D(b) = \tau(b)\chi_b(er)\xi_r(b)$. Keep in A_C the exact frozen CRT scalar, $\mu(e)$ and fixed character values, including zeros. Convention-independently, if $f_{C,D,e}(x)$ is the actual active common character times both residual zero-extended characters at ex, define

$$\mathcal{T}_{C,D,e}(h) = \sum_{x \bmod rab} f_{C,D,e}(x) e(hx/(rab)).$$

Poisson multiplies it by $H_0/(q_e q_r q_a q_b)$; CRT defines all displayed factors simultaneously, preserving unit phases. Three Gauss sums restore $\sqrt{q_r q_a q_b}$ and central normalization gives $1/V$. The cross factor $\chi_a(br)\overline{\chi_b(ar)\xi_r(ab)}$ gives $R(a,b)$ by reciprocity. This proves the formula with nonunit zeros; changing orientation conjugates the whole character.

On retained smooth frequency dyads define $\mathcal{B}_q$ as V times the normalized block (120), removing its $1/V$ once. Include all common labels, inactive subsets and dyadic coordinates. Its normalized contribution is $\mathcal{B}_q/V$ and physical mass is

$$\mathcal{M}_{\mathrm{phys}} = \sum_q |\mathcal{B}_q| = \left|\sum_q \zeta_q \mathcal{B}_q\right|, \qquad \zeta_q = \begin{cases} |\mathcal{B}_q|/\mathcal{B}_q, & \mathcal{B}_q \neq 0, \\ 0, & \mathcal{B}_q = 0. \end{cases} \tag{121}$$

First labels thus stay in one joint sum, without an extra count.

The powerful zero-frequency argument of ğ7.2.6, now with full product norm V, gives $O(H_0 U^\epsilon \mathcal{H}_J)$ after divisor allocations and central $1/V$, retaining every mask.

The nominal first frequency length is

(122) is (76) under $(A,R,E,m) = (A_0, R_1, E_1, M)$: $K_0 = 2A_0 - c - d + R_1 + E_1 - M$.

Retain whole dyads through each outer box's maximum scale with enclosure $\delta_1 < \xi$. Discarded genuine arguments are $\ge U^{\xi/4}$ after fixed support errors. Use $|\mathfrak{g}(a,h)| \le \sqrt{q_a}$ and polynomial counts, fixing raw cost before Schwartz order. For every fixed $L > 0$,

$$(123)E_D \le \mathcal{D}(\Phi)\{\mathcal{M}_{\rm phys}/V + C\mathcal{H}_J H_0(U^{\epsilon}+U^{-L})\}.$$

Proposition 10.2 (Initial energy reduced to its actual source). *Under Lemma 10.1, one Ψ, positive slot mesh, finite profile orders and J give*

$$(124)E_X \le 2\mathcal{D}(\Phi)\mathcal{M}_{\rm phys}/V + C\mathcal{D}(\Phi)\mathcal{H}_J(1+|t|+\max_i|v_i|)^J U^{M+\epsilon}.$$

After fixing arithmetic and dictionary, thresholds are uniform over live subsets, natural masks and initial scales. Descendants need satisfy no affine condition.

Proof. Combine (119), (123) and Lemma 10.1. Initial supports eventually give $V, H_0^{-1} \le U^{M_{\max}+1}$, the needed coarse caps. Relative norm twists/derivatives have polynomial seminorm cost. Take finite maxima over subsets/anchors; allocate $\epsilon/2$ to comparison/remainder and absorb logarithms in the rest. □

10.3 Complete kernels and allocation normalizers On a retained dyad the nonseparated kernel contains

$$\omega_K(q_h/U^K)\widehat{\Psi}(H_0 q_h/(q_e q_r q_a q_b))(q_a q_b)^{-1/2}.$$

Multiply by common log-row and whole-column cutoffs, shared by both rectangles. For the resulting compact logarithmic kernel,

$$(125)\mathcal{K}(\mathbf{x}) = \int_{\mathbb{R}^d} \widehat{\mathcal{K}}(\mathbf{u}) e^{2\pi i \mathbf{u}\cdot\mathbf{x}}\, d\mathbf{u}, \qquad \int (1+\|\mathbf{u}\|)^J |\widehat{\mathcal{K}}(\mathbf{u})|\, d\mathbf{u} < \infty.$$

Bounded Euler derivatives of $\widehat{\Psi}$ and integration by parts of order $> J+d$ give live-column-independent seminorm control. Separate the second full kernel/roots in the same order.

Extend (120) off coprimality using separate Gauss sums, fixed bicharacter and original masks. Insert complete $\sum_{s|(a,b)}\mu(s)$ before factorwise estimates; artificial primes gain no common-support restrictions. Expand the ray factor and apply joint Cauchy after (125), giving

$$(126)\sum_h \omega_K(q_h/U^K)\left|U^{-(A_0-c)/2}\sum_{s|a} A_C(a)\tau'_C(a) q_a^{it}\mathfrak{g}(a,h)\right|^2,$$

and similarly on the right. Set

$$B_c = (3c-5d-R_1)_+/6, \qquad B_d = (3d-5c-R_1)_+/6,$$

$$(127)S_C = p+A_0-c+R_1+E_1+B_c, \quad S_D = p+A_0-d+R_1+E_1+B_d, \quad O_0 = M-A_0-R_1/2-E_1.$$

By (MF3.56), $O_0 + (S_C+S_D)/2 \le M$. A fixed divisor width s_0 leaves column allowance $A_0 - c + R_1 + E_1 + B_c - s_0$; one common-label count $p+s_0$ gives S_C. Keep that count inside joint weights.

Let j index actual first columns, divisors and ray characters in a common box, with chosen common ideal C_j. An allocation $B = (B_i, B_1, B_2)$ extracts each factor's complete C_j-part, $\prod_i B_i B_1 B_2 = C_j$. Freeze supplying slots; survivors keep coefficients. Divide both rectangle scales of each plain by the same q_{B_k} and puncture all residual factors at complete support, even if exponents cancel.

Let F_B be the actual frozen coefficient with profiles, $M_B = \prod_{i\,\text{frozen}} M_i$ its bound. Retain $F_B = 0$ before enlargement. For slot support bound b and ambient count N, allocation Cauchy gives

$$(128)\lambda_{j,B} = \begin{cases} |\rho_j|^2(\max(1,b))^N \dfrac{\#\,\mathrm{Alloc}(C_j)}{q_{C_j}} M_B^2, & F_B \ne 0, \\ 0, & F_B = 0. \end{cases}$$

Let I_B be the surviving slots. The actual child volume and exact central scalar are

$$V_{j,B} = \frac{X_1 X_2}{q_{B_1} q_{B_2}} \prod_{i\in I_B} P_i, \qquad \frac{V_{j,B}}{V} = \frac{1}{q_{C_j}} \prod_{i\notin I_B} \frac{q_{p_i}}{P_i}.$$

Factoring F_B, the original normalized sum is exactly $F_B\sqrt{V_{j,B}/V}$ times the $V_{j,B}^{-1/2}$-normalized child. Since $q_{p_i}/P_i \le b$, its squared scalar is $\le (\max(1,b))^N M_B^2/q_{C_j}$, yielding (128). Actual child volume divides its energy; the first allocation weight stays undivided.

The scalar ρ_j retains the first Poisson coefficient. For original complex block scalar a_j, including (121)'s phase, an asymmetric split is

$$\rho_{C,j} = a_j U^{-O_0 - S_C/2}, \qquad \rho_{D,j} = U^{-S_D/2}.$$

Five geometry coordinates vary by $\le \log_U 2$ per box; the positive-part allowance varies by at most $3/2$ times this width. Cross-label Cauchy costs $U^{5\log_U 2}$, a constant, after freezing common coordinates—not the square of the label count.

For extracted datum s, let β_s be (118)'s residual coefficient without F_B, τ_s its whole character and V_s its actual volume. Define

$$(129)\quad G(s;\Omega,L) = \frac{1}{V_s}\sum_{h\in\mathcal{O}} \Omega(q_h/L)\left|\sum_u \beta_s(u)\tau_s(u)q_u^{it}\mathfrak{g}(u,h)\right|^2.$$

The side budget is $\mathcal{L} = \sum_{j,B}\lambda_{j,B}G(s_{j,B};\Omega,L_j)$, with fixed annular majorant on $[1/8,2]$, including all columns/allocations.

The column window has an independent Mellin coordinate: left/conjugate-right raw heights are $t+2\pi(u-v)$, $t+2\pi(u+v)$. Window densities have common finite weighted L^1 norms by integration by parts; Minkowski integrates the polynomial-height $\mathcal{L}$ bound through both full windows. Bounded row-profile parameters and partition scalars contract the joint norm.

10.4 Corrected MAIN amplification Set $M = 1$. For each selected source cell, use its exact rational parameters from (S4) for the seed closure and (NF4) for the later closure. Thus the earlier LOW face (B_0,Λ_0), the new face (B,Λ), its slope q, cell endpoint b, width reserve e ($e=\mu$ in (S4)), and return reserve r define the amplifier by

$$t = B - \Lambda(b-e), \qquad a = \frac{q\{b+(t-B_0)/\Lambda_0\}+r}{1-6qB_0/\Lambda_0}, \qquad \sigma = 3a.$$

The rational gates in those certificates give a positive denominator and

$$0 < a < 1/75. \qquad (130)$$

There are finitely many selected cells. Choose the common slot mesh below half their minimum positive a, and keep that cell's amplifier fixed across rectangles, derivatives and descendants.

Use the literal prime pool

$$\mathcal{P} = \{\mathfrak{p} : U^a/2 < q_{\mathfrak{p}} \le U^a,\ \mathfrak{p}\notin S_{\mathrm{bad}},\ \mathfrak{p} \text{ in the prescribed identity ray class}\}.$$

Fixed-extension Chebotarev in the abelian ray-class extension L/K gives $\pi_{\{1\}}(x) \sim |\operatorname{Gal}(L/K)|^{-1}\operatorname{Li}(x)$; subtract at $x, x/2$. A primary reference is Thorner–Zaman, *A unified and improved Chebotarev density theorem*, Algebra Number Theory 13 (2019), 1039–1068, arXiv:1803.02823v3, Theorem 1.1. For fixed L, a possible exceptional real zero is < 1 and contributes $o(x/\log x)$. Thus $\#\mathcal{P} = U^{a+o(1)} \ge U^{a-\ell}$ for each fixed $\ell > 0$, uniformly over finite cells, with ray-dependent threshold and no new zero-free assumption. Eventually the pool avoids all slots.

Use the corrected whole coefficient $\widetilde{b}$ of (CA4), retaining the squarefree ordinary-product projector and the same common puncture. The local Gauss identity (8) gives (CA3); averaging its exact pool identity gives

$$F_b(h) = \frac{1}{\#\mathcal{P}}\sum_{p\in\mathcal{P}} F_{\widetilde{b}}(p^6 h). \qquad (131)$$

This is (CA5), applied before the positive norm to the complete centered coefficient, with all masks, natural zeros and profile heights retained.

The actual retained dyad has scale $K \le K_0 + \delta$. The identity

$$K + (d + K_0 - K - c)_+ + 2\sigma = \max(K + 2\sigma, d + K_0 - c + 2\sigma)$$

gives the MAIN radius

$$L_{\mathrm{main}} = U^{K_0+(d-c)_+ +6a+\delta+r}. \qquad (134)$$

Reserve $r > 0$, $U^r \geq 2$, pays smooth-ball doubling. For sector constant one, $\delta = \xi/2 + \log_U 128$. These derive from the original retained annulus. Negative exponents still define positive radii; test actual emptiness and never count a nonempty bounded range below one.

Proposition 10.3 (MAIN amplification of the actual affine source). *For the actual nonnegative weights (128), put* $\mathcal{M}(s) = G_{\widetilde{b}}(s; \Phi_{\mathrm{ball}}, L_{\mathrm{main}})$*, using the corrected whole coefficient in every transformed row. Then*

$$\mathcal{L} \leq \frac{56}{\#\mathcal{P}} \frac{15/2+6a}{a/2} \sum_{j,B} \lambda_{j,B} \mathcal{M}(s_{j,B}). \qquad (136)$$

The threshold precedes live labels and weights, orientations and heights.

Proof. Apply Jensen to (131), before any source estimate. Push its terms through $(h,p) \mapsto p^6 h$. If $q_p \geq U^{a/2}$ and the output norm is at most U^{C+6a}, their product divides that output, so there are at most $(C+6a)/(a/2)$ preimages. Formula (134) gives the radius majorant. Support $[1/8, 2]$ meets at most seven dyadic bands. The seven-band factor is at most the conservative constant 56. The full pool average in (131) is independent of frequency.

Initially $V \leq U^2$, $H_0^{-1} \leq 1$, and actual columns/pools have norms $\leq U^3$ eventually. Retained scales satisfy

$$8L_j \leq U^{3\max(2,0)+1/2+1} = U^{15/2} \qquad (0 \leq \xi \leq 1).$$

Nominal scale is $\leq V^3 H_0^{-1}$, then $U^{\xi/2}$ and factor ≤ 32 absorbed by U. Thus $C = 15/2$ works uniformly. Padded zero columns justify support admission only and add no source mass.

Multiply by $\lambda_{j,B} \geq 0$ and sum, taking one threshold over the finite selected cells and live subsets. The same pool and factor $56/\#\mathcal{P}$ occur once outside the joint sum. □

10.5 Second transform and normalized source classes Terms retain the whole corrected Gauss source of (131). MAIN has nominal density $U^{-a+o(1)}$ from the prime average. Put

$$a_0 = A_0 - c, \qquad J = d - c + (K_0 - K), \qquad g = J_+ + 6a.$$

Carry radius padding unchanged and restore live slots before squaring. For complete $B(u)$,

(137) is (MF3.60) under $Z = U$, $\mathcal{G} = \mathfrak{g}$, $\tau_1 = \tau$, $K_2 = \widehat{\Phi}_2$, with the same full $B(u)$, central $U^{-a_0/2}$ and density $U^{-\ell}$.

Take $\ell = a$ after multiplying both sides by nominal main density, keeping prime-count loss separate. The identity includes exact central powers and roots.

The genuine $j = 0$ diagonal is (11). The separate artificial zero Gauss row is supported on sixth powers, added only at final enlargement and never amplified. Its count retains $-s_0$ when $s \mid u$. Their terminal bounds are (MF3.57)–(MF3.61).

For $j \neq 0$, retain whole genuine dyads before extracting complete $u = C_2 a, v = D_2 b$. Put

$$c_2 = \log_U q_{C_2}, \quad d_2 = \log_U q_{D_2}, \quad \bar{b}_2 = (c_2+d_2)/2, \quad p_2 = \log_U q_{\mathrm{rad}(C_2D_2)}, \quad g_2 = \log_U q_{(C_2,D_2)}.$$

Apply (12) only on $(a,b) = (ab, C_2D_2) = 1$. At equal non-sixfold multiplicities split divided frequency into units/nonunits, widths t_2, V_2, latter radical V_{id}. Then (13)–(14) give

$$j = (C_2, D_2) V_{\mathrm{id}} h', \qquad |F(C_2, D_2; j)| \leq U^{g_2 - t_2}.$$

Extend the partition scalar by zero before enlargement. All extracted primes puncture residual factors; plain fifth power plus slot prime retains zero-extended sixth power.

The nominal child widths are

$$\begin{aligned} m' &= 2a_0 - K - g - g_2 - V_2, \\ q' &= R_1 + E_1 + t_2 + V_2, \\ M' &= m' + q' = M + J - g - g_2 + t_2. \end{aligned} \qquad (138)$$

For the actual main ball, writing all its radius padding as ω, this is

$$(139) M' = M - (c-d)_+ - 6a - \omega - g_2 + t_2.$$

Actual row width is $m' + \delta_2$, $0 \le \delta_2 < \xi$, and natural width $M' + \delta_2$ retains q' despite cancellations. An upper norm below one is empty; otherwise use its nonnegative width. The fixed unit remains in the literal h' character.

The first projector divides both common parts; second support counts cost $U^{p_2 - s_0 + o(1)}$. The full ledger is

$$(K + g - a_0) - \ell + (p_2 - s_0) + (g_2 - t_2) + (a_0 - \bar{b}_2).$$

Subtract it from the individual first allowance, with the MAIN normalization just displayed. The exact inner allowance is

$$(140) M' + \bar{b}_2 - p_2 + B_c + \ell.$$

The MAIN allowance includes density a once. Here $\bar{b}_2$ is paired average; after positive Cauchy, HIGH uses actual chosen-side c_2 or d_2 and corresponding frozen slots. Estimate sides separately before restoring the pair.

Remove residual coprimality by its complete Möbius sum and allocate each of its primes with

$$\mathbf{1}_{p \mid \prod n_i} = \sum_{\varnothing \neq J} (-1)^{|J|+1} \prod_{i \in J} \mathbf{1}_{p \mid n_i}.$$

Selected plains are pl with unrestricted l and identical rectangle division; selected slots freeze, with distinct primes in one slot giving zero. Artificial primes gain no genuine frequency restrictions.

The extraction check is (MF3.67)–(MF3.69): for pre-total α_j, extracted central length r_j, frozen ratio error e_j and pre/post negative lengths $\pi_{0,j}, \pi_j$, nonempty rectangles satisfy

$$|e_j| \le \theta_N, \quad 0 \le \pi_{0,j} \le \pi_j \le \theta_N, \quad r_{\mathrm{clip},j} = r_j - (\pi_j - \pi_{0,j}) \ge 0.$$

On a lower-width-t_- divisor dyad, divisibility gives $r_j + \omega_j \ge t_-$; disjoint entrance-slot subsets give $|e_j + \omega_j| \le \theta_N$. Hence

(141) is (MF3.69), with these same actual extraction and clipping parameters.

It pays divisor count and both central scalars. Form positive norms before splitting/clipping; at most four rectangle pairs occur.

Define classes by restricting the exact second sum: zero frequency, discarded whole dyads, fixed-inducing rows and natural nonexceptional rows. Only the last receives buffered zero bins from the child's finite whole-character dictionary and conjugates. Threshold-crossing bins are LOW; higher bins HIGH. Keep pulled-back weights, units, both whole-product Mellin powers, natural holes and both projector allocations. Only then bound the completed positive norms.

This reduces MAIN to LOW, HIGH, fixed-inducing, diagonal and tail sources. The global source proof applies the earlier global face to LOW, the actual marked row count to HIGH, and (CA13) to centered terminals.

Fixed-inducing rows retain predicate, forced valuations, masks and positive physical-radius factor. After freezing slots, their plains have one character, puncture, norm power and D. Equal products cancel leading lattice terms even after extraction, exactly as (MF3.73)–(MF3.80). Apply affine terminal reserves to this source; no enlarged zero predicate removes it.

10.6 Uniform analytic closure Choose the selected cell, mesh, strict scalar reserves and support padding before live scales, mesh before dictionary. Then choose one H_N over subsets/four profile pairs, $\theta_N = H_N / \log U$, without per-slot error. Choose seminorm/height orders backwards through kernels and functional equations before labels/heights.

Integrate retained heights against (125)'s full densities. For external cutoffs, fix height degree before tail order dominating it and raw counts. Discard genuine Poisson tails before separation using arguments $\ge U^{\xi/4}$, $|\mathfrak{g}| \le \sqrt{q_u}$, $|F| \le q_u q_v$, obtaining arbitrary decay for independent profile heights.

Contract (121), (128), (136), (140) in order. This pays first labels and amplifier density once, preserving all zeros/masks, and completes the literal-energy bridge to MAIN LOW and HIGH.

12.1 The literal row and its normalization Use $K = \mathbb{Q}(\sqrt{-3})$, $\mathcal{O} = \mathbb{Z}[\omega]$, the fixed bad set and natural zero-extended sextic ideal character χ_z. Before scale, fix η, η_0, ray modulus $\mathfrak{q}$ and a unit-quotient subgroup containing all global units. For ray character ϑ with Hecke lift η_ϑ, retain actual relative coefficient $\nu_\vartheta = \eta_\vartheta \overline{\eta_0}$ with both zero sets. Let $\mathcal{R}$ be the identity class.

For an element m, an anchor A, and a row z, put

$$c_z(I) = \eta(I)\chi_{m^6Az}(I), \qquad L_z(W, X; t) = \sum_{I\neq 0} c_z(I) q_I^{it} W(q_I/X).$$

All ideal characters mean their actual coefficients. For finite slots $\mathcal{J}$, put $P_j = U^{w_j}$ and

$$\mathcal{P}_j = \{\mathfrak{p} : \mathfrak{p} \text{ prime}, q_{\mathfrak{p}} \le 2P_j,\ \mathfrak{p} \in \mathcal{R}\}, \tag{158}$$
$$b_j(I) = \nu_{\vartheta_j}(I) W(q_I/P_j)(q_I/P_j)^{-\sigma_j + iv_j},$$
$$A_j(z) = P_j^{-1/2} \sum_{\mathfrak{p}\in\mathcal{P}_j} b_j(\mathfrak{p}) c_z(\mathfrak{p}) q_{\mathfrak{p}}^{it}. \tag{159}$$

Here $\operatorname{supp} W \subset [1, 2]$, so smaller pool primes contribute zero. Pools may overlap; retain fixed masks, relative coefficients and phases. The global source proof applies its exact collision decomposition to the full original marked pools.

For two annular profiles p_1, p_2, the normalized positive row is

$$\mathscr{R}_z = (X_1X_2)^{-1/2} L_z(p_1, X_1; t) L_z(p_2, X_2; t) \prod_{j\in\mathcal{J}} A_j(z). \tag{160}$$

Total central normalization is $(X_1X_2\prod_j P_j)^{-1/2}$, without prime-count or logarithmic factors.

In the initial energy $m = m_{\mathrm{bad}}$ and $A = 1$. Its row restriction is

$$\mathrm{keep}_{\eta,Q}(z): \quad z \neq 0 \quad \text{and} \quad z \text{ is not a fixed-inducing row for } (\eta, Q, m_{\mathrm{bad}}, 1).$$

The exceptional family is the incumbent's fixed-inducing family. Put $Q' = \mathrm{internalQ}(Q, \eta_0)$, retaining the relative anchor's fixed holes, and

$$\mathcal{E}_\eta(\Phi; p_1, p_2) = \sum_{z\in\mathcal{O}:\mathrm{keep}_{\eta,Q'}(z)} |\mathscr{R}_z|^2 \Re\Phi(q_z/U). \tag{161}$$

Independent plain heights are included by the unit-modulus norm powers in the common profile controls of Section 1.3; the single displayed t abbreviates that family. Retain all six units, sixth-power repetitions and root-relative holes.

Assume upper-supported Schwartz Φ, $\Re\Phi \ge 0$, $\operatorname{supp}\Phi \subset (-\infty, b_\Phi]$, allowing positivity at zero. Let

$$\mathcal{D}(\Phi) = 16(1+\pi)^2 \max_{0\le a\le 2} \sup_x |x|^a |\Phi(x)|.$$

For finite $\mathcal{F} \subset \mathbb{N}^2$, $\mathcal{S}_{\mathcal{F}}(p)$ is the product of the two maximum Schwartz-seminorm controls. Its square has the energy's quartic profile homogeneity; both supports lie in $[1/4, 9/4]$.

12.2 Opening the source without erasing its coefficients The product in (160) has a finite coefficient expansion

$$a(I) = \sum_{I_1I_2 \prod_j \mathfrak{p}_j = I} p_1(q_{I_1}/X_1) p_2(q_{I_2}/X_2) \prod_j b_j(\mathfrak{p}_j),$$

Keep $\eta(I) q_I^{it}$ outside $a(I)$. Open its square into actual pairs I, J; at common primes put $i = v_{\mathfrak{p}}(I)$, $j = v_{\mathfrak{p}}(J)$ and

$$k_{\mathfrak{p}} = (i + 5j) \bmod 6. \tag{164}$$

For $i, j > 0$, split common primes into active $k_{\mathfrak{p}} \neq 0$ and principal $k_{\mathfrak{p}} = 0$, giving

$$\chi_z(I_{\mathrm{com}})\overline{\chi_z(J_{\mathrm{com}})} = \mathbf{1}_{(z,\mathfrak{r}_{\mathrm{pr}})=1} \prod_{\mathfrak{p}\ \text{active}} \chi_{\mathfrak{p}}(z)^{k_{\mathfrak{p}}}. \tag{165}$$

Exponent zero retains its principal character's zero at the prime.

Let a, b generate coprime residual ideals, r the active radical, and $\rho = \prod \chi_{\mathfrak{p}}^{k_{\mathfrak{p}}}$. Expand the principal mask over $e \mid \mathfrak{r}_{\mathrm{pr}}$; set $K_e = K/q_e$. For the product f of both residual finite characters, ρ and fixed masks, choose a factoring modulus M_f and define

$$G_f(h) = \sum_{x \bmod M_f} f(x) e_K(hx/M_f).$$

With the Fourier convention of the arithmetic chapter, additive Poisson gives

$$(166) \sum_z \chi_z(I)\overline{\chi_z(J)}\Phi(q_z/K) = \sum_{e \mid \mathfrak{r}_{\mathrm{pr}}} \mu(e) f(e) \frac{K_e}{q_{M_f}} \sum_h G_f(h) \widehat{\Phi}(K_e q_h / q_{M_f}).$$

Proof. Substitute $z = ex$, then apply additive Poisson in each class modulo M_f. Fixed coefficients multiply both sides unchanged; $f(e) = 0$ when e shares an active/residual prime. Retain every frequency, including nonunits. This exact identity precedes estimates and applies to the multi-slot convolution.

Canonical Gauss factorization and sextic reciprocity give residual columns

$$(167) \ell_e(n) = \chi_{er}((n))\rho(n), \qquad r_e(n) = \chi_{er}((n))\overline{\rho(n)}.$$

The fixed phase $\mathfrak{R}(a, b)$ has ray-character expansion mass ≤ 256; the separated product is

$$\rho(e)\mathfrak{R}(a, b)\ell_e(a)\overline{r_e(b)}.$$

Retain masks at er, the fixed conductor and residual ideals. The reverse orientation exchanges columns and conjugates whole coefficients.

Apply the same finite transform to the allocated canonical residual columns. Their common ideals C_2, D_2 have equal support; put

$$g_2 = \prod_{\mathfrak{p} \mid C_2 D_2} \varpi_{\mathfrak{p}}^{\min(v_{\mathfrak{p}} C_2, v_{\mathfrak{p}} D_2)},$$

Correlation vanishes unless frequency is $g_2 h$. Expand principal factors over subsets $\mathcal{U}$ of common support, retaining holes at g_2 and $\prod_{\mathfrak{p} \in \mathcal{U}} \mathfrak{p}$. Local orthogonality saves at least $(\log_U q_{C_2} + \log_U q_{D_2})/3$ on nonzero support. Retain completed Gauss coefficients and the generator discrepancy as one of six unit labels.

The actual joint row is

$$(168) w = u_0 \epsilon\, e\, n_{\pm}\, g_2\, v_{\mathcal{U}} h.$$

Here u_0 is the parent anchor, ϵ a unit, e the inactive projector, $n_{\pm}$ the uncancelled numerator below, and $g_2 v_{\mathcal{U}} h$ the second frequency. Pre-partition valuations determine $\mathcal{U}$. The whole row character is a fixed finite ray character times χ_w with original moving puncture; reverse orientation conjugates it wholly. Zero selection uses that whole character.

12.3 The two actual allocations For tuple indices $\mathcal{J} \sqcup \{1, 2\}$, extraction of C uniquely splits coordinates $B_i V_i$ with $\prod B_i = C$, B_i supported on C and V_i outside it. A prime slot is live ($B_j = 1$) or wholly frozen ($V_j = 1$). Its frozen coefficient is

$$(169) F_B(C, R) = \mathbf{1}_{(C,R)=1} \prod_{j : B_j \neq 1} \nu_j(B_j) W_j(q_{B_j}/P_j).$$

Absorb fixed-annulus powers into W_j. Substitution dilates the plain scales to X_i/q_{B_i} and punctures at RC; summing admissible allocations recovers the exact coefficient, including dependent slots.

Lemma 12.2 (Width conserved by two allocations). *Suppose there are at most N_0 slots, the fixed annular lower endpoint is $a_0 > 0$, and both successive frozen coefficients are nonzero. Let C_1, C_2 be the ideals extracted in the two allocations. For every $\zeta_w > 0$, if $w_{\min} > 0$ and $N_0 w_{\min} \leq \zeta_w/2$, then, for all sufficiently large U,*

$$(170) s - \zeta_w \leq \log_U q_{C_1} + \log_U q_{C_2} + \sum_{\substack{j \text{ live after both allocations} \\ w_j \geq w_{\min}}} w_j.$$

The scale threshold is uniform in both allocations and their characters and heights.

Proof. Nonzero (169) gives $q_{B_j} \geq a_0 P_j$, while their product divides q_C. Thus frozen width is $\leq \log_U q_C - N_0 \log_U \min(1, a_0)$. Apply twice, identifying live slots with original indices. The total support

loss $-2N_0 \log_U \min(1, a_0)$ is eventually $\le \zeta_w/2$; remaining tiny widths cost $\le N_0 w_{\min} \le \zeta_w/2$. Remove zero allocations before maximization so the support implication applies. □

If final projector extraction replaces plain scales X_i by X_i/U^{r_i} and formally removes logarithmic volume Δ, then

$$(171)\ \sum_{i=1}^{2} \max(0, \log_U X_i - r_i) + \sum_{j \text{ live}} w_j = \sum_{i=1}^{2} \max(0, \log_U X_i) + \sum_j w_j - \left[\Delta + \sum_i (-\log_U X_i)_+ - \sum_i (r_i - \log_U X_i)_+\right].$$

Apply $x_+ = x + (-x)_+$ to raw volume conservation. The bracket is the actual clipped reduction, nonnegative on nonzero support; retain it even when a short plain scale is replaced by one.

12.4 The actual first numerator At active primes, reciprocity in (167) gives powers $1 + k_{\mathfrak{p}}$ or $7 - k_{\mathfrak{p}}$. Replace sixth powers by their coprimality masks and define

$$e_+(i,j) = \begin{cases} 0, & k = 0 \text{ or } k = 5, \\ 1 + k, & \text{otherwise}, \end{cases} \qquad e_-(i,j) = \begin{cases} 0, & k = 0 \text{ or } k = 1, \\ 7 - k, & \text{otherwise}, \end{cases} \qquad k = (i + 5j) \bmod 6.$$

Put $n_\pm = \prod_{\mathfrak{p}} \varpi_{\mathfrak{p}}^{e_\pm(i,j)}$. Retain the unit, projector e, and phases $\prod \mathfrak{R}(\varpi_{\mathfrak{p}}, n)^k$ or $\prod \mathfrak{R}(\varpi_{\mathfrak{p}}, n)^{6-k}$; finite ray expansion makes these fixed characters, without extra moving width.

Lemma 12.3 (Valuation budget). *For every $i, j \ge 1$, $4e_\pm(i,j) \le 5(i + j)$. Hence, if C, D are the actual first common ideals,*

$$(172)\ q_{(n_\pm)}^4 \le (q_C q_D)^5.$$

Writing $L_c = \log_U q_{(n_\pm)}$, $c = \log_U q_C$, $d = \log_U q_D$, R for the active radical width and R_0 for its cancelled part,

$$(173)\ 4L_c \le 5(c + d), \qquad L_c \le 5(R - R_0), \qquad 0 \le R \le \min(c, d), \quad R_0 \ge 0.$$

Proof. Since $e_\pm \le 5$, only $i + j < 4$ needs checking: $(i,j) = (1,1), (1,2), (2,1)$ give $(e_+, e_-) = (0,0), (0,2), (2,0)$. Multiply $4e_\pm \le 5(i + j)$ over common primes; their parts divide C, D, even without equal support. Surviving exponents are ≤ 5 and supported on the active radical minus its cancelled part, proving (173). Both orientations are included without directional divisibility. □

Multiplicity of actual joint rows. Fix a polynomial norm bound U^{B_0} on all four extracted common ideals and on $(w)M_0$, with $w \ne 0$ as in (168), and missing first radical $NM_0 \le U^m$. The number of actual labels giving that row is $O_{\epsilon, B_0}(U^{m+\epsilon})$.

Every first common prime occurs in the inactive projector, the numerator $n_\pm$, or M_0, hence in $(w)M_0$. The second common support is visible through $g_2 \mid w$; its valuations fix the unit/nonunit partition, then (168) uniquely recovers the remaining frequency. Thus, after M_0 is fixed, four polynomially bounded ideals supported on $(w)M_0$ determine all labels up to fixed unit and ray choices.

For completeness, if $NH \le U^B$, Rankin's bound gives, for fixed $s > 0$, at most $U^{B_0 s} \prod_{\mathfrak{p} \mid H} (1 - (N\mathfrak{p})^{-s})^{-1}$ ideals of norm at most U^{B_0} supported on H. For every fixed $\delta > 0$, sufficiently large primes satisfy $-\log(1 - (N\mathfrak{p})^{-s}) \le \delta \log N\mathfrak{p}$; the remaining finite prime product is constant. The displayed product is therefore $O_{s,\delta}(U^{B\delta})$. Choose s and δ small enough for the four ideals together to cost U^ϵ. Ideal counting gives $O(U^m)$ choices for M_0. This proves the claimed multiplicity, with all valuations, units, masks and both frequency partitions retained.

14 The compensated probe, its detector, and the return to ordinary characters

Use finite-order, trivial-infinite-type ideal L-functions over $K = \mathbb{Q}(\sqrt{-3})$, with every integral modulus and finite conductor. Let β be the supremum of $1/2$ and positive-half-plane zero real parts, excluding the principal pole. Primitive reduction below preserves this supremum; attainment is not assumed.

One compensated probe serves every stage: Collet's one-slot 29/33 construction and the later disjoint-slot signed inclusion–exclusion. Levy's weighted numerator changes its nonprincipal Poisson estimate while retaining the principal signal.

14.1 The actual arithmetic coefficients and the two representations Use the arithmetic chapter's primary generators, sextic symbols, angular α, ray phases G, R, normalized Gauss sums γ_j and additive measure. All residue-character powers, including zero/sixth, vanish at nonunits; rows keep six units. Ideal columns avoid fixed finite S, containing primes over six, primitive target conductor, ray moduli and a sufficiently large small-prime cutoff, chosen before scale/tuples/rows.

Let b_* generate the product of the prime ideals in S. Choose a primitive residue character ξ mod b_*, quadratic at odd residue characteristic and cubic at the prime of norm four. Put

$$\tau = q_{b_*}^{-1/2} g_\xi(b_*, 1), \qquad \Xi(A) = \xi(A)\chi_A(b_*), \qquad |\tau| = 1.$$

The finite identities proved in the arithmetic chapter give

$$\gamma_2(c)^3 = \mu(c)\alpha(c), \quad \gamma_1(c)\gamma_2(c) = \mu(c)\alpha(c)G(c), \quad \chi_b(a) = R(a,b)\chi_a(b) \quad ((a,b)=1).$$

These fix coefficient phases. Set

$$V_G(r) = \frac{e^{-(\log r)^2/4}}{2\sqrt{\pi}}, \qquad \widehat{V_G}(t) = e^{t^2},$$

and choose nonnegative, nonzero, smooth annular W_0, W_1. For $A = cn^3$, where c is squarefree and c, n may share primes, define

$$T_{m,a,\eta}(t) = \sum_{c \text{ sf},n} \gamma_2(c)\overline{\alpha(A)G(A)}\eta(A)\Xi(A)^{-1}R(A,a)\chi_A(m)q_c^{-1/2-t}q_n^{-1-3t}, \tag{187}$$

$$I_\eta(X,Y,Z) = \frac{1}{Y}\sum_a \frac{W_1(q_a/Y)\chi_a(b_*)}{\tau\xi(a)\sqrt{q_{b_*}q_aX}} \sum_m \xi(m)q_a^{-1/2}g_{\chi_a}(a,-m)W_0\left(\frac{q_m}{q_{b_*}q_aX}\right)\frac{1}{2\pi i}\int_{(4)} Z^t e^{t^2} T_{m,a,\eta}(t)\,dt. \tag{188}$$

The a columns include prime powers and full zero-extended imprimitive Gauss sums. Annuli make $a, m \neq 0$ finite; all other sums/integrals converge absolutely. Define $I_{\eta;D}$ by inserting $\mathbf{1}_{D|cn^3}$ on the whole theta index.

Choose finitely many distinct positive ℓ_i, $P_i = Z^{\ell_i}$, and common smooth nonnegative nonzero w supported in $(1,2)$. Each slot uses all primes outside S; distinct lengths make supports eventually disjoint. They may be as fine as any prescribed positive mesh with prescribed total $\ell = \sum_i \ell_i$. Write $w_i = w$ to index slots, $p_J = \prod_{i\in J} p_i$, and define

$$I_\eta^{\mathrm{comp}}(Z) = \sum_{(p_i)} \prod_i w_i(q_{p_i}/P_i) \sum_{J\subseteq\{1,\dots,r_{\mathrm{slot}}\}} (-1)^{|J|} q_{p_J}^{-3/2}\overline{\eta(p_{J^c})} I_{\eta;p_{J^c}}(X/q_{p_J}, Y/q_{p_J}, Zq_{p_{J^c}}). \tag{189}$$

Weights are row-independent and the coefficient is $\overline{\eta}$. One slot gives divisibility minus rescaled subtraction; retain all collisions with c, n and later rows.

For the physical representation, expand $\overline{G}$ on its fixed ray group, partition a into classes σ for $R(A,a)$, set $Q_0 = q_{b_*}XY$ and take $\Omega = 1$ on the product annuli. Mellin inversion of W_0 expresses (188) as a finite sum of

$$\frac{Q_0^{-1/2}}{2\pi}\int_{\mathbb{R}} \widehat{W_0}(iv) \sum_{m\neq 0} \Omega(q_m/Q_0)\xi(m)(q_m/Q_0)^{-iv} A_{m,\sigma,v}(Y)B_{m,\sigma}(Z)\,dv, \tag{190}$$

where B is a completed theta sum, with precisely the coefficients

$$\mathcal{C}_V(N;m,\nu) = \sum_{c \text{ sf},n} \frac{\overline{\alpha(cn^3)}\gamma_2(c)\nu(cn^3)\chi_{cn^3}(m)}{\sqrt{q_c}\,q_n} V(q_cq_n^3/N),$$

and

$$A_{m,\sigma,v}(Y) = Y^{-3/2}\sum_{a\in\sigma}\frac{\chi_a(b_*)}{\tau\xi(a)}W_1(q_a/Y)(q_a/Y)^{-1+iv}g_{\chi_a}(a,-m). \tag{191}$$

The identity $q_m/(q_{b_*}q_aX) = (q_m/Q_0)/(q_a/Y)$ gives $Y^{-3/2}$.

Let $\widetilde{W_0}(|y|^2)$ be the planar Fourier transform of $W_0(|z|^2)$ and $\mathscr{M}(z) = \int_0^\infty \widetilde{W_0}(r)r^{z-1}\,dr$, holomorphic for $\Re z > 0$ with arbitrary vertical decay on compact positive substrips. Poisson gives

$$I_\eta(X,Y,Z)=\frac{1}{(2\pi i)^3}\int_{(3)}\int_{(3)}\int_{(2)}X^{1/2-z}Z^{s+z-1}Y^{w-1}e^{(s+z-1)^2}\mathscr{M}(z)\widehat{W}_1(w)\\ \times\sum_{u}^{(6)}q_u^{-z}\overline{\xi(u)}\frac{\zeta_K^S(6z)L_K^S(w,\psi_u)}{L_K^S(s,\eta\overline{\psi_u})}H_{\eta,u}(s,w,z)\,dz\,dw\,ds. \tag{192}$$

Its finite convolution is

$$\mathcal{F}(a,A,H)=\sum_{\substack{d \bmod a\\ H\equiv b_*Ad \bmod a}}\chi_a(d)g_{\xi\chi_A}\left(b_*A,\frac{H-b_*Ad}{a}\right).$$

Expanding both Gauss sums in the full transform modulo ab_*A gives $q_a^{1/2}\mathcal{F}$. Planar dilation X/q_A leaves physical factor $X^{1/2}/(\sqrt{q_{b_*}}q_A)$. The primitive factor at S kills frequencies meeting S, including zero. Write others uniquely $H=ur^6$, with primary r and the unit retained in u; summing r gives $\zeta_K^S(6z)$ without a factor six. Mellin inversion and $s=t+1-z$ yield (192). On its initial lines a majorant is

$$C_S\sum_{u,c,n,a,r}q_u^{-2}q_c^{-5/2}q_n^{-7}q_a^{-2}q_r^{-12},$$

times rapidly decreasing heights, justifying absolute interchange. CRT gives $g_{\xi\chi_A}(b_*A,K)=\xi(A)\chi_A(b_*)g_\xi(b_*,K)g_{\chi_A}(A,K)$; $\Xi(A)^{-1}$ and $\chi_a(b_*)/(\tau\xi(a))$ cancel fixed-prime factors, leaving $\overline{\xi(u)}$. The displayed phase identities cancel cross-prime terms, including shared primes of a,A.

14.2 The local table, including all shared primes The complete local table for (192) and numerator averaging uses, at $p\notin S$,

$$Q=q_p,\quad j=v_p(u)\in\{0,\dots,5\},\quad \rho=\chi_p(u/p^j),\quad \mathfrak{a}=\overline{\alpha(p)}^3\eta(p)^3,\quad \mathfrak{b}=\overline{\alpha(p)}^2\eta(p)^2\overline{\chi_p(4)},\quad \mathfrak{b}^3=\mathfrak{a}^2,\\ V=Q^{-6z},\quad R_p=\mathfrak{a}^2Q^{4-6s-6z},\quad W^\circ=\rho Q^{-w},\quad W=\chi_p(u)Q^{-w},\quad D=\eta(p)\overline{\chi_p(u)}Q^{-s}.$$

Thus $D=W=0$ for $j>0$. Set

$$P_p^*=\frac{1}{1-R_p}\left\{\frac{R_p(1-Q^{-1})-\eta(p)(Q-1)Q^{-s-w}V^{\mathbf{1}_{j\le1}}}{1-V}+J_j\right\},\\ P_p=(1-V)^{-1}+\mathbf{1}_{j=0}\frac{W^\circ}{1-W^\circ}+P_p^*,\qquad H_p=\frac{P_p(1-V)(1-W)}{1-D},\quad H_{\eta,u}=\prod_{p\notin S}H_p, \tag{193}$$

where

j	J_j
0	$-\eta(p)\rho^{-1}Q^{-s}+W^\circ R_p$
1	$\eta(p)Q^{-s-w}$
2	$\mathfrak{a}\rho^{-3}Q^{3/2-3s}$
3	$-\eta(p)\mathfrak{b}\rho^{-2}Q^{2-3s-w}+\mathfrak{b}^2\rho^{-4}Q^{2-4s}$
4	$-\eta(p)\mathfrak{a}\rho^{-3}Q^{5/2-4s-w}$
5	$-\mathfrak{a}^2Q^{3-6s}$.

For $t=e+3l$, $e\in\{0,1\}$, the local convolution is

$$C_p(t,k,j')=\sum_{\substack{d \bmod p^k\\ p^{j'}\equiv p^td \bmod p^k}}\chi_p(d)^kg_{\chi_p^t}\left(p^t,\frac{p^{j'}-p^td}{p^k}\right).$$

Lifting a prime Gauss sum gives

$$g_{\chi_p^h}(p^t,K)=\begin{cases}Q^{t-1}g_{\chi_p^h}(p,K/p^{t-1}), & p^{t-1}\mid K,\\ 0, & p^{t-1}\nmid K.\end{cases}$$

For $t=0$ it is one at $k=0$, $\mathbf{1}_{j'=0}$ at $k>0$. For $t>0, k=0$ it is the source Gauss sum; for $k=1, j'\geq 1$ it is $\chi_p(-1)^{-1}g_{\chi_p}(p,1)g_{\chi_p^{t-1}}(p^t,p^{j'-1})$, while $j'=0$ has no solution. For $k\geq 2$ the last lift $d=d_0+p^{k-1}h$ sums a nontrivial additive character and vanishes. Surviving $t>0$ valuations are

t	$k=0$	$k=1$
$6r+6$	$j'\geq t-1$	$j'=t$
$6r+1$	$j'=t-1$	$j'\geq t$
$6r+3$	$j'=t-1$	$j'=t$
$6r+4$	$j'=t-1$	$j'=t$.

After CRT phase cancellation, with $j'=j+6m$ and $\omega_p=\chi_p(-1)$, the summand is

$$(-1)^e\gamma_1(p)^{-e}\eta(p)^e(\mathfrak{a}\overline{\gamma_3(p)})^l\rho^{k-t}\omega_p^{tk-el-l(l-1)/2}C_p(t,k,j+6m)Q^{-(s+1/2)e-(1+3s)l-wk-6zm}.$$

The first two strict tails sum to $R_p(1-Q^{-1})/((1-R_p)(1-V))$ and $-\eta(p)(Q-1)Q^{-s-w}V^{\mathbf{1}_{j\leq 1}}/((1-R_p)(1-V))$. Finite Gauss identities give equality terms $J_j/(1-R_p)$; $t=0$ gives the initial terms of P_p. This covers every valuation.

Put $c_p=\overline{\eta(p)}Q^s$, $h_p=Q^{-w}$, and

$$G_p=\frac{(c_p-h_p)(1-V)(1-W)P_p^*-h_p(1-VW)}{1-D}. \tag{194}$$

The local effect of the two cases in (189) is the identity

$$\overline{\eta(p)}Q^{s+z-1}P_p^*-Q^{z-w-1}P_p=\frac{(1-D)Q^{z-1}G_p}{(1-V)(1-W)}.$$

Consequently its exact Poisson correction is

$$\mathcal{H}_{\eta,u,Z}=\sum_{(p_i)}\prod_i w_i(Q_i/P_i)Q_i^{z-1}G_{p_i}\prod_{p\notin S\cup\{p_i\}}H_p. \tag{195}$$

For $p\nmid u$ put $v=\chi_p(u)$, $E_p=P_p^*+D$. Substituting $c_pD=v^{-1}$ gives the quotient-free identity

$$G_p+v^{-1}H_p=\frac{(1-V)(1-W)}{1-D}\left[(v^{-1}-h_p)\left(\frac{V}{1-V}-D\right)+(c_p+v^{-1}-h_p)E_p\right]. \tag{196}$$

Define $\Delta_p=G_p+\overline{\psi_u(p)}H_p$ also at row primes. Expanding $G_p=-\overline{\psi_u(p)}H_p+\Delta_p$ gives all main/error choices in (195), with main slot exactly $-\sum_p w_i(Q/P_i)Q^{z-1}\overline{\psi_u(p)}$, including zero character values.

The local table also proves normal convergence and $H_{\eta,u}\ll q_u^\epsilon$ on compact real boxes in

$$\Re s\geq 51/100,\quad \Re z\geq 17/50,\quad \Re w\geq -1/100,\quad \Re(s+w)\geq 1+\epsilon_1. \tag{197}$$

Unramified leading defects Q^{-s-w} and $Q^{4-6s-6z}$ are $\leq Q^{-1-\epsilon_1}$ and $Q^{-11/10}$; all others have summable margins. Row-prime defects are negative powers, giving q_u^ϵ after a fixed cutoff; $1-W$ cancels. For $u=1$ also allow $\Re s\geq 87/100$, $\Re w\geq 3/4$, $\Re z\geq 1/8$. Substituting the six local terms in (195) gives $\ll q_u^\epsilon Z^{(17/50)\ell}$ on $(\beta+e,1/2,17/50)$ and $\ll q_u^\epsilon Z^{v\ell}$ on $(2,2,v)$, $v>2$: selected off-row factors are bounded; the first line's exceptional $O(Q^{1/2})$ occurs only at divisor-many primes. Intervening fixed boxes have fixed polynomial height/scale bounds.

14.3 The signed Gram calculation and the physical bound **Lemma 14.1** (Signed physical Gram estimate). *For (191), with $P_a=Y^2/Q_0\geq 1$,*

$$\sum_{q_m\ll Q_0}|A_{m,\sigma,v}(Y)|^2\ll(1+|v|)^J\frac{Q_0}{Y}\left(1+P_a^{1/6}+\frac{P_a^2}{Y}\right)Z^\epsilon.$$

This assertion uses the displayed fixed ray coefficients.

Proof. Expand the signed square. Arithmetic Fourier correlation has $F(u,v;0)=\mathbf{1}_{u=v}\phi(u)$ and, for $u=Cn_1, v=Cn_2$, $(n_1,n_2)=1$,

$$F(Cn_1,Cn_2;Ck)=\chi_{n_1}(k)\overline{\chi_{n_2}(-k)}R(n_1,n_2)L_C(n_1,n_2;k),\qquad |L_C|\leq q_C.$$

Row Poisson contributes $Q_0/(q_u q_v)$, the Gauss pair $q_u q_v F$, giving prefactor Q_0/Y^3; zero frequency costs $O(Q_0/Y)$.

For nonzero frequencies extend the coefficient to residual pairs, setting it zero when a C-prime divides both, then insert full Möbius coprimality. On divisor d, write $n_i = dm_i$; natural zeros force $(d, Ck) = 1$. The extension cancels in the complete divisor sum. After finite ray separation the period norm is $\ll q_C q_k$; its complete mean vanishes unless the sixth-free part of k is supported on C, S, since every other prime gives a nonprincipal complete character sum.

On $q_C \asymp C_0, q_d \asymp D_0, q_C q_k \asymp RP_a$, put $N = Y/(C_0 D_0)$. Two-column lattice Poisson bounds zero-mean sums by

$$(1 + |v|)^J R^{-B} C_0 N^2 \min\{1, (RP_a/N)^A\}.$$

Below period use counting; above period integrate all nonzero dual frequencies by parts. There are $O(D_0 RP_a)$ choices of C, d, k, and the remaining dyads use

$$\sum_{i,j\ge 0} 2^{-i-j} \min\{1, (2^{i+j}/\Lambda)^A\} \ll_A \frac{1 + \log(2+\Lambda)}{\Lambda}, \quad \Lambda = Y/(RP_a), \quad A > 1.$$

Group by $i + j = n$ and split $(n+1)2^{-n}$ at $2^n = \Lambda$. Including Q_0/Y^3 gives $(Q_0/Y)P_a^2/Y$; large B sums R.

For nonzero mean, write $k = er^6$ with sixth-free e supported on C, S. There are divisor-many e and $O(Z^\epsilon (RP_a/C_0)^{1/6})$ roots. Absolute column volume gives $(Q_0/Y)P_a^{1/6}$ with summable weights $C_0^{-1/6+\epsilon} D_0^{-1+\epsilon}$. Add zero frequency; finitely many profile derivatives give $(1 + |v|)^J$. □

We apply the completed marked energy of Lemma 7.1 with its following exact interface. For the displayed $\mathcal{C}_V$, product-form row-independent prime marks of total width z, completed length Z^N, and row range Z^M,

$$M + 2z \le N \quad \Longrightarrow \quad \sum_{0<q_m \ll Z^M} |\mathcal{T}(m)|^2 \ll Z^{M+(N+2z-2M)_+/4+\epsilon}. \tag{198}$$

Lemma 7.1 includes both collision masks, all cusps, joint profiles and full kernel tails for these multiple slots.

Let $X = Z^x, Y = Z^y, M = x + y$ and assume

$$0 < \ell < x \le y, \qquad M + \ell \le 1, \qquad y - 2x + \ell < 0.$$

Cauchy in (190), the Gram lemma and (198) give

$$|I_\eta^{\mathrm{comp}}(Z)| \ll Z^{x/2+(y-x)/12+(1+3\ell-2M)_+/8+\epsilon}. \tag{199}$$

For subset J of width d, the rescaled row/mark/completed exponents are $M' = M - 2d$, $z' = \ell - d$, $N' = 1 + \ell - d$, satisfying $M' + 2z' \le N'$. Outside factor, tuple mass and energy square root add

$$-d + \frac{(1 + 3\ell - 2M + d)_+}{8}.$$

Its slopes $-1, -7/8$ make $d = 0$ maximal. The Gram condition $P_a^2/(Y/q_{p_J}) \ll 1$ follows from $y - 2x + \ell < 0$. Integrate common Mellin profiles and sum finite rays to obtain (199) for every subset.

14.4 The unique principal signal and its nonvanishing numerator Only $u = 1$ is principal in (192): a nonunit sixth-free row has a good prime of valuation $1, \dots, 5$, hence nontrivial tame Kummer inertia. For unit $u \ne 1$, CRT gives primary $n \equiv -2 - 3\omega \pmod{36\mathcal{O}}$ and $n \equiv 1$ at other fixed primes. Then $q_n \equiv 7 \pmod{36}$ and $\chi_n(u) = u^{(q_n-1)/6} = u \ne 1$. Keep all five units in the remainder.

Let $\rho_S = \mathrm{Res}_{s=1} \zeta_K^S(s) > 0$. At $w = 1, z = 1/6$ the table simplifies, with $t = Q^{-1}$, to

$$H_{\eta,p}(s) = \frac{1 - t^2}{1 - R_p}\left(1 + \frac{t(D - R_p)}{1 - D}\right), \qquad D = \eta(p)Q^{-s}, \quad R_p = \overline{\alpha(p)}^6 \eta(p)^6 Q^{3-6s}. \tag{200}$$

In particular, for any fixed $b > 2/3$,

$$|H_{\eta,p}(s) - 1| \ll_b Q^{-2} + Q^{-1-b} + Q^{3-6b} + Q^{2-6b} \qquad (\Re s \ge b).$$

These powers are prime summable. Enlarge the fixed cutoff until their majorant sum is $\le \log(3/2)$; normal convergence and $\prod(1+a_p) \le e^{\sum a_p}$ give

$$H_\eta(s) = \prod_{p \notin S} H_{\eta,p}(s), \qquad |H_\eta(s) - 1| \le \tfrac{1}{2} \quad (\Re s \ge b). \tag{201}$$

Thus H_η is holomorphic and nonzero uniformly in height; $b = 87/100$ suffices for all established endpoints here.

The positive residue scalar is

$$c_S = \frac{\rho_S^2}{6} \widehat{W}_1(1) \mathscr{M}(1/6) > 0.$$

Indeed, Fourier transformation of the radial Riesz kernel, first written as a positive integral of Gaussians, gives for $0 < \sigma < 1$

$$\mathscr{M}(\sigma) = \left(\frac{2\pi}{\sqrt{3}}\right)^{1-2\sigma} \frac{\Gamma(\sigma)}{\Gamma(1-\sigma)} \int_0^\infty W_0(r) r^{-\sigma}\, dr > 0.$$

Annular support/Schwartz decay justify interchange; $1/6$ in c_S is the residue scaling of $\zeta_K^S(6z)$.

On this residue locus (196) gives $G_p/H_p = -1 + O(Q^{-b})$. Define the real nonzero normalizer

$$A(Z) = (-1)^{r_{\mathrm{slot}}} Z^{-\ell/6} \prod_i \sum_p w_i(Q/P_i) Q^{-5/6}.$$

The fixed-field prime ideal theorem gives $|A(Z)| \asymp (\log Z)^{-r_{\mathrm{slot}}}$. Multiplying over fixed disjoint slots gives

$$\mathcal{H}_{\eta,1,Z}(s,1,1/6) = H_\eta(s) Z^{\ell/6} A(Z)(1 + O(Z^{-\mu_P})), \quad 0 < \mu_P < b \min_i \ell_i, \quad \Re s \ge b. \tag{202}$$

Put

$$h = 1+\ell-x, \quad k = \tfrac{5}{6} - \tfrac{x}{3} - \tfrac{\ell}{6}, \qquad J_\eta(Z) = \frac{I_\eta^{\mathrm{comp}}(Z)}{c_S A(Z)}, \quad f_\eta(Z) = \frac{1}{2\pi i} \int_{(2)} Z^{s-k} e^{(s-5/6)^2} \frac{H_\eta(s)}{L_K^S(s,\eta)}\, ds. \tag{203}$$

The residue power is $X^{1/3} Z^{s-5/6} Z^{\ell/6} = Z^{s-k}$. Equation (202) therefore identifies f_η up to power-small error: the compensator gives $Z^{\ell/6}$ and $A(Z)$ has logarithmic size.

14.5 Buffered zero bins and two witnesses at the same zero For fixed finite-ray ν, let $\varphi_u = \nu \psi_u^{\pm 1}$ with its natural zeros, and set

$$M_u(D;W) = D^{-1/2} \sum_n \mu(n) \varphi_u(n) W(q_n/D), \qquad S_u(N;W) = N^{-1/2} \sum_n \varphi_u(n) W(q_n/N).$$

Preceding moments have common annular controls for real powers, scale derivatives and independent polynomial heights. In particular,

$$\sum_{q_u \asymp U}^{(6)} |M_u(U^r;W)|^2 \ll U^{\max\{1,(1+5r)/6\}+\epsilon}, \qquad \sum_{q_u \asymp U}^{(6),\varphi_u^* \ne 1} |S_u(U^m;W)|^4 \ll U^{1+\epsilon}. \tag{204}$$

The long inverse exponent follows from Lemma 4.1 by the exact amplification

$$M_u(D;W) = \sum_{d \mid \mathrm{rad}(a)} \mu(d) \varphi_u(d) q_d^{-1/2} M_{ua^6}(D/q_d;W).$$

This separates squarefree column primes dividing a, retaining shared-prime zeros, with coefficient mass q_a^ϵ. Log-scale differentiation replaces W by $-W/2 - yW'$. Sobolev and the inverse moment bound $\sup_{0<E\le D}$ on balls $H \ge D^{1+\vartheta}$. Average over primary $q_a \le (H/U)^{1/6}$ for $H = \max(2U, D^{1+\vartheta})$; valuations modulo six make $(u,a) \mapsto ua^6$ injective. Its $\gg_S (H/U)^{1/6}$ preimages give $U^{1/6} H^{5/6} (UD)^\epsilon$, hence (204). Row $u = 1$ gives smooth Möbius sums $\ll D^{11/12+\epsilon}$; Mellin inversion with a test nonzero at a proposed zero yields preliminary $\beta \le 11/12$, before $7/8$ or marked plain moments.

Fix finite conjugation-stable Θ containing all target numerator/denominator/ray/phase twists. For nonunit sixth-free u, $\mathcal{X}(u) = \{\nu\psi_u^{\pm1} : \nu \in \Theta\}$ is nonprincipal of conductor $O_S(q_u)$ with redundant zeros. Put $a_* = 51/100$, small $e > 0$, $I = \lceil 1/e \rceil + 2$. Let $B_j(u)$ maximize a_* and zero real parts of $\mathcal{X}(u)$ up to height $3^j T$. Some $i < I$ has $B_{i+1} - B_i \le e$; round B_i down on the e mesh:

$$a \le B_i < a + e, \qquad B_{i+1} < a + 2e. \tag{205}$$

Labels (i, a) are finite. Above the floor there is one actual zero $\rho = \sigma + i\gamma$, $a \le \sigma < a + e$, $|\gamma| \le 3^i T$, supplying both witnesses. Put $\delta = 2a - 1$.

We record the analytic consequence of the buffer. The ordinary Hecke functional equation in this normalization is

$$\Lambda(s, \varphi) = (3q_{\mathfrak{f}})^{s/2}(2\pi)^{-s}\Gamma(s)L_K(s, \varphi), \qquad \Lambda(s, \varphi) = \epsilon_\varphi \Lambda(1 - s, \bar{\varphi}), \quad |\epsilon_\varphi| = 1.$$

Euler convergence and the functional equation give polynomial conductor/height growth. Continue the Euler logarithm on a zero-free disk about $2+it$. Borel–Carathéodory bounds it by $O_e(\log(2q_{\mathfrak{f}}(3+|t|)^2))$ on a smaller disk; Euler convergence bounds it on radius $49/100$. Three circles with another buffer reduces the logarithm to a power < 1. Exponentiating either sign gives

$$|L_K(s, \varphi)| + |L_K(s, \varphi)^{-1}| \ll_{e,\epsilon} \{2q_{\mathfrak{f}}(3 + |t|)^2\}^{\epsilon}. \tag{206}$$

Use radii $2-a-2e, 2-a-4e, 2-a-6e$; Cauchy gives logarithmic derivatives. For principal characters apply this to $(s-1)\zeta_K(s)/(s+1)$; to the right of $\beta + e$ it holds at all heights. Deleted factors cost U^{ϵ} on fixed positive half-planes and $U^{v+\epsilon}$ on $\Re s \ge -v$, by summing $O(q_p^{-\sigma})$ beyond a fixed cutoff and the divisor bound.

Mellin inversion and reflection now give

$$|M_u(U^r)|^2 \ll U^{\delta r + \epsilon_0}, \qquad |S_u(U^m)|^2 \ll U^{\delta \min(m, 1-m) + \epsilon_0}. \tag{207}$$

For prime tests apply $-L'/L$, divide by $\log q_p$ and subtract prime powers costing $O(P^{1/2+\epsilon})$ before normalization:

$$\left| P^{-1/2} \sum_p \overline{\psi_u(p)} w(q_p/P)(q_p/P)^{it} \right| \ll P^{\delta/2 + O(e) + \epsilon} U^{\epsilon}(1 + T)^A. \tag{208}$$

The buffer (205) covers the witness neighborhood; distinct norm twists keep independent heights.

Lemma 14.2 (Same-zero witnesses). *Fix $t_0 \in [1, 3/2]$ for a nonfloor bin before selecting its witnesses. Each row has a presentation, a real power $\sigma \in [a, a + e)$, a common height and dyadic lengths U^r, U^m such that*

$$r \le t_0 + o(1), \quad r + m \ge t_0 - o(1), \quad 0 \le m \le \tfrac{1}{2} + O(\epsilon_0),$$

$$|M_u(U^r)|^2 \gg U^{\delta r - O(\epsilon_0)}, \qquad |S_u(U^m)|^2 \gg U^{\delta m - O(\epsilon_0)}.$$

After fixing presentations and dyads, the base profiles are common.

Proof. Choose $V_{\le} = 1$ on $[0, 1]$, zero on $[2, \infty)$, and put $C_\varphi(s) = \sum_l \mu(l)\varphi(l)V_{\le}(q_l/U^{t_0})q_l^{-s}$. At the actual zero ρ shift

$$\frac{1}{2\pi i} \int_{(2)} U^{40z}\Gamma(z)L_K^S(\rho + z, \varphi)C_\varphi(\rho + z)\, dz$$

to $\Re z = -1/4$. The zero residue vanishes; polynomial growth, gamma decay and small height exponent make the new integral $o(1)$, since U^{-10} dominates ideal counting. On the absolute line the unit contributes $1 + o(1)$ and $\mu * 1 = \mathbf{1}_1$ cancels $1 < q_{lk} \le U^{t_0}$. The remaining sum has magnitude $1 + o(1)$ and product norm $\gg U^{t_0}$. Truncate smoothly at U^{41} with error $o(1)$, then dyadically partition and Fourier-separate the product weight to select one pair/frequency with

$$|M_u(U^r)S_u(U^m)|^2 \gg U^{\delta(r+m) - O(\epsilon_0)}.$$

Both factors inherit the same character, real power and height from the convolution at ρ. Equation (207) and $\delta \ge 1/50$ force $m \le 1/2 + O(\epsilon_0)$; division by each upper bound gives both lower bounds. The inverse cutoff $V_{\le}(U^r y/U^{t_0})$ has uniform annular transition derivatives. Dyads and rapidly decaying separating-frequency tails fit the loss. □

14.7 The weighted numerator and its conductor allocation For each main/error term of (195), define its actual numerator by full w inversion:

$$\mathcal{N}_u(Y;s,z)=\frac{1}{2\pi i}\int_{(3)} Y^{w-1}\widehat{W}_1(w)L_K^S(w,\psi_u)\mathcal{H}^{\mathrm{term}}_{\eta,u,Z}(s,w,z)\,dw. \tag{216}$$

Bound the w-independent denominator afterwards. The following coefficient calculation admits a plain moment.

Write $t=Q^{-w}$. Away from the row let $v=\chi_p(u)$ and $d_p=\eta(p)v^{-1}Q^{-s}$. Direct expansion of (193) gives $H_p=H_{p,0}+H_{p,1}t+H_{p,2}t^2$, where

$$\begin{aligned}
H_{p,0}&=\frac{1-R_p/Q-d_p(1-V)}{(1-R_p)(1-d_p)},\\
H_{p,1}&=\frac{\eta(p)Q^{-s}(1-QV)-vV+vR_p/Q}{(1-R_p)(1-d_p)},\\
H_{p,2}&=\frac{v\eta(p)(Q-1)Q^{-s}V-v^2R_p(1-V)}{(1-R_p)(1-d_p)}.
\end{aligned} \tag{217}$$

On $\Re s=a+16e,\Re z=z_0=17/50$ these imply

$$|H_{p,0}-1|+\sum_{k=1}^{2}|H_{p,k}|Q^{k(a-1)}\ll Q^{-1-\epsilon_e}. \tag{218}$$

The largest linear exponent is $-1-16e$; other candidates $a-1-6z_0, 2-4a-6z_0, 4-6a-6z_0, -a-6z_0$ are <-1 for $a\ge 51/100$. For quotient-free error estimates write $E_p=E_0+E_1t$ with

$$E_0=\frac{R_p}{1-R_p}\left(\frac{1-Q^{-1}}{1-V}-d_p\right),\qquad E_1=\frac{-\eta(p)(Q-1)Q^{-s}V/(1-V)+vR_p}{1-R_p}.$$

Insertion in (196) proves

$$\sum_{k=0}^{3}|\Delta_{p,k}|Q^{k(a-1)}\ll Q^{-a}. \tag{219}$$

The constant is $\ll Q^{-a}+Q^{-6z_0}+Q^{4-5a-6z_0}\ll Q^{-a}$. Each additional t weighs $Q^{a-1}<1$, and $1-6z_0<-1$.

At a row prime $p^j\|u$, $1\le j\le 5$, write $J_j=A_j+B_jt$. For $j\ge 2$ the exact coefficients are

$$\begin{aligned}
H_0&=\frac{1-R_p/Q+(1-V)A_j}{1-R_p}, & H_1&=\frac{-\eta(p)(Q-1)Q^{-s}+(1-V)B_j}{1-R_p},\\
G_0&=\frac{c_p\{R_p(1-Q^{-1})+(1-V)A_j\}}{1-R_p}, & G_1&=\frac{-Q+R_p/Q+(1-V)(c_pB_j-A_j)}{1-R_p}, \quad G_2=-H_1.
\end{aligned} \tag{220}$$

For $j=1$ replace these by

$$H_0=\frac{1-R_p/Q}{1-R_p},\quad H_1=\frac{\eta(p)Q^{-s}(1-QV)}{1-R_p},\quad G_0=\frac{c_pR_p(1-Q^{-1})}{1-R_p},\quad G_1=\frac{R_p/Q-QV}{1-R_p},\quad G_2=-H_1.$$

At row primes $\Delta_p=G_p$. The primitive conductor's good-prime exponent is one. Allocate $\min(k,j-1)$ powers of shift Q^k to $h_r\mid\Lambda_u=q_u/q_{\mathrm{rad}(u)}$, the rest to h_0. With $\delta=2a-1$,

$$\sum_k|H_{p,k}|Q^{-k/2}Q^{\delta(k-j+1)_+/2}\le 1+CQ^{-c_e},\quad \sum_k|\Delta_{p,k}|Q^{-(k+1)/2}Q^{\delta(k-j+1)_+/2}\ll 1. \tag{221}$$

For $j\ge 2$, $G_1\asymp Q$ cancels against Q^{-1} and its shift lies in Λ_u. At $j=2$, quadratic error is $Q^{1-a}Q^{-3/2}Q^{\delta/2}=Q^{-1}$. Remaining nontail constant/linear exponents are $-\delta$ at $j=2$; $-3\delta/2,-\delta$ at $j=3$; $-3\delta/2$ at $j=4$; $-5\delta/2$ at $j=5$. The explicit $j=1$ terms give negative powers. At unramified error slots, removing normalizer $P_i^{z_0-1/2}$ leaves

$$P_i^{-1/2}\sum_{q_p\asymp P_i}q_p^{-a}\ll P_i^{1/2-a}\le 1.$$

Products of these inequalities and divisor bounds at the row primes prove the complete mass bound

$$\sum_{h_0,h_r} |c_u(h_0,h_r)|h_0^{\delta/2} \ll U^{\epsilon}. \qquad (222)$$

Include all local main/error choices after normalizers, monomial shifts and fixed-prime restoration in c_u; keep moving coefficients out of seminorms.

Inverting monomial shift $h = h_0h_r$ gives $Y^{-1/2}h^{-1/2}S_{\psi_u}(Y/h)$. Expand fixed Euler deletions, reflect primitively and restore original zeros with bounded fixed Euler mass, only shortening scales. Central amplitude is preserved, giving

$$\epsilon_u S_{\overline{\psi_u}}(\mathfrak{c}_u h/Y; W_1^{\vee}), \quad \mathfrak{c}_u = \frac{3q_{\mathfrak{f}_u}}{(2\pi)^2} \asymp_S q_{\mathrm{rad}(u)}, \quad \widehat{W_1}^{\vee}(v) = \widehat{W_1}(1-v)\frac{\Gamma(v)}{\Gamma(1-v)}.$$

Since $q_{\mathrm{rad}(u)}h_r \le q_u$, dual scale is $\ll_S (U/Y)h_0$. Annular support forces $h \le CY$, keeping lengths bounded. Move the reflected Mellin line just left of zero across its simple gamma pole: the profile is bounded at zero and rapidly decreasing at infinity, with summable normalized annuli. Varying conductor changes scale only.

For $m = \log_U Y > 1/2$, $h_0 \asymp U^t$, reflected length is $\le U^{1-m+t+o(1)}$. Take the common-profile scale supremum before the fourth moment. Hölder with count U^R gives $U^{1/4+3R/4}$, selected primes give $U^{-qz(t)/2}$ after fourth root, and (222) gives $U^{-\delta t/2}$. Thus old capacities

$$z_O(t) = \max\{0, \min(1, 2m-1-2t)/(6\kappa)\}, \qquad z_A = \frac{2\min(m,1)-1}{6\kappa},$$

gives

$$\sum_{u \text{ in class}} |\mathcal{N}_u| \ll Y^{-1/2}Z^{\ell(z_0-1/2)+\ell q}U^{1/4+3R/4-qz_A/2+\epsilon}(1+T)^A. \qquad (223)$$

Indeed $z_A - z_O(t) \le t/(3\kappa)$ and $-\delta t/2 + qt/(6\kappa) \le 0$ for $q \le \delta/2$, $\kappa \ge 7/10$. Use marked moments for positive capacities and unmarked for zero. Bounded-at-zero profiles assign subunit scales length zero.

14.12 Height order, Mellin continuation, and the global contradiction **Theorem 14.12** (Generic common-reserve return). *Fix $87/100 < \theta < 1$, a real normalization k, and physical geometry with $M = 1-\ell$, $0 < \ell < 1/5$, and $k + x/2 + (y-x)/12 = 11/12 - \ell/4 = \theta$. Suppose that the complete source and conductor estimates, with one mesh and all reserves chosen before the primitive target, give (237) with a common $\sigma > 0$, and that the physical bound (199) gives $J_\eta(Z) \ll_{\eta,\epsilon} Z^{\theta-k+\epsilon}$ for every $\epsilon > 0$. Suppose also that the signal is (203), with the absolute vertical convergence stated there, and that (201) gives a holomorphic nonzero numerator on $\Re s > 87/100$. Then $\beta \le \theta$.*

Proof. Suppose $\mu = \beta - \theta > 0$. Before the target, choose a fraction of feedback margin, small row extension ζ, moment losses, capacity decrements and amplitude mesh. Obtain a common moment mesh; fix positive distinct slots fine enough for every $d \ge d_0$, with total ℓ, fixing $\mu_P > 0$. Choose contour/detector buffers below remaining margins and $y/4, h/24, \mu_P, 1/20$.

Then fix target, dictionary, S and finite Sobolev/smooth/moment orders. Maximize degrees and choose $\tau_\eta > 0$, $T = 3Z^{\tau_\eta}$, paying all heights and small relative to d_0, ℓ_i for witnesses/prime derivatives. Independent external coordinates are

$$\Im(s+z), \qquad \Im z, \qquad \Im w.$$

The Gaussian decays in the first coordinate and Mellin weights arbitrarily in the other two; allocate each part of the zero-free height buffer. Joining reciprocals stay buffered or global. Before tail-dependent separation, correction is $\ll Z^B U^{\epsilon}$ with fixed B, so omitted/joining pieces are $O_N(Z^{B'}T^{-N})$ with B' fixed before N. Choose N last, preserving mesh, slots and real margins. The numerator's absolute-line w inversion and reflected-profile tails were paid before moments; its height remains independent of witnesses.

The common source-to-probe remainder hypothesis is

$$J_\eta(Z) - f_\eta(Z) \ll_\eta Z^{\beta-k-\sigma}. \qquad (237)$$

The geometry has $M = 1-\ell$ and $\ell < 1/5$, so (199), divided by its logarithmic normalizer, gives

$$J_\eta(Z) \ll_{\eta,\epsilon} Z^{\theta-k+\epsilon}, \qquad k + x/2 + (y-x)/12 = 11/12 - \ell/4 = \theta.$$

Take $\epsilon < \mu/4$, then common $0 < \Delta < \min(\mu/4, \sigma/2)$ with $\beta - \Delta > 87/100$. Thus $f_\eta \ll_\eta Z^{\beta-k-\Delta}$ at infinity. Shifting (203) to any $A > 2$, absolute Euler convergence/Gaussian decay give $f_\eta = O_{\eta,A}(Z^{A-k})$ at zero. Hence

$$\mathfrak{F}_\eta(s) = \int_0^\infty f_\eta(Z) Z^{-s+k} \frac{dZ}{Z} \qquad (238)$$

is holomorphic on $\Re s > \beta - \Delta$, since endpoint bounds remain integrable with every fixed logarithmic power. Fourier inversion in $\log Z$ on a line right of one gives

$$\mathfrak{F}_\eta(s) = e^{(s-5/6)^2} \frac{H_\eta(s)}{L_K^S(s,\eta)}. \qquad (239)$$

Gaussian/endpoint bounds make both Fourier functions integrable. Meromorphic identity extends (239); by (201) every denominator zero, of any multiplicity, would be a pole, contradicting holomorphy.

Common Δ precedes every primitive target. Supremum approximation supplies a zero with real part $> \beta - \Delta/2$; its arbitrarily large finite conductor/height affects constants, cutoff and τ_η, but not the gap. This contradiction proves $\beta \le \theta$. □

Apply this theorem separately with the geometry and complete analytic estimates of the first return and the final return. The value of θ and the source premises are fixed separately in each application.

14.13 Primitive reduction and every presented Dirichlet character **Theorem 14.13** (Ordinary transfer). *Let $87/100 < \theta < 1$. If the full finite-order Hecke family has $\beta \le \theta$, then every presented Dirichlet character is nonvanishing for $\Re s > \theta$, with its principal pole excluded.*

Proof. Let a Hecke character in modulus $\mathfrak{m}$ be induced by the primitive η^* of conductor $\mathfrak{f} \mid \mathfrak{m}$. Absolute Euler products on $\Re s > 1$ give

$$L_{\mathfrak{m}}(s,\eta) = L(s,\eta^*) \prod_{\substack{p \mid \mathfrak{m} \\ p \nmid \mathfrak{f}}} (1 - \eta^*(p) q_p^{-s}).$$

Continuation extends this identity away from the principal pole. Each deleted factor is nonzero for $\Re s > 0$ since $|\eta^*(p) q_p^{-s}| < 1$. Thus primitive reduction and fixed-S restoration preserve zeros. Reflected numerator bounds (221)–(222) still retain repeated row powers/holes.

For any $q \ge 1$ and complex Dirichlet $\chi \bmod q$, norm pullback $\eta_\chi(\mathfrak{a}) = \chi(N\mathfrak{a})$ has finite order in modulus $q\mathcal{O}$ and is trivial on units. Let χ_{-3} be the field character and present $\chi\chi_{-3}$ at a common modulus. Euler factors give

$$L_K(s,\eta_\chi) = L(s,\chi) L(s,\chi\chi_{-3}). \qquad (240)$$

Split primes give two norm-p factors with $\chi_{-3}(p) = 1$; inert primes give a norm-p^2 factor equal to the product with signs $\pm$. Prime three gives one factor and $\chi_{-3}(3) = 0$. At $p \mid q$ all corresponding factors are deleted. This proves (240) absolutely, then by continuation.

For $\theta < \Re s < 1$, both Dirichlet factors are holomorphic, so a zero contradicts Hecke nonvanishing. For $\Re s \ge 1$ use ordinary Dirichlet nonvanishing, excepting the principal pole; treat nonprincipal $s = 1$ directly even if its companion is principal. Thus every presented modulus and finite height satisfy

$$\Re s > \theta, \qquad \neg(\chi = 1 \text{ and } s = 1) \quad \Longrightarrow \quad L(s,\chi) \ne 0.$$

Modulus-one principal gives zeta. The conclusion uses the strict half-plane; it asserts nothing on its boundary. □

Chapter 25. Squarefree preprocessing and ordinary recovery

Final route, coefficient scope, and uniform choices

This note fixes the route used by Chapters 25–30 of the complete proof. The accepted earlier theorem, weakened as stated above, supplies the fixed input

$$\theta_I = 0.87481891, \qquad \kappa_I = 0.74963782, \qquad K_I = 6\kappa_I = 4.49782692.$$

The new endpoint is not used in proving any member of the source family.

The actual coefficient class

The ordinary source theorem concerns two factorized smooth ordinary ideal sums, with the literal characters, natural zeros, common punctures, fixed-inducing exclusions, finite profile seminorms, and height conventions of the inherited source identities. It includes all shorter plain lengths under its stated total affine cap.

The temporary ordinary restriction is $\mu^2(n_1n_2)$. Retain also the squarefree projector on the complete marked product during common classification. The canonical marked prime pools may overlap; their exact collision decomposition is proved below. Both restrictions remain in the transformed coefficient. They are removed at completed positive norms by the divisor identities and the finite slot-cardinality reconstruction.

The allowed new operations are the two exact divisor kernels, actual extracted common punctures, outside character scalars of modulus at most one, and the particular finite Newton correction of the amplifier. Bounded pointwise multiplication of a coefficient is not substituted for a quadratic-norm argument.

Elementary centering is the final premise

The final route uses the contour

$$\Re s_i = \frac{501}{1000}, \qquad \alpha = \frac{499}{1000}, \qquad L_{\mathrm{ref}}(b) = \frac{49}{100} - K_I b.$$

The centered terminal gate is

$$\frac{5}{6} + \frac{499}{1000} L_{\mathrm{ref}}(b) - T - 9a > 0.$$

The exact divisor transport below proves this saving; the global source theorem uses it in (G5). The finite source schedule uses this elementary saving and density $\nu \geq 1/2$.

One mesh works through the whole finite induction

Every application of an earlier source theorem occurs at an actual ambient scale $U' = U^Q$, where the source certificate and common geometry give a uniform strictly positive lower bound for Q. In particular, the proof uses $Q \geq F - T/2 - o(1) > 0$.

For a path γ in the finite source graph, let $q_\gamma > 0$ be the product of uniform lower bounds for these scale exponents along the path. The empty path has $q_\gamma = 1$. At every source node choose positive local thresholds for the marked slot mesh, amplifier-pool separation, greedy-selection error, allowed height exponent, and logarithmic support erosion, inside the strict reserves of that node. Then choose the corresponding outer thresholds smaller than the minimum of q_γ times those local thresholds, over all the finitely many paths and nodes.

That minimum is positive. If a slot or height has exponent τ in the original scale, its exponent at a child scale is τ/Q_γ; the preceding choice makes it smaller than the child threshold. The same calculation applies to support erosion. In particular every amplifier pool lies above each marked slot support at every recursive source invocation. The mesh choices precede even the ambient finite slot type. Once that type and the fixed profiles are chosen, the collision constants, finite controls and polynomial height degrees are fixed before moving arithmetic labels and heights. The eventual scale and large-height thresholds follow the required degrees.

The displayed 10^{-12} mesh and padding values in the exact certificate are upper accounting allowances. Actual local and outer parameters are chosen finer after the finite positive reserves and transform constants are fixed. Thus the pullback just described requires no additional numerical inequality and does not assume that an unmeasured analytic loss equals 10^{-12}.

Squarefree total-product preprocessing for a two-plain source

Status and purpose

This note proves exact coefficient identities and weighted mass bounds. It also specifies the restricted coefficient class that must remain intact through the two source transforms, and proves its elementary extraction law. The source coefficient is the finite convolution (160), with its exact transforms (166)–(168). The present section proves the two divisor identities and the restriction retained by those transforms.

The central distinction is timing. The squarefree-product restriction is retained while the first common ideals are extracted. Its divisor expansion is used only after the complete transformed positive row norm has been reconstructed and a previously proved natural or initial moment is to be invoked.

Throughout, ideals are good integral ideals over the same fixed Eisenstein field. A presented multiplicative character retains all of its natural zeros. The fixed bad-prime set may be enlarged once, before scale, to contain all prime ideals of norm below 16. This is already an allowed operation in the supplied physical probe.

1. The exact two-variable projector

For nonnegative integers i,j define

$$f(i,j) = \mathbf{1}_{i+j\le 1}.$$

Its two-variable generating series is 1+X+Y. The ordinary unconstrained pair has series 1/((1-X)(1-Y)). Define the multiplicative two-ideal coefficient h by

$$h_p(X,Y) = (1-X)(1-Y)(1+X+Y) = 1 - X^2 - XY - Y^2 + X^2Y + XY^2. \tag{1}$$

Then, for all ideals a,b,

$$\mu^2(ab) = \sum_{d_1|a}\sum_{d_2|b} h(d_1,d_2). \tag{2}$$

Here mu-squared(ab) is one exactly when both a,b are squarefree and coprime. Equation (2) follows prime by prime: multiplication of (1) by 1/((1-X)(1-Y)) gives 1+X+Y. In particular it is a finite, exact divisor identity, with no analytic convergence issue.

The only nonzero local h values, apart from h_p(0,0)=1, are

Exponents of d1,d2	h_p
(2,0)	-1
(1,1)	-1
(0,2)	-1
(2,1)	1
(1,2)	1

Let k be the convolution inverse of h in the two-ideal divisor algebra:

$$k_p(X,Y) = h_p(X,Y)^{-1}. \tag{3}$$

Its coefficients are integers, k(1,1)=1, and it has no local monomials of total degree one. Thus

$$1 = \sum_{d_1|a}\sum_{d_2|b} k(d_1,d_2)\mu^2((a/d_1)(b/d_2)). \tag{4}$$

Both (2) and (4) are coefficient identities for each finite pair (a,b).

2. Subpower normalized mass, in both directions

Fix sigma>1/2 and put z_p=(Np)^(-sigma). The absolute local h series is

$$1 + 3z_p^2 + 2z_p^3.$$

The geometric expansion of (3) gives the absolute coefficient bound

$$\sum_{i,j\ge 0} |k_p(i,j)| z_p^{i+j} \le \frac{1}{1 - 3z_p^2 - 2z_p^3}. \tag{5}$$

For Np>=16, the denominator is bounded positively, already at sigma=1/2: 3/16+2/64=7/32<1. Both Euler products converge absolutely when sigma>1/2, because the degree-two prime sum has exponent 2sigma>1. This uses only the supplied absolute ideal Euler convergence to the right of one. The finitely many excluded primes remain natural zeros.

Consequently, for each epsilon>0 and each fixed B0,

$$\sum_{Nd_1Nd_2\le U^{B_0}} \frac{|h(d_1,d_2)|+|k(d_1,d_2)|}{\sqrt{Nd_1Nd_2}} \ll_{\epsilon,B_0} U^{\epsilon}. \tag{6}$$

Indeed choose sigma=1/2+epsilon/B0 when B0>0 and apply Rankin's inequality to the convergent products. The B0=0 case is bounded. Divisor weights of any fixed order are permitted by reducing the Rankin margin first; the resulting local series still has no degree-one term and its degree-two coefficients are bounded constants.

3. Exact transfers of centrally normalized products

For a natural presented character psi define

$$S_\psi(X;W)=X^{-1/2}\sum_a \psi(a)W(Na/X),$$

and

$$T_\psi(X_1,X_2;W_1,W_2)=(X_1X_2)^{-1/2}\sum_{a,b}\mu^2(ab)\psi(ab)W_1(Na/X_1)W_2(Nb/X_2).$$

Equations (2) and (4) imply, exactly,

$$T_\psi(X_1,X_2)=\sum_{d_1,d_2}\frac{h(d_1,d_2)\psi(d_1d_2)}{\sqrt{Nd_1Nd_2}}S_\psi(X_1/Nd_1)S_\psi(X_2/Nd_2), \tag{7}$$

$$S_\psi(X_1)S_\psi(X_2)=\sum_{d_1,d_2}\frac{k(d_1,d_2)\psi(d_1d_2)}{\sqrt{Nd_1Nd_2}}T_\psi(X_1/Nd_1,X_2/Nd_2). \tag{8}$$

The fixed annular supports make the sums finite. Each scale is only shortened. If the character has a common puncture R, multiplicativity with zeros gives the identical formulas using psi_R everywhere. No nonunit value is replaced by one.

An arbitrary fixed product of the original marked prime polynomials may multiply every term of (7) or (8). The factor psi(d1d2), though row dependent, has modulus at most one and is outside the child two-plain polynomial. Minkowski in the same positive row Hilbert space, followed by (6), therefore transfers a U^(Q+epsilon) second-moment bound in either direction, provided that every shortened child is admitted. There is no arbitrary moving character in this operation: the child character is exactly psi, and the divisor character is a contracting row scalar.

Every divisor summand in (7) and (8) shortens each ordinary factor. The global domain (G1) admits all such shorter pairs under the same total affine cap. The common-mask transport at the moving puncture $\mathfrak{R}d_1d_2$, proved below, pays its radical small-power cost uniformly before that puncture and the heights. Choose both this loss and the loss in (6) inside the finite strict reserves.

The same formulas hold for the centered difference of equal-product rectangles. Each divisor pair divides both rectangles by the same norms, so their product scales remain equal. At a fixed-inducing terminal one retains that difference until cancellation has been applied. The ordinary equal-product bound is the minimum of the lattice absolute bound $O(\sqrt{T})$ and the centered bound $O(\sqrt{T}/L)$, including all positive subunit scales. Thus, for $0\le\alpha\le1$, it is $O(\sqrt{T}L^{-\alpha})$. After the same divisor extraction it costs $(Nd_1Nd_2)^{-(1-\alpha)}$. Take $\alpha=499/1000$, for which $1-\alpha=501/1000>1/2$, and apply the exact forward-kernel mass. The finite Newton correction is transported through the same complete centered coefficient, with its fixed finite mass. This proves the elementary terminal saving without treating bounded pointwise multiplication as contraction of a quadratic norm.

4. Exact extraction closure of the restriction

For fixed extracted ordinary factors A,B and variable quotients a,b,

$$\mu^2(ABab)=\mu^2(AB)\mu^2(ab)\mathbf{1}_{(ab,AB)=1}. \tag{9}$$

If AB is not squarefree, the whole allocated coefficient is zero. If AB is squarefree, the remaining coefficient is again the squarefree total-product projector, together with the additional common puncture

rad(AB). This is a statement about the literal original ordinary factors, and it is independent of the choice of common slot allocation.

Repeated first and second common extractions replace A,B by their accumulated extracted factors. Equation (9) applies to the accumulated product and retains its cross-prime condition. It follows that the restricted ordinary coefficient class is closed under these exact divisor extractions. The ordinary quotients remain squarefree and mutually coprime until (7) is deliberately invoked at the final positive moment norm.

Row Poisson summation itself is linear in the column coefficients and does not change their mu-squared factor. The first complete common support extraction and the subsequent full artificial-coprimality Möbius identity therefore remain exact on this restricted class. Common Fourier separation touches only the smooth norm profiles, leaving (9) as an arithmetic restriction. No restriction may be discarded during that separation or before common multiplicities are classified.

In both transforms (166) and (137), retain this same restriction on the surviving ordinary indices and the complete common puncture. Apply a prior moment only after reconstructing its completed positive row norm, as in (140)–(141).

5. Marked-pool collision reduction

The canonical marked prime pools may overlap. Let M be the live marked product and K its frozen part. During the common classification retain both $\mu^2(n_1n_2)$ and $\mu^2(KM)$. On a tuple of prime ideals, $\mu^2(M)$ is the indicator that its entries are distinct. For the finite edge set $\mathcal{A} = \{(i,j) : i \neq j\}$,

$$\mathbf{1}_{\text{injective}}(\mathfrak{p}_\bullet) = \prod_{(i,j)\in\mathcal{A}} (1 - \mathbf{1}_{\mathfrak{p}_i=\mathfrak{p}_j}) = \sum_{E\subseteq\mathcal{A}} (-1)^{|E|} \prod_{(i,j)\in E} \mathbf{1}_{\mathfrak{p}_i=\mathfrak{p}_j}. \tag{10a}$$

For each equality pattern, its connected components determine the frozen blocks. Every singleton keeps its full original pool and its literal row character. For a nonsingleton block, sum the bounded coefficients by Cauchy with their normalized square masses. The annular prime norm bounds make each such mass bounded by a constant depending on its fixed block size and coefficient bound. There are finitely many edge patterns. Consequently, if every completed full-singleton-subfamily norm is at most E, finite Cauchy bounds the completed projected energy by $C_{N,B}E$, with N the ambient finite slot count and B a fixed coefficient bound. This constant is independent of scale, live labels and heights. Strong induction on the surviving slot count applies the same decomposition to the collision terms and recovers the unrestricted marked source from the retained-projector source estimates for every surviving subset. All source calls use full singleton pools and the original coefficients.

On a surviving marked term, $(K, M) = 1$ and $\mu^2(KM) = \mu^2(K)\mu^2(M)$. Together with the ordinary projector this bounds every complete column valuation by two. Thus every nonempty genuine first common pair has

$$(i,j) \in \{(1,1),(1,2),(2,1),(2,2)\}. \tag{10}$$

Artificial Möbius conductors keep their separate provenance. The sixth-power amplifier terms retain their original error payment.

6. First common definitions and their shared payment

Let C, D_c be the complete first common ideals, with widths $c = \log_U NC$, $d = \log_U ND_c$. Let π be their full common radical width, R the active radical width from (164)–(165), and L_c, L_d the numerator widths from (172). The quantities $R_{0,c}, R_{0,d}$ are the widths canceled when the corresponding local numerator exponent is zero. Set

$$B_c = (3c - 5d - R)_+/6, \quad B_d = (3d - 5c - R)_+/6, \qquad \Sigma = (c + d - 2\pi - R - B_c - B_d)/2 \geq 0. \tag{RC1}$$

Nonnegativity is (MF3.56). The first-transform outside exponent O and the two side allowances satisfy, after unit row normalization,

$$O + (S_c + S_d)/2 = 1 - \Sigma. \tag{RC2}$$

This is the common-support payment (127)–(129), retaining the same first allocation weights. Refine by the displayed common/numerator widths before Cauchy; fixed dyadic refinements cost only U^ϵ. For nonnegative sector majorants, $\sqrt{N_cN_d} \leq \sum_{\alpha,\beta} \sqrt{N_{c,\alpha}N_{d,\beta}}$. Subtract Σ once from each side before pairing. Fix an earlier face (B, Λ).

7. Exact raw radius on this restricted class

Let A,b,c,D now denote the nonnegative logarithmic widths of first common primes of types (1,1),(1,2),(2,1),(2,2), respectively. To avoid confusion, write C,Dc for the two total first common lengths in this paragraph. The exact local rules give

$$\begin{aligned} C &= A+b+2c+2D, & D_c &= A+2b+c+2D, & & \\ R &= b+c, & L_C &= 2c, & L_{D_c} &= 2b, \\ R_{0,C} &= b, & R_{0,D_c} &= c, & \Sigma &= D. \end{aligned} \tag{12}$$

Both B positive parts vanish: 3C-5Dc-R and its reverse are sums of nonpositive local contributions. Hence

$$\tau_C = L_C - R - (C-D_c)_+ = -(D_c-C)_+, \quad Q_C = F-(D_c-C)_+, \tag{13}$$

and similarly with C,Dc exchanged. In particular

$$Q_C \le F, \qquad |C-D_c| \le (C+D_c)/3. \tag{14}$$

Thus C+Dc<=u implies F-u/3<=Q<=F.

The individual RAW coefficient simplifies to

$$W_C = -b-D-l\{C-B(b-c)_+\}. \tag{15}$$

For 0<B<=1, C>=b implies W_C<=0, individually. This statement does not eliminate a need to pay a HIGH partner's remaining coefficient; the paired HIGH and exceptional schedules still have to be checked.

The nonpositive bound (15) is used without any further frozen-slot deduction. The complete erasure cost, including both floors, is proved in (G10)–(G13).

Chapter 26. Exact corrected amplification

Exact amplification with a corrected squarefree coefficient

Status and scope

This is a new exact amplifier lemma for the restricted source class in the squarefree preprocessing section above. It uses the local Gauss identity (8), the first transform (120), and the second transform (137). It gives a single MAIN norm. The affine source face follows after the source inequalities and admission gates below are established.

The required condition is squarefreeness at amplifier primes. A column whose two ordinary factors have squarefree coprime product may still have valuation two at another prime contributed by a retained marked slot. The amplifier pool avoids all marked slot supports.

Fix a positive amplifier width `a`, the literal pool

$$\mathcal{P} = \{\mathfrak{p} : U^a/2 < N\mathfrak{p} \le U^a,\ \mathfrak{p} \text{ in the fixed permitted ray}\}, \qquad P = \#\mathcal{P}. \tag{CA1}$$

The finite bad set fixed in Section 1.2 is omitted. Fix the pool before the centered rectangles, live labels and Mellin heights. The inherited fixed-extension prime estimate gives `P=U^{a+o(1)}`. All norms and supports in the source are bounded by a fixed power of `U`. Consequently there is a fixed integer `R`, depending on that power and `a`, for which every supported column satisfies

$$r(u) := \#\{\mathfrak{p} \in \mathcal{P} : \mathfrak{p} \mid u\} \le R. \tag{CA2}$$

Increase the eventual threshold so that `P>2R+2`.

1. The identity at every frequency

Suppose `v_p(u)` is zero or one for every pool prime. If `p` does not divide `u`, the primitive Gauss change of variable at `p` gives

$$\mathfrak{g}(u, p^6 h) = \mathfrak{g}(u, h).$$

If `p` divides `u` once, the local primitive Gauss sum at the frequency `p^6h` is zero. If `p` also divides `h`, the local factor at `h` is already zero. Thus, including all nonunit and zero frequencies,

$$\mathfrak{g}(u, p^6 h) = \mathbf{1}_{p \nmid u} \mathfrak{g}(u, h). \tag{CA3}$$

Let `b(u)` be the complete residual coefficient, including the full centered difference, common puncture, artificial divisibility projector, whole ray character and norm twist. Define

$$\widetilde{b}(u) = \frac{P}{P - r(u)} b(u), \qquad F_b(h) = \sum_u b(u) \mathfrak{g}(u, h). \tag{CA4}$$

Summing (CA3) over the entire pool yields the exact identity

$$\boxed{F_b(h) = \frac{1}{P} \sum_{p \in \mathcal{P}} F_{\widetilde{b}}(p^6 h).} \tag{CA5}$$

The coefficient on the right is independent of the individual prime `p`.

For a nonnegative annular majorant `Omega`, Cauchy's inequality gives

$$\sum_h \Omega(Nh/L) |F_b(h)|^2 \le \frac{1}{P} \sum_{p,h} \Omega(Nh/L) |F_{\widetilde{b}}(p^6 h)|^2. \tag{CA6}$$

The exact pushforward through `(h,p) -> p^6h` in the sixth-power amplification identity (131) gives: each nonzero output of polynomial norm has at most a fixed number of such prime preimages, because their product divides its norm. The annular majorant in (CA6) is supported away from zero. Any separate zero-frequency terminal is kept with its original support and terminal bound; no sparse pushforward count is asserted at zero. The smooth MAIN radius therefore majorizes the right side, with one density factor `P^{-1}=U^{-a+o(1)}` and a constant depending only on the fixed cells. Multiply by the original nonnegative allocation weights and sum; those weights are unchanged. The enlarged MAIN norm is transformed a second time with the whole coefficient `tilde b` still intact.

Use the bound `|tilde b/b|<=2` for diagonal and absolute terminal estimates. For the other completed positive norms use the exact finite divisor representation below.

2. Exact finite divisor representation

For integers `0<=r<=R`, Newton interpolation gives

$$\frac{P}{P - r} = \sum_{j=0}^{R} c_j(P) \binom{r}{j}, \qquad c_0(P) = 1, \qquad c_j(P) = \frac{j!}{(P-1)(P-2)\cdots(P-j)}. \tag{CA7}$$

Terms with `j>r` vanish. The formula follows from the finite difference identity

$$\Delta^j \frac{P}{P - r}\bigg|_{r=0} = \frac{j!}{(P-1)\cdots(P-j)},$$

and finite Newton interpolation.

After a complete second common extraction, let `r0` be the number of pool primes in the frozen extracted part. The remaining column is punctured at that part, so its pool count adds disjointly. The exact shifted version is

$$\frac{P}{P - r_0 - r} = \sum_{j=0}^{R} c_j^{(r_0)}(P) \binom{r}{j}, \qquad c_j^{(r_0)}(P) = \frac{P j!}{(P - r_0)(P - r_0 - 1)\cdots(P - r_0 - j)}. \tag{CA8}$$

Here `R` can be enlarged once to cover all accumulated extractions. On the supported range `r0+r<=R`, all denominators are positive. For use in Mellin inversion, the finite polynomial on the right of (CA8) may be extended to all nonnegative `r`: the smooth supports ensure that only the range where it agrees with the desired multiplier contributes to the original sum.

For a column squarefree on the pool,

$$\binom{r(u)}{j} = \sum_{\substack{d\mid u,\ d\ \text{squarefree}\\ p\mid d\Rightarrow p\in\mathcal{P},\ \omega(d)=j}} 1. \tag{CA9}$$

The corresponding divisors are kept as coefficient restrictions through first common classification and both row transforms. Equation (CA9) does not justify enlarging the first common multiplicity class.

Support of the actual Newton correction. Write $P = \#\mathcal{P}$, the actual integer pool cardinality, and let Ψ_z be the literal multiplicative row weight with accumulated puncture $\mathfrak{R}$. Keep the selected marked product M inside the correction:

$$\frac{P}{P - r_{\mathcal{P}}(DMIJ)}\,\mu^2((A_1I)(A_2J))\Psi_z(IJ)\big(\text{the same centered profile}\big).$$

Terms with $\Psi_z(M) = 0$ or $\Psi_z(IJ) = 0$ contribute zero. On each surviving term, $D \mid \mathfrak{R}$ gives $(D, MIJ) = 1$. Since the amplifier pool avoids the marked support,

$$r_{\mathcal{P}}(DMIJ) = r_{\mathcal{P}}(D) + r_{\mathcal{P}}(IJ).$$

The support norm bound and prime lower norm bound give a fixed residual depth J_0. The actual pool budget

$$2\big(r_{\mathcal{P}}(D) + J_0 + 1\big) \le P$$

then makes every supported denominator positive and the correction at most two. Apply the finite shifted Newton expansion on these surviving terms, retaining the same puncture, marked product, smooth profiles and independent heights in both rectangles. Natural character zeros remain in each summand.

3. Restoration at completed nonexceptional row norms

When a previously proved natural or initial affine moment is to be applied to a complete positive row norm, expand (CA8) and allocate each prime of `d` among the two ordinary factors. The pool avoids slots, so no marked prime has to be selected in this operation. There are at most `2^j` allocations. Exact extraction divides both rectangles' first and second scales by the same assigned ideals, respectively, and contributes the central scalar

$$\frac{\psi(d)}{\sqrt{Nd}}, \qquad |\psi(d)| \le 1. \tag{CA10}$$

The remaining two ordinary factors still have squarefree coprime product and the accumulated common puncture. Thus their restriction can subsequently be restored using the finite divisor kernel `h` from the squarefree transfer lemma. All character zeros are retained. The whole row character is the same presented character; (CA10) is a contracting row scalar, not a replacement character.

For `P>=2(R+r0+1)`, the coefficients obey

$$c_j^{(r_0)}(P) \le 2^{j+1} j! P^{-j}.$$

There are at most `P^j/j!` squarefree choices of `d`, each with `Nd >= (U^a/2)^j`. Hence the total central absolute coefficient mass, including allocations, is bounded by

$$\sum_{j=0}^{R} c_j^{(r_0)}(P) \sum_{\substack{d\ \text{as in (CA9)}\\ \omega(d)=j}} \frac{2^j}{\sqrt{Nd}} \le 2\sum_{j=0}^{R}(4\sqrt{2}\,U^{-a/2})^j = O_R(1). \tag{CA11}$$

Minkowski in the same positive row Hilbert space now applies. This is a uniform finite mass calculation, with no new prime-label count after the completed norm. The separate `h` restoration costs only `U^epsilon` as proved in the transfer lemma. Shortened plain factors must satisfy the relevant earlier affine admission or its proved natural fallback; the coefficient identity does not erase those gates.

One Newton bound for the complete row family. Use the same pool $\mathcal{P}$, with $P = \#\mathcal{P} > 0$, for the finite row family. The accumulated divisor D_u, the ordinary factors, puncture, character and surviving slot subset may depend on the row u. Choose the common depth J_0 from their polynomial norm caps. For $S \subseteq \mathcal{P}$, $j = |S| \le J_0$, and $A \subseteq S$, let $\mathfrak{d}_{S,A}$ be the ordinary divisor pair assigned by the exact Newton expansion, writing $N\mathfrak{d}_{S,A} = N\mathfrak{d}_1\, N\mathfrak{d}_2$. Its scalar coefficient, including the natural character values, has absolute value at most

$$w_{S,A} = \frac{2^{j+1} j!}{P^j \sqrt{N\mathfrak{d}_{S,A}}}.$$

The pool budget above proves this for every D_u simultaneously. Since $N\mathfrak{d}_{S,A} \ge 1$,

$$\sum_{\substack{S\subseteq\mathcal{P}\\ |S|\le J_0}} \sum_{A\subseteq S} w_{S,A} \le M_{J_0} := \sum_{j=0}^{J_0} 2^{2j+1} j!.$$

Indeed, there are $\binom{P}{j} \le P^j$ subsets of size j and 2^j assignments per subset.

Write the literal normalized corrected row as $H_u = \sum_{S,A} c_{S,A}(u) F_{S,A}(u)$, keeping its original marked factor and both nonzero-row gates inside $F_{S,A}$. For any nonnegative original row weights ω_u, weighted Cauchy followed by interchange of these finite sums gives

$$\sum_u \omega_u |H_u|^2 \le M_{J_0} \sum_{S,A} w_{S,A} \sum_u \omega_u |F_{S,A}(u)|^2 \le M_{J_0}^2 E$$

whenever every displayed complete cofactor-family sum is at most E. Apply the ordinary singleton-source and HIGH count estimates to these same simultaneous cofactor families, with their actual moving punctures and common Mellin height. The finite factor $M_{J_0}^2$ is independent of the scale and moving row data.

Ratio restrictions of the full normalized family. Retain the complete nonnegative effective weight in each normalized quadratic energy. Exact reindexing by the original row label shows that restriction to any subtype or ratio fiber decreases this energy. Let E_{full} bound the full energy in both orientations, with the stated polynomial Mellin-height envelope. Use this same bound on every ratio fiber. The completed ratio contraction then gives

$$\left\| \sum_q \mathrm{phase}_q\, \mathrm{block}_q \right\| \le C\, \#\mathcal{B}_Z\, E_{\mathrm{full}},$$

where $\mathcal{B}_Z$ is the original finite set of ratio bins. The constant is chosen before the moving row and label families. For the actual complete family, the polynomial caps on its radial scale, frequency and source radius imply $\#\mathcal{B}_Z \le C_\epsilon Z^\epsilon$. Thus this contraction costs one arbitrarily small scale power and retains both rays, residual projectors, phases and all original outside weights. It introduces no source-label cardinality factor.

Selection of the actual source family. For an original first column j, let $\mathcal{A}_j$ be its complete first-allocation set and write its complete nonnegative cost as

$$T_j = \sum_{B\in\mathcal{A}_j} |a_{j,B}|^2 K_{j,B}.$$

The cost $K_{j,B}$ includes all subsequent blocks, residual-ideal weights, windows, profiles and physical scalars. If $T_j > 0$, choose an actual B_j maximizing the entire product $|a_{j,B}|^2 K_{j,B}$. Then $a_{j,B_j} \ne 0$, $K_{j,B_j} > 0$, and

$$T_j \le \#\mathcal{A}_j\, |a_{j,B_j}|^2 K_{j,B_j}.$$

The zero fibers contribute zero. Retain the previous allocation-Cauchy factor in the outside weight. Bound each of the two resulting allocation cardinality factors by its ideal-divisor small-power estimate.

The full first-column code consists of its source label, seed, four dyads and two ray characters. For a fixed source-label/seed pair, its fiber therefore has cardinality at most $\#\mathcal{D}_Z\, \#\widehat{\mathcal{R}}^{\,2}$, where $\mathcal{D}_Z$ is the actual dyad envelope and $\widehat{\mathcal{R}}$ is the fixed finite ray-character group. The polynomial bounds on the source radius, volume and reciprocal normalizer give $\#\mathcal{D}_Z \le C_\epsilon Z^\epsilon$. Choosing a column of maximal complete cost in each nonzero fiber thus leaves a family with injective source-label/seed key, at this uniform small-power cost. Keep every actual corrected second block at its chosen first allocation. Its modified coefficient, child character, first seed, mask and original singleton slots are the literal ones of the transformed source. All subsequent joint estimates use this family.

Forced labels and their actual allocations. For a complete allocation B of a common ideal C, extraction sends

$$A_i \longmapsto A_iB_i, \qquad D \longmapsto DC, \qquad \mathfrak{R} \longmapsto \mathfrak{R}C.$$

The forced divisor seed remains present, and the invariants $A_1A_2 \mid \mathfrak{R}$, $D \mid \mathfrak{R}$ propagate. Fix a squarefree nonzero forcing ideal L. In the original finite row set $\mathcal{U}$, keep the rows for which L belongs to the actual residual pool, using a membership indicator in a common finite union of those pools. For each nonzero normalized mother row $H_{u,L}$, choose an actual complete allocation with nonzero frozen coefficient and positive seed-one child norm. Its complete forced ideal $C_{u,L}$ satisfies $L \mid C_{u,L} \mid L^2$, and its exact scalar is $|a_B|^2/\operatorname{rawReduction}(B)$. Finite Cauchy and the ideal-divisor small-power bound give one $K_{N,\delta} > 0$, before L, the row family and all its moving data, such that

$$|H_{u,L}|^2 \le K_{N,\delta}(NL)^{\delta-1}\mathcal{C}_{u,L}|G_{u,L}|^2.$$

Here $\mathcal{C}_{u,L}$ is the full profile-dilation cost and $G_{u,L}$ retains the actual child volume, puncture and heights. For nonnegative original weights ω_u, summation gives

$$\sum_{u\in\mathcal{U}} \omega_u|H_{u,L}|^2 \le K_{N,\delta}(NL)^{\delta-1} \sum_{u\in\mathcal{U}_{L,+}} \omega_u\mathcal{C}_{u,L}|G_{u,L}|^2.$$

The live set $\mathcal{U}_{L,+}$ is a subset of the original rows. Fix the missing ideal, ray labels and common divisor pair before applying the earlier source to this family. Since $L \mid C_{u,L} \mid L^2$, the prime supports of $C_{u,L}$ and L agree. The excluded-character mask therefore has the same coefficients as the common mask with $C_{u,L}$ replaced by L. This is one fixed puncture for the joint source estimate. Its constants are uniform in L. Partition this same fixed-L family into LOW and HIGH, preserving its original characters, correction factors, heights, masks, weights and first outside allocation scalar. After the joint estimates, sum over the full residual union with its retained inverse-norm weight.

For the two ordinary allocated ideals put $d = N(B_1B_2) \ge 1$. Every live child slot scale is at least one, and the literal child data satisfy

$$V_{\mathrm{child}} \le V/d, \qquad L_{\mathrm{child}} \ge L_0/d, \qquad \mathcal{S}_{\mathrm{child}} \le \mathcal{S},$$

where $L_0 > 0$ bounds the four original raw factors below and $\mathcal{S}$ is the original slot-control factor. For $0 \le \alpha < 1/2$,

$$\mathcal{S}_{\mathrm{child}}V_{\mathrm{child}} \max(1, L_0/d)^{-2\alpha} \le \mathcal{S}V \max(1, L_0)^{-2\alpha}.$$

Indeed $\max(1, L_0) \le d\max(1, L_0/d)$, and $d^{2\alpha-1} \le 1$. All slot controls are nonnegative; multiplying once more by $\mathcal{S}_{\mathrm{child}} \le \mathcal{S}$ gives the same inequality with squared slot controls when required by the energy bound.

The full volume through the three extractions. Let $b \ge 1$ bound the fixed annular supports and let the original marked set have at most N elements. For a contribution with nonzero complete frozen coefficient, denote the two common ideals and the complete forced ideal by $\mathfrak{C}_1, \mathfrak{C}_2, \mathfrak{C}_f$, and their actual allocations by $B^{(1)}, B^{(2)}, B^{(3)}$. The exact raw-scale identities give

$$V_3 = \frac{X_1X_2\prod_{i\in T} P_i}{\prod_{k=1}^{3} \operatorname{rawReduction}(B^{(k)})} \le \frac{b^{3N}V_0}{N\mathfrak{C}_1\, N\mathfrak{C}_2\, N\mathfrak{C}_f}, \qquad V_0 = X_1X_2\prod_{i\in T} P_i.$$

Every ordinary corner contracts separately, and the surviving singleton product is the exact image subset of the original marked set. For $Z > 1$, the logarithmic volume allowance is therefore $3N\log b/\log Z$. It is absorbed into the chosen fixed loss by an eventual threshold depending on N. The scalar normalization and the four separate corner budgets retain their original factors.

The surviving original slot widths. The canonical marked profiles have lower support endpoint one. Keep the three nonzero frozen coefficients in the preceding extraction. Let T_0 be the original marked set, $F \subseteq T_0$ the exact image of the final live slots, and $P_i = Z^{w_i}$, with $w_i \ge 0$. For a cutoff $\omega_* \ge 0$, write $F_+ = \{i \in F : w_i \ge \omega_*\}$ and $\ell = \sum_{i\in F_+} w_i$. The frozen support bounds give

$$\sum_{i\in T_0} w_i - \log_Z(N\mathfrak{C}_1) - \log_Z(N\mathfrak{C}_2) - \log_Z(N\mathfrak{C}_f) - N\omega_* \le \ell.$$

Thus, if $\beta - \delta \le \sum_{i\in T_0} w_i \le \beta$ and $\delta + N\omega_* \le 2\varepsilon_w$, then

$$0 \le \ell \le \beta, \qquad \beta - 2\varepsilon_w \le \ell + \log_Z(N\mathfrak{C}_1) + \log_Z(N\mathfrak{C}_2) + \log_Z(N\mathfrak{C}_f).$$

For every collision singleton subset $S \subseteq F$, put $s_+ = \sum_{i\in S\cap F_+} w_i$. Nonnegativity gives

$$0 \le s_+ \le \ell, \qquad \sum_{i\in S} w_i + (\ell - s_+) \le \sum_{i\in F} w_i.$$

These are the slot coordinates used with the full-volume bound above.

The common-ideal allocation payment. Let $\mathfrak{C} \neq 0$ be an actual common ideal, $\mathcal{A}_{\mathfrak{C}}$ its complete allocation set, a_B the frozen coefficient and V_B the child volume. Put $d_B = N(B_1B_2)$ and $R_{\mathfrak{C}} = N\operatorname{rad}\mathfrak{C}$. The ordinary squarefree projector makes the actual child row zero unless B_1B_2 is squarefree: every restored ordinary product contains this frozen divisor. Retain this zero term in the original allocation sum. On the surviving set $\mathcal{A}^{\mathrm{sf}}_{\mathfrak{C}}$, the squarefree divisor $B_1B_2 \mid \mathfrak{C}$ divides $\operatorname{rad}\mathfrak{C}$, so $1 \le d_B \le R_{\mathfrak{C}}$. The exact raw-scale identity and frozen support give

$$|a_B|^2V_B^2 \le \frac{\mathcal{F}^2}{(N\mathfrak{C})^2}V^2, \qquad \max(1,L_0) \le d_B\max(1,L_0/d_B).$$

Thus, for $0 \le \alpha < 1/2$,

$$|a_B|^2V_B^2\big[\max(1,L_0/d_B)^{-\alpha}\mathcal{S}_B\big]^2 \le \frac{\mathcal{F}^2d_B^{2\alpha}}{(N\mathfrak{C})^2}V^2\big[\max(1,L_0)^{-\alpha}\mathcal{S}\big]^2.$$

Cauchy with the complete original allocation cardinality gives

$$\#\mathcal{A}_{\mathfrak{C}}\sum_{B\in\mathcal{A}^{\mathrm{sf}}_{\mathfrak{C}}}\frac{\mathcal{F}^2d_B^{2\alpha}}{(N\mathfrak{C})^2} \le (\#\mathcal{A}_{\mathfrak{C}})^2\mathcal{F}^2\frac{R_{\mathfrak{C}}^{2\alpha}}{(N\mathfrak{C})^2}.$$

The original frozen coefficients, profile-dilation cost, forcing seed, first outside weight and row selector remain in the corresponding completed norm.

The fixed-modulus forcing count. Keep the fixed nonzero inducing modulus Q throughout the exceptional row count. Split the squarefree forcing ideal as $F = F_{\mathrm{good}}F_Q$, according as its prime factors are coprime to Q or divide Q. The second factor divides $\operatorname{rad}Q$. At each prime of F_{good}, fixed induction forces the row valuation into the residue class four modulo six. The sixth-power parameterization therefore gives, on $N(z) \le C_rZ^M$,

$$\#\{z \text{ in the exceptional row set}\} \le C_QC_r^{1/6}Z^{M/6}(NF_{\mathrm{good}})^{-2/3} \le C_Q(N\operatorname{rad}Q)^{2/3}C_r^{1/6}Z^{M/6}(NF)^{-2/3}.$$

Here $C_Q = 768\,6^{\omega(Q)}$, with the fixed ray and unit sectors retained. The second inequality follows from $NF \le NF_{\mathrm{good}}\,N\operatorname{rad}Q$. The modulus is fixed before the source constant, so its radical factor is included in that constant. The profile orders, source exponents and both independent Mellin heights retain their stated quantifier order.

Paired windows and the full forcing sum. Let $\mathfrak{C},\mathfrak{D}$ be the two actual common ideals with the same prime support, and put $R = N\operatorname{rad}\mathfrak{C} = N\operatorname{rad}\mathfrak{D}$. Taking square roots of the two masked allocation sums gives

$$\mathcal{W}_{\mathfrak{C}}\mathcal{W}_{\mathfrak{D}} \le \#\mathcal{A}_{\mathfrak{C}}\#\mathcal{A}_{\mathfrak{D}}\frac{R^{2\alpha}}{N\mathfrak{C}\,N\mathfrak{D}}\mathcal{F}^2E_L\mathcal{H}_{J,t}^2.$$

Here $\mathcal{H}_{J,t}$ is the retained height envelope times its Mellin profile moment. Each window uses its own residual puncture and the same parent budget. For at most N marked slots, the divisor bound uses all $N+2$ slots and gives, for each fixed $\eta > 0$,

$$\#\mathcal{A}_{\mathfrak{C}}\#\mathcal{A}_{\mathfrak{D}} \le A_{N,\eta}(N\mathfrak{C}\,N\mathfrak{D})^{\eta}.$$

The second common radical payment. Write $\mathcal{R}$ for the actual row radius, F_ν for the common Fourier-frequency ideal and $\mathcal{N}$ for the literal common normalizer. The radial majorant uses $H = \max(1, 4\mathcal{R}/NF_\nu)$. A nonzero physical block supplies a nonzero retained row, so $H = 4\mathcal{R}/NF_\nu$ on that block. A zero block contributes zero. At a common prime of norm q, put $c = v_{\mathfrak{p}}(\mathfrak{C})$, $d = v_{\mathfrak{p}}(\mathfrak{D})$, and $m = \min(c, d)$. The canonical partition gives exponents u, v equal to one in its equal-valuation unit and nonunit cases, respectively, and zero otherwise. The forcing exponent is $f = 1$ in the $(1, 1)$ nonunit case and zero otherwise. Thus the prime exponents in $\mathcal{N}, NF_\nu, NF$ are $m - u, m + v, f$. The local factor of

$$\frac{\mathcal{N}}{N\mathfrak{C}\, N\mathfrak{D}\, (NF_\nu)^{1/6}(NF)^{2/3}}$$

is $q^{-\kappa}$, where

$$\kappa = c + d - \frac{5}{6}m + u + \frac{1}{6}v + \frac{2}{3}f \geq 2.$$

For the four possible types, its lower bounds are respectively $2, 13/6, 13/6, 5/2$. Multiplying over the common primes gives

$$\frac{\mathcal{N}H^{1/6}}{N\mathfrak{C}\, N\mathfrak{D}\, (NF)^{2/3}} \leq (4\mathcal{R})^{1/6}R^{-2}.$$

Together with the paired window, the radical factor is $R^{2\alpha-2}$. The retained projectors give $N\mathfrak{C}\, N\mathfrak{D} \leq R^4$, so the allocation-cardinality loss costs at most $R^{4\eta}$. For $\alpha = 499/1000$ and $\eta = 1/10000$, $2\alpha + 4\eta - 2 = -626/625 < -1$. The remaining divisor and finite type multiplicities are bounded by an arbitrarily small power of R; choose that power within this strict surplus. The resulting inverse-radical weight is summed with the ideal harmonic bound, with its arbitrarily small scale loss. All original outside, dyadic and profile factors stay in the sum.

The literal corrected seed budget has the exact form $E_L = (NL)^{\varepsilon_L - 1}E_1$. Use the norm cap $NL \leq Z^{B_L}$ for the full forcing pool. For the actual second residual pool take $B_L = 4$, independently of the number N of marked slots. Indeed, the four initial raw scales are at most Z. The product of the two stored ordinary factor norms is at most a fixed support constant times Z^2. The complete marked product has norm at most $2^N Z^{\sum_i w_i} \leq 2^N Z$. Thus every stored initial column product has norm at most $C_N Z^3$, and hence at most Z^4 after an eventual threshold depending on N. Each quotient-child column reconstructs a stored parent column after multiplication by its actual common allocation. Its norm therefore contracts through both extractions. The actual sector reconstruction gives $L \mid J$, with $\mathfrak{D}J$ in that stored pool and $\mathfrak{D} \neq 0$. Hence $NL \leq NJ \leq N(\mathfrak{D}J) \leq Z^4$ throughout the padded residual union, including zero-coefficient terms. Only the threshold depends on N; the exponent used to choose the earlier source losses is fixed. For the joint source call, write the common puncture as $\mathfrak{R}_* L d_1 d_2$, where $\mathfrak{R}_*$ depends only on the fixed character anchor, fixed bad-prime ideals and initial puncture. The Newton assignment partitions one set $S \subseteq \mathcal{P}$, with $\#S \leq J_0$, so

$$d_1 d_2 = \prod_{\mathfrak{p} \in S} \mathfrak{p}, \qquad N(d_1 d_2) \leq Z^{J_0 a} \quad \text{if } N\mathfrak{p} \leq Z^a \quad (\mathfrak{p} \in \mathcal{P}).$$

For any fixed $\eta_* > 0$, eventually $N\mathfrak{R}_* \leq Z^{\eta_*}$. Consequently,

$$N(\mathfrak{R}_* L d_1 d_2) \leq Z^{4 + J_0 a + \eta_*}.$$

The exponent is fixed before the ambient marked-slot count. Thus the radical loss for this common puncture can be chosen within the fixed source reserve. The moving missing ideal retains its own radical factor in the weighted missing-ideal sum.

The full ideal harmonic bound now implies, for $\delta > 0$,

$$\sum_L \|\mu(L)\| \frac{(NL)^{\varepsilon_L}}{NL} \leq H_{B_L, \delta} Z^{\delta + B_L \varepsilon_L} \qquad (\varepsilon_L \geq 0).$$

The constant and threshold precede the forcing pool and ε_L. Multiplication by the displayed paired-window bound retains its complete allocation, profile and height costs.

Deleted marked scales remain in the exact raw reduction factor. Apply the Newton and two-ideal expansions at this completed positive norm, with their central factors and outside character scalars in the same row Hilbert space.

Uniform admission at the moving puncture. For each extracted divisor pair (d_1, d_2), use precisely $\mathfrak{R}d_1d_2$, the original keep predicate, the same outer height and both rectangles shortened by the same divisors. The global affine cap admits each positive shortening. Expanding the supplementary holes in the common mask transports the completed ordinary bound from its fixed character anchor. Uniformly over every surviving full marked subset and every such shortening, for each fixed $\delta > 0$ the cost is

$$C_\delta\big(N \operatorname{rad}(\mathfrak{R}d_1d_2)\big)^\delta.$$

The constant precedes the puncture, live characters and heights. The previously established polynomial norm bound on the puncture absorbs this factor in the original small power after δ is chosen inside the fixed reserve. The mesh precedes the ambient finite slot type. Collision constants may depend on its later fixed cardinality; finite profile controls and polynomial height degrees precede live rows and heights, and the final large-height threshold follows those degrees.

A fixed positive radial lower bound. Fix a node allowance $e > 0$. Let t, b be the original plain and marked width bounds. Write $Q = F - \delta - \rho$ in the common geometry. For the actual retained row, the two radius allowances r, η and frequency loss ω give

$$Q = x_2 = F - \delta - \rho, \qquad \rho = \omega + r - \eta, \qquad \operatorname{normValue}(z) \le U^Q.$$

Choose $r = \eta$, so that $\rho = \omega$ and the existing frequency-loss guard applies to this same radius. Let c be the sum of the three common-ideal norm logarithms, and let f be the removed marked width. Denote the annular, clipping and rounding allowance by ℓ. For a nonzero positive corner, the supported clipped-scale bound and nonnegative Newton norm logarithms give $0 \le t - c + f + \ell$. Hence, using $f \le b$, $c \le t + b + \ell$. The actual geometry also gives $0 \le \delta \le c$. With $0 \le B \le 1$ and the chosen guard

$$\ell + B\rho + 2e \le BF - t - b,$$

we obtain

$$BQ = BF - B\delta - B\rho \ge BF - c - B\rho \ge 2e.$$

This first implies $Q > 0$, and then $BQ \le Q$ gives $Q \ge 2e$. The other branch is an exactly zero positive corner. Apply this argument separately to both corners. Thus the parent-source lower endpoint $M_{\min} = e$ is fixed from the node allowance before the marked-slot cardinality; its eventual scale threshold may depend on that cardinality.

Common scales. For each positive corner put $x_i(u) = \log_Z \max(1, X_i(u))$, and let $Q(u) \in [M_{\min}, M_{\max}]$ be its radial exponent. Partition the original rows by the three integer parts of $(x_1(u), x_2(u), Q(u))/\omega$, with fixed $\omega > 0$. In a nonempty cell let u_1, u_2, u_3 be the three upper endpoints. Then the common caps $C_i = Z^{u_i}$ and $M = \min(M_{\max}, u_3)$ satisfy, for every row in that cell,

$$X_i(u) \le C_i, \quad u_1 + u_2 \le x_1(u) + x_2(u) + 2\omega, \quad M_{\min} \le M \le M_{\max}, \quad Q(u) \le M \le Q(u) + \omega.$$

The rounded affine admission includes the displayed 2ω allowance. Increasing its radial exponent from $Q(u)$ to M preserves the downward face. Thus each cell has common admitted caps and a common admitted radial scale. The polynomial geometry bounds all three coordinates in a fixed interval, so the number of cells is bounded independently of Z and of the original row family. The weighted squared norm is exactly the sum over these cells; their collection pays this fixed count and at most one additional ω in the radial exponent.

In each nonempty cell, choose an original row attaining the largest nonnegative weight in that cell, including its balanced first mass and normalized allocation coefficient. The common radial and missing-ideal upper exponents differ from those of this same row by at most the scheduled cell widths. Apply the combined coefficient, missing-ideal, radial and erasure price to this attaining row. The resulting bound therefore retains the full scalar compensation. Empty cells contribute zero. Sum the priced estimates over the finite cells and surviving slot subsets, with their fixed cardinality factors.

The shortening scales may vary with the original row u. Choose common positive caps C_i before all functions $0 < Y_i(u) \le X_i(u) \le C_i$, with the same affine parent length budgets. If the profile support is bounded above by $b > 0$, put $a = 1/\max(1, b)$. A plain factor with scale below a vanishes by its support. Otherwise replacing its scale by $\max(a, Y_i(u))$ leaves it unchanged. The norm comparison therefore holds

on every original row and retains its character, ideal mask, numerator, slots and height. The common logarithmic windows cost

$$\prod_{i=1}^{2}\bigl(1+2\log(\max(1,C_i)/a)\bigr).$$

These factors are uniform over the row-dependent shortenings.

For the forward two-ideal expansion take the union, over the original finite rows, of their actual supported divisor-pair pools. Pad each row's sum by its own membership indicator in this one union. Every pair is good and inherits its norm cap from a witness row. If $b^2X_1(u)X_2(u)\le U^B$, the arbitrary-finite kernel mass bound gives $C_\delta U^\delta$ for this union. Finite-prime hole deletion contributes its fixed mass. Weighted Cauchy now applies to the whole row family at once: its restored squared norm is bounded by the square of these masses, hence $C_\delta U^{2\delta}$, times the one shortened-family budget. The polynomial caps on the C_i, the divisor norms and the puncture absorb the two logarithmic windows and these small powers after their parameters are chosen within the fixed reserve. For the centered row, first use $|F_X-F_Y|^2\le 2|F_X|^2+2|F_Y|^2$. For each positive corner restrict the original key family to the keys where that corner is nonzero. Its actual supported terms then give that corner's clipped-length bounds and earlier affine capacity budget. Apply the one-corner ordinary source to the X family with (C_1^X,C_2^X), and separately to the Y family with (C_1^Y,C_2^Y). The zero terms vanish, and the original weights, characters, masks, heights and singleton subsets are retained in both families. The source constants are uniform before these restrictions. Retain the corresponding explicit window budgets E_X,E_Y. With the common restoration mass $\mathcal{R}$, weighted Cauchy gives

$$\sum_u \omega_u|F_X(u)-F_Y(u)|^2\le 2\mathcal{R}^2(E_X+E_Y).$$

The equal products give the same central normalization for both corners. Choose the larger constants, degrees and eventual thresholds, and take the union of the finite profile controls. These choices are uniform. Both rectangles retain their original outside scalars.

4. Centered fixed-inducing terminals retain their reference saving

Use the exact finite Newton expansion (CA8)–(CA9). Since $0\le r_0\le R$, increase the eventual threshold to $P\ge 4R+4$. After common puncture extraction its shifted coefficients satisfy, for $0\le j\le R$ and $P\ge 2(R+r_0+1)$,

$$0\le c_j^{(r_0)}(P)\le 2^{j+1}j!P^{-j}.$$

There are at most P^j pool selections and at most 2^j assignments of those selected divisors to the two ordinary factors. Since the product norm of an extracted divisor pair is at least one, the absolute weighted mass of the finite Newton kernel, at every positive exponent, is at most

$$C_R=\sum_{j=0}^{R}2^{2j+1}j!. \tag{CA12}$$

This deliberately uniform bound suffices. Convolve that finite kernel with the exact forward two-ideal kernel (1)–(2). At exponent $\sigma=1-\alpha=501/1000$, (6) and (CA12) bound its total divisor mass by $O_R(U^\epsilon)$, uniformly over the common puncture.

Apply the ordinary equal-product centered estimate (MF3.78), interpolated with the absolute lattice bound as in the squarefree preprocessing section. For equal product scale T and minimum scale $L>0$, it is $O(\sqrt{T}L^{-\alpha})$, for $\alpha=499/1000$, including subunit scales. A divisor pair of product norm d divides both rectangles by the same norms, so its centrally normalized contribution is

$$d^{-1/2}\sqrt{T/d}\,(L/d)^{-\alpha}=\sqrt{T}L^{-\alpha}d^{-(1-\alpha)}. \tag{CA13}$$

Its extracted character is the same natural character, with an outside scalar of modulus at most one. The common puncture, Newton polynomial, profiles, heights and equal-product subtraction are identical for both rectangles. Sum (CA13) with the exact kernel mass at exponent .501. Taking the better absolute bound when the minimum scale is below one retains the relative terminal saving

$$U^{-(499/1000)r+\epsilon}, \tag{CA14}$$

where r is the nonnegative remaining reference exponent. With common width v, this gives the saving $\alpha(L_{\mathrm{ref}}-v)_+$ used in (G5). All finite profile controls and independent polynomial heights come from the same ordinary centered estimate before the divisor sum.

5. Other terminal sectors and the remaining source proof

The second Poisson true diagonal is already a sum of coefficient squares. On supported columns `P/(P-r)<=2`, so its diagonal estimate changes by at most a fixed factor four. The same direct coefficient bound applies to the coarse terminal absolute estimates. The artificial zero Gauss row retains its exact support; no zero is amplified into a nonzero contribution by (CA5).

An absolute envelope for Mellin inversion. Retain the actual finite second family, its corrected coefficients, both ray characters and every residual ideal in the stored pool. Let l index its labels, let w_l be its scalar weight, let $o_l > 0$ be its physical outside factor, and let V_l be its input volume. Write $P_{00} = \|W_1\|_{0,0}\|W_2\|_{0,0}$. The finite coefficient expansion and the lattice bound give

$$\sum_I |\widetilde{b}_l(I)| \le C_m P_{00} V_l.$$

Here C_m depends on the fixed slot cardinality, coefficient and support bounds. The correction has modulus at most two on the full finite tuple pool, including its zero-column terms. If the physical row norm is at most $X \ge 1$ and every stored ideal has norm at most Y, the full residual pool has cardinality at most $128Y$. Summing the nonnegative row majorant over the two ray characters gives at most $q^2 D_0 X$, where q is the fixed ray-character cardinality and D_0 is the diagonal control of the fixed ball profile. Consequently the absolute second budget $\mathcal{A}$ satisfies

$$\mathcal{A} \le 128 q^2 D_0 C_m^2 P_{00}^2 XY \sum_l |w_l| o_l V_l^2.$$

This follows by the triangle inequality for the complete coefficient sum, followed by the two finite counts. It retains the volume squared for cancellation with the actual physical normalization. For $U > 1$, if

$$X \le U^{B_r}, \quad Y \le U^{B_p}, \quad \#\{l\} \le C_l U^{B_l}, \quad |w_l| o_l V_l^2 \le C_w U^{B_w}, \qquad B_r \ge 0,$$

then

$$\mathcal{A} \le 128 q^2 D_0 C_m^2 C_l C_w P_{00}^2 U^{B_r + B_p + B_l + B_w}.$$

The same bound holds for each normalized second sum, in either orientation and at all real Mellin variables. All masks, natural character zeros and original finite rows are retained. The stored ideal pool has the exponent $B_p = 4$ established above.

Completing the Mellin integrals. Let $H_0 \ge 1$ denote the external height factor and put $T = (1 + |t|)(1 + |u|)H_0$. Retain the original volume and the matched bin factors E/H and EH in the two orientations. After their scalar payments, write the bounded-range estimate as $C_s \mathcal{F} U^q$ for $T \le U^\tau$, with $\tau > 0$, and the corresponding absolute estimate as $C_c \mathcal{F} U^A$. Choose an integer $J \ge 0$ with $\tau J \ge A - q$. For $T > U^\tau$, one has $U^A \le T^J U^q$; the bounded range has $T^J \ge 1$. Both ranges therefore satisfy

$$\text{budget} \le (C_s + C_c)\mathcal{F} T^J U^q.$$

The finite Mellin moments of order J are controlled by finitely many seminorms of the original smooth profiles. Minkowski's inequality and Mellin inversion now give the physical norm bound, with a factor H_0^J and the original weighted sum $\sum_k E(k)/(1 + \text{dyadicScale}(k_1))^d$. The bin compensator stays paired with its own orientation until this contraction. The polynomial exponents and J may depend on the fixed slot type; they precede the moving profiles, rows and Mellin variables.

The MAIN zero and tail terms. Let V be the original volume and $A_V = \log_U V$. For a node (B, Λ, h) with allowance $e > 0$, fix a width cell with upper endpoint b, rounding width δ_w, and $b - e - \delta_w \le \sum_i w_i \le b$. Put $t = B - \Lambda(b - e)$. The original affine admission gives

$$A_V \le t + b + \Lambda \delta_w.$$

The parent-budget certificate, $B_0 \le 1$, and $\delta_w \le e/(16(\Lambda + 1))$ give

$$t + b + 4e < B_0(1 - 6a) \le 1 - 6a, \qquad A_V + 6a + 3e \le 1.$$

The actual MAIN radius has exponent $R+6a+\omega$, where ω is the frequency and support allowance. The balanced-side and common-radical identity pays its base exponent by A_V-1. The reciprocal amplifier-pool cardinality contributes $-a$; the annular band count is at most seven. Both allocation-cardinality payments are retained. After the remaining small counting losses ℓ, the zero-term exponent is therefore at most

$$A_V-1+5a+\omega+\ell\le -a \qquad \text{when } \omega+\ell\le 3e.$$

The tail with its original saving $s>0$ has exponent at most $-a-s$. The previously displayed profile and height factors remain outside these powers of U.

All MAIN radius, raw/natural width, central normalization and common allocation formulas are unchanged. The resulting source certificate checks MAIN HIGH, MAIN LOW, RAW admission, fixed and tiny-frequency sectors, the squarefree centered terminal, and each shortened-factor fallback. The finite Newton multiplier and the squarefree restriction are expanded only at the explicit completed norm stage described above; fixed-inducing terminals retain equal-product cancellation in (CA13).

Chapter 27. Restricted HIGH and every exceptional sector

Restricted HIGH counting, root completion, and exceptional sectors

Use the density interval

$$\frac{1}{2}\le \nu \le \frac{3}{4}. \tag{LD1}$$

Use inverse Theorem 7.3, plain Theorem 7.4, and the common-zero witness Lemma 14.2. The ordinary and complete marked squarefree projectors give exactly the four first-common types (10).

1. Coefficients and the exact witness calculation

Write

$$d=5-2\nu,\quad w=5-6\nu,\quad \rho=1-\frac{10\nu}{3d},\quad \phi=\frac{\nu w}{d},$$

$$A_\nu=\frac{14\nu w}{115-54\nu},\quad B_\nu=\frac{2\nu w^2}{3d(115-54\nu)},\quad C_\nu=\frac{16\nu w^2}{3d(185-138\nu)}.$$

All denominators are positive on (LD1). Define

$$G_\nu(S,Q)=\min\{\phi S, A_\nu S+B_\nu Q, C_\nu Q\}. \tag{LD2}$$

Normalize Q=1, first take amplitude parameter p=1/2, and use exactly the inverse and ordinary witnesses of Lemma 14.2. Their available marked widths are

$$h_I=\min(S,(1-r)/2),\qquad h_P=\min(S,2(1-2m)/9).$$

The count bounds, when invoked, are

$$E\le 1-\nu\min(r+S,(1+r)/2),\quad E\le 1-\nu\min(2m+S,(14m+2)/9).$$

For a target xi put

$$r_*=\max(\xi-S,2\xi-1),\quad y_*=\max(\xi-S,(9\xi-2)/7),\quad t_*=r_*+y_*/2.$$

The equation

$$1-\nu\xi=\frac{1}{6}+(5/6-\nu)t_* \tag{LD3}$$

has the following three branches:

$$\xi = \begin{cases} (10+3wS)/(15-6\nu), & 0 \le S \le s_1, \\ (70+2w+14wS)/(115-54\nu), & s_1 \le S \le s_2, \\ (70+16w)/(185-138\nu), & S \ge s_2, \end{cases}$$

$$s_1 = \frac{2w}{3(45-26\nu)}, \qquad s_2 = \frac{7w}{185-138\nu}. \tag{LD4}$$

These are rational identities, not new analytic inputs. Their joins agree, 0<=s1<=s2, and the plain cap becomes active before the inverse cap. Substitution gives

$$\rho - (1-\nu\xi) = G_\nu(S,1). \tag{LD5}$$

2. Every witness admission remains strict

On (LD1), the inverse second gate is checked as follows:

$$\xi \ge \frac{10}{3(5-2\nu)} \ge \frac{5}{6}, \qquad r_* \ge \frac{22}{29} > \frac{3}{4}. \tag{LD6}$$

For completeness, the three expressions for ξ in (LD4) give the last bound directly. If $0 \le S \le 7/58$, clearing their positive denominators gives $\xi_i - S \ge 22/29$ for each branch. If $S \ge 7/58$, each branch satisfies $\xi_i \ge 51/58$. Taking their minimum and then $r_* = \max(\xi - S, 2\xi - 1)$ proves (LD6) on the whole density interval. The inverse moment is invoked only when $r \ge r_*$. Its second capacity therefore satisfies

$$2r + 8h_I \le 4 - 2r, \qquad 3 - (2r + 8h_I) \ge \frac{15}{29}. \tag{LD7}$$

The first inverse gate and ordinary marked gate receive the same fixed width erosions described after (LD8). The unused second inverse margin (LD7) uniformly absorbs the witness enclosure error.

The largest xi is its saturated value

$$\xi_{\mathrm{sat}} = \frac{150-96\nu}{185-138\nu} < 1,$$

because 35-42nu>0 on (LD1). Hence (LD3), w>0, and (LD5) imply

$$1 < t_* \le \frac{5}{5-2\nu} \le \frac{10}{7} < \frac{3}{2}. \tag{LD8}$$

This is an admitted cutoff for the same-zero detector. If the inverse witness is longer than one, its original unmarked long bound is monotone in r since 5/6-nu>0, and r<=t* pays it. If the ordinary witness exceeds one half only by its admitted enclosure, its unmarked bound suffices because xi<1. Thus no witness outside a moment's domain has been used.

The passage from p=1/2 to 0<=p<=1/2 is unchanged: subtract nu*(1-2p)*S from the count exponent, and use 2p*h+(1-2p)*S>=h for every selected width h<=S. All marked subsets remain whole original slots with the canonical pools and coefficients. Consequently the natural common-zero count is

$$\rho Q + \nu S - G_\nu(S,Q) + \varepsilon. \tag{LD9}$$

3. Uniform coefficient bounds and full sixth-root completion

Direct differentiation shows rho decreases on (LD1), so

$$\frac{2}{7} \le \rho \le \frac{7}{12}, \qquad 0 \le \phi < \frac{13}{50}. \tag{LD10}$$

The second bound holds for every real nu with positive d: after clearing d, its strict difference is the positive quadratic 300nu^2-276nu+65, whose discriminant is -1824. The bound applies to the whole interval (LD1). The natural plain capacity remains at most 9/2; the density ν does not replace that capacity.

Put $x = \nu - 1/2 \in [0, 1/4]$ and $D = 3(5-2\nu)(185-138\nu) > 0$. The exact identity

$$D - 40\big(16\nu(5-6\nu)^2\big) = 112 + 2495x + 1092x(1/4-x) + 23040x^2(1/4-x) > 0$$

proves the uniform coefficient bound

$$0 < C_\nu < \frac{1}{40}. \tag{LD11}$$

Clearing positive denominators also gives $0 < A_\nu < \phi$ and $0 < B_\nu < C_\nu$. Thus G is monotone, ϕ-Lipschitz in marked width and C_ν-Lipschitz in row width.

For the two root sums use $y = 3/4 - \nu \geq 0$. Direct expansion gives

$$175D\big(6(\rho - C_\nu) - 279/175\big) = 59283/4 + 1191798y + 1734588y^2 + 604800y^3,$$
$$350D\big((\rho - 1/6)/2 - C_\nu - 83/2100\big) = 19761/4 + 193516y + 233196y^2 + 201600y^3.$$

Both right sides are positive. Hence the small-root series exponent and large-root tail reserve satisfy

$$6(\rho - C_\nu) > \frac{279}{175} > 1, \tag{LD12}$$

$$(\rho - 1/6)/2 - C_\nu > \frac{83}{2100} > 0. \tag{LD13}$$

These estimates hold uniformly on the full interval (LD1). Here is the full root summation, with the same masks and independent heights. Write each actual row uniquely as $\zeta u B^6$, retaining its six-unit label and sixth-free core, and put $n = NB$. At total raw width $D \in [D_{\min}, D_{\max}]$, $D_{\min} > 0$, its residual core width is $q = D - 6\log_U n$. For $n \leq U^{D/12}$, $q \geq D/2$. Choose the core mesh at most $\mu D_{\min}/2$ before roots or heights. Since $(U^q)^{w_j/q} = U^{w_j}$, the actual slot scales and common profiles are unchanged. The fixed positive core lower width permits one common height exponent, finite seminorm set and eventual threshold.

The exact deletion of root and moving-hole primes from a slot costs $O(P_j^{-1/2}\log U)$: subtract precisely its terms with a prime dividing the hole. After the previously assigned tiny-slot cutoff this tends to zero uniformly for polynomially bounded holes. The core character and relative coefficients remain the same; only their natural zero factors change. The finite positive maximum-product comparison cost is at most $4^{|J|}$, a constant after the slot system is fixed.

Let g be the row-width Lipschitz bound of the gain. Then $G(L,q) \geq G(L,D) - 6g\log_U n$, so a small-root contribution is at most

$$U^{\rho D + \nu L - G(L,D) + \epsilon} n^{-6(\rho - g)}. \tag{RT1}$$

The ideal sum converges when $6(\rho - g) > 1$, with no additional root cardinality. For $n > U^{D/12}$, use the unmarked core count and the pointwise marked envelope. Partial summation of its ideal root series saves $(\rho - 1/6)D/2$. Since $G(L,D) \leq gD$, this pays the full gain whenever $(\rho - 1/6)/2 - g > 0$. Empty subunit residual windows contribute zero. The fixed finite nonprincipal exceptional cores cost their marked envelope and are covered by $U^{(\rho - g)D/2} \geq 1$.

For the three-facet gain take $g = C_\nu$, using (LD12)–(LD13). For the later six-facet gain take $g = C_{\max}$, using (HF13)–(HF14). Fixed-inducing cores retain their separate count $U^{Q/6+L}$; tiny nonprincipal cores retain U^{Q+L}. Their complete payments are (LD15)–(LD17). This proves full-gain completion for each gain used in the source graph.

4. The restricted first-common cone is individually favorable

Let A,b,c,D be the total logarithmic widths of genuine common primes of types (1,1),(1,2),(2,1),(2,2), respectively. The total first lengths are

$$C = A + b + 2c + 2D, \qquad D_c = A + 2b + c + 2D.$$

The exact first-local rules give R=b+c, L_C=2c, R0_C=b, Sigma=D, and both B positive parts vanish. The complete common form before the density charge is

$$V_C = (1 - \rho)(C - D_c)_+ - (1 + \rho)R + \rho L_C + R_{0,C},$$

one has the global identity

$$V_C - \Sigma = -\min(b,c) - \rho(b - c)_+ - D \leq 0. \tag{LD14}$$

This holds for every 0<=rho<=1. It does not require primewise ordering. The reverse side has the same identity with b,c exchanged.

For an ordinary HIGH side, conserved second-extraction width pays the following exact Lipschitz payment. If $L + c + b_2 \geq S$ and $Q + 2c + 2b_2 \geq F$, monotonicity, clipping, and the width Lipschitz constants give

$$\nu b_2 + G_\nu(L, Q) \geq G_\nu((S - c)_+, (F - 2c)_+) \geq G_\nu(S, F) - (\phi + 2C_\nu)c. \tag{LD14a}$$

To prove the first inequality, put back second common width b_2: its possible gain cost is at most $(\phi + 2C_\nu)b_2 \leq \nu b_2$. The second inequality puts back the first common width c and uses the same constants. This argument includes all clipped regimes. The bound is

$$\phi + 2C_\nu < \frac{13}{50} + \frac{9}{100} = \frac{7}{20} < \nu.$$

Equivalently use eta=nu-phi-9/100>1/10. Equations (LD14) and this inequality make the residual ordinary HIGH common excess individually nonpositive after the three no-common scalar gates H+G_ nu(S,F)>0 have been certified.

5. Fixed-inducing and tiny-radius raw sectors remain present and paid

For a fixed-inducing core replace the ordinary core count by its actual exponent Q/6+L. The same inner allowance (140) and common payment (RC2) then give the excess

$$-H + V_C - \Sigma - \nu(C + b_2) - (\rho - 1/6)Q + (1 - \nu)L. \tag{LD15}$$

Use (LD14), the original geometric relation 2(C+b2)+Q>=F, and L<=s. With u=C+b2, the fixed-inducing reserve is at least

$$H + \nu u + (\rho - 1/6)Q - (1 - \nu)s.$$

It is paid by the two sufficient scalar gates

$$H + \frac{\nu F}{2} - (1 - \nu)s > 0, \qquad H + (\rho - 1/6)F - (1 - \nu)s > 0. \tag{LD16}$$

The count Q/6+L retains the actual fixed cores, their units, and every sixth-power repetition. In a fixed nonprincipal tiny raw ball Q<Qmin, use the original absolute count Q+L. Its sufficient gate is

$$H + \frac{\nu(F - Q_{\min})}{2} - (1 - \rho)Q_{\min} - (1 - \nu)s > 0. \tag{LD17}$$

Choose Qmin>0 before all live labels. On the complementary positive radius, (LD9)–(LD13) apply uniformly. If a LOW partner can have positive natural excess Delta, subtract max(Delta,0) from each special reserve; one cannot replace that deduction by bare positivity. If the admitted RAW route has already made every LOW individually nonpositive, no such positive partner remains. Ordinary HIGH partners are individually nonpositive by Section 4. Two special sectors are each paid by their own scalar gates.

Chapter 28. The global ordinary source theorem

A global affine source family and a finite acyclic construction

1. Statement of the inductive family

Retain the literal row character, all natural zero sets, fixed-inducing exclusions, ray factors, two ordinary ideal sums, and canonical marked prime polynomials (158)–(160). Canonical marked prime pools may overlap. All heights and smooth profiles have the same finite seminorm conventions of Section 1.3 and (MF3.40).

Write `G(B,Lambda,h)` for the following **ordinary** initial moment:

$$\mathcal{E} \ll U^{1+\epsilon}\mathcal{H}_J \quad \text{whenever} \quad m_1, m_2 \geq 0, \quad 0 \leq s \leq h, \quad m_1 + m_2 + \Lambda s \leq B. \tag{G1}$$

There are no individual lower or upper plain edges in this family. Empty and bounded subunit annuli are treated with the inherited clipping convention. Initial actual nonempty profiles under (G1) have polynomial norm, since the construction keeps `B<=1`.

The first input is the already proved natural theorem `G(1,K,h0)`, where `K=6(2 theta_I-1)` and any required fixed `h0` is permitted by its natural capacity inequality. Every later `G` will use one strictly earlier `G`, together with the natural theorem (MF3.40) and the inherited marked HIGH count. Its exact source certificate records the earlier parent ID, its rational affine data and its entire width interval. The construction is a finite directed acyclic graph.

2. Restricted preprocessing and global completion

First prove the current moment with $\mu^2(l_1 l_2)$ on the ordinary product and the retained squarefree projector on the complete marked product. The finite collision decomposition (10a) and slot-cardinality induction then recover the unrestricted marked source. The exact forward and inverse kernels, central masses, and extraction law are proved in (1)–(9) of the squarefree preprocessing section.

The inverse kernel expresses the desired ordinary row as a sum of restricted rows, with absolute central mass `U^epsilon`. Each of its two plain scales is only shortened. Because the new statement (G1) contains every shorter pair under the total cap, every such row is admitted. Its extracted character is a contracting scalar in the same positive row Hilbert space. This transfers the completed restricted theorem to the ordinary theorem (G1); no pointwise comparison between uncentered coefficient norms is used.

The forward restriction is kept through the first common extraction, amplification and the second row transform. It is restored only at a completed non-fixed-inducing norm to which an earlier moment applies. Fixed-inducing terminals retain the whole restricted centered coefficient. The exact extraction law preserves the restriction plus a common puncture; a non-squarefree extracted ordinary product contributes zero before positivity. Thus all genuine first common multiplicities belong to `(1,1),(1,2),(2,1),(2,2)`.

3. Initial short plains and an adjustable reference

For a source width cell with upper width `b`, choose one fixed reference

$$L(b) = \frac{49}{100} - Kb > 0. \tag{G2}$$

If either original plain is shorter than `L(b)`, restore the restricted row by its forward kernel and use the natural reflection (MF3.44) on the other ordinary factor. The first factor remains no longer than `L(b)`. After reflection, whole-factor conjugation and the inherited profile control, the other factor is no longer than `1/2+o(1)`. The natural capacity is therefore at most

$$L(b) + \frac{1}{2} + Kb + o(1) = \frac{99}{100} + o(1) < 1. \tag{G3}$$

The same reasoning applies to every shifted forward-kernel summand: the first factor is only shorter and the reflected bound for the second is still one half. All character zeros and central factors are those of natural reflection (MF3.44). One reflects only after forming an ordinary factorized product; the squarefree projector itself is not treated as a separable smooth coefficient.

It remains to consider original plains at least `L(b)`. If their actual total is `Pact`, use the centered reference rectangle

$$Y_1 = U^{L(b)}, \qquad Y_2 = U^{Pact - L(b)}. \tag{G4}$$

Here `Pact>=2L(b)`, so both reference lengths are at least `L(b)` and their product agrees exactly with that of the original rectangle. Its initial energy is bounded by the same restoration/reflection argument (G3). The first exact source identity is then applied to the entire restricted centered difference.

An earlier global theorem (G1), when called at a completed transformed RAW norm, admits either rectangle directly from its total-cap inequality.

4. Elementary centered terminal bound

Apply the equal-product ordinary centered estimate and its exact divisor transport proved in the preprocessing section, with $\alpha = 499/1000$ and mass exponent $1-\alpha = 501/1000$. Both rectangles retain the same character, puncture and Newton multiplier. The common leading term cancels before absolute values; all finite profile seminorms and independent heights are kept. This gives the relative saving $U^{-\alpha r+\epsilon}$ at every fixed-inducing terminal.

The terminal expression with common width `v` is bounded by

$$A-\frac{5}{6}-\frac{2}{3}v-\alpha(L(b)-v)_+.$$

Since `0<alpha<2/3`, its maximum over `v>=0` is at `v=0`. Including the amplifier and support costs in (131)–(141), the sufficient centered terminal gate is therefore

$$\boxed{\frac{5}{6}+\alpha L(b)-T-9a>0,\quad T=t+b.} \tag{G5}$$

The final schedule checks this width-dependent gate directly at both endpoints of every source interval. The divisor transport uses the absolutely convergent mass at exponent 501/1000. All fixed-character leading terms are retained until the centered subtraction is made.

5. Exact common geometry

Let `A,B12,C21,D` be the nonnegative log-widths of the four permitted first common types. Then

$$\begin{aligned} &c=A+B_{12}+2C_{21}+2D, && d=A+2B_{12}+C_{21}+2D, && \\ &R=B_{12}+C_{21}, && L_c=2C_{21}, && R_{0,c}=B_{12}, \\ &B_c=B_d=0, && \Sigma=D. && \end{aligned} \tag{G6}$$

Put `delta=(d-c)_+=(B12-C21)_+`. The actual reconstructed RAW norm is

$$Q=F-\delta,\qquad F=1-6a. \tag{G7}$$

The reverse orientation has the same formulas with `B12,C21` exchanged. In particular `delta<=B12`, `delta<=c`, and `2delta<=d`. The complete common ideal divides an actual original column, so `d<=T+o(1)`. Thus

$$Q\ge F-T/2-o(1). \tag{G8}$$

The exact certificate includes `F-T/2>0` with reserve. Earlier source moments are therefore called at a scale `U'=U^Q` with a uniformly positive exponent. Their own natural reflection and reference arguments are expressed in that scale. Tiny individual actual rows inside this ball do not change the ambient source scale. The separate small-core HIGH sectors are still checked.

The first centered payment. Put $A_0=\log_Z V$ for the original parent volume, $p=\log_Z N\,\mathrm{rad}(\mathfrak{C})$, and $L=49/100-Kb$. Let $e\ge 0$ be the first projector exponent. With the common widths c,d,R of (G6), define

$$\Pi=\frac{R}{3}-\frac{e}{6}-\frac{2c}{3}-\frac{4d}{3}+\frac{5}{6}(d-c)_+.$$

The four nonnegative type widths in (G6) give $\Pi+p\le -2c/3$, in either orientation. The actual first MAIN radius has logarithm

$$2A_0-1+R+e-c-d+(d-c)_++6a+\omega,$$

where ω is the retained frequency and radius reserve. After the original balanced normalization, one prime average and all explicit remaining small losses ℓ, its centered exponent is

$$A_0-\frac{5}{6}+4a+\frac{5}{6}\omega+\ell+\Pi-2\alpha(L-c)_+.$$

Here the clipped reference length is exactly $\max(1,Z^L/N\mathfrak{C})=Z^{(L-c)_+}$, including subunit ratios. For $0\le\alpha\le 2/3$ and $c\ge 0$,

$$-\frac{2c}{3}-2\alpha(L-c)_+\le-\alpha L.$$

Write h_H for the prescribed HIGH source gain. Choose the small errors so that $A_0\le T+\delta$ and $\delta+5\omega/6+\ell\le a$. The exact schedule checks the charged form of (G5),

$$T+9a+h_H<\frac{5}{6}+\alpha L.$$

Consequently the displayed exponent is at most $-h_H-4a-p$. The remaining first-key and projector sum retains its original balanced scalar and is bounded by the ideal harmonic sum against Z^{-p}, with an arbitrarily small scale loss. The four amplification widths left above absorb that loss and the original first allocation, band and fixed-cell factors.

6. Pure RAW LOW with both floors and a finite upper slot cap

Let the strictly earlier theorem be `G(B0,Lambda0,h0)`, with `B0<=1` and `Lambda0>1`. Fix a LOW threshold `q`, a positive small reserve `r`, and choose

$$a = \frac{q\{b + (t - B_0)/\Lambda_0\} + r}{1 - 6qB_0/\Lambda_0}, \qquad g = a - q\{b + (t - B_0F)/\Lambda_0\} = r. \tag{G9}$$

The denominator, amplifier range and padding reserves are explicit certificate gates. Let `f` be the first frozen marked width, so that `0<=f<=min(c,b)`. The first quotient caps are

$$P_1 = t - c + f, \quad s_1 = b - f, \quad z_0 = \frac{B_0F - t}{\Lambda_0}, \quad z = \frac{B_0Q - P_1}{\Lambda_0} = z_0 + \frac{c - B_0\delta - f}{\Lambda_0}. \tag{G10}$$

Require the global scalar guards

$$B_0F - t - b > 0, \qquad z_0 \le h_0F, \qquad qh_0 \le 1. \tag{G11}$$

The first implies `z>0` in every genuine common geometry, since `c-B0 delta>=0` and `f<=b`. Retain LOW slots only up to `min(z,h0 Q)`; subsequent common or artificial extractions only shorten the two plains and remove slots. The earlier global theorem applies to every resulting positive completed row norm. Greedy mesh overshoot is charged to the fixed small losses.

For the actual joint row (168), the first allowance (MF3.55), the second allowance (140), and the shared first payment (RC2) give the complete RAW excess before erasure:

$$-a - 2R - E - B_c + L_c + R_{0,c} + g_2 - t_2 - b_2 - \Sigma. \tag{RC3}$$

Here $E \ge 0$ is the inactive projector width; the second extraction has $b_2 - g_2 + t_2 \ge 0$, as follows from its divisor and central normalizers in (138)–(141). The source theorem is applied to the whole row (168), with its fixed ray character and puncture, after regrouping actual source labels by that row. The multiplicity bound in the literal row section pays the missing first radical. Using (G6), (RC3) is at most $-a - B_{12} - D$. The entire positive-part erasure cost is therefore bounded by

$$E_c \le -a - B_{12} - D + q\max\{0, b - f - z, b - f - h_0Q\} + o(1). \tag{G12}$$

This maximum retains both the zero-erasure floor and the prior's upper marked-width cap. Its three branches are paid separately:

$$\begin{aligned} E_{c,0} &\le -a, \\ E_{c,z} &= -g - B_{12} - D - \frac{q}{\Lambda_0}(c - B_0\delta) - q(1 - 1/\Lambda_0)f \le -g, \\ E_{c,h} &= -g + q(z_0 - h_0F) + qh_0\delta - B_{12} - D - qf \le -g. \end{aligned} \tag{G13}$$

For the last inequality use `delta<=B12` and (G11). Consequently each orientation has the individual LOW reserve `min(a,g)`, before the scheduled small losses. The third branch handles a normalized slot width above `h0` in the given common geometry.

The three branches prove every earlier-face call on its full global domain, including shortening and cap branches.

7. HIGH and the remaining gates

Use the three-branch marked HIGH gain (LD2) with

$$\rho = 1 - \frac{10\nu}{3(5 - 2\nu)}, \quad \phi = \frac{\nu(5 - 6\nu)}{5 - 2\nu},$$

$$A_\nu = \frac{14\nu(5 - 6\nu)}{115 - 54\nu}, \quad D_\nu = \frac{2\nu(5 - 6\nu)^2}{3(5 - 2\nu)(115 - 54\nu)}, \quad C_\nu = \frac{16\nu(5 - 6\nu)^2}{3(5 - 2\nu)(185 - 138\nu)}. \tag{G14}$$

The inverse admissions and full-root completion on `1/2<=nu<=3/4` are proved in (LD3)–(LD13) and the root summation following them. In this restricted first common class the balanced common expression is

$$V_c - \Sigma = -\min(B_{12}, C_{21}) - \rho(B_{12} - C_{21})_+ - D \le 0. \tag{G15}$$

It therefore pays `nu*c` individually before marked-width losses. The exact certificate checks `nu>phi+2Cnu` on the full density interval and the three regular gates, for `H=1-rho-nu*T-a(5-6rho)`,

$$H + \phi S > 0, \quad H + A_\nu S + D_\nu F > 0, \quad H + C_\nu F > 0. \tag{G16}$$

The fixed-inducing and tiny-core HIGH sectors are retained with the individual common rate `nu`:

$$\begin{aligned} H + \nu F/2 - (1-\nu)b &> 0, \\ H + (\rho - 1/6)F - (1-\nu)b &> 0, \\ H + \nu(F - r_0)/2 - (1-\rho)r_0 - (1-\nu)b &> 0, \qquad r_0 = 1/50. \end{aligned} \tag{G17}$$

Here `F,S` in a certificate include its explicit downward padding. Other true-diagonal, tiny-frequency and reference-extraction guards are `1-T-5a`, `1-r0-T-4a`, `1-r0-T-(9/2)a`, and `L(b)-7a`, each with its stated positive fee. The corrected amplifier has only MAIN, by the exact corrected-amplification identity (CA6).

All quantities are affine in the width-cell endpoint `b` when `q` and the two source faces are fixed. After multiplying (G16)–(G17) by their positive density denominators, they are polynomials of degree at most three in `nu`. Exact Bernstein positivity at both width endpoints therefore proves the full continuous rectangle of cell widths and densities. Subdivision in density, when needed, remains an exact finite certificate.

8. Acyclic inheritance and uniform completion

A new affine face may be weaker than its selected parent below their exact rational crossing. On that lower interval use the parent theorem directly; its cap must cover the entire inheritance interval. Above the crossing, certify the source gates at both width endpoints. This proves the new face on its complete interval `[0,h]`, including every smaller marked width. Each parent is an earlier node, never a proposed current or later face.

Choose all arithmetic-independent width meshes, positive amplifier lower bounds, row/radius losses, fixed-character contour orders, profile orders and height exponents over this finite graph before the moving slot labels and target height. The common mesh bounds every marked slot width below each fixed amplifier lower bound, before the finite slot type is chosen. The original Fourier-window, ray, unit, puncture, first allocation and positive row norm conventions are those of (MF3.40), (120)–(141), and (158)–(168). The all-height completion is applied to the original marked initial energy. Write N for the ambient slot cardinality, $H = 1 + |t| + \text{height} \ge 1$, and $\Delta_S = \text{diagonalControl}(\Phi)\ \text{control}_S(p)^2$. The elementary absolute bound gives

$$E \le C_c \Delta_{S_c} H^{j_c} U^{4+2N\mu},$$

where μ is the fixed marked-width mesh. The bounded-height physical estimate and reference reduction give

$$H \le U^\tau \quad \Longrightarrow \quad E \le C_s \Delta_{S_s} H^{j_s} U^{1+\varepsilon}$$

for some $\tau > 0$. Choose an integer $n \ge 0$ with $\tau n \ge 3 + 2N\mu - \varepsilon$, and put $J = \max(j_s, j_c + n)$, $S = S_s \cup S_c$. If $H > U^\tau$, then $U^{4+2N\mu} \le H^n U^{1+\varepsilon}$. Thus both cases give

$$E \le (C_s + C_c)\Delta_S H^J U^{1+\varepsilon}.$$

The source mesh and frequency allowance are fixed before the slot type. After that type and the arithmetic anchors are fixed, the bounded-height cutoff, finite controls, degrees and constants precede the eventual scale threshold and all moving profiles, characters and heights. Finite minima of the positive cutoffs handle all original slot subsets and width cells. The energy, its puncture, keep predicate, reference rectangle and radial scale remain the original ones.

The finite rational gates below and the preceding analytic arguments prove each new admitted initial moment.

The numerical certificate's printed width, radius and analytic accounting caps are each at most 10^{-12}. They are upper allowances for the actual losses chosen below. First take the minimum of the finitely many strict rational reserves, after normalizing the cleared density denominators. Then choose the actual support buffers, Fourier losses, greedy slot overshoot and asymptotic error exponents small enough that

their total contribution is below those allowances and that minimum reserve. Such a choice exists by continuity and finiteness. In a width cell, q, a and the reference length are chosen from its fixed upper endpoint, before any moving prime labels or heights.

The centering parameter used by this theorem and its checker is exactly $\alpha = 499/1000$.

Chapter 29. The complete finite rational source schedule

Exact finite source schedule

1. Final admitted source moments

The following are proved ordinary initial source moments. They retain the literal row character, natural zero sets, fixed-inducing exclusions, all-height profile conventions and marked polynomial conventions of the literal row (160)–(161). Canonical marked prime pools may overlap. The source bound is

$$\mathcal{E} \ll U^{1+\epsilon}\mathcal{H}_J \quad \text{if} \quad m_1, m_2 \geq 0, \quad 0 \leq s \leq h, \quad m_1 + m_2 + \Lambda s \leq B. \tag{S1}$$

In particular every shorter plain pair and every smaller marked width is included. No lower marked-width edge or individual plain floor is part of this source contract.

Role	Exact source ID	B	Λ	q	h
Physical numerator	d16_39000	0.96934050457	3.328741896069	0.572847762211	0.054998999999
Marked HIGH count	d16_42000	0.964090187378	3.198839752287	0.553741151397	0.054998999999

The 64 nonnatural nodes listed below and the proved natural input give these two conclusions.

2. Premise, induction rule and exact constants

The sole initial source premise is

$$\theta_I = 0.87481891, \qquad \kappa_I = 2\theta_I - 1 = 0.74963782, \qquad K = 6\kappa_I = 4.49782692,$$

and the inherited natural theorem $G(1, K, 0.055)$ in the global convention (S1). The analytic induction rule is the global theorem (G1)–(G17), using divisor transfer (1)–(9), corrected amplification (CA1)–(CA14), and HIGH counting (LD1)–(LD17). The finite schedule uses elementary centered saving $\alpha = 499/1000$, from the contour $501/1000$.

Every decimal in this appendix is an exact rational number. The fixed accounting parameters are

$$r = 10^{-8}, \quad e_H = 10^{-7}, \quad \mu = \tau = p = \eta = 10^{-12}, \quad r_0 = 1/50, \quad \alpha = 499/1000. \tag{S2}$$

Here μ is the width mesh allowance, τ the tiny-slot allowance, p the radius allowance, and η the analytic allowance. The actual analytic choices are made sufficiently small to stay below these allowances, after the finite positive reserves have been measured.

For a row of the table with data B, Λ, q, h and parent B_0, Λ_0, h_0, define the exact crossing

$$b_\times = \frac{B_0 - B}{\Lambda_0 - \Lambda}. \tag{S3}$$

The verifier checks $0 \le b_\times \le \min(h, h_0)$ and $\Lambda < \Lambda_0$. For $0 \le s \le b_\times$, the new face is no larger than the parent face, so the parent theorem proves it. On the remaining interval $[b_\times, h]$, fix q as in the table, and at each width-cell upper endpoint b use

$$\begin{aligned} t &= B - \Lambda(b - \mu), & T &= t + b, & S &= b - \mu - \tau, \\ a &= \frac{q\{b + (t - B_0)/\Lambda_0\} + r}{1 - 6qB_0/\Lambda_0}, & F &= 1 - 6a, & F_p &= F - p, \\ L_{\mathrm{ref}} &= \frac{49}{100} - Kb, & z_0 &= \frac{B_0 F - t}{\Lambda_0}. \end{aligned} \tag{S4}$$

Thus $a - q\{b + (t - B_0F)/\Lambda_0\} = r$ exactly. All scalar quantities are affine in b.

The scalar certificate checks the positivity of

$$\begin{gathered} 1 - 6qB_0/\Lambda_0,\ S,\ a,\ 1/75 - a,\ F - T/2 - 4p, \\ B_0F - t - b - 4p, \quad 1 - qh_0, \\ r - (qB_0/\Lambda_0 + qh_0 + 4)p - 10\eta, \quad a - 10p - 10\eta, \quad L_{\mathrm{ref}}, \\ 1 - L_{\mathrm{ref}} - 1/2 - Kb - 10p, \\ 5/6 + \alpha L_{\mathrm{ref}} - T - 9a - e_H, \\ 1 - T - 5a - e_H, \quad 1 - r_0 - T - 4a - e_H, \\ 1 - r_0 - T - (9/2)a - e_H, \quad L_{\mathrm{ref}} - 7a - e_H, \end{gathered} \tag{S5}$$

and the weak guard $h_0F - z_0 \ge 0$. Equality in the latter is permitted because the capped LOW branch retains the positive reserve in (S5). Both width endpoints are checked.

For $q \le \nu \le 3/4$, put

$$\begin{aligned} \rho &= 1 - \frac{10\nu}{3(5 - 2\nu)}, & \phi &= \frac{\nu(5 - 6\nu)}{5 - 2\nu}, \\ A_\nu &= \frac{14\nu(5 - 6\nu)}{115 - 54\nu}, & D_\nu &= \frac{2\nu(5 - 6\nu)^2}{3(5 - 2\nu)(115 - 54\nu)}, \\ C_\nu &= \frac{16\nu(5 - 6\nu)^2}{3(5 - 2\nu)(185 - 138\nu)}, & H &= 1 - \rho - \nu T - a(5 - 6\rho). \end{aligned} \tag{S6}$$

The six density gates are

$$
\begin{gathered}
H+\phi S>e_H, \qquad H+A_\nu S+D_\nu F_p>e_H, \qquad H+C_\nu F_p>e_H,\\
H+\nu F_p/2-(1-\nu)b>e_H,\\
H+(\rho-1/6)F_p-(1-\nu)b>e_H,\\
H+\nu(F_p-r_0)/2-(1-\rho)r_0-(1-\nu)b>e_H.
\end{gathered}
\tag{S7}
$$

The verifier also proves the common-width rate $\nu-\phi-2C_\nu>0$ on the entire interval $[1/2,3/4]$. All scheduled q's lie in that interval. Multiplication by the positive denominators gives polynomials of degree at most three in ν. Exact Bernstein positivity at both b endpoints proves the full continuous width-density region. This is a continuum certificate; no density mesh is used in the proof.

3. Complete acyclic parameter table

The table is in dependency order. Each nonnatural parent appears strictly earlier. The natural row is the only premise. The six columns below are the complete numerical inputs needed to reproduce the source certificate.

Source ID	Parent ID	B	Λ	q	h
natural	Natural premise	1	4.49782692	—	0.055
g1_14000	natural	0.999728337095	4.477808881865	0.746594775892	0.029031830945
g2_14000	g1_14000	0.999594902255	4.468137794316	0.745036095459	0.02928436799
g3_16000	g2_14000	0.998823044164	4.41636472517	0.737187559984	0.030553384982
g4_16000	g3_16000	0.998440012556	4.391616116947	0.733210163288	0.031189418935
g5_18000	g4_16000	0.997436711696	4.331747952564	0.724157400173	0.032632588034
g6_18000	g5_18000	0.996942367778	4.303350744585	0.719597099949	0.033351088663
g7_20000	g6_18000	0.995776809752	4.241113549589	0.710258669042	0.034819680228
g8_20000	g7_20000	0.995208400142	4.211810110132	0.70555065255	0.035550781757
g9_20000	g8_20000	0.994935006182	4.197944457713	0.703317368738	0.035896662305
g9_22000	g8_20000	0.993898618067	4.14861000466	0.696148985463	0.037010817529
g10_20000	g9_20000	0.994804310728	4.191367962573	0.702256856613	0.036060700065
g10_22000	g9_22000	0.993266628247	4.119077756608	0.691397158393	0.037738686793
g11_20000	g10_20000	0.994741992764	4.188245023928	0.701752989642	0.036138585534
g11_24000	g10_22000	0.991817177568	4.055302505813	0.681991012399	0.039182743845
g12_22000	g11_20000	0.993683145537	4.137770984837	0.694394223566	0.037282481135
g12_24000	g11_24000	0.99112521825	4.025739050372	0.677222297508	0.039903896403
g13_22000	g12_22000	0.993174584155	4.114139074471	0.69058794895	0.037864981859
g13_24000	g12_24000	0.990798325415	4.011960981086	0.674992876714	0.040240282621
g14_22000	g13_22000	0.992932619959	4.103028665954	0.688794396301	0.038138921239
g14_26000	g13_24000	0.989389157004	3.955320276699	0.666819652435	0.041483679205
g15_24000	g14_22000	0.991675436417	4.048135087056	0.680815207376	0.039365150407
g15_26000	g14_26000	0.988724970081	3.929254639921	0.662597284058	0.042114349434
g16_24000	g15_24000	0.991076914713	4.022653080472	0.676701859858	0.039986846052
g16_28000	g15_26000	0.987041667591	3.866371669216	0.653500312943	0.043484255538
g17_24000	g16_24000	0.990794489129	4.010768775632	0.674778285668	0.040277012093
g17_28000	g16_28000	0.9862556267	3.837720344138	0.6488403297	0.044172852358
g18_26000	g17_24000	0.989406842387	3.955032836539	0.666751225982	0.041498424982
g18_30000	g17_28000	0.984404340698	3.773326953405	0.639576460655	0.045556254073
g19_26000	g18_26000	0.9887529443	3.929379290092	0.662595564572	0.042119111023
g19_30000	g18_30000	0.983549291104	3.744274316547	0.634824648122	0.046251281505
g20_26000	g19_26000	0.988447215192	3.917514278132	0.660667730523	0.042406535592
g20_30000	g19_30000	0.983157119711	3.731089626457	0.632659212628	0.046567475664
d1_27000	g20_26000	0.987640126735	3.886935287304	0.656240492195	0.043073258508
d1_31000	g20_30000	0.982201211907	3.699630048938	0.628164141304	0.047234056529
d2_28000	d1_27000	0.986568011422	3.847553285526	0.650405110191	0.043947667353
d2_32000	d1_31000	0.980956443971	3.659765312852	0.622307764492	0.048097948757
d3_29000	d2_28000	0.985344081992	3.8040695482	0.643920667613	0.044914128421
d3_32500	d2_32000	0.979976585481	3.629051065709	0.617602569731	0.04878594477

Source ID	Parent ID	B	Λ	q	h
d4_30000	d3_29000	0.984023104139	3.758696083224	0.637142847271	0.045919661693
d4_33000	d3_32500	0.979113180459	3.602471114575	0.613577421569	0.049374470046
d5_31000	d4_30000	0.982631740443	3.712471468246	0.630238005469	0.046939764667
d5_34000	d4_33000	0.977860651719	3.564825264904	0.608110688495	0.050176812956
d6_32000	d5_31000	0.981183111871	3.665885657775	0.623284086622	0.047963168636
d6_34500	d5_34000	0.976856579435	3.535230090578	0.603586734106	0.050834011243
d7_32500	d6_32000	0.980108998561	3.63215533704	0.618084793667	0.048722743123
d7_35500	d6_34500	0.975502114374	3.496250704498	0.597915939364	0.051662290537
d8_33000	d7_32500	0.979202261989	3.604218839864	0.613836558219	0.049343482376
d8_36000	d7_35500	0.974436908062	3.46617658788	0.593307531891	0.052328493983
d9_34000	d8_33000	0.977929649618	3.56595586436	0.608268400998	0.050160409949
d9_36500	d8_36000	0.973496789612	3.440059281266	0.589358458856	0.052899909767
d10_34500	d9_34000	0.976916338897	3.536083748735	0.60369906868	0.050824121205
d10_37500	d9_36500	0.972122320321	3.402690702652	0.583975777317	0.053683456432
d11_35500	d10_34500	0.975557513138	3.49697622457	0.598007471598	0.051655367964
d11_38000	d10_37500	0.971029173485	3.373491761056	0.579503754483	0.054326688134
d12_36000	d11_35500	0.974490243582	3.466842950752	0.593389683168	0.052322908519
d12_38500	d11_38000	0.970056014935	3.347887196314	0.575636335117	0.054883739096
d13_37000	d12_36000	0.973070357376	3.427670025232	0.587700845567	0.053150754524
d13_39500	d12_38500	0.968618636223	3.310830868376	0.570323433367	0.054998999999
d14_37500	d13_37000	0.971961487485	3.397636928241	0.583092034117	0.053814209892
d14_40500	d13_39500	0.966952592645	3.26886140552	0.564180496021	0.054998999999
d15_38500	d14_37500	0.970487094978	3.358580552439	0.577434117401	0.054635093269
d15_41000	d14_40500	0.965705536457	3.237999482351	0.559400373933	0.054998999999
d16_39000	d15_38500	0.96934050457	3.328741896069	0.572847762211	0.054998999999
d16_42000	d15_41000	0.964090187378	3.198839752287	0.553741151397	0.054998999999

The first complete return and the new natural cost

Final return from the global affine source family

1. Endpoint, fixed input, and selected source moments The global source theorem proved above gives the following improved ordinary-character conclusion:

$$\boxed{c = \frac{1252241}{10000000} = 0.1252241, \qquad \theta = 1 - c = \frac{8747759}{10000000} = 0.8747759.} \tag{FR1}$$

For every presented Dirichlet character χ, $L(s,\chi) \neq 0$ on $\Re s > \theta$, with the principal pole at $s = 1$ excluded. The proof retains the ordinary character family, all natural zeros, the original same-zero detector, the compensated physical probe, and the full reciprocal Mellin return. The arithmetic restriction used to prove the new source moments has already been removed by the completed source theorem before these moments are used here.

The fixed earlier input is the established endpoint $\theta_I = 0.87481891$, hence

$$\kappa_I = 2\theta_I - 1 = \frac{37481891}{50000000}, \qquad K_I = 6\kappa_I = \frac{112445673}{25000000}. \tag{FR2}$$

The minimal source construction contains 64 derived faces and the natural premise $G(1, K_I, 11/200)$. Every derived face uses a strictly earlier face. The final return uses the following two members of that finite construction; their displayed decimals are exact.

Label	Exact source ID	B	Λ	Proved upper slot width
C	`d16_42000`	0.964090187378	3.198839752287	0.054998999999
P	`d16_39000`	0.969340504570	3.328741896069	0.054998999999

Each is the ordinary moment $G(B,\Lambda,s_{\max})$: it admits all $m_1,m_2\ge 0$, all $0\le s\le s_{\max}$, and $m_1+m_2+\Lambda s\le B$, with the inherited coefficient, common-profile, puncture, keep, marked-pool, and independent-height conventions. All source constants, width cells and admission domains are fixed using (FR2), independently of the contrary zero supremum introduced at the end of the proof. In particular, no source moment uses (FR1) as an input.

For a conservative exact return certificate, use only the subdomains

$$\frac{2}{5}\le m_i\le\frac{11}{25},\qquad \frac{17}{500}\le s\le s_+,\quad s_+=\begin{cases}11/250 & \text{for } C,\\ 43/1000 & \text{for } P.\end{cases} \tag{FR3}$$

They lie strictly inside the global width caps. We may also replace either intercept by $B-j/4000$, $0\le j\le 8$, retaining the same slope and domain. These weaker faces follow by inclusion. The physical numerator uses P itself; the same-zero count is the admitted finite minimum of the old natural count and the faces just listed, including their constant-width branches.

2. One complete same-zero count on every admitted window Write $0\le\delta\le 3/4$, $0\le p\le 1/2$, and $V=5/6-\delta>0$. For an affine face (B,Λ), put

$$\begin{aligned}
&D=3-p-4p/\Lambda, && A=2-4p/\Lambda, && C_0=1-2pB/\Lambda,\\
&U=(1-p)C_0, && J=VD+\delta(1-p)A, && N=(1-\delta)J+V\delta U,\\
&T=J+\delta U, && R_n=AT+(2pB/\Lambda-p)J, && M_n=DT-R_n.
\end{aligned} \tag{FR4}$$

The proposed count is $R=N/J$, with common detector and crossing lengths

$$t_*=T/J,\qquad r_*=R_n/(DJ),\qquad m_*=M_n/(DJ),\qquad t_*=r_*+m_*. \tag{FR5}$$

For a constant-width branch, let $\eta=10^{-7}$ and $z_*=s_+-\eta$, replace (D,A,C_0) in (FR4) by $(3-p,2,1-2pz_*)$, and replace $2pB/\Lambda$ in R_n by $2pz_*$. This branch is continued by $z(m)=\min(z_*,(B-2m)/\Lambda)$, so that its plain bound is continuous at the cap.

The always available old count has

$$\begin{aligned}
&D_O=3-p-\frac{2p}{3\kappa}, && C_O=2(1-p)\left(1-\frac{p}{3\kappa}\right),\\
&J_O=VD_O+\delta C_O, && N_O=(1-\delta)J_O+\frac{\delta C_O V}{2}, \qquad R_O=N_O/J_O.
\end{aligned} \tag{FR6}$$

Here only κ in (FR6) is allowed to vary. The tail point for every improved count is fixed at

$$m_H=\frac{1-K_I(17/500)}{2}. \tag{FR7}$$

For clarity, the complete buffered admission tests used by the exact checker are specified here. Set $m_-=2/5$, $m_+=11/25$, $s_-=17/500$. On an affine branch define

$$\begin{aligned}
m_L&=\max\left(m_-+\eta,\frac{B-\Lambda(s_+-\eta)}{2}+\eta\right),\\
m_U&=\min\left(m_+-\eta,m_H-\eta,\frac{B-\Lambda(s_-+\eta)}{2}-\eta\right).
\end{aligned}$$

On a capped branch use $m_L=m_-+\eta$ and $m_U=\min(m_+-\eta,m_H-\eta,(B-\Lambda z_*)/2-\eta)$. All the following polynomials must be strictly positive on the whole closed admission window:

$$\begin{gathered}
D,\ A,\ C_0,\ J,\quad T-J,\quad (12499/10000)J-T,\\
R_n-(23/37)DJ,\quad DJ-R_n,\quad M_n-m_LDJ,\quad m_UDJ-M_n,\\
N-\{1-2\delta m_H-2\delta ps_-\}J,\quad UJ_p-U_pJ,\\
m_+-m_H-\eta,\qquad B-2m_H-\Lambda(s_-+\eta)-2\eta.
\end{gathered} \tag{FR8}$$

The capped branch also requires $(z_*-s_--\eta)DJ>0$. In particular, (FR8) verifies the whole source continuation through the source-to-natural tail join; it checks more than the isolated crossing point.

Here is the actual witness argument. Fix one buffered nonfloor fixed-amplitude row class and choose $t=t_*$ before selecting its witnesses. Lemma 14.2 gives one common-zero inverse/plain pair at one height

and presentation, with $r \le t_* + o(1)$, $r + m \ge t_* - o(1)$, and $0 \le m \le 1/2 + o(1)$. The three relevant count lines are

$$I(r) = 1 - \delta[p + (1-p)r], \qquad P(m) = 1 - 2\delta m - 2\delta p z(m), \qquad L(r) = \frac{1}{6} + (5/6 - \delta)r. \tag{FR9}$$

Direct substitution of (FR4) gives $I(r_*) = P(m_*) = L(t_*) = N/J$. If $r_* \le r \le 1$, the admitted short inverse line decreases with r. If $r < r_*$, then $m \ge m_* - o(1)$; the source plain line decreases, with slopes -2δ on the capped part and $-2\delta(1 - 2p/\Lambda) < 0$ on the affine part. The source bounds remain admitted to m_H by (FR8). For $m \ge m_H$, the natural width $(1 - 2m)/K_I$ gives a decreasing plain count up to $m = 1/2$; the explicit tail polynomial in (FR8) pays its value at m_H. For larger m, the unmarked plain bound and the inherited endpoint loss apply. Finally, for $r > 1$, $L(r)$ increases and $r \le t_* + o(1)$, so the long inverse line also gives $N/J + o(1)$. A plain witness below m_- forces $r > r_*$, and is already covered by the inverse case.

This proves the same-zero class bound $U^{N/J+\epsilon}$, with the original finite label, height and profile costs. All terms use that same zero and its witness heights. All actual whole-slot selections and support erosions are paid from the positive buffers in (FR8) and the final loss budget.

Every improved admission window is extended from its lower p edge through $p = 1/2$. The derivative test in (FR8) implies $\partial_p(N/J) \le 0$. New candidates can therefore appear as p increases but cannot disappear. Their admitted finite minimum with R_O, denoted R_*, remains antitone, including at window boundaries.

3. Exact physical geometry and a fixed conductor join The geometry is fixed as follows.

Quantity	Exact rational value
theta	8747759/10000000
h	799737/1000000
ell	1256723/7500000
x	5517391/15000000
y	6969163/15000000
κstar	3747759/5000000
d_0	1/200
d_+	7997370001/10000000000
d_s	174229075/221229831
d_k	69691630000000/87285687845457

These values satisfy

$$\ell = 4c - \frac{1}{3} = \frac{11}{3} - 4\theta, \quad x = 1 + \ell - h, \quad y = h - 2\ell, \quad x + y = 1 - \ell, \quad \kappa_* = 2\theta - 1. \tag{FR10}$$

Set $s_P = 1/25$, $d_0 = 1/200$, $d_+ = h + 10^{-10}$. Define

$$C_O(\kappa, d) = \frac{\min(d, 2y - d)}{6\kappa}, \qquad C_N(d) = \min\left(s_P d, \frac{2y - (2 - B_P)d}{\Lambda_P}\right), \tag{FR11}$$

and choose the conductor join and new-capacity kink once and for all by

$$d_s = \frac{2y}{1 + 6\kappa_* s_P}, \qquad d_k = \frac{2y}{2 - B_P + \Lambda_P s_P}. \tag{FR12}$$

The exact reserves below prove $d_0 < y < d_s < d_k < d_+ < 2y$, with $C_O(\kappa_*, d_s) = s_P d_s = C_N(d_s)$. Use the natural physical estimate on $[d_0, d_s]$ and the new estimate on $[d_s, d_+]$. The join d_s is fixed at κ_*; it is not recomputed when the actual contrary parameter $\kappa = 2\beta - 1$ is substituted later.

The complete normalized return exponent at a capacity C is

$$E_\kappa(d, C; \delta, p) = \frac{1}{2} + \frac{13h}{75} - \frac{y}{2} - \frac{\ell}{2} - \frac{9}{100}d + \frac{\delta}{2} + \delta p\left(\ell - \frac{C}{2}\right) + \frac{3}{4}dR_*(\kappa, \delta, p). \tag{FR13}$$

At fixed (δ, p), this expression is affine in d on each of the four intervals cut by y, d_s, d_k. The exact return thus needs only the six labeled conductor/capacity edges $O_0, O_y, O_s, N_s, N_+, N_k$, with the join retained on both source branches. These are the six edges recorded in the certificate. Although different valid counts may certify different endpoints, their common minimum R_* is independent of d. Endpoint interpolation therefore uses one valid count envelope on every complete conductor interval.

4. Exact continuous return certificate Put $r_0 = 10^{-9}$. For each endpoint and admitted branch $R_b = N_b/J_b$, form

$$F_b(\delta, p) = J_b\{\theta - r_0 - E_{\mathrm{base}}(d, C; \delta, p)\} - \frac{3}{4} d N_b, \tag{FR14}$$

where E_{base} is (FR13) with the final count term removed and every varying old parameter is set to κ_*. The polynomials have degree at most $(2, 3)$. On a closed box $[a, b] \times [c, d]$, substitute $\delta = a + (b-a)u$, $p = c + (d-c)v$. If the resulting power coefficients are A_{kl}, the tensor Bernstein coefficients are exactly

$$\beta_{ij} = \sum_{k \le i,\ l \le j} A_{kl} \frac{\binom{i}{k}}{\binom{2}{k}} \frac{\binom{j}{l}}{\binom{3}{l}}, \qquad 0 \le i \le 2, \quad 0 \le j \le 3. \tag{FR15}$$

The nonnegative tensor Bernstein basis sums to one on the full closed square. Strict positivity of all twelve coefficients proves (FR14) continuously, including every edge and corner. Admission guards are checked in the same way on the larger box whose upper p edge is $1/2$. The old count requires only its positive denominator.

The complete rational receipt contains 212 closed return rectangles and 2,544 return coefficients. The following table summarizes their strict lower bounds; the full rectangle table appears at the end of this section. Every displayed integer lower bound is in units of 10^{-18}.

Conductor edge	Number of rectangles	Lower bound for every return coefficient
old_small	35	86018277460362112
old_peak	35	58288142713806320
old_switch	35	4245652310513802
new_switch	35	4245652310513802
new_top	37	745062308695
new_kink	35	523643385986201

The least exact coefficient is

$$\frac{623916364894193443745561517570741564 7}{837401593950926431248171899289600000000000 0} > 0. \tag{FR16}$$

Every conductor edge has exact total rectangle area $3/8$, no positive-area overlap, and a complete closed cover of $[0,3/4]\times[0,1/2]$. The construction starts from the retained exact closed cover and only bisects its rectangles, preserving containment; the checker verifies exact area and pairwise overlap. Thus there is no inference from a sampled numerical grid. Dividing (FR14) by $J_b > 0$, using $R_* \le R_b$, and interpolating along the conductor intervals gives

$$\begin{aligned} E_{\kappa_*}(d, C_O(\kappa_*, d); \delta, p) &\le \theta - r_0 \quad (d_0 \le d \le d_s), \\ E_{\kappa_*}(d, C_N(d); \delta, p) &\le \theta - r_0 \quad (d_s \le d \le d_+). \end{aligned} \tag{FR17}$$

5. Every reflected shift and shortening is admitted For $d_s \le d \le d_+$, let

$$n_0(d) = 1 - y/d, \quad b(d) = 2y/d - 1, \quad z_d(t) = \min\left(s_P, \frac{B_P - 2n_0(d) - 2t}{\Lambda_P}\right). \tag{FR18}$$

Use the free-shift cutoff $t_f = 1/400$. The exact domain guards prove, for $0 \le t \le t_f$, that $2/5 < n_0(d) + t < 11/25$ and $17/500 < z_d(t) < 43/1000$, with buffer η. The selected equal plains satisfy the affine source inequality; all shorter plains and all smaller marked subsets are in the global source theorem. For the narrower subdomain used by the checker, the following conservative natural margins also provide a direct shortening completion:

$$1 - 2n_{\max} - \frac{9}{2}\frac{17}{500} > \eta, \qquad 1 - \frac{2}{5} - \frac{B_P}{2} - \left(\frac{9}{2} - \frac{\Lambda_P}{2}\right) s_P > \eta, \tag{FR19}$$

where $n_{\max} = 1 - y/d_+ + t_f$. The first covers a marked width falling below $17/500$. The second covers one plain falling below $2/5$, including simultaneous shortening and zero slots. These follow from the old natural moment with the conservative coefficient $9/2$.

For small shifts, $z_d(0) - z_d(t) \le 2t/\Lambda_P$. If $q = \delta p \le \delta/2$, the coefficient damping pays the fourth-root width loss, with a uniform proportional reserve:

$$\frac{\delta t}{2} - \frac{q}{2}\{z_d(0) - z_d(t)\} \ge \delta t\left(\frac{1}{2} - \frac{1}{2\Lambda_P}\right) \ge \frac{\delta t}{500}. \tag{FR20}$$

The final comparison is strict whenever $\delta t > 0$; the zero endpoints retain the fixed return reserve from (FR17).

For $t \ge t_f$, use the clipped old width $z_O(t) = \max(0, (b - 2t)/(9/2))$. On its positive branch, with $z_0 = \min(s_P, (b - (1 - B_P))/\Lambda_P)$,

$$\frac{t}{2} - \frac{1}{4}(z_0 - z_O(t)) - \frac{t}{500} = \left(\frac{7}{18} - \frac{1}{500}\right) t + \frac{b}{18} - \frac{z_0}{4}.$$

Because $3 < \Lambda_P < 9/2$, the last two terms decrease on the affine part and increase on the capped part. Their minimum is at $b = 1 - B_P + \Lambda_P s_P$. The sufficient uniform reserve is therefore

$$\left(\frac{7}{18} - \frac{1}{500}\right) t_f + \frac{1 - B_P}{18} + \left(\frac{\Lambda_P}{18} - \frac{1}{4}\right) s_P > \eta. \tag{FR21}$$

On the clipped branch $b \le 2t$, the corresponding bound is

$$\frac{t}{2} - \frac{z_0}{4} - \frac{t}{500} \ge \left(\frac{1}{2} - \frac{1}{500} - \frac{1}{2\Lambda_P}\right) t + \frac{1 - B_P}{4\Lambda_P} > 0. \tag{FR22}$$

Thus every large shift, including its dyadic tails and Fourier seminorm losses, is paid. This minimization holds for all relevant $b \ge 0$; it imposes no artificial upper bound $b < 1/6$. Lower conductor rows retain the original natural shift argument.

6. Complementary rows and actual-parameter feedback The following table gives every complement reserve used by the exact checker. All entries are exact positive fractions. The domain tests subtract $\eta = 10^{-7}$; the floor test retains a saving $1/200$. The floor saving is a chosen positive loss allowance, so changing an earlier convenient value $1/100$ requires no change to the underlying floor estimate.

Reserve	Exact value
Gram_x_above_ell	200263/1000000
Gram_y_above_x	120981/1250000
Gram_ell_below_one_fifth	243277/7500000
Gram_2x_above_y_plus_ell	517391/5000000
conductor_small_before_peak	6894163/15000000
conductor_peak_before_switch	357216457432849/1106149155000000
conductor_switch_before_top	26966063765699831/2212298310000000000
new_capacity_at_switch	402081518181/25000000000000
capacity_positive_at_top	314895601000696421371/3000000000000000000000
physical_plain_lower	2513421/250000000
physical_plain_upper	4427617241014997/239921100030000000
physical_width_lower	313627054406993802026834179 3/7986354174208224127820700000000
physical_width_upper	29999/10000000
small_width_shortening	937862173149997/239921100030000000
short_plain_shortening	95224281819/50000000000000
large_shift	30434281819/450000000000000
large_shift_clipped	578856732121181/1664370948034500
small_shift_cost	2328741896069/1000000000000
prime_supply	1142349997/119960550015
capacity_domain	3884541997/30000000000
incumbent	4301/50000000
feedback_kappa	247759/5000000
feedback_d	2002629999/10000000000
feedback_y	530837/15000000
floor	13165423799/3000000000000
small_row	5607653/150000000
principal_w	63542419/3000000000
principal_z	16579/200000000
holomorphic_box	247759/10000000
source_lambda	328741896069/1000000000000
source_intercept	3065949543/100000000000

Here the Gram tests are $x > \ell$, $y > x$, $\ell < 1/5$, and $2x > y + \ell$. Prime supply is $\ell/d_+ > 1/5$. The floor is

$$F_{\mathfrak{fl}}(d) = \frac{4}{25}x - \frac{3}{20} - \frac{51}{100}y + \frac{67}{100}d - \frac{3}{20}\ell,$$

and the floor reserve is $\theta - (4 + y - x)/6 - 1/200 - F_{\mathfrak{fl}}(d_+)$. Since the floor increases with d, this covers all floor rows. The small-row reserve is $-1/20 - [13h/75 - y/2 + (63/50)d_0]$, including the inherited nonprincipal unit sector. The principal contour reserves are $y/20 - (1 + h)/1000 - 1/4000$ and $h/600 - 1/1000 - 1/4000$. The holomorphic box is preserved by $\theta > 17/20$. These are the original complementary estimates at the new fixed geometry.

All source faces, count windows and the conductor join d_s are now held fixed. For $\kappa_* \le a \le \kappa \le \kappa_I$, the old count satisfies $R_O(\kappa) - R_O(a) \le (5/216)(\kappa - a)$. For completeness, put

$$D_\kappa = 3\kappa(3 - p) - 2p, \quad C_\kappa = \frac{2(1 - p)(3\kappa - p)}{D_\kappa}, \quad S_\kappa = V + \delta C_\kappa.$$

Then $D_\kappa, D_a \ge 17/4$, and direct subtraction gives

$$C_\kappa - C_a = \frac{6p(1 - p)^2(\kappa - a)}{D_\kappa D_a} \le \frac{\kappa - a}{18}, \qquad R_O(\kappa) - R_O(a) = \frac{\delta V^2}{2S_\kappa S_a}(C_\kappa - C_a) \le \frac{5}{216}(\kappa - a). \quad \text{(FR23)}$$

Every improved candidate is constant in κ, and its admission is independent of κ. Selecting a minimizing candidate at a therefore proves the same upper difference bound for R_*.

The resulting count cost in (FR13) is at most $5d(\kappa-\kappa_*)/288$, which is less than $(\kappa-\kappa_*)/50$ because $d \le d_+ < 1$. New physical capacity is fixed. On old rows its additional cost is

$$\frac{\delta p \min(d, 2y-d)}{12}(\kappa_*^{-1}-\kappa^{-1}) \le \frac{25}{784}(\kappa-\kappa_*),$$

using $y < 1/2$, $\delta p \le 3/8$, and $\kappa\kappa_* \ge (7/10)^2$. Since $5/288 + 25/784 < 1/8$, setting $\mu = \beta - \theta > 0$ and $\kappa - \kappa_* = 2\mu$ in (FR17) gives

$$\begin{aligned} E_\kappa - \beta &\le -r_0 - \frac{24}{25}\mu \quad \text{on new rows,} \\ E_\kappa - \beta &\le -r_0 - \frac{3}{4}\mu \quad \text{on old rows.} \end{aligned} \tag{FR24}$$

At the fixed join, the old capacity may decrease as κ increases. That change is exactly the cost just bounded. No equality between old and new capacity at the actual κ is assumed, and the source admission on the new interval is unaffected.

7. The complete reciprocal Mellin contradiction Let β be the zero supremum of the full finite-order Eisenstein Hecke family retained in the earlier chapters. If $\beta \le \theta$, the Hecke conclusion already holds. Otherwise the previously proved endpoint (FR2) gives

$$\theta < \beta \le \theta_I, \qquad \kappa_* < \kappa = 2\beta - 1 \le \kappa_I < 3/4. \tag{FR25}$$

This is exactly the range required by the source, count and feedback arguments. Let $k = (4+y-x)/6$. The geometry identities (FR10) give

$$\theta - k = \frac{x}{2} + \frac{y-x}{12}. \tag{FR26}$$

The original signed Gram and marked-energy bounds for the common compensated probe, together with its nonzero prime normalizer, imply for a common $0 < \omega < (\beta-\theta)/4$ that

$$J_\eta(Z) \ll_\eta Z^{\theta-k+\omega}. \tag{FR27}$$

Equations (FR17)–(FR24), all shifted-product estimates, and the strict complementary reserves imply a common $\sigma_0 > 0$ such that

$$J_\eta(Z) - f_\eta(Z) \ll_\eta Z^{\beta-k-\sigma_0}. \tag{FR28}$$

To specify the quantifiers, choose the front, frequency, scalar, mass, count, source, selection, height and zero-buffer losses after the positive difference $\beta - \theta$, and before scale, with their total smaller than both the retained fixed reserves and a fixed fraction of $\beta - \theta$. All source meshes are fixed before the target and the finite slot system. Once that system is fixed, its least positive width is fixed; choose the principal normalizer loss and the principal contour buffer inside the displayed margins. The original far-right contour gives arbitrary saving for large rows, and $d_+ = h + 10^{-10}$ covers the intervening central rows. The finite minimum of all remaining positive exponent gaps supplies σ_0. Thresholds and constants may depend on the primitive target; the exponent gaps do not. Independent-height integrals, finite profile seminorms and all natural-zero factors are unchanged.

The signal remains the exact reciprocal Mellin integral

$$f_\eta(Z) = \frac{1}{2\pi i}\int_{(a)} Z^{s-k} e^{(s-5/6)^2} \frac{H_\eta(s)}{L_K^S(s,\eta)}\, ds, \qquad a > 1. \tag{FR29}$$

The retained principal Euler-product proof gives a holomorphic, nonvanishing H_η on $\Re s > \theta$; fixed deletions can ensure $|H_\eta - 1| \le 1/2$ there. The fixed-ray prime ideal theorem gives a nonzero normalizer with subpower reciprocal. The Gaussian and original Mellin weights give vertical convergence. Absolute-line shifts imply $f_\eta(Z) = O_{\eta,A}(Z^{A-k})$ as $Z \to 0$ for every fixed A.

Put $m = \min(\beta - \theta - \omega, \sigma_0) > 0$. Equations (FR27)–(FR28) imply $f_\eta(Z) \ll_\eta Z^{\beta-k-m}$ as $Z \to \infty$. Consequently

$$\mathcal{F}_\eta(s) = \int_0^\infty f_\eta(Z) Z^{k-s} \frac{dZ}{Z} \tag{FR30}$$

is locally uniformly holomorphic on $\Re s > \beta - m$. Integrability at infinity follows from the displayed exponent, and at zero from choosing A larger than the real parts of the compact set under consideration. The same bounds with fixed powers of $\log Z$ justify differentiation under the integral.

On a line right of one, absolute Fourier–Mellin inversion gives

$$\mathcal{F}_\eta(s) = e^{(s-5/6)^2} \frac{H_\eta(s)}{L_K^S(s,\eta)}. \tag{FR31}$$

This meromorphic identity continues to $\Re s > \beta - m$. Its numerator is nonzero there, since $\beta - m \geq \theta + \omega > \theta$. A denominator zero of any multiplicity would therefore be a pole, contrary to the holomorphy of (FR30). The gap m is common to all primitive targets. Lemma 9.1 supplies a primitive zero with real part greater than $\beta - m/2$, inside this forbidden half-plane. This contradiction proves $\beta \leq \theta$.

Primitive reduction and restoration of the fixed Euler factors preserve zeros on $\Re s > 0$. For every presented ordinary Dirichlet character χ, the original norm-pullback identity is

$$L_K(s, \eta_\chi) = L(s,\chi)L(s,\chi\chi_{-3}). \tag{FR32}$$

Both ordinary factors are holomorphic on $\theta < \Re s < 1$. A zero of either would contradict the Hecke nonvanishing just proved. The classical boundary nonvanishing and absolute Euler convergence handle the line $\Re s = 1$ and the half-plane to its right, with the principal pole excluded. This proves (FR1) for every ordinary Dirichlet character, and in particular for the Riemann zeta function. No assertion is made on the new boundary line itself.

The exact improvement over the immediately preceding endpoint is

$$\frac{1252241}{10000000} - \frac{12518109}{100000000} = \frac{4301}{100000000} = 0.00004301 > 0. \tag{FR33}$$

The exact closed cover for this first return is given in the certificate appendix. Its endpoint is used only after the ordinary source theorem, the complete same-zero count, the shifted physical numerator, and the Mellin contradiction just proved. It therefore supplies the later natural moment $G(1, 11243277/2500000, 11/200)$.

Chapter 31. Feeding proved ordinary moments into the complete HIGH count

Feeding proved global ordinary sources into the complete HIGH count

Scope

Use the natural inverse moment, the same-zero detector and strictly earlier global ordinary moments.

Fix a finite list of already proved moments

$$G(B_i, \Lambda_i, h_i): \qquad m_1 + m_2 + \Lambda_i s \leq B_i Q, \quad m_1, m_2 \geq 0, \quad 0 \leq s \leq h_i Q. \tag{HF1}$$

Use only faces satisfying $22/23 \leq B_i \leq 1,\ \Lambda_i > 1,\ 0 \leq h_i < 1$. The natural source may be included. The actual final coefficient and row restrictions, canonical fine marked pools, common profiles, heights, holes and fixed-inducing exclusions are those of the global source theorem (G1). Every face in (HF1) is fixed before the moving rows or witness lengths.

Let $Q > 0$ be the ambient row exponent, $S \geq 0$ the available marked width and $1/2 \leq \nu \leq 3/4$ the HIGH density. Put

$$w = 5 - 6\nu, \quad d = 5 - 2\nu, \quad t_\nu = 3d = 10 + w, \qquad \xi_0 = \frac{10}{t_\nu}, \quad \rho = 1 - \nu\xi_0, \quad \phi = \frac{\nu w}{d}. \tag{HF2}$$

Here ξ below is a LENGTH EXPONENT, not a normalized density: the target unweighted counting exponent at amplitude $p = 1/2$ is $Q - \nu\xi$.

1. The actual witness thresholds

The natural inverse moment permits selected width $z_I = \min(S, (Q-r)/2)$. Therefore a target $r + z_I \geq \xi$ is guaranteed once

$$r \geq r_*(\xi) := \max(\xi - S,\, 2\xi - Q). \tag{HF3}$$

For the plain witness put $y = 2m$. The earlier source (HF1) permits

$$z_P = \min\left(S, hQ, \frac{BQ - y}{\Lambda}\right).$$

The target $y + z_P \geq \xi$ is guaranteed once

$$y \geq y_*(\xi) := \max\left(\xi - S,\ \xi - hQ,\ \frac{\Lambda\xi - BQ}{\Lambda - 1}\right). \tag{HF4}$$

This formula retains both the total marked-width cap and the affine source cap. There is no lower plain or marked edge to restore. When a literal witness has $y > BQ$, the inherited unmarked natural plain estimate supplies the required count whenever $\xi \leq BQ$. Witnesses above the natural upper range retain their inherited long unmarked treatment; they are not inserted into (HF1).

The same-zero detector has the exact cutoff equation

$$6\nu\xi + w\{r_*(\xi) + y_*(\xi)/2\} = 5Q. \tag{HF5}$$

Its left side is continuous and strictly increasing in ξ. Let $\xi_{B,\Lambda,h}(S, Q)$ be its unique solution.

The admissions are uniform. At $\xi = \xi_0 Q$, each threshold in (HF3)–(HF4) is at most $\xi_0 Q$, because $B \geq 22/23 > 20/21 \geq \xi_0$. Thus $\xi \geq \xi_0 Q$. At $\xi = BQ$, the inverse-cap and plain-affine branches give

$$6\nu B + w(5B/2 - 1) \geq 5,$$

because

$$\frac{5 + w}{6\nu + 5w/2} \leq \frac{22}{23} \leq B.$$

The first inequality has the exact cleared remainder

$$\frac{22}{23}(6\nu + 5w/2) - (5 + w) = \frac{60}{23}(3/4 - \nu) \geq 0.$$

Hence $\xi \leq BQ$, which justifies the unmarked fallback above. Also $\xi < Q$: the inverse-cap/plain-cap branch at $\xi = Q$ has left side greater than $5Q$ when $h < 1$. The same-zero cutoff $r_* + y_*/2 = (5Q - 6\nu\xi)/w$ therefore lies strictly above Q and at most $5Q/d \leq 10Q/7 < 3Q/2$. The plain threshold is positive at this root. Indeed, if $y_* \leq 0$, then $r_* \leq 2\xi - Q$ at a possible solution with $\xi \geq \xi_0 Q$, and the left side of (HF5) is at most $(10 - 6\nu)\xi - wQ < 5Q$, a contradiction.

Moreover $r_* \geq 2\xi_0 Q - Q \geq 2Q/3$. On every invoked inverse branch, $2r + 8z_I \leq 4Q - 2r \leq 8Q/3$, leaving the same positive second inverse-gate reserve $Q/3$. The original small width and witness enclosure erosions fit inside these uniform reserves.

2. Six explicit rational facets

Define the improvement over the unmarked density exponent by

$$\mathcal{G}_{B,\Lambda,h}(S, Q) = \nu\{\xi_{B,\Lambda,h}(S, Q) - \xi_0 Q\}. \tag{HF6}$$

The two maxima in (HF3)–(HF4) give exactly six branches. Put

$$D_{13} = \Lambda t_\nu - 10, \qquad D_{23} = \Lambda(10 + 3w) - (10 + 2w).$$

Both denominators are positive for $\Lambda > 1$. Then $\mathcal{G}_{B,\Lambda,h}$ is the minimum of the following six quantities:

Inverse branch	Plain branch	Exact gain
Available slots	Available slots	ϕS
Available slots	Parent width cap	$(2\phi/3)S + (\phi h/3)Q$
Available slots	Parent affine cap	$\dfrac{2\nu w(\Lambda-1)}{D_{13}}S + \dfrac{\nu w(Bt_\nu-10)}{t_\nu D_{13}}Q$
Inverse cap	Available slots	$\dfrac{\nu w}{10+3w}S + \dfrac{2\nu w^2}{t_\nu(10+3w)}Q$
Inverse cap	Parent width cap	$\left\{\dfrac{2\nu w^2}{t_\nu(10+3w)} + \dfrac{\nu w h}{10+3w}\right\}Q$
Inverse cap	Parent affine cap	$\dfrac{\nu w\{2w(\Lambda-1)+Bt_\nu-10\}}{t_\nu D_{23}}Q$

These are identities obtained by solving six linear equations for ξ. No witness branch, cap or complement is omitted. For the natural face $B = 1, \Lambda = 9/2$, with an inactive width cap, the usual three NR branches are recovered after removing redundant facets. Using the actually proved natural cost $K = 4.4973108$ is permitted directly in the same six-facet formula.

For a finite proved family define

$$\mathcal{G}_*(S,Q) = \max_i \mathcal{G}_{B_i,\Lambda_i,h_i}(S,Q). \tag{HF7}$$

For each fixed face, the entire detector argument, including its cutoff, proves the full row-class count below. Taking the minimum of this finite list of complete bounds gives (HF7). Equivalently, the face can be selected using (S,Q,ν) before invoking the detector and before its inverse or ordinary witness is produced. Every selected theorem is strictly earlier, and its cutoff remains fixed throughout the detector and witness argument.

3. Count and weighted mass at the original zero

At marked amplitude $p = 1/2$, the preceding inverse and plain bounds give the count exponent

$$\#\mathrm{HIGH} \ll U^{\rho Q - \mathcal{G}_*(S,Q)+\epsilon}. \tag{HF8}$$

For $0 \le p \le 1/2$, the same deduction gives

$$\#\mathrm{HIGH} \ll U^{\rho Q + (1-2p)\nu S - \mathcal{G}_*(S,Q)+\epsilon}. \tag{HF9}$$

Indeed each retained marked width $z \le S$ satisfies $2pz + (1-2p)S \ge z$, in both the inverse and plain Markov estimates. The detector stays at the original zero and retains its original finite witness and independent common height. Multiplying by the original amplitude-bin weight gives the complete marked weighted-mass exponent

$$\boxed{\rho Q + \nu S - \mathcal{G}_*(S,Q) + \epsilon.} \tag{HF10}$$

All statements keep the same original eligible presentations and fixed-inducing exclusions as the inherited HIGH count.

4. Common-width losses and full sixth-root completion

Every facet has nonnegative coefficients in S, Q. Nonnegativity of the affine-parent row coefficients follows from $B \ge \xi_0$. The marked-width coefficients are at most ϕ. Every row coefficient is at most

$$C_{\max}(\nu) = \nu(1-\xi_0) = \nu + \rho - 1. \tag{HF11}$$

For the latter assertion, the corresponding normalized root of each linear branch at $S = 0$ is at most one. This follows directly from $B \le 1,\ h < 1,\ \Lambda > 1,\ w > 0$. More explicitly, if that root is N/D, then in either inverse branch the three values of $D - N$, in the order of the plain branches, are

$$\frac{w}{2}, \qquad \frac{w(1-h)}{2}, \qquad \frac{w(\Lambda - B)}{2(\Lambda - 1)}.$$

All are nonnegative. Thus each gain, its minimum of facets, and the finite maximum (HF7) are monotone, ϕ-Lipschitz in S, and $C_{\max}$-Lipschitz in Q. The first and either slot-independent facet also give $\mathcal{G}_* \le \min(\phi S, C_{\max} Q)$.

The existing restricted first-common calculation therefore still pays the full loss of gain:

$$\nu - \phi - 2C_{\max} = \frac{\nu(8\nu - 10/3)}{5 - 2\nu} \ge \frac{1}{12} > 0 \quad (1/2 \le \nu \le 3/4). \tag{HF12}$$

For an exact check of its uniform margin,

$$\nu - \phi - 2C_{\max} - \frac{1}{12} = \frac{(2\nu - 1)(48\nu + 5)}{12(5 - 2\nu)} \ge 0.$$

The original common relations $L + c + b_2 \ge S$ and $Q + 2c + 2b_2 \ge F$, together with the individually favorable restricted common form, suffice. Fixed-inducing and tiny-core HIGH sectors use (LD15)–(LD17), with their actual core counts; (HF10) is used only on its eligible ordinary cores.

The complete sixth-root argument also retains explicit positive reserves:

$$6(\rho - C_{\max}) = 6(1-\nu) \geq \frac{3}{2} > 1, \tag{HF13}$$

$$\frac{\rho - 1/6}{2} - C_{\max} = \frac{5}{12} - \nu + \frac{5\nu}{3(5-2\nu)} \geq \frac{1}{42} > 0. \tag{HF14}$$

The last margin has the exact factorization

$$\frac{\rho - 1/6}{2} - C_{\max} - \frac{1}{42} = \frac{(3-4\nu)(55-42\nu)}{84(5-2\nu)} \geq 0.$$

The complete root summation (RT1) now applies with g=Cmax, using (HF13) for the small-root series and (HF14) for the large-root tail. The proof retains natural zeros, all six units, supplementary root holes and the separate fixed-inducing terminal exclusions.

5. Finite source certification requirement

In a new source proof replace the regular HIGH gate by

$$H + \mathcal{G}_*(S, F_p) > e_H, \qquad H = 1 - \rho - \nu T - a(5 - 6\rho). \tag{HF15}$$

LOW, true diagonal, centered terminal, natural reference and special HIGH gates remain explicit. Each HIGH source ID in (HF7) is an additional earlier dependency of the new node.

For an exact proof, choose one earlier HIGH face on each finite width-density rectangle and verify all six positive-denominator-cleared facet inequalities there. Each is affine in the width endpoint and polynomial in ν after clearing denominators. Exact endpoint/ Bernstein positivity proves the rectangle. If different faces are chosen at the two width endpoints, positivity does NOT automatically interpolate; split the width interval or use a common face first.

The finite rational source graph must be certified before the new source face is used in a further HIGH or LOW calculation.

The selected finite ordinary source moments

1. Statement and literal coefficient scope

We now construct the finite graph of ordinary source moments used in the final return. The coefficient is the literal two-ordinary-factor row (160), with the fixed marked pools and natural zeros. The final selected graph and exact constants are printed below; its two selected faces are used by the following complete return chapter. The analytic ingredients are the complete source identities, divisor transport, centered terminal, and LOW/HIGH estimates proved above.

For a fixed face write

$$G(B, \Lambda, h): \quad \mathcal{E} \ll U^{Q+\epsilon}\mathcal{H}_J, \qquad m_1, m_2 \geq 0, \quad 0 \leq s \leq hQ, \quad m_1 + m_2 + \Lambda s \leq BQ. \tag{NF1}$$

Here the ambient row scale is U^Q; the normalized exponent is Q=1 when no rescaling is displayed. Equivalently, at an independent ambient scale V the bound is V^(1+epsilon), with source lengths normalized to V. The row character, natural zeros, finite rays, keep predicate, supplementary holes, ordinary two-factor coefficient, common profiles and independently bounded polynomial heights are exactly those of G1. Every canonical marked pool is fixed before moving rows and heights; the retained-projector collision reconstruction includes overlapping pools. The final ordinary source has no lower individual plain-length or marked-width edge.

Theorem NF. Every face in the selected 222-node graph printed below is an admitted moment (NF1). Its LOW and HIGH dependencies are strictly earlier faces. The only natural premise in the graph is

$$G(1, K, 11/200), \qquad K = 11243277/2500000 = 4.4973108, \tag{NF2}$$

which was proved from the earlier endpoint c=0.1252241 in the first complete return above. The natural premise precedes all 221 derived nodes. Every LOW and HIGH ancestor used here is printed below and precedes its child.

2. Fixed data and the source cell

For each node use exactly the rational parameters in the selected graph. Fix its earlier LOW parent (B_0, Λ_0, h_0), earlier HIGH parent when present, and the printed accounting schedule (e, r, f). At width-cell upper endpoint b, put

$$t = B - \Lambda(b - e), \quad T = t + b, \quad S = b - 2e,$$

$$a = \frac{q\{b + (t - B_0)/\Lambda_0\} + r}{1 - 6qB_0/\Lambda_0}, \qquad F = 1 - 6a, \quad F_p = F - e. \tag{NF4}$$

These are the actual parameters checked in the gates below.

3. Complete LOW domain and induction

For a child and its LOW parent let

$$b_\times = (B_0 - B)/(\Lambda_0 - \Lambda). \tag{NF8}$$

The final graph verifies $\Lambda<\Lambda0$ and $0\le b\times\le\min(h,h0)$. For $0\le s\le b\times$ the child line $B-\Lambda s$ is no greater than its earlier parent line, and the parent cap covers s. Thus that complete lower-width interval is already proved. On $b\times\le b\le h$ use the original source proof with (NF4).

The temporary two-plain squarefree restriction is imposed exactly as in Chapter 25. The forward and inverse divisor kernels have subpower central mass, and the restoration shortens both ordinary factors. The extraction law retains the full common puncture. Therefore only the original four first-common types 11,12,21,22 occur. With their notation the genuine RAW radius is $Q=F-(B12-C21)+$, and

$$Q \ge F - T/2 - o(1). \tag{NF9}$$

The exact certificate keeps this uniformly positive. The earlier theorem is applied at the actual rescaled ambient norm $U\hat{}Q$, including both the actual and equal-product reference rectangles; its globally downward closed source domain admits every later shortening.

The LOW proof retains both erasure floors. Its three branches are exactly G12–G13: the zero-erasure branch costs at most $-a$; the affine parent-cap branch costs at most $-r$; and the parent's upper-width-cap branch costs at most $-r$ after retaining the weak guard $h0F-z0\ge0$, where $z0=(B0F-t)/\Lambda0$. The upper-width-cap branch uses this non-strict guard. The positive reserves in the other gates pay the smoothing and greedy slot loss. The natural zero factors in the RAW character are never removed by division.

The independent scalar replay checks every one of the following at $b\times$ and h; every expression is affine in b:

$$\begin{gathered}
1 - 6qB_0/\Lambda_0 > 0, \quad S > 0, \quad a > 0, \quad 1/75 - a > 0, \\
F - T/2 - 4e > 0, \quad B_0F - t - b - 4e > 0, \quad 1 - qh_0 > 0, \\
r - (qB_0/\Lambda_0 + qh_0 + 4)e - 10e > 0, \quad a - 20e > 0, \\
L_{\mathrm{ref}} = 49/100 - Kb > 0, \quad 1 - L_{\mathrm{ref}} - 1/2 - Kb - 10e > 0, \\
5/6 + (499/1000)L_{\mathrm{ref}} - T - 9a - f > 0, \\
1 - T - 5a - f > 0, \quad 1 - 1/50 - T - 4a - f > 0, \\
1 - 1/50 - T - (9/2)a - f > 0, \quad L_{\mathrm{ref}} - 7a - f > 0, \\
h_0F - z_0 \ge 0.
\end{gathered} \tag{NF10}$$

The centered terminal uses the same elementary contour 501/1000 and saving 499/1000. The equal-product double residue is canceled before absolute values. The corrected amplifier retains the full restricted centered coefficient at fixed-inducing terminals. The exact corrected-amplifier identity is applied to the whole centered coefficient.

4. All HIGH facets and exceptional sectors

For $q\le\nu\le3/4$ put $w\nu=5-6\nu$, $t\nu=15-6\nu$, $\rho\nu=1-10\nu/t\nu$, and

$$H_\nu = 1 - \rho_\nu - \nu T - a(5 - 6\rho_\nu). \tag{NF11}$$

The earlier HIGH face has the six gains proved in HF3–HF10. In their original order they are

$$\begin{gathered}\frac{3\nu w_\nu}{t_\nu}S, \qquad \frac{2\nu w_\nu}{t_\nu}S+\frac{\nu w_\nu h_H}{t_\nu}F_p,\\ \frac{2\nu w_\nu(\Lambda_H-1)}{\Lambda_H t_\nu-10}S+\frac{\nu w_\nu(B_H t_\nu-10)}{t_\nu(\Lambda_H t_\nu-10)}F_p,\\ \frac{\nu w_\nu}{10+3w_\nu}S+\frac{2\nu w_\nu^2}{t_\nu(10+3w_\nu)}F_p,\\ \left\{\frac{2\nu w_\nu^2}{t_\nu(10+3w_\nu)}+\frac{\nu w_\nu h_H}{10+3w_\nu}\right\}F_p,\\ \frac{\nu w_\nu\{2w_\nu(\Lambda_H-1)+B_H t_\nu-10\}}{t_\nu\{\Lambda_H(10+3w_\nu)-(10+2w_\nu)\}}F_p.\end{gathered} \tag{NF12}$$

The final graph checks Hν plus **each** gain in (NF12) exceeds f on the full closed density interval. Thus its minimum is sufficient. A single earlier HIGH parent remains fixed on a child's whole width interval. Every HIGH parent satisfies 22/23≤BH≤1, ΛH>1, 0≤hH<1.

For original nodes without HIGH feedback, the original three regular gains G14–G16 are replayed instead. The rule for each node is determined by whether it has an earlier HIGH parent. The original special sectors are retained with r0=1/50:

$$\begin{gathered}H_\nu+\nu F_p/2-(1-\nu)b>f,\\ H_\nu+(\rho_\nu-1/6)F_p-(1-\nu)b>f,\\ H_\nu+\nu(F_p-r_0)/2-(1-\rho_\nu)r_0-(1-\nu)b>f.\end{gathered} \tag{NF13}$$

These are the actual fixed-inducing, small-core and tiny-core bounds, not uses of the regular count outside its eligible family. They include every common root and natural zero in their original orientation.

The earlier HF proof already establishes the complete root and common-width reserves

$$\nu-\phi-2C_{\max}\geq 1/12, \quad 6(\rho_\nu-C_{\max})\geq 3/2, \quad (\rho_\nu-1/6)/2-C_{\max}\geq 1/42. \tag{NF14}$$

The checked parent domains preserve these inequalities. Thus the new choice of source constants neither truncates sixth-power completion nor loses the large-root tail. The original whole-source restoration and coefficient-specific physical transport remain applicable.

5. Exact continuous and computational proof

Multiplying by the explicitly positive density denominators in (NF12) and (NF13) produces polynomials of degree at most three in ν. For example, set

$$P_H(\nu)=(a-\nu T)t_\nu+(10-60a)\nu.$$

The third regular inequality has the exact cleared polynomial

$$\begin{gathered}P_H(\Lambda_H t_\nu-10)+2\nu w_\nu(\Lambda_H-1)St_\nu+\nu w_\nu(B_H t_\nu-10)F_p\\ -ft_\nu(\Lambda_H t_\nu-10)>0.\end{gathered} \tag{NF15}$$

If p(q+(3/4−q)x)=Σa_jxˆj has degree n≤3, its exact Bernstein coefficients are b_i=Σ_{j≤i}a_j binom(i,j)/binom(n,j). Positivity of all b_i implies p>0 on the closed interval. Both width endpoints are checked. Affinity in b then proves the complete width-density rectangle, including all boundary values. Exact rational bisection is permitted if a polynomial's initial Bernstein presentation has a negative coefficient; no point sampling is substituted for positivity.

Use the exact rational parameters in the complete selected graph below. For each node, form every cleared scalar and density polynomial above and test its Bernstein coefficients at both width endpoints. Both parent relations are included in the printed graph, so the induction has no unlisted source premise.

The two permitted per-node accounting schedules are

Schedule	e (each support, tiny-slot, radius and analytic allowance)	r	f
original	10^−12	10^−8	10^−7
tight	10^−17	10^−12	10^−10

These are fixed positive rational budgets for the actual losses. Once the finite graph and its minimum positive normalized reserve are fixed, choose every actual smoothing, height, mesh, Fourier, profile and tail loss smaller than the relevant allowances and that minimum. The finite positive lower bounds in (NF9) permit pulling the meshes back through every dependency path. This fixes a common permitted outer mesh before the moving arithmetic data and target height.

Induction over the finite acyclic graph now proves all faces in (NF1). The exact ordinary restoration is completed at each node before its moment enters a later LOW or HIGH calculation. This proves Theorem NF.

Complete selected source graph

All decimals below are exact rationals. Both parent columns are dependencies. There are 222 nodes, including the natural premise, and 221 derived nodes. Each parent precedes its child. O means $e = 10^{-12}, r = 10^{-8}, f = 10^{-7}$; T means $e = 10^{-17}, r = 10^{-12}, f = 10^{-10}$.

ID	Earlier LOW	Earlier HIGH	B	Lambda	q	Cap	Budget
natural	—	—	1	4.4973108	—	0.055	O
g1_14000	natural	—	0.999728337095	4.477808881865	0.746594775892	0.029031830945	O
g2_14000	g1_14000	—	0.999594902255	4.468137794316	0.745036095459	0.02928436799	O
g3_14000	g2_14000	—	0.999529803492	4.463458465136	0.744281511009	0.029406485102	O
d1_15000	g3_14000	—	0.999160571842	4.437915516591	0.740398227149	0.030034563047	O
d2_15000	d1_15000	—	0.998979353436	4.425621858467	0.738418617533	0.030352855066	O
d3_16000	d2_15000	—	0.998526782073	4.396190384595	0.733933259727	0.031074755403	O
d4_16000	d3_16000	—	0.998305148906	4.382076743662	0.73166196806	0.031437171018	O
d5_17000	d4_16000	—	0.997806629136	4.351604636136	0.727033114831	0.032176781955	O
d6_17000	d5_17000	—	0.997563469218	4.337040689065	0.724689897985	0.032547808962	O
d7_17000	d6_17000	—	0.99744582902	4.330063533905	0.723566154511	0.032725474844	O
d8_17000	d7_17000	—	0.997389116831	4.326717090516	0.72302691939	0.032810663544	O
d9_18000	d8_17000	—	0.996945441116	4.301321094119	0.719238029225	0.033412291804	O
hf1_18000	d9_18000	d9_18000	0.99672936873	4.289147546159	0.717279384318	0.033719481145	O
hf2_18000	hf1_18000	hf1_18000	0.996625328299	4.283331355402	0.716342414852	0.033866370851	O
hf3_18000	hf2_18000	hf2_18000	0.996575370429	4.280550216348	0.715894196565	0.033936591505	O
hf4_19000	hf3_18000	hf3_18000	0.996107700343	4.255238388437	0.712142385557	0.034527842718	O
hf5_19000	hf4_19000	hf4_19000	0.995882099985	4.243209054717	0.710205709078	0.034829541873	O
hf6_19000	hf5_19000	hf5_19000	0.995773892149	4.237481015354	0.709282624823	0.034973164796	O
hf7_20000	hf6_19000	hf6_19000	0.995250827005	4.210584516618	0.705298937632	0.03559664186	O
hf8_20000	hf7_20000	hf7_20000	0.994999413199	4.197848916322	0.703247759178	0.035913866151	O
hf9_20000	hf8_20000	hf8_20000	0.994879250807	4.191805939032	0.702273428495	0.036064361467	O
hf10_20000	hf9_20000	hf9_20000	0.99482194504	4.188935102552	0.70181033895	0.036135843944	O
hf11_21000	hf10_20000	hf10_20000	0.994296235347	4.163260665204	0.698046419081	0.036721446668	O
hf12_21000	hf11_21000	hf11_21000	0.994044763586	4.15114487246	0.696093431603	0.037021283019	O
hf13_21000	hf12_21000	hf12_21000	0.99392506662	4.145415673306	0.695168917434	0.037163056127	O
hf14_22000	hf13_21000	hf13_21000	0.993341719307	4.118217994941	0.691182423061	0.037779250731	O
hf15_22000	hf14_22000	hf14_22000	0.993063736241	4.105433270088	0.689119844696	0.038093734712	O
hf16_22000	hf15_22000	hf15_22000	0.992931914267	4.099410332475	0.688146977768	0.03824189353	O
hf17_23000	hf16_22000	hf16_22000	0.992315106324	4.071930830049	0.68413754839	0.038857977104	O
hf18_23000	hf17_23000	hf17_23000	0.992022432465	4.059062212086	0.682058959125	0.03917274804	O
hf19_24000	hf18_23000	hf18_23000	0.99130064954	4.028192423722	0.677524714812	0.039864751485	O
hf20_24000	hf19_24000	hf19_24000	0.990959343997	4.013799589186	0.675197222069	0.040214962673	O
hf21_24000	hf20_24000	hf20_24000	0.990798743487	4.007072252125	0.674107674548	0.040378700074	O
hf22_24000	hf21_24000	hf21_24000	0.990723321394	4.003923795898	0.67359741305	0.040455333676	O
hf23_25000	hf22_24000	hf22_24000	0.990077625687	3.977541258072	0.669809745484	0.041031369151	O
hf24_25000	hf23_25000	hf23_25000	0.989774091932	3.965281527808	0.667822948365	0.041328286815	O
hf25_25000	hf24_25000	hf24_25000	0.989631953208	3.959572136647	0.666896432582	0.041466606877	O
hf26_26000	hf25_25000	hf25_25000	0.988926394765	3.931849506857	0.662913807865	0.042068799432	O

ID	Earlier LOW	Earlier HIGH	B	Lambda	q	Cap	Budget
hf27_26000	hf26_26000	hf26_26000	0.988596125754	3.919022727362	0.660831253508	0.042378100024	O
hf28_27000	hf27_26000	hf27_26000	0.987773250694	3.887835855811	0.656308351216	0.043057452934	O
hf29_27000	hf28_27000	hf28_27000	0.987389549473	3.873475154569	0.653972325597	0.043402354825	O
hf30_28000	hf29_27000	hf29_27000	0.986513545378	3.8414719366	0.649336924806	0.044094993118	O
hf31_28000	hf30_28000	hf30_28000	0.986106993006	3.826802385042	0.6469448432	0.044446083805	O
hf32_29000	hf31_28000	hf31_28000	0.985192390682	3.794564884978	0.642290688484	0.045138232893	O
hf33_29000	hf32_29000	hf32_29000	0.984770022896	3.779855703275	0.639885490763	0.045489205967	O
hf34_29000	hf33_29000	hf33_29000	0.984575692548	3.773125909625	0.638782863509	0.045649928802	O
hf35_30000	hf34_29000	hf34_29000	0.983732461943	3.744486776747	0.634735336593	0.046250643205	O
hfd1_30000	hf35_30000	hf35_30000	0.983345336653	3.731473928414	0.632600050951	0.046560406497	O
hfd2_30500	hfd1_30000	hfd1_30000	0.982781163598	3.712754005416	0.629859768541	0.046963721777	O
hfd3_31000	hfd2_30500	hfd2_30500	0.982128539481	3.691417541341	0.626693906859	0.04742770084	O
hfd4_31500	hfd3_31000	hfd3_31000	0.981428622487	3.668886738299	0.6233348932	0.047918525836	O
hfd5_32000	hfd4_31500	hfd4_31500	0.98070055431	3.64581392482	0.619889261176	0.048420745598	O
hfd6_32000	hfd5_32000	hfd5_32000	0.980370106018	3.635423616051	0.618171022473	0.048667246738	O
hfd7_32500	hfd6_32000	hfd6_32000	0.979804946823	3.617851768494	0.615634958594	0.049037936909	O
hfd8_33000	hfd7_32500	hfd7_32500	0.979125927981	3.597023089626	0.61256406237	0.049484402919	O
hfd9_33000	hfd8_33000	hfd8_33000	0.978819509534	3.587688258235	0.611013711509	0.04970564193	O
hfd10_33500	hfd9_33000	hfd9_33000	0.978251705516	3.570571696674	0.608560693423	0.050063096856	O
hfd11_34000	hfd10_33500	hfd10_33500	0.977558486421	3.549943534308	0.605528254125	0.05050233955	O
hfd12_34000	hfd11_34000	hfd11_34000	0.977247443492	3.540748883153	0.603994468703	0.05072011157	O
hfd13_34500	hfd12_34000	hfd12_34000	0.97666394312	3.52367483789	0.601556071949	0.051074234096	O
hfd14_34500	hfd13_34500	hfd13_34500	0.976402999446	3.516080692664	0.600285744531	0.051254121523	O
hfd15_35000	hfd14_34500	hfd14_34500	0.975835299743	3.499715244101	0.597969382119	0.05159038214	O
hfd16_35000	hfd15_35000	hfd15_35000	0.975582191037	3.492456227587	0.596752083824	0.051762334998	O
hfd17_35500	hfd16_35000	hfd16_35000	0.975011157491	3.476230034848	0.594462939819	0.052094218112	O
hfd18_36000	hfd17_35500	hfd17_35500	0.974290739357	3.456002433325	0.591506671298	0.052519542295	O
hfd19_36000	hfd18_36000	hfd18_36000	0.973970684391	3.447071234237	0.59000283812	0.052731013724	O
hfd20_36500	hfd19_36000	hfd19_36000	0.973356670357	3.430099495213	0.587595344622	0.053078501843	O
hfd21_36500	hfd20_36500	hfd20_36500	0.973085321609	3.422637616638	0.586334523985	0.05325534842	O
hfd22_37000	hfd21_36500	hfd21_36500	0.972485691643	3.406294983996	0.5840371495	0.053586901491	O
hfd23_37000	hfd22_37000	hfd22_37000	0.972221535637	3.39913063001	0.582823182389	0.053756779143	O
hfd24_37500	hfd23_37000	hfd23_37000	0.971618318473	3.3829124746	0.580550503438	0.054084415412	O
hfd25_38000	hfd24_37500	hfd24_37500	0.97085814833	3.362704874854	0.577605351897	0.054505430243	O
hfd26_38000	hfd25_38000	hfd25_38000	0.97052522138	3.353906246758	0.576106840654	0.054714220194	O
hfd27_38500	hfd26_38000	hfd26_38000	0.969878141446	3.336958424757	0.573717169116	0.054998999999	O
hfd28_39000	hfd27_38500	hfd27_38500	0.969086121998	3.316446771585	0.570723367576	0.054998999999	O
near_g1_s22	hfd27_38500	hfd26_38000	0.969595880287	3.329601584001	0.572459738081	0.054998900000	O
near_g1_s24	hfd28_39000	hfd28_39000	0.968741508719	3.307573478665	0.569203024136	0.054998900000	O

ID	Earlier LOW	Earlier HIGH	B	Lambda	q	Cap	Budget
near_g2_s22	near_g1_s22	hfd27_38500	0.969472744386	3.326399659468	0.571911697535	0.054998900000	O
near_g2_s23	near_g1_s22	hfd27_38500	0.969219972657	3.319855423243	0.571047449562	0.054998900000	O
near_g2_s24	near_g1_s24	near_g1_s24	0.968591750349	3.303727757550	0.568542832967	0.054998900000	O
near_g3_s23	near_g2_s22	hfd27_38500	0.969166575597	3.318469007242	0.570809729630	0.054998900000	O
near_g3_s25	near_g2_s24	near_g2_s23	0.968270645266	3.295515409091	0.567394304180	0.054998900000	O
near_g4_s23	near_g3_s23	hfd27_38500	0.969033339078	3.315026290121	0.570219531498	0.054998900000	O
near_g4_s24	near_g3_s23	near_g3_s23	0.968778613874	3.308473874365	0.569355289036	0.054998900000	O
near_g4_s25	near_g3_s25	near_g2_s24	0.968131273476	3.291959856623	0.566782922512	0.054998900000	O
near_g5_s24	near_g4_s23	near_g4_s23	0.968720922087	3.306985426201	0.569099689595	0.054998900000	O
near_g5_s26	near_g4_s25	near_g4_s24	0.967813002360	3.283872228334	0.565656651889	0.054998900000	O
near_g6_s24	near_g5_s24	near_g4_s23	0.968585184306	3.303500502907	0.568501352641	0.054998900000	O
near_g6_s25	near_g5_s24	near_g4_s23	0.968328812394	3.296947881826	0.567638254688	0.054998900000	O
near_g6_s26	near_g5_s26	near_g4_s25	0.967675086336	3.280376240015	0.565054535826	0.054998900000	O
near_g7_s25	near_g6_s24	near_g4_s23	0.968270152871	3.295444133760	0.567379606644	0.054998900000	O
near_g7_s27	near_g6_s26	near_g6_s25	0.967355787369	3.272313829340	0.563933433932	0.054998900000	O
near_g8_s25	near_g7_s25	near_g6_s24	0.968133428615	3.291956266589	0.566779858426	0.054998900000	O
near_g8_s26	near_g7_s25	near_g6_s24	0.967875280040	3.285400127685	0.565917424960	0.054998900000	O
near_g8_s27	near_g7_s27	near_g6_s26	0.967217655071	3.268834445446	0.563333198617	0.054998900000	O
near_g9_s26	near_g8_s25	near_g6_s24	0.967816296200	3.283897695840	0.565658584607	0.054998900000	O
near_g9_s28	near_g8_s27	near_g8_s26	0.966896596881	3.260778508627	0.562214045109	0.054998900000	O
near_g10_s26	near_g9_s26	near_g8_s25	0.967678884239	3.280414511918	0.565058681958	0.054998900000	O
near_g10_s27	near_g9_s26	near_g8_s25	0.967418966176	3.273855080971	0.564196942541	0.054998900000	O
near_g10_s28	near_g9_s28	near_g8_s27	0.966757941500	3.257307824619	0.561614313838	0.054998900000	O
near_g11_s27	near_g10_s26	near_g8_s25	0.967359786280	3.272357201287	0.563938464526	0.054998900000	O
near_g11_s29	near_g10_s28	near_g10_s27	0.966434991168	3.249254952469	0.560496517784	0.054998900000	O
near_g12_s27	near_g11_s27	near_g10_s26	0.967221755234	3.268880373211	0.563338677046	0.054998900000	O
near_g12_s28	near_g11_s27	near_g10_s26	0.966960069533	3.262317737533	0.562477642957	0.054998900000	O
near_g12_s29	near_g11_s29	near_g10_s28	0.966295759952	3.245791544802	0.559897033079	0.054998900000	O
near_g13_s28	near_g12_s27	near_g10_s26	0.966900722291	3.260825076321	0.562219640711	0.054998900000	O
near_g13_s30	near_g12_s29	near_g12_s28	0.965970895277	3.237741146891	0.558780488201	0.054998900000	O
near_g14_s28	near_g13_s28	near_g12_s27	0.966762092583	3.257355036325	0.561620027871	0.054998900000	O
near_g14_s29	near_g13_s28	near_g12_s27	0.966498635891	3.250789152281	0.560759693257	0.054998900000	O
near_g14_s30	near_g13_s30	near_g12_s29	0.965831078243	3.234284678635	0.558181184861	0.054998900000	O
near_g15_s29	near_g14_s28	near_g12_s27	0.966439136844	3.249302032211	0.560502213342	0.054998900000	O
near_g15_s31	near_g14_s30	near_g14_s29	0.965504304646	3.226236858253	0.557065895168	0.054998900000	O
near_g16_s29	near_g15_s29	near_g14_s28	0.966299908209	3.245838689575	0.559902757726	0.054998900000	O
near_g16_s30	near_g15_s29	near_g14_s28	0.966034687307	3.239269766124	0.559043149898	0.054998900000	O
near_g16_s31	near_g15_s31	near_g14_s30	0.965363915306	3.222787576083	0.556466796784	0.054998900000	O
near_g17_s30	near_g16_s29	near_g14_s28	0.965975035477	3.237788094612	0.558786175096	0.054998900000	O

ID	Earlier LOW	Earlier HIGH	B	Lambda	q	Cap	Budget
near_g17_s32	near_g16_s31	near_g16_s30	0.965035243319	3.214742558764	0.555352792887	0.054998900000	O
near_g18_s30	near_g17_s30	near_g14_s28	0.965835330569	3.234334402478	0.558187271830	0.054998900000	O
near_g18_s31	near_g17_s30	near_g16_s29	0.965568242001	3.227759914860	0.557328043237	0.054998900000	O
near_g18_s32	near_g17_s32	near_g16_s31	0.964894284621	3.211300446733	0.554753890987	0.054998900000	O
near_g19_s31	near_g18_s30	near_g16_s29	0.965508487376	3.226284864241	0.557071777794	0.054998900000	O
near_g19_s33	near_g18_s32	near_g18_s31	0.964563716387	3.203258250691	0.553641171680	0.054998900000	O
near_g20_s31	near_g19_s31	near_g16_s29	0.965368178272	3.222837556364	0.556472947105	0.054998900000	O
near_g20_s32	near_g19_s31	near_g18_s30	0.965099329661	3.216260220742	0.555614470316	0.054998900000	O
near_g20_s33	near_g19_s33	near_g18_s32	0.964422199241	3.199823486632	0.553042479238	0.054998900000	O
near_g21_s49	near_g1_s22	near_g2_s22	0.969472726138	3.326397668936	0.571911422249	0.054998900000	T
near_g21_s50	near_g4_s23	near_g2_s22	0.968924641326	3.312220918364	0.569790124318	0.054998900000	T
near_g21_s51	near_g5_s24	near_g4_s23	0.968686957675	3.306111929954	0.568845616821	0.054998900000	T
near_g21_s52	near_g5_s24	near_g2_s22	0.968585271796	3.303501257044	0.568501467049	0.054998900000	T
near_g21_s53	near_g6_s24	near_g6_s24	0.968424066978	3.299371517830	0.567896749934	0.054998900000	T
near_g21_s54	near_g7_s25	near_g2_s22	0.968184939214	3.293268004062	0.566952556714	0.054998900000	T
near_g21_s55	near_g8_s25	near_g5_s24	0.968022796250	3.289137326692	0.566347729311	0.054998900000	T
near_g21_s57	near_g9_s26	near_g7_s25	0.967678905491	3.280413576135	0.565058563776	0.054998900000	T
near_g21_s58	near_g10_s26	near_g9_s26	0.967515830180	3.276288774145	0.564454811962	0.054998900000	T
near_g21_s59	near_g11_s27	near_g7_s25	0.967273875958	3.270190953854	0.563510765501	0.054998900000	T
near_g21_s60	near_g12_s27	near_g9_s26	0.967110057053	3.266070094774	0.562907691347	0.054998900000	T
near_g21_s61	near_g13_s28	near_g7_s25	0.966866957015	3.259978485824	0.561964078319	0.054998900000	T
near_g21_s64	near_g15_s29	near_g10_s26	0.966352853957	3.247153498501	0.560074957322	0.054998900000	T
near_g21_s65	near_g17_s30	near_g13_s28	0.965835366195	3.234333849119	0.558187204689	0.054998900000	T
near_g21_s66	near_g20_s31	near_g18_s30	0.965039414015	3.214788851815	0.555358467161	0.054998900000	T
near_g21_s67	near_g20_s33	near_g20_s32	0.964089731988	3.191782604847	0.551930848624	0.054998900000	T
near_g22_s49	near_g21_s49	hfd26_38000	0.969419229822	3.325007191901	0.571673090338	0.054998900000	T
near_g22_s50	near_g21_s50	near_g21_s49	0.968877401975	3.311002677100	0.569581008719	0.054998900000	T
near_g22_s53	near_g21_s53	near_g21_s49	0.968354341158	3.297586689220	0.567589718874	0.054998900000	T
near_g22_s55	near_g21_s55	near_g21_s49	0.967975123266	3.287923556580	0.566138467121	0.054998900000	T
near_g22_s57	near_g21_s57	near_g21_s52	0.967619540989	3.278909582251	0.564799197707	0.054998900000	T
near_g22_s58	near_g21_s58	near_g21_s51	0.967445532061	3.274512044945	0.564148145928	0.054998900000	T
near_g22_s59	near_g21_s59	near_g21_s54	0.967236743202	3.269255198608	0.563349204036	0.054998900000	T
near_g22_s60	near_g21_s60	near_g21_s51	0.967062145193	3.264865582557	0.562699278159	0.054998900000	T
near_g22_s64	near_g21_s64	near_g21_s57	0.966315714405	3.246229223437	0.559914809154	0.054998900000	T
near_g22_s65	near_g21_s65	near_g21_s61	0.965775345546	3.232850821421	0.557929797330	0.054998900000	T
near_g22_s66	near_g21_s66	near_g21_s65	0.964898597740	3.211348806496	0.554759774914	0.054998900000	T
near_g22_s67	near_g21_s67	near_g21_s66	0.963947815110	3.188357536618	0.551332622880	0.054998900000	T
near_g23_s49	near_g22_s49	hfd24_37500	0.969396431979	3.324414865924	0.571571079621	0.054998900000	T
near_g23_s50	near_g22_s50	near_g22_s49	0.968856868639	3.310473329652	0.569490126046	0.054998900000	T

ID	Earlier LOW	Earlier HIGH	B	Lambda	q	Cap	Budget
near_g23_s51	near_g22_s50	near_g22_s49	0.968755410916	3.307861806079	0.569145676280	0.054998900000	T
near_g23_s57	near_g22_s57	near_g22_s49	0.967594218141	3.278268298766	0.564688233604	0.054998900000	T
near_g23_s59	near_g22_s58	near_g22_s53	0.967311322662	3.271125936247	0.563671754699	0.054998900000	T
near_g23_s61	near_g22_s60	near_g22_s55	0.966936922090	3.261722428769	0.562265249057	0.054998900000	T
near_g23_s65	near_g22_s65	near_g22_s59	0.965749716837	3.232217825336	0.557819723233	0.054998900000	T
near_g23_s66	near_g22_s66	near_g22_s64	0.964838496778	3.209881928452	0.554504071955	0.054998900000	T
near_g23_s67	near_g22_s67	near_g22_s66	0.963887225356	3.186896562588	0.551077261979	0.054998900000	T
near_g23_s68	near_g22_s67	near_g22_s67	0.963558298572	3.178999175297	0.550050673458	0.054998900000	T
near_g24_s49	near_g23_s49	hfd23_37000	0.969386945549	3.324168435540	0.571528390971	0.054998900000	T
near_g24_s50	near_g23_s50	near_g23_s50	0.968847912094	3.310242465208	0.569450516508	0.054998900000	T
near_g24_s51	near_g23_s51	near_g23_s49	0.968702423595	3.306498749821	0.568911570651	0.054998900000	T
near_g24_s59	near_g23_s59	near_g23_s51	0.967253497295	3.269668354838	0.563419960589	0.054998900000	T
near_g24_s61	near_g23_s61	near_g23_s57	0.966882802263	3.260365160093	0.562030702078	0.054998900000	T
near_g24_s66	near_g23_s66	near_g23_s65	0.964812552546	3.209248956478	0.554394000237	0.054998900000	T
near_g24_s67	near_g23_s67	near_g23_s65	0.963861781463	3.186283284460	0.550969583231	0.054998900000	T
near_g24_s68	near_g23_s68	near_g23_s67	0.963392384197	3.175022733550	0.549354811198	0.054998900000	T
near_g25_s51	near_g24_s51	hfd25_38000	0.968680008119	3.305922353438	0.568811940428	0.054998900000	T
near_g25_s52	near_g24_s51	near_g24_s49	0.968577672810	3.303294975281	0.568465575630	0.054998900000	T
near_g25_s66	near_g24_s66	near_g24_s59	0.964802261525	3.208997927249	0.554349384912	0.054998900000	T
near_g25_s67	near_g24_s67	near_g24_s61	0.963851853156	3.186044022457	0.550926734549	0.054998900000	T
near_g25_s68	near_g24_s68	near_g24_s66	0.963321995933	3.173337509610	0.549059348063	0.054998900000	T
near_g26_s51	near_g25_s51	hfd26_38000	0.968670079697	3.305667092635	0.568768274855	0.054998900000	T
near_g26_s52	near_g25_s52	near_g24_s49	0.968523530672	3.301905768140	0.568226819395	0.054998900000	T
near_g26_s53	near_g25_s52	near_g24_s49	0.968421464632	3.299292042025	0.567882442879	0.054998900000	T
near_g26_s68	near_g25_s68	near_g25_s67	0.963291785509	3.172614534583	0.548932895858	0.054998900000	T
near_g26_s69	near_g25_s68	near_g25_s67	0.963181865635	3.169988015303	0.548591879188	0.054998900000	T
near_g27_s52	near_g26_s52	hfd25_38000	0.968500721000	3.301320737179	0.568125545749	0.054998900000	T
near_g27_s53	near_g26_s53	near_g24_s49	0.968353704595	3.297557576250	0.567584201578	0.054998900000	T
near_g27_s68	near_g26_s68	near_g25_s66	0.963279180581	3.172312940437	0.548879758596	0.054998900000	T
near_g27_s70	near_g26_s69	near_g25_s67	0.963012019864	3.165936633315	0.548001273281	0.054998900000	T
near_g28_s52	near_g27_s52	hfd25_38000	0.968490826188	3.301066993060	0.568081909573	0.054998900000	T
near_g28_s53	near_g27_s53	near_g24_s49	0.968324322000	3.296805827059	0.567454883433	0.054998900000	T
near_g28_s54	near_g27_s53	near_g26_s51	0.968221845907	3.294188295990	0.567110171499	0.054998900000	T
near_g28_s71	near_g27_s70	near_g25_s67	0.962939824583	3.164216356823	0.547699454760	0.054998900000	T
near_g29_s53	near_g28_s53	near_g24_s49	0.968311583998	3.296479995594	0.567398821416	0.054998900000	T
near_g29_s54	near_g28_s54	near_g24_s49	0.968164868545	3.292733759277	0.566859712084	0.054998900000	T
near_g29_s72	near_g28_s71	near_g25_s67	0.962798689407	3.160858881964	0.547230515727	0.054998900000	T
near_g30_s54	near_g29_s54	near_g24_s49	0.968140180139	3.292103759762	0.566751259983	0.054998900000	T
near_g30_s55	near_g29_s54	near_g24_s50	0.968037440162	3.289486174492	0.566406733666	0.054998900000	T

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near_g30_s72	near_g29_s72	near_g25_s67	0.962738747040	3.159434155041	0.546980322634	0.054998900000	T
near_g30_s73	near_g29_s72	near_g27_s68	0.962627788448	3.156801677691	0.546639054678	0.054998900000	T
near_g31_s54	near_g30_s54	hfd25_38000	0.968130351194	3.291852991271	0.566707191296	0.054998900000	T
near_g31_s55	near_g30_s55	near_g24_s49	0.967982428791	3.288085438761	0.566165353718	0.054998900000	T
near_g31_s72	near_g30_s72	near_g25_s66	0.962713737745	3.158839950807	0.546875456467	0.054998900000	T
near_g31_s73	near_g30_s73	near_g25_s67	0.962555438185	3.155085888612	0.546337381821	0.054998900000	T
near_g32_s56	near_g31_s55	near_g24_s50	0.967855556785	3.284860288864	0.565716404654	0.054998900000	T
near_g32_s73	near_g31_s73	near_g31_s72	0.962524459258	3.154351548125	0.546208578249	0.054998900000	T
near_g33_s56	near_g32_s56	near_g24_s49	0.967800854158	3.283470960109	0.565476797504	0.054998900000	T
near_g33_s73	near_g32_s73	near_g25_s66	0.962512109261	3.154058856095	0.546156255166	0.054998900000	T
near_g34_s57	near_g33_s56	near_g31_s54	0.967673636288	3.280245219427	0.565028081698	0.054998900000	T
near_g35_s57	near_g34_s57	near_g29_s53	0.967618636710	3.278851885347	0.564787823542	0.054998900000	T
near_g36_s58	near_g35_s57	near_g29_s53	0.967491410923	3.275634115668	0.564340238491	0.054998900000	T
near_g37_s58	near_g36_s58	near_g28_s52	0.967436469761	3.274245781992	0.564100653008	0.054998900000	T
near_g38_s58	near_g37_s58	near_g24_s49	0.967413145687	3.273656623119	0.563998537102	0.054998900000	T
near_g39_s58	near_g38_s58	near_g24_s49	0.967403070550	3.273402169816	0.563954607164	0.054998900000	T
alt_focused_g1_C	near_g33_s73	near_g33_s73	0.96236728488060	3.15063185213785	0.54570649935321	0.05499890000000	T
alt_focused_g1_P	near_g39_s58	near_g39_s58	0.96739676975432	3.27324305194549	0.56392916380335	0.05499890000000	T
alt_focused_g2_C	alt_focused_g1_C	alt_focused_g1_C	0.96230568251169	3.14917544130912	0.54545054061067	0.05499890000000	T
alt_focused_g2_P	alt_focused_g1_P	near_g38_s58	0.96739447187175	3.27318502444400	0.56391870494007	0.05499890000000	T
alt_focused_g3_C	alt_focused_g2_C	alt_focused_g2_C	0.96227948182933	3.14855623588510	0.54534167948918	0.05499890000000	T
alt_focused_g3_P	alt_focused_g2_P	alt_focused_g2_P	0.96739250152685	3.27313526952874	0.56391113411921	0.05499890000000	T
alt_focused_g4_C	alt_focused_g3_C	alt_focused_g3_C	0.96226831325612	3.14829232960439	0.54529530355548	0.05499890000000	T
alt_focused_g4_P	alt_focused_g3_P	near_g39_s58	0.96739119188673	3.27310219934594	0.56390590273686	0.05499890000000	T
alt_focused_g5_C	alt_focused_g4_C	alt_focused_g4_C	0.96226354319698	3.14817962398374	0.54527550684579	0.05499890000000	T
alt_focused_g6_C	alt_focused_g5_C	alt_focused_g4_C	0.96226151753982	3.14813176381578	0.54526708734661	0.05499890000000	T
alt_focused_g7_P	alt_focused_g4_P	alt_focused_g1_P	0.96739031091242	3.27307995389766	0.56390239081147	0.05499890000000	T

The final parameterized return

The earlier bound is now $\theta_I = 8747759/10000000$; the natural cost is $11243277/2500000$. All count faces and join points are fixed before the contrary zero supremum.

1. The literal same-zero witness bounds

For $0 \le \delta \le 3/4$, $0 \le p \le 1/2$, put $V = 5/6 - \delta$. The three original witness lines are

$$I(r) = 1 - \delta\{p + (1-p)r\}, \quad P(m) = 1 - 2\delta m - 2\delta p z(m), \quad L(r) = \frac{1}{6} + Vr. \tag{NR1}$$

Here p is the actual prime-amplitude class of the row. For each complete admissible count, choose its detector cutoff before the inverse or plain witnesses are produced at that same zero and common height. For an affine source (B, Λ), define

$$\begin{aligned} &D = 3 - p - 4p/\Lambda, && A = 2 - 4p/\Lambda, && C_0 = 1 - 2pB/\Lambda, \\ &U = (1-p)C_0, && J = VD + \delta(1-p)A, && \\ &N = (1-\delta)J + V\delta U, && T = J + \delta U, && \\ &R_n = AT + (2pB/\Lambda - p)J, && M_n = DT - R_n. && \end{aligned} \tag{NR2}$$

Then $R = N/J$, $t_* = T/J$, $r_* = R_n/(DJ)$, and $m_* = M_n/(DJ)$ satisfy $r_* + m_* = t_*$ and $I(r_*) = P(m_*) = L(t_*) = R$. For a capped branch put $z_* = s_+ - 10^{-7}$, use $(D, A, C_0) = (3 - p, 2, 1 - 2pz_*)$, and replace $2pB/\Lambda$ in R_n by $2pz_*$. The continuation is $z(m) = \min(z_*, (B - 2m)/\Lambda)$.

The count candidates are the natural count (FR6), the affine and capped branches of C, P, and the weaker faces $(B - j/4000, \Lambda)$, $0 \le j \le 8$. The latter follow by literal domain inclusion. Keep the conservative local domains

$$2/5 \le m_i \le 11/25, \quad 17/500 \le s \le 11/250 \quad \text{for } C,$$

and replace the upper width by $43/1000$ for P. Both are strictly below their proved global caps. Set $m_H = (1 - K_I(17/500))/2$.

Every improved count is used only where all the fourteen strict polynomial admission tests (FR8) hold, plus the capped-width test when applicable. Those tests retain the full inverse range, the plain crossing, source and natural tail joins, the original upper cutoff, the source's lower and upper widths, and the positive denominators. Thus a witness beyond the crossing is covered continuously: the short inverse decreases up to $r = 1$; the source plain decreases through the cap and affine portions to m_H; the natural plain decreases from that join to $m = 1/2$; longer plains retain the inherited unmarked treatment. The long inverse increases on $1 < r \le t_*$, where the detector's original upper witness bound gives $L(r) \le L(t_*)$. Both witness cases retain that one zero, the common height and the original profiles.

On each improved admission box its upper p edge is $1/2$. The derivative test $UJ_p - U_pJ > 0$ makes the count antitone in p. Taking the finite minimum with the always admitted natural count gives $R_*(\kappa, \delta, p)$; new improved candidates may appear as p increases but do not disappear. This preserves the required amplitude monotonicity.

2. Geometry and all conductor intervals

Set

$$\theta = 1 - c, \quad \ell = 4c - \tfrac{1}{3}, \quad x = 1 + \ell - h, \quad y = h - 2\ell, \quad \kappa_* = 1 - 2c. \tag{NR3}$$

The required Gram geometry is $x > \ell$, $y > x$, $\ell < 1/5$, $2x > y + \ell$. It supplies the same original reciprocal signal and nonzero prime normalizer. Let $d_0 = 1/200$, $d_+ = h + 10^{-10}$, and $s_P = 1/25$. The natural and new physical capacities are

$$C_O(\kappa, d) = \frac{\min(d, 2y - d)}{6\kappa}, \qquad C_N(d) = \min\left(\frac{d}{25}, \frac{2y - (2 - B_P)d}{\Lambda_P}\right). \tag{NR4}$$

Fix, once,

$$d_s = \frac{2y}{1 + 6\kappa_*/25}, \qquad d_k = \frac{2y}{2 - B_P + \Lambda_P/25}. \tag{NR5}$$

Check $d_0 < y < d_s < d_k < d_+ < 2y$ and the exact identity $C_O(\kappa_*, d_s) = C_N(d_s) = d_s/25$. On $[d_0, d_s]$ use the natural physical bound and on $[d_s, d_+]$ use P. The join is not recomputed at the contrary zero supremum.

For each capacity the complete return exponent is

$$E_\kappa = \frac{1}{2} + \frac{13h}{75} - \frac{y}{2} - \frac{\ell}{2} - \frac{9}{100}d + \frac{\delta}{2} + \delta p\left(\ell - \frac{C}{2}\right) + \frac{3d}{4}R_*(\kappa, \delta, p). \tag{NR6}$$

At fixed (δ, p) it is affine in d on the intervals cut by y, d_s, d_k. The six endpoints $O_0, O_y, O_s, N_s, N_+, N_k$ suffice. Different count candidates can certify different endpoint boxes because their common finite minimum R_* is independent of d. Interpolation therefore uses one valid count envelope on the full conductor interval.

For each endpoint and admitted branch N_b/J_b, certify

$$F_b = J_b\{\theta - 10^{-9} - E_{\mathrm{base}}\} - \tfrac{3}{4}dN_b > 0. \tag{NR7}$$

The polynomial has bidegree at most $(2, 3)$. Rational tensor Bernstein coefficients on a complete finite closed rectangular cover prove it continuously, including edges and corners. Positivity at sample points alone would not suffice. All corresponding admission polynomials are certified on their larger upward p boxes. Exact area and pairwise interior-disjointness checks certify the full rectangle cover. Dividing by the positive J_b then gives $E_{\kappa_*} \le \theta - 10^{-9}$ on every complete ordinary conductor interval.

3. All shortenings, shifts and complementary sectors

For $d_s \le d \le d_+$, the actual equal-plain length and width are

$$n_0(d) = 1 - y/d, \quad z_d(t) = \min\left(\tfrac{1}{25}, \frac{B_P - 2n_0(d) - 2t}{\Lambda_P}\right). \tag{NR8}$$

Check the domains with margin 10^{-7} for $0 \le t \le t_f = 1/400$. Every shorter pair and smaller marked subset lies in the global source. The natural estimates cover additional shortenings below the narrower return domain by the two exact inequalities (FR19). The small-shift width difference is at most $2t/\Lambda_P$; its cost is paid by $\delta t(1/2 - 1/(2\Lambda_P)) \ge \delta t/500$. The zero endpoints retain the fixed return reserve.

For $t \ge t_f$ retain the clipped natural width with denominator $9/2$. The complete nonclipped and clipped reserves are precisely (FR21) and (FR22), which the certificate recomputes as positive rational numbers. Their minimization is over the full relevant parameter range, so no artificial upper bound on the shifted ordinary length is imposed. The resulting proportional damping also pays the complete shift tails and Fourier seminorm losses.

The exact scalar receipt separately checks the original floor, small-row and nonprincipal unit bounds, prime supply $\ell/d_+ > 1/5$, the positive physical capacity, both principal-contour reserves, and the holomorphic box $\theta > 17/20$. The floor is increasing in the conductor parameter and is checked at d_+. These checks preserve the fixed-inducing, imprimitive and excluded natural-zero conventions of the inherited theorem; no exceptional row is inserted into an improved nonprincipal count.

4. Substitution of the actual zero supremum

All improved faces and their admission boxes are independent of κ. Only the old count and natural physical capacity vary. For $\kappa_* \le a \le \kappa \le \kappa_I$, the exact subtraction identity (FR23) gives

$$R_O(\kappa) - R_O(a) \le \tfrac{5}{216}(\kappa - a).$$

Selecting a minimizing candidate at a proves the same upper difference bound for R_*. The count cost in (NR6) is at most $5d(\kappa - \kappa_*)/288$. On old physical rows the extra capacity cost is at most $25(\kappa - \kappa_*)/784$, because $y < 1/2$, $\delta p \le 3/8$, $\kappa\kappa_* \ge (7/10)^2$. The exact inequalities $d_+ < 1$ and $5/288 + 25/784 < 1/8$ are retained.

Let β be the contrary Hecke zero supremum and put $\mu = \beta - \theta > 0$, $\kappa = 2\beta - 1$. The independently earlier theorem bounds $\theta < \beta \le \theta_I$. Therefore the complete return gives

$$E_\kappa - \beta \le -10^{-9} - \tfrac{24}{25}\mu \quad \text{on new rows}, \qquad E_\kappa - \beta \le -10^{-9} - \tfrac{3}{4}\mu \quad \text{on old rows}. \tag{NR9}$$

The fixed join is valid: no equality of old and new capacities at the moving κ is assumed. All analytic loss parameters are chosen after the finite positive reserves and μ, then before scale, heights and prime pools. This is the same acyclic quantifier order as the source construction.

5. Complete Mellin contradiction and character scope

With $k = (4 + y - x)/6$, the geometry gives $\theta - k = x/2 + (y - x)/12$. The signed Gram estimate in Section 14.3 for the same compensated probe yields $J_\eta(Z) \ll Z^{\theta-k+\omega}$ for a common $0 < \omega < (\beta - \theta)/4$. Equations (NR7)–(NR9), the shift completion and the complementary sectors give a common $\sigma_0 > 0$ with $J_\eta(Z) - f_\eta(Z) \ll Z^{\beta-k-\sigma_0}$. Choose a positive Mellin gap smaller than the available fixed and contrary-supremum reserves. The complete Mellin transform of the probe error is then holomorphic in $\Re s > \beta - m$, uniformly over the primitive target family. Its identity on $\Re s > 1$ is the original

$$\mathcal{F}_\eta(s) = e^{(s-5/6)^2} \frac{H_\eta(s)}{L_K^S(s, \eta)}. \tag{NR10}$$

Absolute all-height Fourier–Mellin completion gives this identity before continuation. The numerator argument in Section 14.4, with the same literal coefficients and cutoffs, proves that H_η is nonzero in the relevant region. A denominator zero of any positive multiplicity would therefore be a pole, contradicting the holomorphy just established. The original zero-supremum lemma produces a primitive zero inside that region if $\beta > \theta$. Hence $\beta \le \theta$.

Primitive reduction and restoration of fixed Euler factors preserve zeros on $\Re s > 0$. For every ordinary presented Dirichlet character the original norm-pullback factorization is

$$L_K(s, \eta_\chi) = L(s, \chi)L(s, \chi\chi_{-3}).$$

A zero of either ordinary factor in $\theta < \Re s < 1$ contradicts the proved Hecke nonvanishing. Classical boundary nonvanishing and absolute Euler convergence cover $\Re s \ge 1$, excluding the principal pole. The theorem therefore has the full original presented-character scope; it does not make an assertion on the boundary $\Re s = \theta$.

7. Final certified instance after finite source refinement

Use the two global ordinary source moments proved by the frozen finite graph printed above. Their coefficients are the following exact terminating decimals:

Role	Source node	B	Λ	Proved width cap
Common-zero count C	`alt_focused_g6_C`	0.96226151753982	3.14813176381578	0.05499890000000
Physical numerator P	`alt_focused_g7_P`	0.96739031091242	3.27307995389766	0.05499890000000

The selected source graph contains 222 nodes, including the natural premise, and is closed under both parent relations. Its 221 derived steps use the complete LOW/HIGH, exceptional-sector and common-root conditions proved above. All new nodes retain the old proved natural cost $K_I = 11243277/2500000$; no improved zero-free conclusion is used to prove one of its own source premises.

Choose

$$\boxed{c_{\mathrm{new}} = \frac{1252270103}{10000000000} = 0.1252270103}, \qquad h = \frac{8001009617}{10000000000} = 0.8001009617. \tag{NR11}$$

The complete exact geometry is

$$\begin{aligned} \theta &= \frac{8747729897}{10000000000}, & \ell &= \frac{1256810309}{7500000000}, \\ x &= \frac{2204842477}{6000000000}, & y &= \frac{13948546379}{30000000000}, \\ \kappa_* &= \frac{3747729897}{5000000000}, & d_s &= \frac{348713659475}{442459138146}, \\ d_k &= \frac{1743568297375000}{2181624163581537}, & d_+ &= \frac{4000504809}{5000000000}. \end{aligned} \tag{NR12}$$

The fixed return reserve remains $r_0 = 10^{-9}$. The complete finite return certificate is printed in Certificate B. It contains **224 closed rectangles and 2,688 exact degree-$(2,3)$ Bernstein coefficients**. The least return coefficient is

$$\frac{1484520979921542150931518129915517660 09}{5186216782535828843621832223258312704000000000000} > 0. \tag{NR13}$$

All 49 distinct upward admission windows pass. The least corresponding admission coefficient is

$$\frac{3115707242734356667079458895336944054522553048606 9}{2356498276148164678960068941578240000000000000000000000} > 0. \tag{NR14}$$

All 27 separately listed domain and complementary-sector reserves pass. Their least value is $29103/5000000000 > 0$. The independently written return replay reproduces every coefficient, all admission and complement checks, and the complete cover. It also checks the exact selected source IDs and coefficients against the frozen original and new source-node bytes.

Consequently Sections 1–5 apply with every premise discharged by the retained analytic source theorems and the new finite certificates. This yields

$$\boxed{L(s,\chi) \neq 0 \qquad \big(\Re s > 1 - 0.1252270103\big)} \tag{NR15}$$

for every ordinary presented Dirichlet character of every fixed modulus, excluding the principal pole.

Certificate A: first return

Every box is an exact rational rectangle. The return reserve is 10^{-9}. Each final column is an integer $m > 0$ such that every relevant Bernstein coefficient is at least $m/10^{20}$. They are obtained by taking the exact rational minimum and rounding down after multiplication by 10^{20}. The return columns cover all twelve tensor Bernstein coefficients; the admission columns cover the full stated upward admission windows. The formulas in the proof reconstruct the coefficients exactly.

Edge	Count branch	Closed rectangle	Admission rectangle	Return m	Admission m
old_small	old	[0,3/8] × [0,1/2]	[0,3/8] × [0,1/2]	40873814386024128103	123362466039334736067
old_small	old	[3/8,3/4] × [0,1/4]	[3/8,3/4] × [0,1/4]	17590847917074339222	12105623301966736803 3
old_small	old	[3/8,9/16] × [1/4,3/8]	[3/8,9/16] × [1/4,3/8]	28270862054230285039	12064731668367238483 9
old_small	old	[3/8,15/32] × [3/8,7/16]	[3/8,15/32] × [3/8,7/16]	34302914415579039286	12171209779431781024 9
old_small	C cap	[3/8,27/64] × [7/16,1/2]	[3/8,27/64] × [7/16,1/2]	44116681875863620133	531604393186981245
old_small	old	[27/64,15/32] × [7/16,15/32]	[27/64,15/32] × [7/16,15/32]	325519199449397710 53	116507607020608293293
old_small	P cap	[27/64,57/128] × [15/32,31/64]	[27/64,57/128] × [15/32,1/2]	41870292187057731281	68107769803543896
old_small	C cap	[27/64,111/256] × [31/64,1/2]	[27/64,111/256] × [31/64,1/2]	42608184187103693445	122895500742614776
old_small	C−8	[111/256,57/128] × [31/64,63/128]	[111/256,57/128] × [31/64,1/2]	30492009883828844196	27777772562329852
old_small	C−6	[111/256,57/128] × [63/128,1/2]	[111/256,57/128] × [63/128,1/2]	30029998137128845966	34107128284628607
old_small	C−2	[57/128,15/32] × [15/32,31/64]	[57/128,15/32] × [15/32,1/2]	28745214534713404146	15320176019205513
old_small	C	[57/128,117/256] × [31/64,63/128]	[57/128,117/256] × [31/64,1/2]	29385559118605591965	80103371206417244
old_small	C	[57/128,231/512] × [63/128,127/256]	[57/128,231/512] × [63/128,1/2]	29707278448298907469	147379288065556711
old_small	C	[57/128,459/1024] × [127/256,255/512]	[57/128,459/1024] × [127/256,1/2]	29868519807653102873	180881202355114296
old_small	C	[57/128,459/1024] × [255/512,1/2]	[57/128,459/1024] × [255/512,1/2]	29753929212055609748	197596768635511027
old_small	C	[459/1024,231/512] × [127/256,1/2]	[459/1024,231/512] × [127/256,1/2]	29479029046023550650	258943527185101606
old_small	C	[231/512,117/256] × [63/128,127/256]	[231/512,117/256] × [63/128,1/2]	29158716931899743684	306404113244036075
old_small	C	[231/512,465/1024] × [127/256,1/2]	[231/512,465/1024] × [127/256,1/2]	29205312831208450746	337005852015088916
old_small	C	[465/1024,117/256] × [127/256,1/2]	[465/1024,117/256] × [127/256,1/2]	28932780567610310038	415068176845076226
old_small	C	[117/256,15/32] × [31/64,63/128]	[117/256,15/32] × [31/64,1/2]	28298019453127295009	363020815605665393
old_small	C	[117/256,237/512] × [63/128,1/2]	[117/256,237/512] × [63/128,1/2]	28391267894064906204	414923613463534945
old_small	C	[237/512,477/1024] × [63/128,127/256]	[237/512,477/1024] × [63/128,1/2]	28344751659473399714	389150823895527196
old_small	C	[237/512,951/2048] × [127/256,1/2]	[237/512,951/2048] × [127/256,1/2]	28256629695189154742	402081871019762827
old_small	C	[951/2048,477/1024] × [127/256,1/2]	[951/2048,477/1024] × [127/256,1/2]	28122287484117643078	389150823895527196
old_small	C	[477/1024,15/32] × [63/128,1/2]	[477/1024,15/32] × [63/128,1/2]	27854491025387339148	363020815605665393
old_small	old	[15/32,9/16] × [3/8,7/16]	[15/32,9/16] × [3/8,7/16]	25092781519543112955	109831541150750948025
old_small	old	[15/32,33/64] × [7/16,15/32]	[15/32,33/64] × [7/16,15/32]	27907061406207985096	110538916185652823905
old_small	C	[15/32,63/128] × [15/32,31/64]	[15/32,63/128] × [15/32,1/2]	26608475023959556265	141120875300025068
old_small	C	[15/32,123/256] × [31/64,63/128]	[15/32,123/256] × [31/64,1/2]	27229411138682959706	254928595227677654
old_small	C	[15/32,123/256] × [63/128,1/2]	[15/32,123/256] × [63/128,1/2]	26795144702635715371	254928595227677654
old_small	P	[123/256,63/128] × [31/64,63/128]	[123/256,63/128] × [31/64,1/2]	26554539373400198399	142753022046347773
old_small	P	[123/256,63/128] × [63/128,1/2]	[123/256,63/128] × [63/128,1/2]	26132101915020828985	142753022046347773
old_small	old	[63/128,33/64] × [15/32,1/2]	[63/128,33/64] × [15/32,1/2]	26325678392398377122	105395584656020067221
old_small	old	[33/64,9/16] × [7/16,1/2]	[33/64,9/16] × [7/16,1/2]	22128603995491202041	99406624194915177606
old_small	old	[9/16,3/4] × [1/4,1/2]	[9/16,3/4] × [1/4,1/2]	8601827746036211283	75450782350495619144
old_peak	old	[0,3/8] × [0,1/2]	[0,3/8] × [0,1/2]	18277455069944998681	123362466039334736067
old_peak	old	[3/8,3/4] × [0,1/4]	[3/8,3/4] × [0,1/4]	11889726150104728332	12105623301966736803 3
old_peak	old	[3/8,9/16] × [1/4,3/8]	[3/8,9/16] × [1/4,3/8]	13632257776562979467	12064731668367238483 9

Edge	Count branch	Closed rectangle	Admission rectangle	Return m	Admission m
old_peak	old	[3/8,15/32] × [3/8,7/16]	[3/8,15/32] × [3/8,7/16]	15655321522093045771	121712097794317810249
old_peak	C cap	[3/8,27/64] × [7/16,1/2]	[3/8,27/64] × [7/16,1/2]	19913442788328377422	531604393186981245
old_peak	old	[27/64,15/32] × [7/16,15/32]	[27/64,15/32] × [7/16,15/32]	14866559834800373884	116507607020608293293
old_peak	P cap	[27/64,57/128] × [15/32,31/64]	[27/64,57/128] × [15/32,1/2]	19001824335582330328	68107769803543896
old_peak	C cap	[27/64,111/256] × [31/64,1/2]	[27/64,111/256] × [31/64,1/2]	19289612423879006847	122895500742614776
old_peak	C−8	[111/256,57/128] × [31/64,63/128]	[111/256,57/128] × [31/64,1/2]	13842076339919384600	27777772562329852
old_peak	C−6	[111/256,57/128] × [63/128,1/2]	[111/256,57/128] × [63/128,1/2]	13637174400778430988	34107128284628607
old_peak	C−2	[57/128,15/32] × [15/32,31/64]	[57/128,15/32] × [15/32,1/2]	13144257876251855991	15320176019205513
old_peak	C	[57/128,117/256] × [31/64,63/128]	[57/128,117/256] × [31/64,1/2]	13388580423572225960	80103371206417244
old_peak	C	[57/128,231/512] × [63/128,127/256]	[57/128,231/512] × [63/128,1/2]	13512833302749104266	147379288065556711
old_peak	C	[57/128,459/1024] × [127/256,255/512]	[57/128,459/1024] × [127/256,1/2]	13575524526894962712	180881202355114296
old_peak	C	[57/128,459/1024] × [255/512,1/2]	[57/128,459/1024] × [255/512,1/2]	13524529164682455810	197596768635511027
old_peak	C	[459/1024,231/512] × [127/256,1/2]	[459/1024,231/512] × [127/256,1/2]	13411108195329234904	258943527185101606
old_peak	C	[231/512,117/256] × [63/128,127/256]	[231/512,117/256] × [63/128,1/2]	13287158839686193396	306404113244036075
old_peak	C	[231/512,465/1024] × [127/256,1/2]	[231/512,465/1024] × [127/256,1/2]	13298323182710100604	337005852015088916
old_peak	C	[465/1024,117/256] × [127/256,1/2]	[465/1024,117/256] × [127/256,1/2]	13186174126825052912	415068176845076226
old_peak	C	[117/256,15/32] × [31/64,63/128]	[117/256,15/32] × [31/64,1/2]	12943434385137617673	363020815605665393
old_peak	C	[117/256,237/512] × [63/128,1/2]	[117/256,237/512] × [63/128,1/2]	12963783885257217347	414923613463534945
old_peak	C	[237/512,477/1024] × [63/128,127/256]	[237/512,477/1024] × [63/128,1/2]	12953430046468949367	389150823895527196
old_peak	C	[237/512,951/2048] × [127/256,1/2]	[237/512,951/2048] × [127/256,1/2]	12908583797824062585	402081871019762827
old_peak	C	[951/2048,477/1024] × [127/256,1/2]	[951/2048,477/1024] × [127/256,1/2]	12853542699574429474	389150823895527196
old_peak	C	[477/1024,15/32] × [63/128,1/2]	[477/1024,15/32] × [63/128,1/2]	12743937470625728209	363020815605665393
old_peak	old	[15/32,9/16] × [3/8,7/16]	[15/32,9/16] × [3/8,7/16]	12074214736544987471	109831541150750948025
old_peak	old	[15/32,33/64] × [7/16,15/32]	[15/32,33/64] × [7/16,15/32]	13013661757618144600	110538916185652823905
old_peak	C	[15/32,63/128] × [15/32,31/64]	[15/32,63/128] × [15/32,1/2]	12279554753426154434	141120875300025068
old_peak	C	[15/32,123/256] × [31/64,63/128]	[15/32,123/256] × [31/64,1/2]	12508520018402833801	254928595227677654
old_peak	C	[15/32,123/256] × [63/128,1/2]	[15/32,123/256] × [63/128,1/2]	12311876122171789214	254928595227677654
old_peak	P	[123/256,63/128] × [31/64,63/128]	[123/256,63/128] × [31/64,1/2]	12256967301680072559	142753022046347773
old_peak	P	[123/256,63/128] × [63/128,1/2]	[123/256,63/128] × [63/128,1/2]	12064354085899303839	142753022046347773
old_peak	old	[63/128,33/64] × [15/32,1/2]	[63/128,33/64] × [15/32,1/2]	12280472833569233102	105395584656020067221
old_peak	old	[33/64,9/16] × [7/16,1/2]	[33/64,9/16] × [7/16,1/2]	10635809937954412072	99406624194915177606
old_peak	old	[9/16,3/4] × [1/4,1/2]	[9/16,3/4] × [1/4,1/2]	5828814271380632098	75450782350495619144
old_switch	old	[0,3/8] × [0,1/2]	[0,3/8] × [0,1/2]	739624454898048658	123362466039334736067
old_switch	old	[3/8,3/4] × [0,1/4]	[3/8,3/4] × [0,1/4]	1748153127034117240	121056233019667368033
old_switch	old	[3/8,9/16] × [1/4,3/8]	[3/8,9/16] × [1/4,3/8]	899396546538227019	120647316683672384839
old_switch	old	[3/8,15/32] × [3/8,7/16]	[3/8,15/32] × [3/8,7/16]	751958025151612726	121712097794317810249
old_switch	C cap	[3/8,27/64] × [7/16,1/2]	[3/8,27/64] × [7/16,1/2]	710426355085171774	531604393186981245
old_switch	old	[27/64,15/32] × [7/16,15/32]	[27/64,15/32] × [7/16,15/32]	602034778460907232	116507607020608293293

Edge	Count branch	Closed rectangle	Admission rectangle	Return m	Admission m
old_switch	P cap	[27/64,57/128] × [15/32,31/64]	[27/64,57/128] × [15/32,1/2]	710581061345568744	68107769803543896
old_switch	C cap	[27/64,111/256] × [31/64,1/2]	[27/64,111/256] × [31/64,1/2]	674526062702452374	122895500742614776
old_switch	C−8	[111/256,57/128] × [31/64,63/128]	[111/256,57/128] × [31/64,1/2]	494925079560179964	27777772562329852
old_switch	C−6	[111/256,57/128] × [63/128,1/2]	[111/256,57/128] × [63/128,1/2]	466597842932493798	34107128284628607
old_switch	C−2	[57/128,15/32] × [15/32,31/64]	[57/128,15/32] × [15/32,1/2]	497812645802212089	15320176019205513
old_switch	C	[57/128,117/256] × [31/64,63/128]	[57/128,117/256] × [31/64,1/2]	479348427580702569	80103371206417244
old_switch	C	[57/128,231/512] × [63/128,127/256]	[57/128,231/512] × [63/128,1/2]	473084095917446164	147379288065556711
old_switch	C	[57/128,459/1024] × [127/256,255/512]	[57/128,459/1024] × [127/256,1/2]	470769313098502464	180881202355114296
old_switch	C	[57/128,459/1024] × [255/512,1/2]	[57/128,459/1024] × [255/512,1/2]	463506079517154129	197596768635511027
old_switch	C	[459/1024,231/512] × [127/256,1/2]	[459/1024,231/512] × [127/256,1/2]	458345502951417867	258943527185101606
old_switch	C	[231/512,117/256] × [63/128,127/256]	[231/512,117/256] × [63/128,1/2]	464044647054754696	306404113244036075
old_switch	C	[231/512,465/1024] × [127/256,1/2]	[231/512,465/1024] × [127/256,1/2]	453509026272178423	337005852015088916
old_switch	C	[465/1024,117/256] × [127/256,1/2]	[465/1024,117/256] × [127/256,1/2]	448996649479435797	415068176845076226
old_switch	C	[117/256,15/32] × [31/64,63/128]	[117/256,15/32] × [31/64,1/2]	465855438139279499	363020815605665393
old_switch	C	[117/256,237/512] × [63/128,1/2]	[117/256,237/512] × [63/128,1/2]	440944195553440996	414923613463534945
old_switch	C	[237/512,477/1024] × [63/128,127/256]	[237/512,477/1024] × [63/128,1/2]	452936742966036046	389150823895527196
old_switch	C	[237/512,951/2048] × [127/256,1/2]	[237/512,951/2048] × [127/256,1/2]	439133644501002807	402081871019762827
old_switch	C	[951/2048,477/1024] × [127/256,1/2]	[951/2048,477/1024] × [127/256,1/2]	437404118420188822	389150823895527196
old_switch	C	[477/1024,15/32] × [63/128,1/2]	[477/1024,15/32] × [63/128,1/2]	434188141173433465	363020815605665393
old_switch	old	[15/32,9/16] × [3/8,7/16]	[15/32,9/16] × [3/8,7/16]	760609579846061257	109831541150750948025
old_switch	old	[15/32,33/64] × [7/16,15/32]	[15/32,33/64] × [7/16,15/32]	591768615906262789	110538916185652823905
old_switch	C	[15/32,63/128] × [15/32,31/64]	[15/32,63/128] × [15/32,1/2]	489153754522767257	141120875300025068
old_switch	C	[15/32,123/256] × [31/64,63/128]	[15/32,123/256] × [31/64,1/2]	457635883008722770	254928595227677654
old_switch	C	[15/32,123/256] × [63/128,1/2]	[15/32,123/256] × [63/128,1/2]	424565231051380212	254928595227677654
old_switch	P	[123/256,63/128] × [31/64,63/128]	[123/256,63/128] × [31/64,1/2]	461421362255662201	142753022046347773
old_switch	P	[123/256,63/128] × [63/128,1/2]	[123/256,63/128] × [63/128,1/2]	426540903535987638	142753022046347773
old_switch	old	[63/128,33/64] × [15/32,1/2]	[63/128,33/64] × [15/32,1/2]	449220962043543479	105395584656020067221
old_switch	old	[33/64,9/16] × [7/16,1/2]	[33/64,9/16] × [7/16,1/2]	460719422834628608	99406624194915177606
old_switch	old	[9/16,3/4] × [1/4,1/2]	[9/16,3/4] × [1/4,1/2]	553033995060795969	75450782350495619144
new_switch	old	[0,3/8] × [0,1/2]	[0,3/8] × [0,1/2]	739624454898048658	123362466039334736067
new_switch	old	[3/8,3/4] × [0,1/4]	[3/8,3/4] × [0,1/4]	1748153127034117240	121056233019667368033
new_switch	old	[3/8,9/16] × [1/4,3/8]	[3/8,9/16] × [1/4,3/8]	899396546538227019	120647316683672384839
new_switch	old	[3/8,15/32] × [3/8,7/16]	[3/8,15/32] × [3/8,7/16]	751958025151612726	121712097794317810249
new_switch	C cap	[3/8,27/64] × [7/16,1/2]	[3/8,27/64] × [7/16,1/2]	710426355085171774	531604393186981245
new_switch	old	[27/64,15/32] × [7/16,15/32]	[27/64,15/32] × [7/16,15/32]	602034778460907232	116507607020608293293
new_switch	P cap	[27/64,57/128] × [15/32,31/64]	[27/64,57/128] × [15/32,1/2]	710581061345568744	68107769803543896
new_switch	C cap	[27/64,111/256] × [31/64,1/2]	[27/64,111/256] × [31/64,1/2]	674526062702452374	122895500742614776
new_switch	C−8	[111/256,57/128] × [31/64,63/128]	[111/256,57/128] × [31/64,1/2]	494925079560179964	27777772562329852

Edge	Count branch	Closed rectangle	Admission rectangle	Return m	Admission m
new_switch	C−6	[111/256,57/128] × [63/128,1/2]	[111/256,57/128] × [63/128,1/2]	466597842932493798	34107128284628607
new_switch	C−2	[57/128,15/32] × [15/32,31/64]	[57/128,15/32] × [15/32,1/2]	497812645802212089	15320176019205513
new_switch	C	[57/128,117/256] × [31/64,63/128]	[57/128,117/256] × [31/64,1/2]	479348427580702569	80103371206417244
new_switch	C	[57/128,231/512] × [63/128,127/256]	[57/128,231/512] × [63/128,1/2]	473084095917446164	147379288065556711
new_switch	C	[57/128,459/1024] × [127/256,255/512]	[57/128,459/1024] × [127/256,1/2]	470769313098502464	180881202355114296
new_switch	C	[57/128,459/1024] × [255/512,1/2]	[57/128,459/1024] × [255/512,1/2]	463506079517154129	197596768635511027
new_switch	C	[459/1024,231/512] × [127/256,1/2]	[459/1024,231/512] × [127/256,1/2]	458345502951417867	258943527185101606
new_switch	C	[231/512,117/256] × [63/128,127/256]	[231/512,117/256] × [63/128,1/2]	464044647054754696	306404113244036075
new_switch	C	[231/512,465/1024] × [127/256,1/2]	[231/512,465/1024] × [127/256,1/2]	453509026272178423	337005852015088916
new_switch	C	[465/1024,117/256] × [127/256,1/2]	[465/1024,117/256] × [127/256,1/2]	448996649479435797	415068176845076226
new_switch	C	[117/256,15/32] × [31/64,63/128]	[117/256,15/32] × [31/64,1/2]	465855438139279499	363020815605665393
new_switch	C	[117/256,237/512] × [63/128,1/2]	[117/256,237/512] × [63/128,1/2]	440944195553440996	414923613463534945
new_switch	C	[237/512,477/1024] × [63/128,127/256]	[237/512,477/1024] × [63/128,1/2]	452936742966036046	389150823895527196
new_switch	C	[237/512,951/2048] × [127/256,1/2]	[237/512,951/2048] × [127/256,1/2]	439133644501002807	402081871019762827
new_switch	C	[951/2048,477/1024] × [127/256,1/2]	[951/2048,477/1024] × [127/256,1/2]	437404118420188822	389150823895527196
new_switch	C	[477/1024,15/32] × [63/128,1/2]	[477/1024,15/32] × [63/128,1/2]	434188141173433465	363020815605665393
new_switch	old	[15/32,9/16] × [3/8,7/16]	[15/32,9/16] × [3/8,7/16]	760609579846061257	109831541150750948025
new_switch	old	[15/32,33/64] × [7/16,15/32]	[15/32,33/64] × [7/16,15/32]	591768615906262789	110538916185652823905
new_switch	C	[15/32,63/128] × [15/32,31/64]	[15/32,63/128] × [15/32,1/2]	489153754522767257	141120875300025068
new_switch	C	[15/32,123/256] × [31/64,63/128]	[15/32,123/256] × [31/64,1/2]	457635883008722770	254928595227677654
new_switch	C	[15/32,123/256] × [63/128,1/2]	[15/32,123/256] × [63/128,1/2]	424565231051380212	254928595227677654
new_switch	P	[123/256,63/128] × [31/64,63/128]	[123/256,63/128] × [31/64,1/2]	461421362255662201	142753022046347773
new_switch	P	[123/256,63/128] × [63/128,1/2]	[123/256,63/128] × [63/128,1/2]	426540903535987638	142753022046347773
new_switch	old	[63/128,33/64] × [15/32,1/2]	[63/128,33/64] × [15/32,1/2]	449220962043543479	105395584656020067221
new_switch	old	[33/64,9/16] × [7/16,1/2]	[33/64,9/16] × [7/16,1/2]	460719422834628608	99406624194915177606
new_switch	old	[9/16,3/4] × [1/4,1/2]	[9/16,3/4] × [1/4,1/2]	553033995060795969	75450782350495619144
new_top	old	[0,3/8] × [0,1/2]	[0,3/8] × [0,1/2]	109369076639592191	123362466039334736067
new_top	old	[3/8,3/4] × [0,1/4]	[3/8,3/4] × [0,1/4]	1023810749718844068	121056233019667368033
new_top	old	[3/8,9/16] × [1/4,3/8]	[3/8,9/16] × [1/4,3/8]	326621326507867727	120647316683672384839
new_top	old	[3/8,15/32] × [3/8,7/16]	[3/8,15/32] × [3/8,7/16]	147442843230363382	121712097794317810249
new_top	C cap	[3/8,27/64] × [7/16,1/2]	[3/8,27/64] × [7/16,1/2]	27553311860991508	531604393186981245
new_top	old	[27/64,15/32] × [7/16,15/32]	[27/64,15/32] × [7/16,15/32]	71848895617744116	116507607020608293293
new_top	P cap	[27/64,57/128] × [15/32,31/64]	[27/64,57/128] × [15/32,1/2]	61348123156982969	68107769803543896
new_top	C cap	[27/64,111/256] × [31/64,1/2]	[27/64,111/256] × [31/64,1/2]	14486891073591225	122895500742614776
new_top	C−8	[111/256,57/128] × [31/64,63/128]	[111/256,57/128] × [31/64,1/2]	22606411227953467	27777772562329852
new_top	C−6	[111/256,57/128] × [63/128,1/2]	[111/256,57/128] × [63/128,1/2]	1021792895269533	34107128284628607
new_top	C−2	[57/128,15/32] × [15/32,31/64]	[57/128,15/32] × [15/32,1/2]	45347768718479841	15320176019205513
new_top	C	[57/128,117/256] × [31/64,63/128]	[57/128,117/256] × [31/64,1/2]	23635326152476767	80103371206417244

Edge	Count branch	Closed rectangle	Admission rectangle	Return m	Admission m
new_top	C	[57/128,231/512] × [63/128,127/256]	[57/128,231/512] × [63/128,1/2]	12609479701838613	147379288065556711
new_top	C	[57/128,459/1024] × [127/256,255/512]	[57/128,459/1024] × [127/256,1/2]	7760158063583983	180881202355114296
new_top	C	[57/128,459/1024] × [255/512,1/2]	[57/128,459/1024] × [255/512,1/2]	2165751094174884	197596768635511027
new_top	C	[459/1024,231/512] × [127/256,1/2]	[459/1024,231/512] × [127/256,1/2]	1190692276089324	258943527185101606
new_top	C	[231/512,117/256] × [63/128,127/256]	[231/512,117/256] × [63/128,1/2]	11942240537120626	306404113244036075
new_top	C	[231/512,465/1024] × [127/256,1/2]	[231/512,465/1024] × [127/256,1/2]	526565458002377	337005852015088916
new_top	C	[465/1024,117/256] × [127/256,1/2]	[465/1024,117/256] × [127/256,1/2]	173370639914043	415068176845076226
new_top	C	[117/256,15/32] × [31/64,63/128]	[117/256,15/32] × [31/64,1/2]	23957225267316253	363020815605665393
new_top	C	[117/256,237/512] × [63/128,127/256]	[117/256,237/512] × [63/128,1/2]	11922335465092431	414923613463534945
new_top	C	[117/256,471/1024] × [127/256,1/2]	[117/256,471/1024] × [127/256,1/2]	74506230869529	440339184309688642
new_top	C	[471/1024,237/512] × [127/256,1/2]	[471/1024,237/512] × [127/256,1/2]	131107821824321	414923613463534945
new_top	C	[237/512,477/1024] × [63/128,127/256]	[237/512,477/1024] × [63/128,1/2]	12529859413725835	389150823895527196
new_top	C	[237/512,951/2048] × [127/256,1/2]	[237/512,951/2048] × [127/256,1/2]	399777003733211	402081871019762827
new_top	C	[951/2048,477/1024] × [127/256,1/2]	[951/2048,477/1024] × [127/256,1/2]	650711094687136	389150823895527196
new_top	C	[477/1024,15/32] × [63/128,1/2]	[477/1024,15/32] × [63/128,1/2]	979378185640714	363020815605665393
new_top	old	[15/32,9/16] × [3/8,7/16]	[15/32,9/16] × [3/8,7/16]	232626372055900899	109831541150750948025
new_top	old	[15/32,33/64] × [7/16,15/32]	[15/32,33/64] × [7/16,15/32]	98714119370831098	110538916185652823905
new_top	C	[15/32,63/128] × [15/32,31/64]	[15/32,63/128] × [15/32,1/2]	53435901991439195	141120875300025068
new_top	C	[15/32,123/256] × [31/64,63/128]	[15/32,123/256] × [31/64,1/2]	27133148710935140	254928595227677654
new_top	C	[15/32,123/256] × [63/128,1/2]	[15/32,123/256] × [63/128,1/2]	1869911367546830	254928595227677654
new_top	P	[123/256,63/128] × [31/64,63/128]	[123/256,63/128] × [31/64,1/2]	35884928384740274	142753022046347773
new_top	P	[123/256,63/128] × [63/128,1/2]	[123/256,63/128] × [63/128,1/2]	8711613097965182	142753022046347773
new_top	old	[63/128,33/64] × [15/32,1/2]	[63/128,33/64] × [15/32,1/2]	5754537025164573	105395584656020067221
new_top	old	[33/64,9/16] × [7/16,1/2]	[33/64,9/16] × [7/16,1/2]	51831368033545611	99406624194915177606
new_top	old	[9/16,3/4] × [1/4,1/2]	[9/16,3/4] × [1/4,1/2]	210784768981574248	75450782350495619144
new_kink	old	[0,3/8] × [0,1/2]	[0,3/8] × [0,1/2]	181592904033352050	123362466039334736067
new_kink	old	[3/8,3/4] × [0,1/4]	[3/8,3/4] × [0,1/4]	1120186392586475751	121056233019667368033
new_kink	old	[3/8,9/16] × [1/4,3/8]	[3/8,9/16] × [1/4,3/8]	392800004352350779	120647316683672384839
new_kink	old	[3/8,15/32] × [3/8,7/16]	[3/8,15/32] × [3/8,7/16]	217022037819421625	121712097794317810249
new_kink	C cap	[3/8,27/64] × [7/16,1/2]	[3/8,27/64] × [7/16,1/2]	106934917215358201	531604393186981245
new_kink	old	[27/64,15/32] × [7/16,15/32]	[27/64,15/32] × [7/16,15/32]	134829437341822126	116507607020608293293
new_kink	P cap	[27/64,57/128] × [15/32,31/64]	[27/64,57/128] × [15/32,1/2]	138336411872302526	68107769803543896
new_kink	C cap	[27/64,111/256] × [31/64,1/2]	[27/64,111/256] × [31/64,1/2]	91518184525248395	122895500742614776
new_kink	C−8	[111/256,57/128] × [31/64,63/128]	[111/256,57/128] × [31/64,1/2]	77877232491809431	27777772562329852
new_kink	C−6	[111/256,57/128] × [63/128,1/2]	[111/256,57/128] × [63/128,1/2]	55581952573870872	34107128284628607
new_kink	C−2	[57/128,15/32] × [15/32,31/64]	[57/128,15/32] × [15/32,1/2]	99551439930279097	15320176019205513
new_kink	C	[57/128,117/256] × [31/64,63/128]	[57/128,117/256] × [31/64,1/2]	77474079878642407	80103371206417244
new_kink	C	[57/128,231/512] × [63/128,127/256]	[57/128,231/512] × [63/128,1/2]	66647491104631053	147379288065556711

Edge	Count branch	Closed rectangle	Admission rectangle	Return m	Admission m
new_kink	C	[57/128,459/1024] × [127/256,255/512]	[57/128,459/1024] × [127/256,1/2]	62057628253163121	180881202355114296
new_kink	C	[57/128,459/1024] × [255/512,1/2]	[57/128,459/1024] × [255/512,1/2]	56287165992866994	197596768635511027
new_kink	C	[459/1024,231/512] × [127/256,1/2]	[459/1024,231/512] × [127/256,1/2]	54878532299597932	258943527185101606
new_kink	C	[231/512,117/256] × [63/128,127/256]	[231/512,117/256] × [63/128,1/2]	65112509960733309	306404113244036075
new_kink	C	[231/512,465/1024] × [127/256,1/2]	[231/512,465/1024] × [127/256,1/2]	53782033392433655	337005852015088916
new_kink	C	[465/1024,117/256] × [127/256,1/2]	[465/1024,117/256] × [127/256,1/2]	52997669271374164	415068176845076226
new_kink	C	[117/256,15/32] × [31/64,63/128]	[117/256,15/32] × [31/64,1/2]	76926422367755772	363020815605665393
new_kink	C	[117/256,237/512] × [63/128,1/2]	[117/256,237/512] × [63/128,1/2]	52365345387569536	414923613463534945
new_kink	C	[237/512,477/1024] × [63/128,127/256]	[237/512,477/1024] × [63/128,1/2]	64837183902515143	389150823895527196
new_kink	C	[237/512,951/2048] × [127/256,1/2]	[237/512,951/2048] × [127/256,1/2]	52364338598620154	402081871019762827
new_kink	C	[951/2048,477/1024] × [127/256,1/2]	[951/2048,477/1024] × [127/256,1/2]	52402348657933870	389150823895527196
new_kink	C	[477/1024,15/32] × [63/128,1/2]	[477/1024,15/32] × [63/128,1/2]	52517385624824400	363020815605665393
new_kink	old	[15/32,9/16] × [3/8,7/16]	[15/32,9/16] × [3/8,7/16]	294268238527765384	109831541150750948025
new_kink	old	[15/32,33/64] × [7/16,15/32]	[15/32,33/64] × [7/16,15/32]	157844541847298144	110538916185652823905
new_kink	C	[15/32,63/128] × [15/32,31/64]	[15/32,63/128] × [15/32,1/2]	105901268467209165	141120875300025068
new_kink	C	[15/32,123/256] × [31/64,63/128]	[15/32,123/256] × [31/64,1/2]	78920455271773645	254928595227677654
new_kink	C	[15/32,123/256] × [63/128,1/2]	[15/32,123/256] × [63/128,1/2]	52981560648184049	254928595227677654
new_kink	P	[123/256,63/128] × [31/64,63/128]	[123/256,63/128] × [31/64,1/2]	86675696650692041	142753022046347773
new_kink	P	[123/256,63/128] × [63/128,1/2]	[123/256,63/128] × [63/128,1/2]	58850824478883091	142753022046347773
new_kink	old	[63/128,33/64] × [15/32,1/2]	[63/128,33/64] × [15/32,1/2]	58701347952611842	105395584656020067221
new_kink	old	[33/64,9/16] × [7/16,1/2]	[33/64,9/16] × [7/16,1/2]	101167080878710445	99406624194915177606
new_kink	old	[9/16,3/4] × [1/4,1/2]	[9/16,3/4] × [1/4,1/2]	253142590873154295	75450782350495619144

Domain and complementary reserves

Reserve	Exact value
Gram_x_above_ell	200263/1000000
Gram_y_above_x	120981/1250000
Gram_ell_below_one_fifth	243277/7500000
Gram_2x_above_y_plus_ell	517391/5000000
conductor_small_before_peak	6894163/15000000
conductor_peak_before_switch	357216457432849/1106149155000000
conductor_switch_before_kink	23351430329090437525/2145577552307689580863
conductor_kink_before_top	1139641491908135935457/872856878454570000000000
conductor_top_before_twoy	3884541997/30000000000
physical_plain_lower	2513421/250000000
physical_plain_upper	4427617241014997/239921100030000000
physical_width_lower	313627054406993802026834179 3/798635417420822412782070000000
physical_width_upper	29999/10000000
small_width_shortening	937862173149997/239921100030000000
short_plain_shortening	95224281819/50000000000000
large_shift	30434281819/450000000000000
large_shift_clipped	578856732121181/1664370948034500
small_shift_cost	578856732121181/1664370948034500
prime_supply	1142349997/119960550015
capacity_domain	3884541997/60000000000
incumbent	4301/50000000
feedback_kappa	247759/5000000
feedback_d	2002629999/10000000000
feedback_y	530837/15000000
floor	13165423799/3000000000000
small_row	5607653/150000000
principal_w	63542419/3000000000
principal_z	16579/200000000
holomorphic_box	247759/10000000
source_lambda	328741896069/1000000000000
source_intercept	3065949543/100000000000
source_C_cap	10998999999/1000000000000
source_P_cap	11998999999/1000000000000

Certificate B: final return

Every box is an exact rational rectangle. The return reserve is 10^{-9}. Each final column is an integer $m > 0$ such that every relevant Bernstein coefficient is at least $m/10^{20}$. They are obtained by taking the exact rational minimum and rounding down after multiplication by 10^{20}. The return columns cover all twelve tensor Bernstein coefficients; the admission columns cover the full stated upward admission windows. The formulas in the proof reconstruct the coefficients exactly.

Edge	Count branch	Closed rectangle	Admission rectangle	Return m	Admission m
old_small	old	[0,3/8] × [0,1/2]	[0,3/8] × [0,1/2]	40887061319233500877	123362243007413058271
old_small	old	[3/8,3/4] × [0,1/4]	[3/8,3/4] × [0,1/4]	17603903515501718818	12105612150370652913 5
old_small	old	[3/8,9/16] × [1/4,3/8]	[3/8,9/16] × [1/4,3/8]	28283819022707562814	120647155480156414093
old_small	old	[3/8,15/32] × [3/8,7/16]	[3/8,15/32] × [3/8,7/16]	34315978638693389180	121711907953008468828
old_small	C cap	[3/8,27/64] × [7/16,1/2]	[3/8,27/64] × [7/16,1/2]	44132305232726009312	604579476895132909
old_small	old	[27/64,15/32] × [7/16,15/32]	[27/64,15/32] × [7/16,15/32]	32564401383144314661	116507408361725183424
old_small	P cap	[27/64,57/128] × [15/32,31/64]	[27/64,57/128] × [15/32,1/2]	41885744118342349528	146530341871907977
old_small	C cap	[27/64,111/256] × [31/64,1/2]	[27/64,111/256] × [31/64,1/2]	42623608441941968359	194986236488414512
old_small	P cap	[111/256,57/128] × [31/64,63/128]	[111/256,57/128] × [31/64,1/2]	41511698873405546488	146530341871907977
old_small	C−7	[111/256,57/128] × [63/128,1/2]	[111/256,57/128] × [63/128,1/2]	29862363098556564528	13221767544965760
old_small	old	[57/128,15/32] × [15/32,31/64]	[57/128,15/32] × [15/32,1/2]	31706646430694825289	111384338273004045977
old_small	C	[57/128,117/256] × [31/64,63/128]	[57/128,117/256] × [31/64,1/2]	29225115229848514739	24808346629254572
old_small	C	[57/128,231/512] × [63/128,127/256]	[57/128,231/512] × [63/128,1/2]	29543270273718213529	92612169784935170
old_small	C	[57/128,459/1024] × [127/256,255/512]	[57/128,459/1024] × [127/256,1/2]	29702707145488907963	126369328644933867
old_small	C	[57/128,459/1024] × [255/512,1/2]	[57/128,459/1024] × [255/512,1/2]	29587728851625143167	143210327716711532
old_small	C	[459/1024,231/512] × [127/256,1/2]	[459/1024,231/512] × [127/256,1/2]	29314255533008772089	204182800597629396
old_small	C	[231/512,117/256] × [63/128,127/256]	[231/512,117/256] × [63/128,1/2]	28997522573300093619	251135132414239857
old_small	C	[231/512,465/1024] × [127/256,1/2]	[231/512,465/1024] × [127/256,1/2]	29041961152402101415	281996272550324926
old_small	C	[465/1024,117/256] × [127/256,1/2]	[465/1024,117/256] × [127/256,1/2]	28770845709805131146	359809744503020456
old_small	C	[117/256,15/32] × [31/64,63/128]	[117/256,15/32] × [31/64,1/2]	28143095367091329090	353578373688634072
old_small	C	[117/256,237/512] × [63/128,1/2]	[117/256,237/512] × [63/128,1/2]	28232151638640291819	409658095043544544
old_small	C	[237/512,477/1024] × [63/128,127/256]	[237/512,477/1024] × [63/128,1/2]	28187741051779472708	386886890908367724
old_small	C	[237/512,951/2048] × [127/256,1/2]	[237/512,951/2048] × [127/256,1/2]	28098214957105144739	399721852629315600
old_small	C	[951/2048,477/1024] × [127/256,1/2]	[951/2048,477/1024] × [127/256,1/2]	27964573010072422761	386886890908367724
old_small	C	[477/1024,15/32] × [63/128,1/2]	[477/1024,15/32] × [63/128,1/2]	27698173319514254107	360951006826142837
old_small	old	[15/32,9/16] × [3/8,7/16]	[15/32,9/16] × [3/8,7/16]	25104525834969131810	109831363703226568790
old_small	old	[15/32,33/64] × [7/16,15/32]	[15/32,33/64] × [7/16,15/32]	27918879457794274792	110538724640549347943
old_small	P	[15/32,63/128] × [15/32,31/64]	[15/32,63/128] × [15/32,1/2]	26835669997748067546	142153051589312834
old_small	C	[15/32,123/256] × [31/64,63/128]	[15/32,123/256] × [31/64,1/2]	27079929302380768310	253661328626188160
old_small	C	[15/32,123/256] × [63/128,1/2]	[15/32,123/256] × [63/128,1/2]	26644363937378583533	253661328626188160
old_small	P	[123/256,63/128] × [31/64,63/128]	[123/256,63/128] × [31/64,1/2]	26408317278375737602	142153051589312834
old_small	P	[123/256,63/128] × [63/128,1/2]	[123/256,63/128] × [63/128,1/2]	25984658876260653082	142153051589312834
old_small	old	[63/128,33/64] × [15/32,1/2]	[63/128,33/64] × [15/32,1/2]	26336923255151751698	105395385905799539830
old_small	old	[33/64,9/16] × [7/16,1/2]	[33/64,9/16] × [7/16,1/2]	22139187152437014855	99406433538595033683
old_small	old	[9/16,3/4] × [1/4,1/2]	[9/16,3/4] × [1/4,1/2]	8609792334807298563	75450624069777009095
old_peak	old	[0,3/8] × [0,1/2]	[0,3/8] × [0,1/2]	18274003827611117915	123362243007413058271
old_peak	old	[3/8,3/4] × [0,1/4]	[3/8,3/4] × [0,1/4]	11898570223167192751	12105612150370652913 5
old_peak	old	[3/8,9/16] × [1/4,3/8]	[3/8,9/16] × [1/4,3/8]	13634394387192406617	120647155480156414093

Edge	Count branch	Closed rectangle	Admission rectangle	Return m	Admission m
old_peak	old	[3/8,15/32] × [3/8,7/16]	[3/8,15/32] × [3/8,7/16]	15654603374564944556	121711907953008468828
old_peak	C cap	[3/8,27/64] × [7/16,1/2]	[3/8,27/64] × [7/16,1/2]	19911138026267273121	604579476895132909
old_peak	old	[27/64,15/32] × [7/16,15/32]	[27/64,15/32] × [7/16,15/32]	14865973465382867637	116507408361725183424
old_peak	P cap	[27/64,57/128] × [15/32,31/64]	[27/64,57/128] × [15/32,1/2]	19000337670271918311	146530341871907977
old_peak	C cap	[27/64,111/256] × [31/64,1/2]	[27/64,111/256] × [31/64,1/2]	19287764491653934376	194986236488414512
old_peak	P cap	[111/256,57/128] × [31/64,63/128]	[111/256,57/128] × [31/64,1/2]	18837085876873785300	146530341871907977
old_peak	C−7	[111/256,57/128] × [63/128,1/2]	[111/256,57/128] × [63/128,1/2]	13555092599966017004	13221767544965760
old_peak	old	[57/128,15/32] × [15/32,31/64]	[57/128,15/32] × [15/32,1/2]	14480870549861099295	111384338273004045977
old_peak	C	[57/128,117/256] × [31/64,63/128]	[57/128,117/256] × [31/64,1/2]	13310025110268876744	24808346629254572
old_peak	C	[57/128,231/512] × [63/128,127/256]	[57/128,231/512] × [63/128,1/2]	13432661853118375572	92612169784935170
old_peak	C	[57/128,459/1024] × [127/256,255/512]	[57/128,459/1024] × [127/256,1/2]	13494533348096751673	126369328644933867
old_peak	C	[57/128,459/1024] × [255/512,1/2]	[57/128,459/1024] × [255/512,1/2]	13443378969505267056	143210327716711532
old_peak	C	[459/1024,231/512] × [127/256,1/2]	[459/1024,231/512] × [127/256,1/2]	13330623086327870538	204182800597629396
old_peak	C	[231/512,117/256] × [63/128,127/256]	[231/512,117/256] × [63/128,1/2]	13208296613253161941	251135132414239857
old_peak	C	[231/512,465/1024] × [127/256,1/2]	[231/512,465/1024] × [127/256,1/2]	13218499677507819691	281996272550324926
old_peak	C	[465/1024,117/256] × [127/256,1/2]	[465/1024,117/256] × [127/256,1/2]	13107008743045114516	359809744503020456
old_peak	C	[117/256,15/32] × [31/64,63/128]	[117/256,15/32] × [31/64,1/2]	12867435576485998769	353578373688634072
old_peak	C	[117/256,237/512] × [63/128,1/2]	[117/256,237/512] × [63/128,1/2]	12885924297191741179	409658095043544544
old_peak	C	[237/512,477/1024] × [63/128,127/256]	[237/512,477/1024] × [63/128,1/2]	12876507587407416696	386886890908367724
old_peak	C	[237/512,951/2048] × [127/256,1/2]	[237/512,951/2048] × [127/256,1/2]	12831048482201738889	399721852629315600
old_peak	C	[951/2048,477/1024] × [127/256,1/2]	[951/2048,477/1024] × [127/256,1/2]	12776330785801073017	386886890908367724
old_peak	C	[477/1024,15/32] × [63/128,1/2]	[477/1024,15/32] × [63/128,1/2]	12667369748767750527	360951006826142837
old_peak	old	[15/32,9/16] × [3/8,7/16]	[15/32,9/16] × [3/8,7/16]	12076341646277054322	109831363703226568790
old_peak	old	[15/32,33/64] × [7/16,15/32]	[15/32,33/64] × [7/16,15/32]	13014477490051887625	110538724640549347943
old_peak	P	[15/32,63/128] × [15/32,31/64]	[15/32,63/128] × [15/32,1/2]	12379897404626334451	142153051589312834
old_peak	C	[15/32,123/256] × [31/64,63/128]	[15/32,123/256] × [31/64,1/2]	12435023067888167161	253661328626188160
old_peak	C	[15/32,123/256] × [63/128,1/2]	[15/32,123/256] × [63/128,1/2]	12237850344207917279	253661328626188160
old_peak	P	[123/256,63/128] × [31/64,63/128]	[123/256,63/128] × [31/64,1/2]	12184940524700380438	142153051589312834
old_peak	P	[123/256,63/128] × [63/128,1/2]	[123/256,63/128] × [63/128,1/2]	11991830449282424797	142153051589312834
old_peak	old	[63/128,33/64] × [15/32,1/2]	[63/128,33/64] × [15/32,1/2]	12281344940276200127	105395385905799539830
old_peak	old	[33/64,9/16] × [7/16,1/2]	[33/64,9/16] × [7/16,1/2]	10637908021779109883	99406433538595033683
old_peak	old	[9/16,3/4] × [1/4,1/2]	[9/16,3/4] × [1/4,1/2]	5834740567530160666	75450624069777009095
old_switch	old	[0,3/8] × [0,1/2]	[0,3/8] × [0,1/2]	723288741128724220	123362243007413058271
old_switch	old	[3/8,3/4] × [0,1/4]	[3/8,3/4] × [0,1/4]	1740595683636321691	121056121503706529135
old_switch	old	[3/8,9/16] × [1/4,3/8]	[3/8,9/16] × [1/4,3/8]	886779799933926592	120647155480156414093
old_switch	old	[3/8,15/32] × [3/8,7/16]	[3/8,15/32] × [3/8,7/16]	737316045673225448	121711907953008468828
old_switch	C cap	[3/8,27/64] × [7/16,1/2]	[3/8,27/64] × [7/16,1/2]	693976818429546143	604579476895132909
old_switch	old	[27/64,15/32] × [7/16,15/32]	[27/64,15/32] × [7/16,15/32]	590965415730780769	116507408361725183424

Edge	Count branch	Closed rectangle	Admission rectangle	Return m	Admission m
old_switch	P cap	[27/64,57/128] × [15/32,31/64]	[27/64,57/128] × [15/32,1/2]	695620718933686808	146530341871907977
old_switch	C cap	[27/64,111/256] × [31/64,1/2]	[27/64,111/256] × [31/64,1/2]	658966027871690979	194986236488414512
old_switch	P cap	[111/256,57/128] × [31/64,63/128]	[111/256,57/128] × [31/64,1/2]	660327869946354647	146530341871907977
old_switch	C−7	[111/256,57/128] × [63/128,1/2]	[111/256,57/128] × [63/128,1/2]	453297378481307654	13221767544965760
old_switch	old	[57/128,15/32] × [15/32,31/64]	[57/128,15/32] × [15/32,1/2]	518600646313830761	111384338273004045977
old_switch	C	[57/128,117/256] × [31/64,63/128]	[57/128,117/256] × [31/64,1/2]	466894797626679029	24808346629254572
old_switch	C	[57/128,231/512] × [63/128,127/256]	[57/128,231/512] × [63/128,1/2]	460436386167326001	92612169784935170
old_switch	C	[57/128,459/1024] × [127/256,255/512]	[57/128,459/1024] × [127/256,1/2]	458019771088059074	126369328644933867
old_switch	C	[57/128,459/1024] × [255/512,1/2]	[57/128,459/1024] × [255/512,1/2]	450820219669460011	143210327716711532
old_switch	C	[459/1024,231/512] × [127/256,1/2]	[459/1024,231/512] × [127/256,1/2]	445825717191593493	204182800597629396
old_switch	C	[231/512,117/256] × [63/128,127/256]	[231/512,117/256] × [63/128,1/2]	451720116723922433	251135132414239857
old_switch	C	[231/512,465/1024] × [127/256,1/2]	[231/512,465/1024] × [127/256,1/2]	441152653654294742	281996272550324926
old_switch	C	[465/1024,117/256] × [127/256,1/2]	[465/1024,117/256] × [127/256,1/2]	436801029057563758	359809744503020456
old_switch	C	[117/256,15/32] × [31/64,63/128]	[117/256,15/32] × [31/64,1/2]	454014439676769984	353578373688634072
old_switch	C	[117/256,237/512] × [63/128,1/2]	[117/256,237/512] × [63/128,1/2]	429062096685805090	409658095043544544
old_switch	C	[237/512,477/1024] × [63/128,127/256]	[237/512,477/1024] × [63/128,1/2]	441077064689711535	386886890908367724
old_switch	C	[237/512,951/2048] × [127/256,1/2]	[237/512,951/2048] × [127/256,1/2]	427328262930720278	399721852629315600
old_switch	C	[951/2048,477/1024] × [127/256,1/2]	[951/2048,477/1024] × [127/256,1/2]	425674788910777407	386886890908367724
old_switch	C	[477/1024,15/32] × [63/128,1/2]	[477/1024,15/32] × [63/128,1/2]	422608920076317491	360951006826142837
old_switch	old	[15/32,9/16] × [3/8,7/16]	[15/32,9/16] × [3/8,7/16]	748943617005908766	109831363703226568790
old_switch	old	[15/32,33/64] × [7/16,15/32]	[15/32,33/64] × [7/16,15/32]	582131896216870339	110538724640549347943
old_switch	P	[15/32,63/128] × [15/32,31/64]	[15/32,63/128] × [15/32,1/2]	484360428622673029	142153051589312834
old_switch	C	[15/32,123/256] × [31/64,63/128]	[15/32,123/256] × [31/64,1/2]	446365114287136430	253661328626188160
old_switch	C	[15/32,123/256] × [63/128,1/2]	[15/32,123/256] × [63/128,1/2]	413559834144155494	253661328626188160
old_switch	P	[123/256,63/128] × [31/64,63/128]	[123/256,63/128] × [31/64,1/2]	450404011866024333	142153051589312834
old_switch	P	[123/256,63/128] × [63/128,1/2]	[123/256,63/128] × [63/128,1/2]	416014574654319967	142153051589312834
old_switch	old	[63/128,33/64] × [15/32,1/2]	[63/128,33/64] × [15/32,1/2]	440174648827927657	105395385905799539830
old_switch	old	[33/64,9/16] × [7/16,1/2]	[33/64,9/16] × [7/16,1/2]	452905697618319945	99406433538595033683
old_switch	old	[9/16,3/4] × [1/4,1/2]	[9/16,3/4] × [1/4,1/2]	547721662449656119	75450624069777009095
new_switch	old	[0,3/8] × [0,1/2]	[0,3/8] × [0,1/2]	723288741128724220	123362243007413058271
new_switch	old	[3/8,3/4] × [0,1/4]	[3/8,3/4] × [0,1/4]	1740595683636321691	121056121503706529135
new_switch	old	[3/8,9/16] × [1/4,3/8]	[3/8,9/16] × [1/4,3/8]	886779799933926592	120647155480156414093
new_switch	old	[3/8,15/32] × [3/8,7/16]	[3/8,15/32] × [3/8,7/16]	737316045673225448	121711907953008468828
new_switch	C cap	[3/8,27/64] × [7/16,1/2]	[3/8,27/64] × [7/16,1/2]	693976818429546143	604579476895132909
new_switch	old	[27/64,15/32] × [7/16,15/32]	[27/64,15/32] × [7/16,15/32]	590965415730780769	116507408361725183424
new_switch	P cap	[27/64,57/128] × [15/32,31/64]	[27/64,57/128] × [15/32,1/2]	695620718933686808	146530341871907977
new_switch	C cap	[27/64,111/256] × [31/64,1/2]	[27/64,111/256] × [31/64,1/2]	658966027871690979	194986236488414512
new_switch	P cap	[111/256,57/128] × [31/64,63/128]	[111/256,57/128] × [31/64,1/2]	660327869946354647	146530341871907977

Edge	Count branch	Closed rectangle	Admission rectangle	Return m	Admission m
new_switch	C−7	[111/256,57/128] × [63/128,1/2]	[111/256,57/128] × [63/128,1/2]	453297378481307654	13221767544965760
new_switch	old	[57/128,15/32] × [15/32,31/64]	[57/128,15/32] × [15/32,1/2]	518600646313830761	111384338273004045977
new_switch	C	[57/128,117/256] × [31/64,63/128]	[57/128,117/256] × [31/64,1/2]	466894797626679029	24808346629254572
new_switch	C	[57/128,231/512] × [63/128,127/256]	[57/128,231/512] × [63/128,1/2]	460436386167326001	92612169784935170
new_switch	C	[57/128,459/1024] × [127/256,255/512]	[57/128,459/1024] × [127/256,1/2]	458019771088059074	126369328644933867
new_switch	C	[57/128,459/1024] × [255/512,1/2]	[57/128,459/1024] × [255/512,1/2]	450820219669460011	143210327716711532
new_switch	C	[459/1024,231/512] × [127/256,1/2]	[459/1024,231/512] × [127/256,1/2]	445825717191593493	204182800597629396
new_switch	C	[231/512,117/256] × [63/128,127/256]	[231/512,117/256] × [63/128,1/2]	451720116723922433	251135132414239857
new_switch	C	[231/512,465/1024] × [127/256,1/2]	[231/512,465/1024] × [127/256,1/2]	441152653654294742	281996272550324926
new_switch	C	[465/1024,117/256] × [127/256,1/2]	[465/1024,117/256] × [127/256,1/2]	436801029057563758	359809744503020456
new_switch	C	[117/256,15/32] × [31/64,63/128]	[117/256,15/32] × [31/64,1/2]	454014439676769984	353578373688634072
new_switch	C	[117/256,237/512] × [63/128,1/2]	[117/256,237/512] × [63/128,1/2]	429062096685805090	409658095043544544
new_switch	C	[237/512,477/1024] × [63/128,127/256]	[237/512,477/1024] × [63/128,1/2]	441077064689711535	386886890908367724
new_switch	C	[237/512,951/2048] × [127/256,1/2]	[237/512,951/2048] × [127/256,1/2]	427328262930720278	399721852629315600
new_switch	C	[951/2048,477/1024] × [127/256,1/2]	[951/2048,477/1024] × [127/256,1/2]	425674788910777407	386886890908367724
new_switch	C	[477/1024,15/32] × [63/128,1/2]	[477/1024,15/32] × [63/128,1/2]	422608920076317491	360951006826142837
new_switch	old	[15/32,9/16] × [3/8,7/16]	[15/32,9/16] × [3/8,7/16]	748943617005908766	109831363703226568790
new_switch	old	[15/32,33/64] × [7/16,15/32]	[15/32,33/64] × [7/16,15/32]	582131896216870339	110538724640549347943
new_switch	P	[15/32,63/128] × [15/32,31/64]	[15/32,63/128] × [15/32,1/2]	484360428622673029	142153051589312834
new_switch	C	[15/32,123/256] × [31/64,63/128]	[15/32,123/256] × [31/64,1/2]	446365114287136430	253661328626188160
new_switch	C	[15/32,123/256] × [63/128,1/2]	[15/32,123/256] × [63/128,1/2]	413559834144155494	253661328626188160
new_switch	P	[123/256,63/128] × [31/64,63/128]	[123/256,63/128] × [31/64,1/2]	450404011866024333	142153051589312834
new_switch	P	[123/256,63/128] × [63/128,1/2]	[123/256,63/128] × [63/128,1/2]	416014574654319967	142153051589312834
new_switch	old	[63/128,33/64] × [15/32,1/2]	[63/128,33/64] × [15/32,1/2]	440174648827927657	105395385905799539830
new_switch	old	[33/64,9/16] × [7/16,1/2]	[33/64,9/16] × [7/16,1/2]	452905697618319945	99406433538595033683
new_switch	old	[9/16,3/4] × [1/4,1/2]	[9/16,3/4] × [1/4,1/2]	547721662449656119	75450624069777009095
new_top	old	[0,3/8] × [0,1/2]	[0,3/8] × [0,1/2]	105631986144438732	123362243007413058271
new_top	old	[3/8,3/4] × [0,1/4]	[3/8,3/4] × [0,1/4]	1017101080639320882	121056121503706529135
new_top	old	[3/8,9/16] × [1/4,3/8]	[3/8,9/16] × [1/4,3/8]	325627362747830969	120647155480156414093
new_top	old	[3/8,15/32] × [3/8,7/16]	[3/8,15/32] × [3/8,7/16]	144982458857589082	121711907953008468828
new_top	C cap	[3/8,27/64] × [7/16,1/2]	[3/8,27/64] × [7/16,1/2]	25114157347027893	604579476895132909
new_top	old	[27/64,15/32] × [7/16,15/32]	[27/64,15/32] × [7/16,15/32]	70485754108294095	116507408361725183424
new_top	P cap	[27/64,57/128] × [15/32,31/64]	[27/64,57/128] × [15/32,1/2]	59296130479918706	146530341871907977
new_top	C cap	[27/64,111/256] × [31/64,1/2]	[27/64,111/256] × [31/64,1/2]	12566107999374630	194986236488414512
new_top	P cap	[111/256,57/128] × [31/64,63/128]	[111/256,57/128] × [31/64,1/2]	30924321760494270	146530341871907977
new_top	C−7	[111/256,57/128] × [63/128,1/2]	[111/256,57/128] × [63/128,1/2]	128628323266068	13221767544965760
new_top	old	[57/128,15/32] × [15/32,31/64]	[57/128,15/32] × [15/32,1/2]	27582575736875367	111384338273004045977
new_top	C	[57/128,117/256] × [31/64,63/128]	[57/128,117/256] × [31/64,1/2]	23185445251744143	24808346629254572

Edge	Count branch	Closed rectangle	Admission rectangle	Return m	Admission m
new_top	C	[57/128,231/512] × [63/128,127/256]	[57/128,231/512] × [63/128,1/2]	1223780507379497 2	92612169784935170
new_top	C	[57/128,459/1024] × [127/256,255/512]	[57/128,459/1024] × [127/256,1/2]	7355321993539332	126369328644933867
new_top	C	[57/128,459/1024] × [255/512,1/2]	[57/128,459/1024] × [255/512,1/2]	1801924215430787	143210327716711532
new_top	C	[459/1024,231/512] × [127/256,1/2]	[459/1024,231/512] × [127/256,1/2]	901786560461401	204182800597629396
new_top	C	[231/512,117/256] × [63/128,127/256]	[231/512,117/256] × [63/128,1/2]	11663504298001541	251135132414239857
new_top	C	[231/512,465/1024] × [127/256,1/2]	[231/512,465/1024] × [127/256,1/2]	310130160615080	281996272550324926
new_top	C	[465/1024,117/256] × [127/256,1/2]	[465/1024,117/256] × [127/256,1/2]	26955015891826	359809744503020456
new_top	C	[117/256,15/32] × [31/64,63/128]	[117/256,15/32] × [31/64,1/2]	23642335171470146	353578373688634072
new_top	C	[117/256,237/512] × [63/128,127/256]	[117/256,237/512] × [63/128,1/2]	11711738592676753	409658095043544544
new_top	C	[117/256,471/1024] × [127/256,255/512]	[117/256,471/1024] × [127/256,1/2]	5818345593671389	437623216455715986
new_top	C	[117/256,939/2048] × [255/512,511/1024]	[117/256,939/2048] × [255/512,1/2]	2903953079564266	450175164045343322
new_top	C	[117/256,1875/4096] × [511/1024,1/2]	[117/256,1875/4096] × [511/1024,1/2]	4361425823991	456382092732223362
new_top	C	[1875/4096,939/2048] × [511/1024,1023/2048]	[1875/4096,939/2048] × [511/1024,1/2]	1453353269455186	450175164045343322
new_top	C	[1875/4096,3753/8192] × [1023/2048,1/2]	[1875/4096,3753/8192] × [1023/2048,1/2]	294660207021	453281398812120104
new_top	C	[3753/8192,939/2048] × [1023/2048,2047/4096]	[3753/8192,939/2048] × [1023/2048,1/2]	727412850291380	450175164045343322
new_top	C	[3753/8192,7509/16384] × [2047/4096,4095/8192]	[3753/8192,7509/16384] × [2047/4096,1/2]	363561163416291	451728974034565903
new_top	C	[3753/8192,15015/32768] × [4095/8192,1/2]	[3753/8192,15015/32768] × [4095/8192,1/2]	31096867043	452505359574801551
new_top	C	[15015/32768,7509/16384] × [4095/8192,8191/16384]	[15015/32768,7509/16384] × [4095/8192,1/2]	181828177208905	451728974034565903
new_top	C	[15015/32768,30033/65536] × [8191/16384,1/2]	[15015/32768,30033/65536] × [8191/16384,1/2]	2862435262	452117210092548364
new_top	C	[30033/65536,7509/16384] × [8191/16384,1/2]	[30033/65536,7509/16384] × [8191/16384,1/2]	12284406695	451728974034565903
new_top	C	[7509/16384,939/2048] × [2047/4096,1/2]	[7509/16384,939/2048] × [2047/4096,1/2]	68784752772	450175164045343322
new_top	C	[939/2048,471/1024] × [255/512,1/2]	[939/2048,471/1024] × [255/512,1/2]	1047914201348	437694816511500959
new_top	C	[471/1024,237/512] × [127/256,1/2]	[471/1024,237/512] × [127/256,1/2]	52261126291637	412468160803487098
new_top	C	[237/512,477/1024] × [63/128,127/256]	[237/512,477/1024] × [63/128,1/2]	12430742252495818	386886890908367724
new_top	C	[237/512,951/2048] × [127/256,1/2]	[237/512,951/2048] × [127/256,1/2]	386048491814514	399721852629315600
new_top	C	[951/2048,477/1024] × [127/256,1/2]	[951/2048,477/1024] × [127/256,1/2]	668622645247102	386886890908367724
new_top	C	[477/1024,15/32] × [63/128,1/2]	[477/1024,15/32] × [63/128,1/2]	1028317112460457	360951006826142837
new_top	old	[15/32,9/16] × [3/8,7/16]	[15/32,9/16] × [3/8,7/16]	231878832076422536	109831363703226568790
new_top	old	[15/32,33/64] × [7/16,15/32]	[15/32,33/64] × [7/16,15/32]	98171035717302578	110538724640549347943
new_top	P	[15/32,63/128] × [15/32,31/64]	[15/32,63/128] × [15/32,1/2]	53791280483626808	142153051589312834
new_top	C	[15/32,123/256] × [31/64,63/128]	[15/32,123/256] × [31/64,1/2]	27068695905889741	253661328626188160
new_top	C	[15/32,123/256] × [63/128,1/2]	[15/32,123/256] × [63/128,1/2]	1979066988229465	253661328626188160
new_top	P	[123/256,63/128] × [31/64,63/128]	[123/256,63/128] × [31/64,1/2]	36137433715995733	142153051589312834

Edge	Count branch	Closed rectangle	Admission rectangle	Return m	Admission m
new_top	P	[123/256,63/128] × [63/128,1/2]	[123/256,63/128] × [63/128,1/2]	9145386765990499	142153051589312834
new_top	old	[63/128,33/64] × [15/32,1/2]	[63/128,33/64] × [15/32,1/2]	6171258989081769	105395385905799539830
new_top	old	[33/64,9/16] × [7/16,1/2]	[33/64,9/16] × [7/16,1/2]	52985064044390462	99406433538595033683
new_top	old	[9/16,3/4] × [1/4,1/2]	[9/16,3/4] × [1/4,1/2]	213318624641387934	75450624069777009095
new_kink	old	[0,3/8] × [0,1/2]	[0,3/8] × [0,1/2]	155168926358182400	123362243007413058271
new_kink	old	[3/8,3/4] × [0,1/4]	[3/8,3/4] × [0,1/4]	1083163208102564143	121056121503706529135
new_kink	old	[3/8,9/16] × [1/4,3/8]	[3/8,9/16] × [1/4,3/8]	371024587410291739	120647155480156414093
new_kink	old	[3/8,15/32] × [3/8,7/16]	[3/8,15/32] × [3/8,7/16]	192709176497842149	121711907953008468828
new_kink	C cap	[3/8,27/64] × [7/16,1/2]	[3/8,27/64] × [7/16,1/2]	79574143429056378	604579476895132909
new_kink	old	[27/64,15/32] × [7/16,15/32]	[27/64,15/32] × [7/16,15/32]	113695502521744459	116507408361725183424
new_kink	P cap	[27/64,57/128] × [15/32,31/64]	[27/64,57/128] × [15/32,1/2]	112841766031593060	146530341871907977
new_kink	C cap	[27/64,111/256] × [31/64,1/2]	[27/64,111/256] × [31/64,1/2]	65417259122289469	194986236488414512
new_kink	P cap	[111/256,57/128] × [31/64,63/128]	[111/256,57/128] × [31/64,1/2]	82531678264300717	146530341871907977
new_kink	C−7	[111/256,57/128] × [63/128,1/2]	[111/256,57/128] × [63/128,1/2]	37342071473821079	13221767544965760
new_kink	old	[57/128,15/32] × [15/32,31/64]	[57/128,15/32] × [15/32,1/2]	68600006141248873	111384338273004045977
new_kink	C	[57/128,117/256] × [31/64,63/128]	[57/128,117/256] × [31/64,1/2]	60149683895101241	24808346629254572
new_kink	C	[57/128,231/512] × [63/128,127/256]	[57/128,231/512] × [63/128,1/2]	49097639610054773	92612169784935170
new_kink	C	[57/128,459/1024] × [127/256,255/512]	[57/128,459/1024] × [127/256,1/2]	44390609860488690	126369328644933867
new_kink	C	[57/128,459/1024] × [255/512,1/2]	[57/128,459/1024] × [255/512,1/2]	38716077706138363	143210327716711532
new_kink	C	[459/1024,231/512] × [127/256,1/2]	[459/1024,231/512] × [127/256,1/2]	37520690607662139	204182800597629396
new_kink	C	[231/512,117/256] × [63/128,127/256]	[231/512,117/256] × [63/128,1/2]	47980630453835351	251135132414239857
new_kink	C	[231/512,465/1024] × [127/256,1/2]	[231/512,465/1024] × [127/256,1/2]	36634603324676943	281996272550324926
new_kink	C	[465/1024,117/256] × [127/256,1/2]	[465/1024,117/256] × [127/256,1/2]	36057815857182774	359809744503020456
new_kink	C	[117/256,15/32] × [31/64,63/128]	[117/256,15/32] × [31/64,1/2]	60014355680610843	353578373688634072
new_kink	C	[117/256,237/512] × [63/128,1/2]	[117/256,237/512] × [63/128,1/2]	35635678297434120	409658095043544544
new_kink	C	[237/512,477/1024] × [63/128,127/256]	[237/512,477/1024] × [63/128,1/2]	48111956399456319	386886890908367724
new_kink	C	[237/512,951/2048] × [127/256,1/2]	[237/512,951/2048] × [127/256,1/2]	35832140368667521	399721852629315600
new_kink	C	[951/2048,477/1024] × [127/256,1/2]	[951/2048,477/1024] × [127/256,1/2]	35969033881220601	386886890908367724
new_kink	C	[477/1024,15/32] × [63/128,1/2]	[477/1024,15/32] × [63/128,1/2]	36183252347646437	360951006826142837
new_kink	old	[15/32,9/16] × [3/8,7/16]	[15/32,9/16] × [3/8,7/16]	274171498608030390	109831363703226568790
new_kink	old	[15/32,33/64] × [7/16,15/32]	[15/32,33/64] × [7/16,15/32]	138744928645186642	110538724640549347943
new_kink	P	[15/32,63/128] × [15/32,31/64]	[15/32,63/128] × [15/32,1/2]	90083630456297708	142153051589312834
new_kink	C	[15/32,123/256] × [31/64,63/128]	[15/32,123/256] × [31/64,1/2]	62398119245366419	253661328626188160
new_kink	C	[15/32,123/256] × [63/128,1/2]	[15/32,123/256] × [63/128,1/2]	36843664142116380	253661328626188160
new_kink	P	[123/256,63/128] × [31/64,63/128]	[123/256,63/128] × [31/64,1/2]	70788807980449394	142153051589312834
new_kink	P	[123/256,63/128] × [63/128,1/2]	[123/256,63/128] × [63/128,1/2]	43348520253269043	142153051589312834
new_kink	old	[63/128,33/64] × [15/32,1/2]	[63/128,33/64] × [15/32,1/2]	42511931651830921	105395385905799539830
new_kink	old	[33/64,9/16] × [7/16,1/2]	[33/64,9/16] × [7/16,1/2]	86853290632277365	99406433538595033683

Edge	Count branch	Closed rectangle	Admission rectangle	Return m	Admission m
new_kink	old	[9/16,3/4] × [1/4,1/2]	[9/16,3/4] × [1/4,1/2]	242408766514886994	75450624069777009095

Domain and complementary reserves

Reserve	Exact value
Gram_x_above_ell	1998990383/10000000000
Gram_y_above_x	487388999/5000000000
Gram_ell_below_one_fifth	243189691/7500000000
Gram_2x_above_y_plus_ell	204842477/2000000000
conductor_small_before_peak	13798546379/30000000000
conductor_peak_before_switch	7149579958346918211112212295690730000000000
conductor_switch_before_kink	3565193491209114783878975/32175984905892499387267013 4
conductor_kink_before_top	97564709635414321114331090812081790768500000000
conductor_top_before_twoy	121689497/937500000
physical_plain_lower	1256797809/125000000000
physical_plain_upper	2234201241610573/120015144270000000
physical_width_lower	8003482192813426706240748295 9/19640958143713630685770410000000
physical_width_upper	29999/10000000
small_width_shortening	507914723825573/120015144270000000
short_plain_shortening	2207929527179/1250000000000000
large_shift	196059842393/3750000000000000
large_shift_clipped	28249845426025867/81826998847441500
small_shift_cost	28249845426025867/81826998847441500
prime_supply	566588663/60007572135
capacity_domain	121689497/1875000000
incumbent	29103/5000000000
feedback_kappa	247729897/5000000000
feedback_d	999495191/5000000000
feedback_y	1051453621/30000000000
floor	4258073897/1000000000000
small_row	56237409433/1500000000000
principal_w	635924290099/30000000000000
principal_z	501009617/6000000000000
holomorphic_box	247729897/10000000000
source_lambda	13653997694883/50000000000000
source_intercept	1630484454379/50000000000000
source_C_cap	109989/10000000
source_P_cap	119989/10000000

Attribution and source navigation

Hailey Collet is the sole human author. Her manuscript and code are released under Apache-2.0; inherited licenses and source notices are retained. The probe, theta reflection and initial source architecture originate in OpenAI's manuscripts of 30 September and 5 October 2026. Hailey's reconstruction supplies the standalone same-zero, signed-Gram and continuation exposition. Akash Levy supplied the weighted numerator refinement; DaysSky the affine detector, with related count arithmetic prepared for liqsweep (source repository blueberryvertigo/QuasiRiemannTracker); Zhongpai Gao the affine-width and balanced-source refinements; timbalud the marked asymmetric counts and common-radical allocations. ArviFoxy supplied the two-level affine bootstrap, built on the credited source tree and earlier marked ledger. Its upstream notices are retained. AI assistance is not additional human authorship or independent verification.

The bootstrap is pinned to the two-level affine submitted revision. In particular, `Cycle63Verification.allHecke` supplies the full Hecke bootstrap. This earlier theorem provides the initial input to the dependency chain.

QRH-Catalogue welcomes source-backed contributions and corrections. Its exact Bernstein-basis entry is LEAN-C61-original-fb6d4fd1-Cycle61.Arithmetic.Bernstein.Basis. The entry identifies a reusable source component; enclosing build, axiom and kernel evidence have their separate scopes.